

Chapter 41. Instruction Appendix

41.1. add

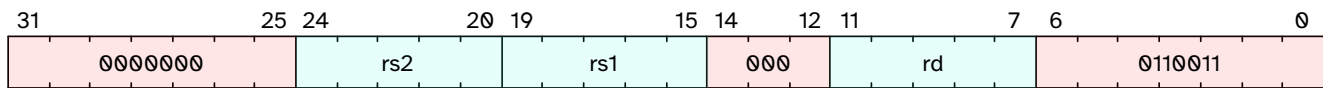
Synopsis

Integer add

Mnemonic

```
rd, rs1, rs2
```

Encoding



Description

Add the value in rs1 to rs2, and store the result in rd. Any overflow is thrown away.

Included in

Extension	Version
I	~> 2.1.0

41.2. add.uw

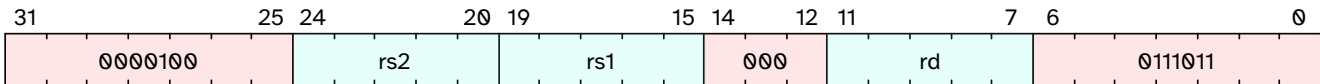
Synopsis

Add unsigned word

Mnemonic

```
rd, rs1, rs2
```

Encoding



Description

This instruction performs an XLEN-wide addition between rs2 and the zero-extended least-significant word of rs1.

Included in

Extension	Version
B	~> 1.0.0
Zba	~> 1.0.0

41.3. addi

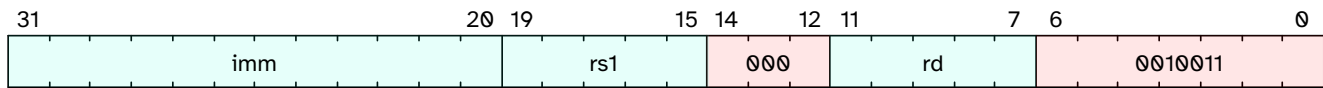
Synopsis

Add immediate

Mnemonic

```
rd, rs1, imm
```

Encoding



Description

Add an immediate to the value in rs1, and store the result in rd

Included in

Extension	Version
I	~> 2.1.0

DRAFT

41.4. addiw

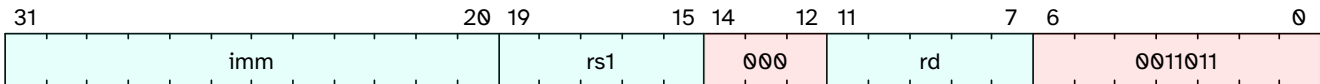
Synopsis

Add immediate word

Mnemonic

```
rd, rs1, imm
```

Encoding



Description

Add an immediate to the 32-bit value in rs1, and store the sign extended result in rd

Included in

Extension	Version
I	~> 2.1.0

41.5. addw

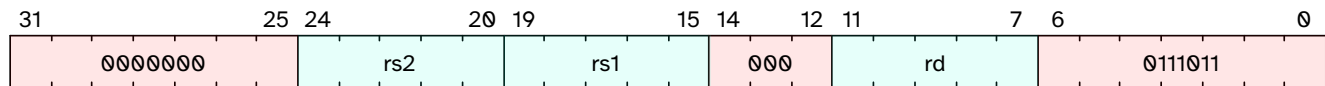
Synopsis

Add word

Mnemonic

```
rd, rs1, rs2
```

Encoding



Description

Add the 32-bit values in rs1 to rs2, and store the sign-extended result in rd. Any overflow is thrown away.

Included in

Extension	Version
I	~> 2.1.0

41.6. aes32dsi

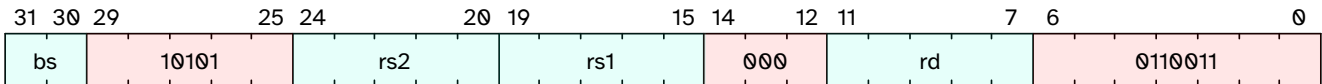
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, bs

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknd	~> 1.0.0

41.7. aes32dsmi

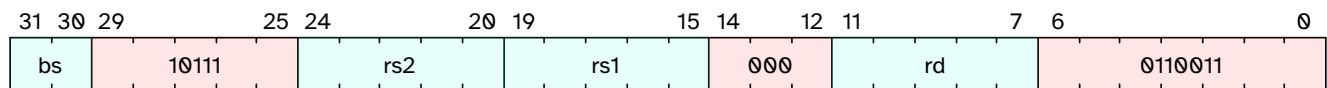
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, bs

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknd	~> 1.0.0

41.8. aes32esi

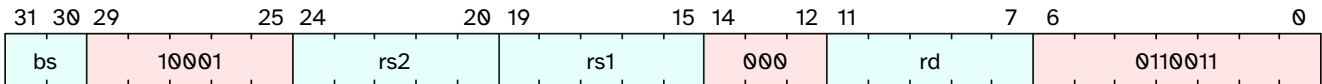
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, bs

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zkne	~> 1.0.0

41.9. aes32esmi

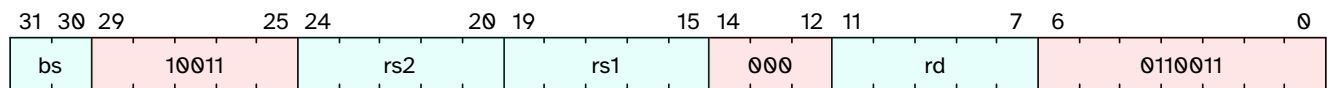
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, bs

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zkne	~> 1.0.0

41.10. aes64ds

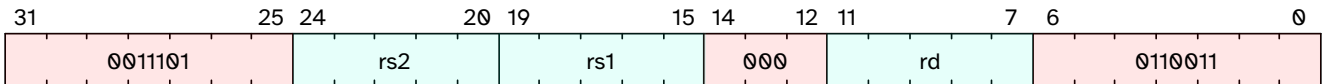
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknd	~> 1.0.0

41.11. aes64dsm

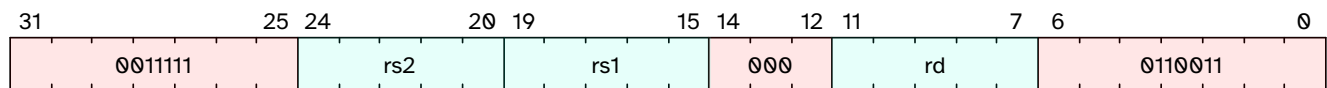
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknd	~> 1.0.0

41.12. aes64es

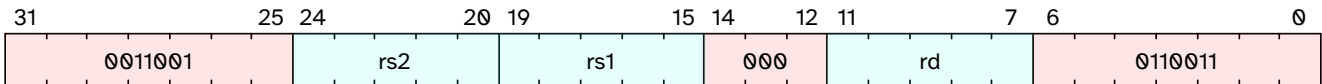
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zkne	~> 1.0.0

41.13. aes64esm

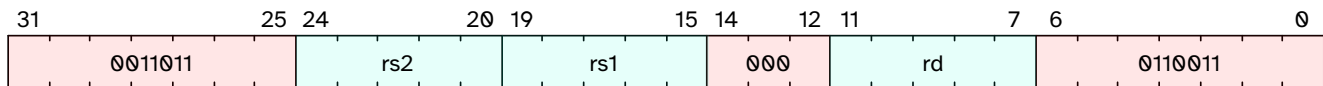
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zkne	~> 1.0.0

41.14. aes64im

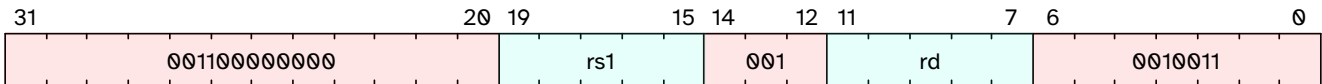
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknd	~> 1.0.0

41.15. aes64ks1i

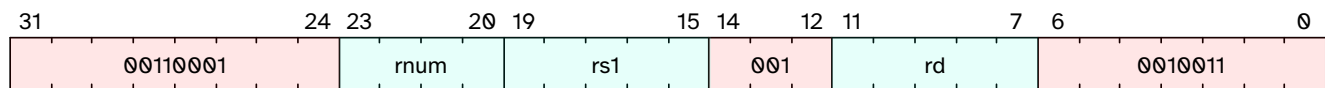
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rnum

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknd	~> 1.0.0
Zkne	~> 1.0.0

41.16. aes64ks2

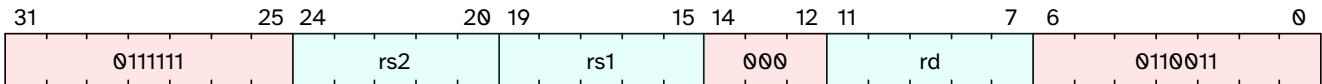
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknd	~> 1.0.0
Zkne	~> 1.0.0

41.17. amoadd.b

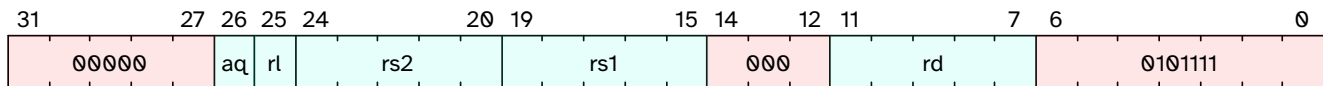
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.18. amoadd.d

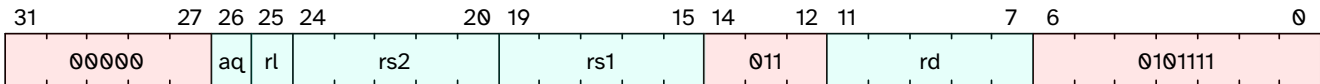
Synopsis

Atomic fetch-and-add doubleword

Mnemonic

```
rd, rs2, (rs1)
```

Encoding



Description

Atomically:

- Load the doubleword at address *rs1*
- Write the loaded value into *rd*
- Add the value of register *rs2* to the loaded value
- Write the sum to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.19. amoadd.h

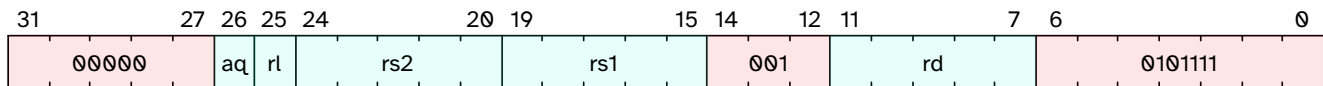
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

DRAFT

41.20. amoadd.w

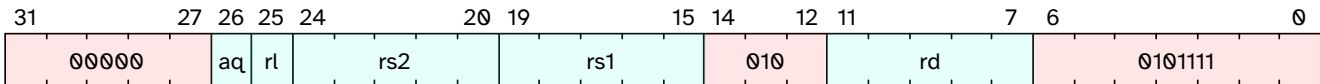
Synopsis

Atomic fetch-and-add word

Mnemonic

```
rd, rs2, (rrs1)
```

Encoding



Description

Atomically:

- Load the word at address *rs1*
- Write the sign-extended value into *rd*
- Add the least-significant word of register *rs2* to the loaded value
- Write the sum to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.21. amoand.b

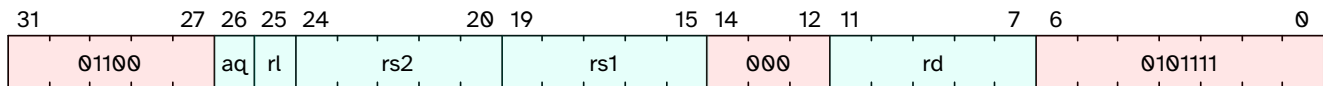
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

DRAFT

41.22. amoand.d

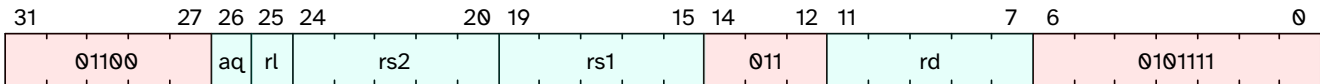
Synopsis

Atomic fetch-and-and doubleword

Mnemonic

rd, rs2, (rrs1)

Encoding



Description

Atomically:

- Load the doubleword at address *rs1*
- Write the loaded value into *rd*
- AND the value of register *rs2* to the loaded value
- Write the result to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.23. amoand.h

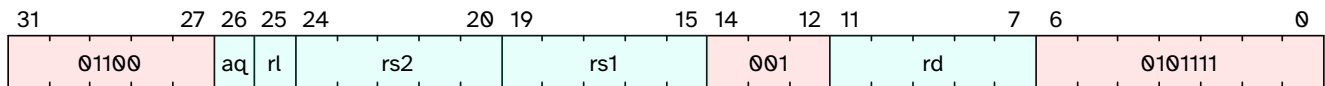
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

DRAFT

41.24. amoand.w

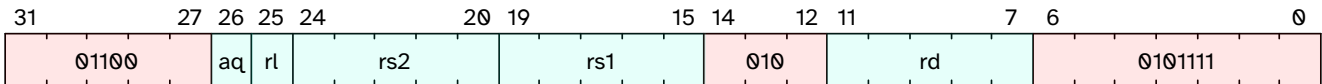
Synopsis

Atomic fetch-and-and word

Mnemonic

rd, rs2, (rrs1)

Encoding



Description

Atomically:

- Load the word at address *rs1*
- Write the sign-extended value into *rd*
- AND the least-significant word of register *rs2* to the loaded value
- Write the result to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.25. amocas.b

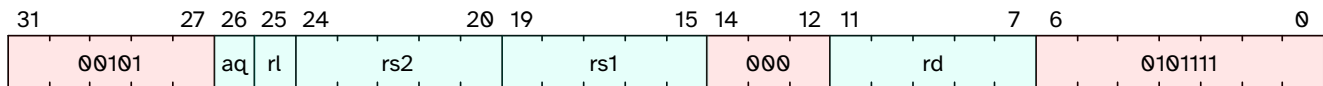
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

DRAFT

41.26. amocas.d

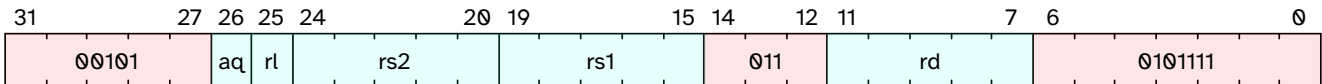
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zacas	~> 1.0.0

41.27. amocas.h

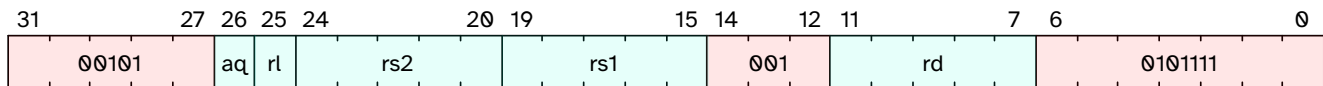
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

DRAFT

41.28. amocas.q

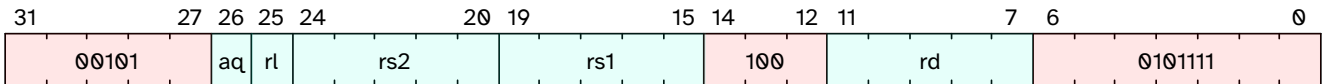
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zacas	~> 1.0.0

41.29. amocas.w

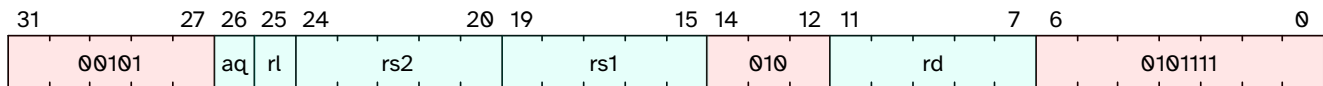
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zacas	~> 1.0.0

DRAFT

41.30. amomax.b

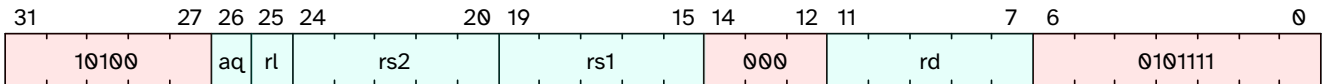
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.31. amomax.d

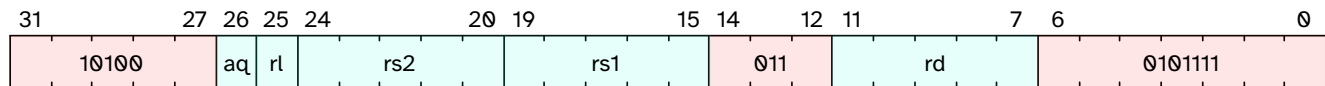
Synopsis

Atomic MAX doubleword

Mnemonic

```
rd, rs2, (rrs1)
```

Encoding



Description

Atomically:

- Load the doubleword at address *rs1*
- Write the loaded value into *rd*
- Signed compare the value of register *rs2* to the loaded value, and select the maximum value
- Write the maximum to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.32. amomax.h

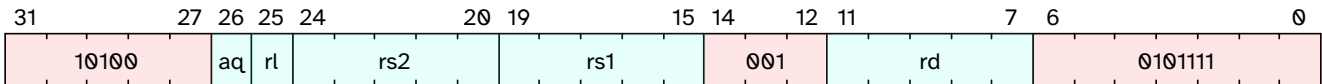
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.33. amomax.w

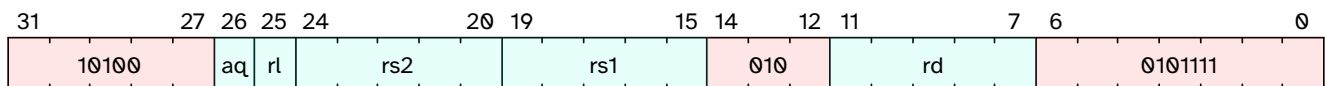
Synopsis

Atomic MAX word

Mnemonic

rd, rs2, (rrs1)

Encoding



Description

Atomically:

- Load the word at address *rs1*
- Write the sign-extended value into *rd*
- Signed compare the least-significant word of register *rs2* to the loaded value, and select the maximum value
- Write the maximum to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.34. amomaxu.b

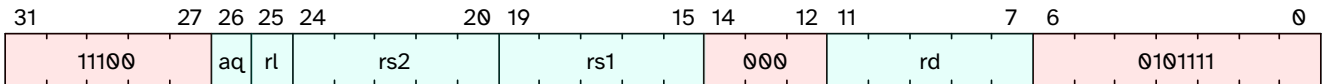
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.35. amomaxu.d

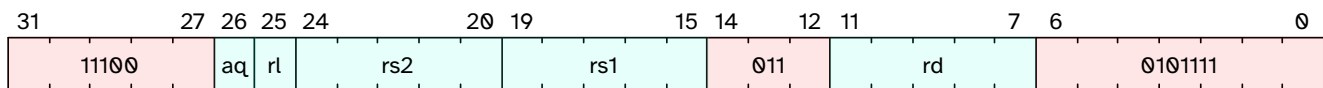
Synopsis

Atomic MAX unsigned doubleword

Mnemonic

rd, rs2, (rrs1)

Encoding



Description

Atomically:

- Load the doubleword at address *rs1*
- Write the loaded value into *rd*
- Unsigned compare the value of register *rs2* to the loaded value, and select the maximum value
- Write the maximum to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.36. amomaxu.h

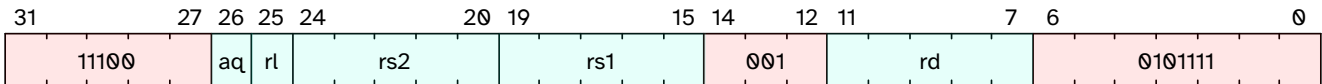
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.37. amomaxu.w

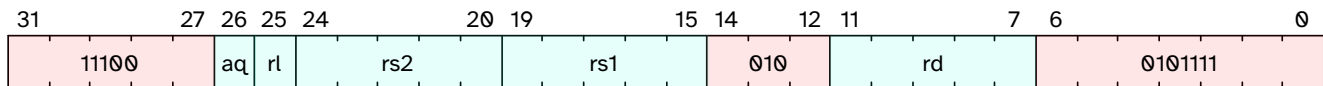
Synopsis

Atomic MAX unsigned word

Mnemonic

rd, rs2, (rrs1)

Encoding



Description

Atomically:

- Load the word at address *rs1*
- Write the sign-extended value into *rd*
- Unsigned compare the least-significant word of register *rs2* to the loaded value, and select the maximum value
- Write the maximum to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.38. amomin.b

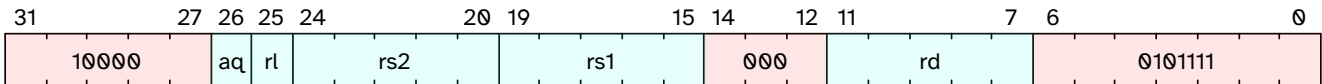
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.39. amomin.d

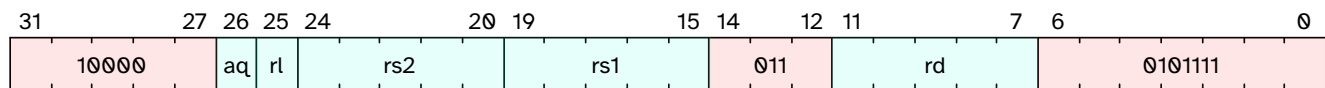
Synopsis

Atomic MIN doubleword

Mnemonic

```
rd, rs2, (rrs1)
```

Encoding



Description

Atomically:

- Load the doubleword at address *rs1*
- Write the loaded value into *rd*
- Signed compare the value of register *rs2* to the loaded value, and select the minimum value
- Write the minimum to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.40. amomin.h

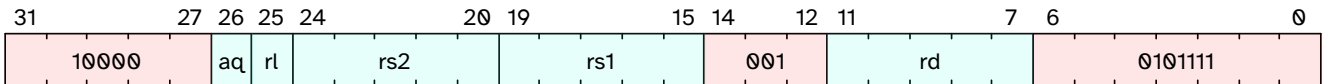
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.41. amomin.w

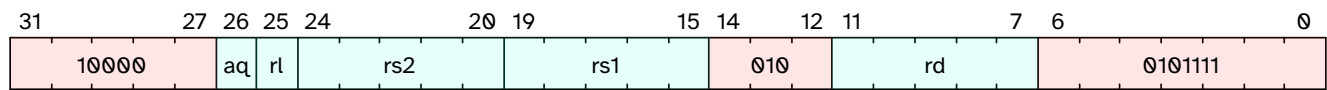
Synopsis

Atomic MIN word

Mnemonic

rd, rs2, (rrs1)

Encoding



Description

Atomically:

- Load the word at address *rs1*
- Write the sign-extended value into *rd*
- Signed compare the least-significant word of register *rs2* to the loaded value, and select the minimum value
- Write the result to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.42. amominu.b

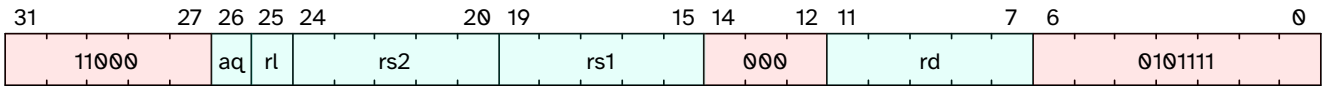
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.43. amominu.d

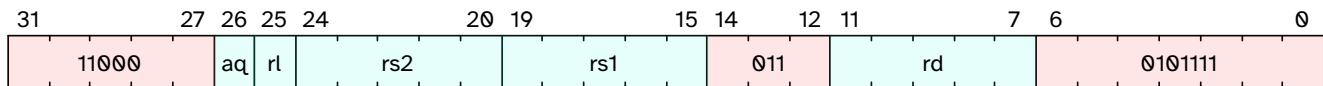
Synopsis

Atomic MIN unsigned doubleword

Mnemonic

rd, rs2, (rrs1)

Encoding



Description

Atomically:

- Load the doubleword at address *rs1*
- Write the loaded value into *rd*
- Unsigned compare the value of register *rs2* to the loaded value, and select the minimum value
- Write the minimum to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.44. amominu.h

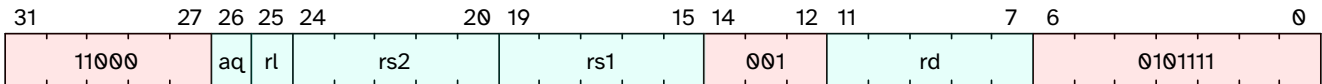
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.45. amominu.w

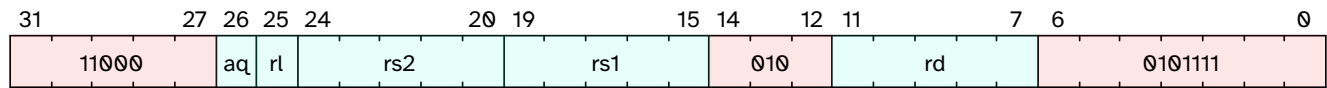
Synopsis

Atomic MIN unsigned word

Mnemonic

rd, rs2, (rrs1)

Encoding



Description

Atomically:

- Load the word at address *rs1*
- Write the sign-extended value into *rd*
- Unsigned compare the least-significant word of register *rs2* to the loaded word, and select the minimum value
- Write the result to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.46. amoor.b

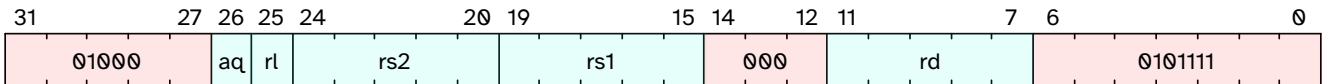
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.47. amoor.d

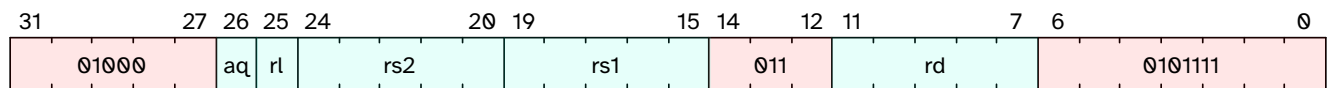
Synopsis

Atomic fetch-and-or doubleword

Mnemonic

rd, rs2, (rrs1)

Encoding



Description

Atomically:

- Load the doubleword at address *rs1*
- Write the loaded value into *rd*
- OR the value of register *rs2* to the loaded value
- Write the result to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.48. amoor.h

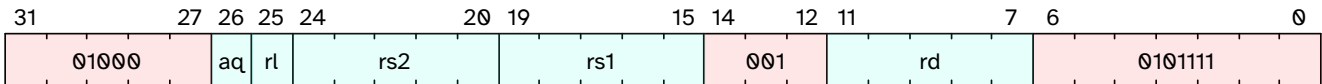
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.49. amoor.w

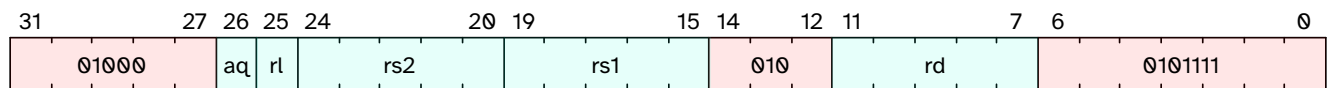
Synopsis

Atomic fetch-and-or word

Mnemonic

```
rd, rs2, (rrs1)
```

Encoding



Description

Atomically:

- Load the word at address *rs1*
- Write the sign-extended value into *rd*
- OR the least-significant word of register *rs2* to the loaded value
- Write the result to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.50. amoswap.b

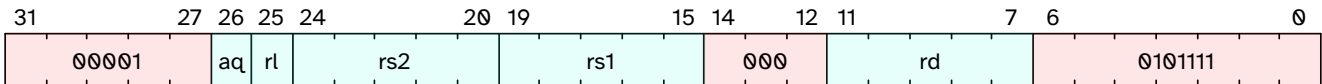
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.51. amoswap.d

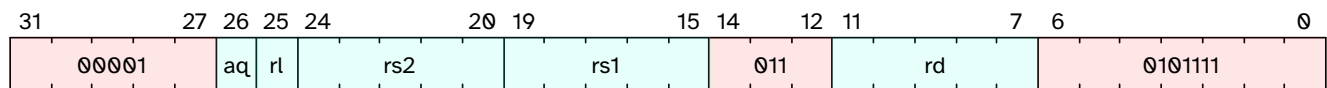
Synopsis

Atomic SWAP doubleword

Mnemonic

rd, rs2, (rrs1)

Encoding



Description

Atomically:

- Load the doubleword at address *rs1*
- Write the value into *rd*
- Store the value of register *rs2* to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.52. amoswap.h

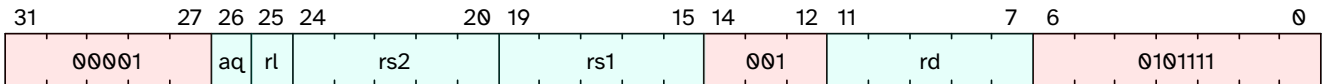
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.53. amoswap.w

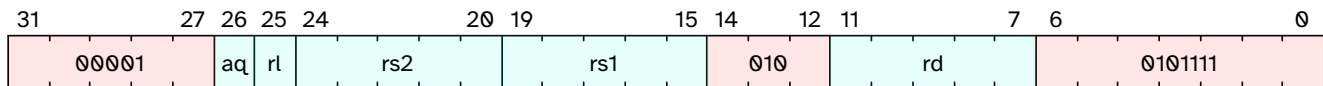
Synopsis

Atomic SWAP word

Mnemonic

rd, rs2, (rrs1)

Encoding



Description

Atomically:

- Load the word at address *rs1*
- Write the sign-extended value into *rd*
- Store the least-significant word of register *rs2* to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.54. amoxor.b

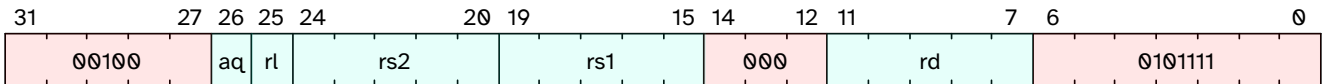
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.55. amoxor.d

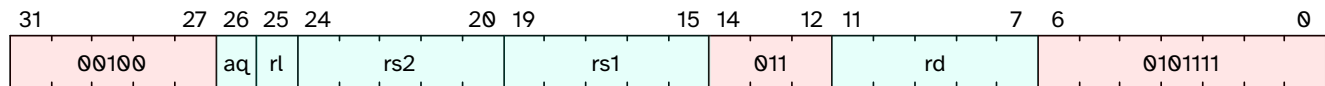
Synopsis

Atomic fetch-and-xor doubleword

Mnemonic

```
rd, rs2, (rrs1)
```

Encoding



Description

Atomically:

- Load the doubleword at address *rs1*
- Write the loaded value into *rd*
- XOR the value of register *rs2* to the loaded value
- Write the result to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.56. amoxor.h

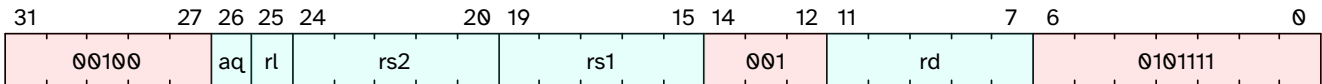
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zabha	~> 1.0.0

41.57. amoxor.w

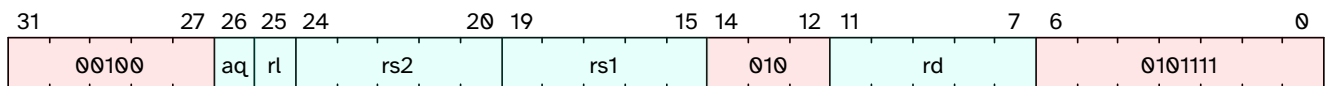
Synopsis

Atomic fetch-and-xor word

Mnemonic

`rd, rs2, (rrs1)`

Encoding



Description

Atomically:

- Load the word at address *rs1*
- Write the sign-extended value into *rd*
- XOR the least-significant word of register *rs2* to the loaded value
- Write the result to the address in *rs1*

Included in

Extension	Version
A	~> 2.1.0
Zaamo	~> 1.0.0

41.58. and

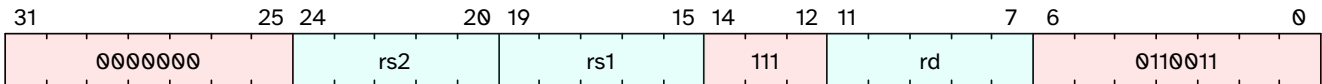
Synopsis

And

Mnemonic

rd, rs1, rs2

Encoding



Description

And rs1 with rs2, and store the result in rd

Included in

Extension	Version
I	~> 2.1.0

41.59. andi

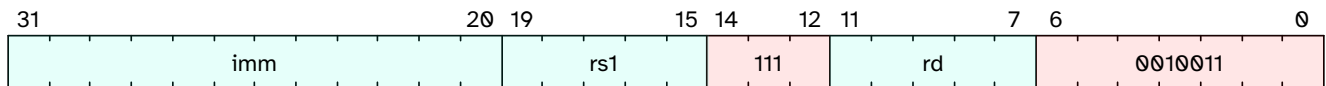
Synopsis

And immediate

Mnemonic

rd, rs1, imm

Encoding



Description

And an immediate to the value in rs1, and store the result in rd

Included in

Extension	Version
I	~> 2.1.0

41.60. andn

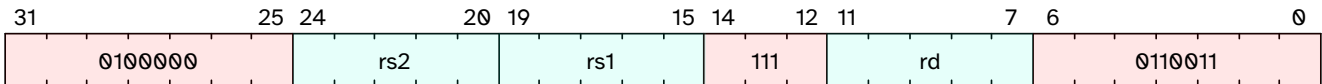
Synopsis

AND with inverted operand

Mnemonic

```
rd, rs1, rs2
```

Encoding



Description

This instruction performs the bitwise logical AND operation between rs1 and the bitwise inversion of rs2.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0
Zbkb	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.61. auipc

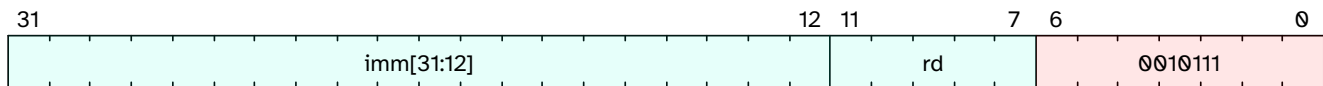
Synopsis

Add upper immediate to pc

Mnemonic

rd, imm

Encoding



Description

Add an immediate to the current PC.

Included in

Extension	Version
I	~> 2.1.0

41.62. bclr

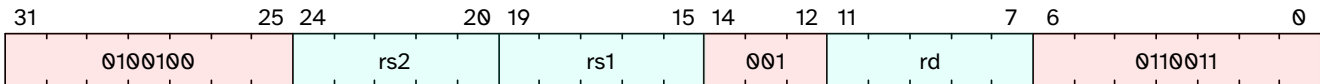
Synopsis

Single-Bit clear (Register)

Mnemonic

```
rd, rs1, rs2
```

Encoding



Description

This instruction returns rs1 with a single bit cleared at the index specified in rs2. The index is read from the lower log2(XLEN) bits of rs2.

Included in

Extension	Version
B	~> 1.0.0
Zbs	~> 1.0.0

41.63. bclri

Synopsis

Single-Bit clear (Immediate)

Mnemonic

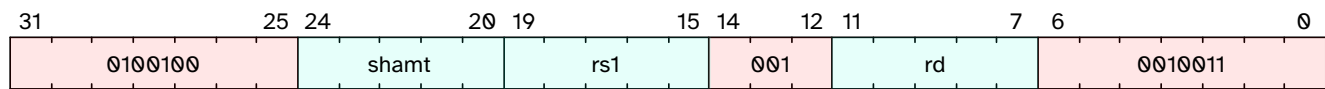
```
rd, rs1, shamt
```

Encoding

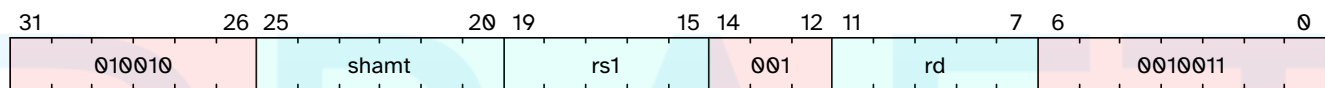


This instruction has different encodings in RV32 and RV64

RV32



RV64



Description

This instruction returns rs1 with a single bit cleared at the index specified in shamt. The index is read from the lower log2(XLEN) bits of shamt. For RV32, the encodings corresponding to shamt[5]=1 are reserved.

Included in

Extension	Version
B	~> 1.0.0
Zbs	~> 1.0.0

41.64. beq

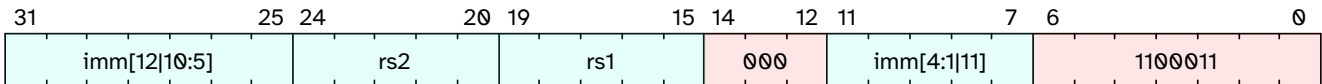
Synopsis

Branch if equal

Mnemonic

rs1, rs2, imm

Encoding



Description

Branch to PC + imm if the value in register rs1 is equal to the value in register rs2.

Raise a MisaignedAddress exception if PC + imm is misaligned.

Included in

Extension	Version
I	~> 2.1.0

41.65. bext

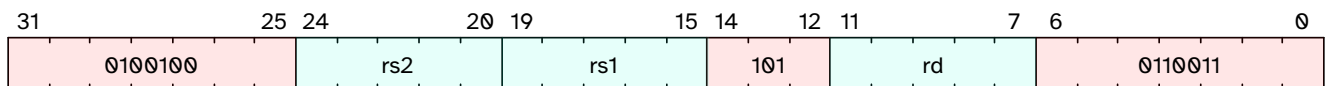
Synopsis

Single-Bit extract (Register)

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction returns a single bit extracted from rs1 at the index specified in rs2. The index is read from the lower $\log_2(\text{XLEN})$ bits of rs2.

Included in

Extension	Version
B	~> 1.0.0
Zbs	~> 1.0.0

41.66. bexti

Synopsis

Single-Bit extract (Immediate)

Mnemonic

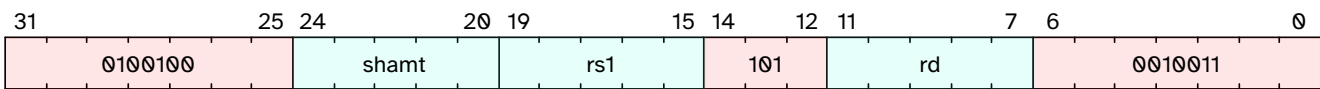
rd, rs1, shamt

Encoding

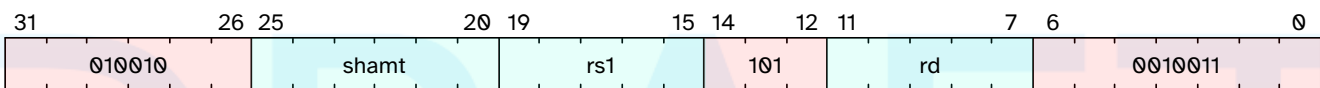


This instruction has different encodings in RV32 and RV64

RV32



RV64



Description

This instruction returns a single bit extracted from rs1 at the index specified in rs2. The index is read from the lower log2(XLEN) bits of shamt. For RV32, the encodings corresponding to shamt[5]=1 are reserved.

Included in

Extension	Version
B	~> 1.0.0
Zbs	~> 1.0.0

41.67. bge

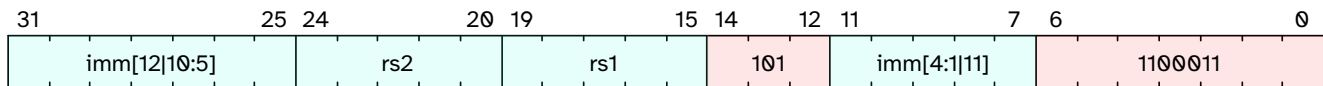
Synopsis

Branch if greater than or equal

Mnemonic

```
rs1, rs2, imm
```

Encoding



Description

Branch to PC + imm if the signed value in register rs1 is greater than or equal to the signed value in register rs2.

Raise a `MisalignedAddress` exception if PC + imm is misaligned.

Included in

Extension	Version
I	~> 2.1.0

41.68. bgeu

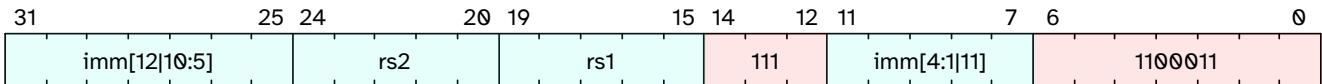
Synopsis

Branch if greater than or equal unsigned

Mnemonic

rs1, rs2, imm

Encoding



Description

Branch to PC + imm if the unsigned value in register rs1 is greater than or equal to the unsigned value in register rs2.

Raise a MisaignedAddress exception if PC + imm is misaligned.

Included in

Extension	Version
I	~> 2.1.0

41.69. binv

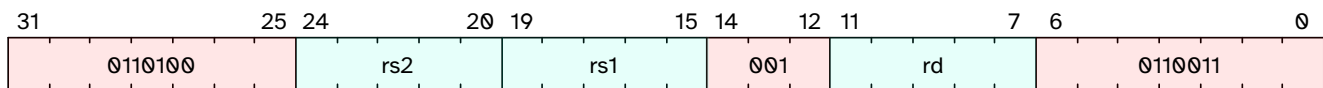
Synopsis

Single-Bit invert (Register)

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction returns rs1 with a single bit inverted at the index specified in rs2. The index is read from the lower log2(XLEN) bits of rs2.

Included in

Extension	Version
B	~> 1.0.0
Zbs	~> 1.0.0

41.70. binvi

Synopsis

Single-Bit invert (Immediate)

Mnemonic

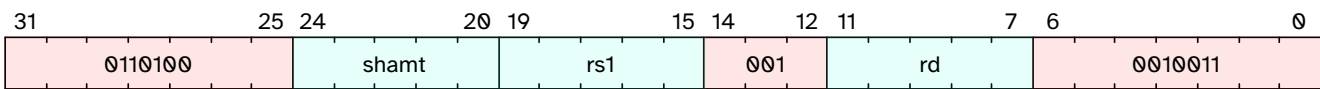
rd, rs1, shamt

Encoding

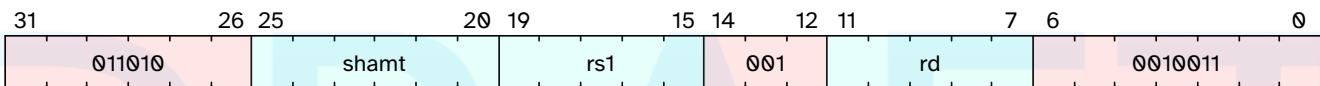


This instruction has different encodings in RV32 and RV64

RV32



RV64



Description

This instruction returns rs1 with a single bit inverted at the index specified in shamt. The index is read from the lower log2(XLEN) bits of shamt. For RV32, the encodings corresponding to shamt[5]=1 are reserved.

Included in

Extension	Version
B	~> 1.0.0
Zbs	~> 1.0.0

41.71. blt

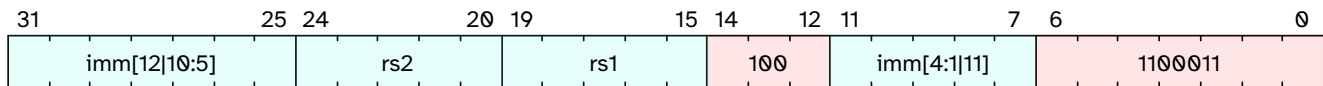
Synopsis

Branch if less than

Mnemonic

```
rs1, rs2, imm
```

Encoding



Description

Branch to PC + imm if the signed value in register rs1 is less than the signed value in register rs2.

Raise a MisalignedAddress exception if PC + imm is misaligned.

Included in

Extension	Version
I	~> 2.1.0

41.72. bltu

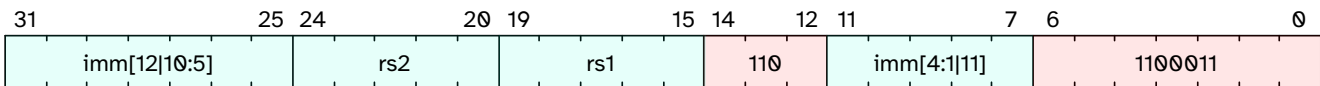
Synopsis

Branch if less than unsigned

Mnemonic

```
rs1, rs2, imm
```

Encoding



Description

Branch to PC + imm if the unsigned value in register rs1 is less than the unsigned value in register rs2.

Raise a `MisalignedAddress` exception if PC + imm is misaligned.

Included in

Extension	Version
I	~> 2.1.0

41.73. bne

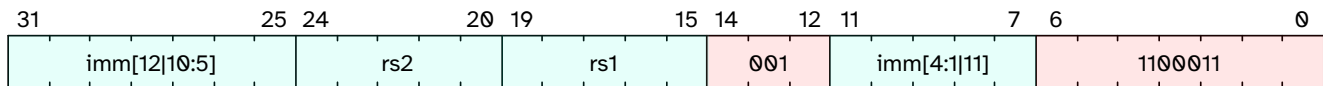
Synopsis

Branch if not equal

Mnemonic

```
rs1, rs2, imm
```

Encoding



Description

Branch to PC + imm if the value in register rs1 is not equal to the value in register rs2.

Raise a MisaignedAddress exception if PC + imm is misaligned.

Included in

Extension	Version
I	~> 2.1.0

41.74. brev8

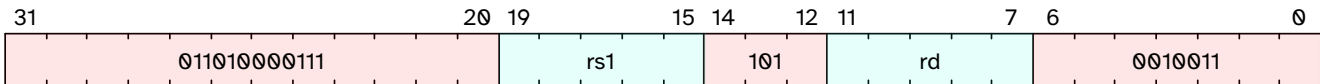
Synopsis

Reverse bits in bytes

Mnemonic

rd, rs1

Encoding



Description

This instruction reverses the order of the bits in every byte of a register.

Included in

Extension	Version
Zbkb	~> 1.0.0

41.75. bset

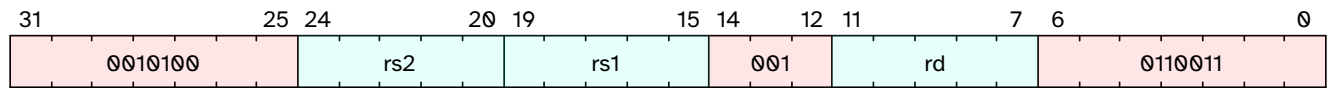
Synopsis

Single-Bit set (Register)

Mnemonic

```
rd, rs1, rs2
```

Encoding



Description

This instruction returns rs1 with a single bit set at the index specified in rs2. The index is read from the lower $\log_2(\text{XLEN})$ bits of rs2.

Included in

Extension	Version
B	~> 1.0.0
Zbs	~> 1.0.0

41.76. bseti

Synopsis

Single-Bit set (Immediate)

Mnemonic

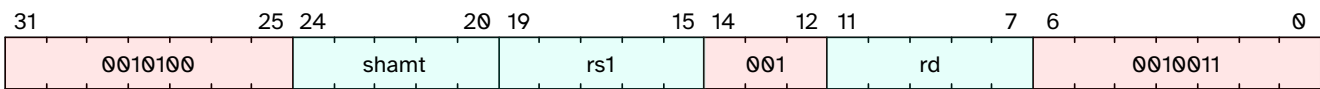
```
rd, rs1, shamt
```

Encoding

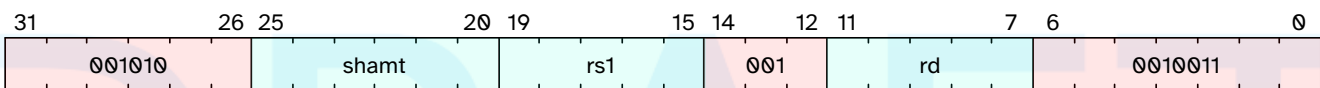


This instruction has different encodings in RV32 and RV64

RV32



RV64



Description

This instruction returns rs1 with a single bit set at the index specified in shamt. The index is read from the lower $\log_2(\text{XLEN})$ bits of shamt. For RV32, the encodings corresponding to shamt[5]=1 are reserved.

Included in

Extension	Version
B	~> 1.0.0
Zbs	~> 1.0.0

41.77. c.add

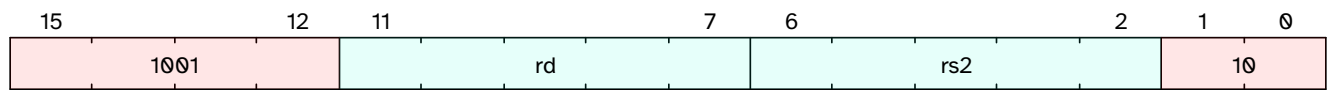
Synopsis

Add

Mnemonic

rd, rs2

Encoding



Description

Add the value in rs2 to rd, and store the result in rd. C.ADD expands into add rd, rd, rs2.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.78. c.addi

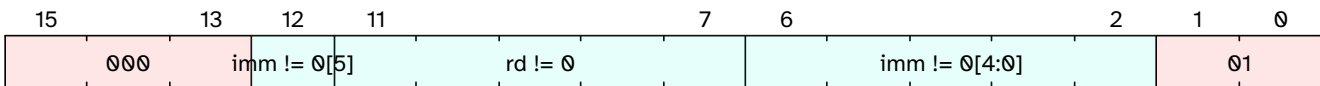
Synopsis

Add a sign-extended non-zero immediate

Mnemonic

```
rd, imm
```

Encoding



Description

C.ADDI adds the non-zero sign-extended 6-bit immediate to the value in register rd then writes the result to rd. C.ADDI expands into addi rd, rd, imm. C.ADDI is only valid when rd ≠ x0 and imm ≠ 0. The code points with rd=x0 encode the C.NOP instruction; the remaining code points with imm=0 encode HINTs.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.79. c.addi16sp

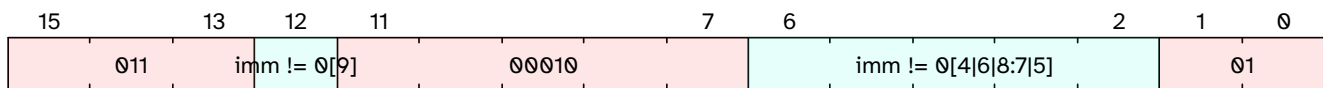
Synopsis

Add a sign-extended non-zero immediate

Mnemonic

imm

Encoding



Description

C.ADDI16SP adds the non-zero sign-extended 6-bit immediate to the value in the stack pointer (sp=x2), where the immediate is scaled to represent multiples of 16 in the range (-512,496). C.ADDI16SP is used to adjust the stack pointer in procedure prologues and epilogues. It expands into addi x2, x2, nzimm[9:4]. C.ADDI16SP is only valid when nzimm ≠ 0; the code point with nzimm=0 is reserved.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.80. c.addi4spn

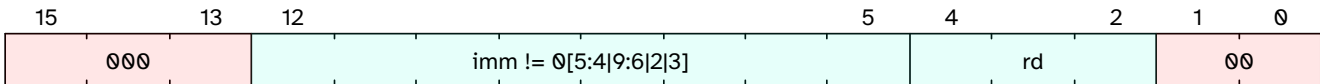
Synopsis

Add a zero-extended non-zero immediate, scaled by 4, to the stack pointer

Mnemonic

```
rd, imm
```

Encoding



Description

Adds a zero-extended non-zero immediate, scaled by 4, to the stack pointer, x2, and writes the result to rd'. This instruction is used to generate pointers to stack-allocated variables. It expands to addi rd', x2, nzuimm[9:2]. C.ADDI4SPN is only valid when nzuimm ≠ 0; the code points with nzuimm=0 are reserved.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.81. c.addiw

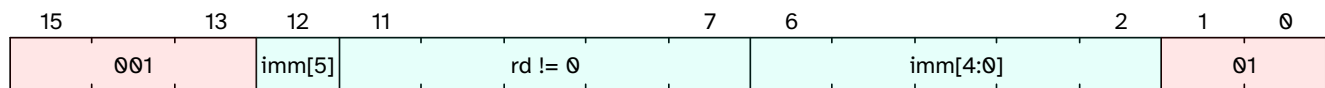
Synopsis

Add a sign-extended non-zero immediate

Mnemonic

```
rd, imm
```

Encoding



Description

C.ADDIW is an RV64C/RV128C-only instruction that performs the same computation as C.ADDI but produces a 32-bit result, then sign-extends result to 64 bits. C.ADDIW expands into `addiw rd, rd, imm`. The immediate can be zero for C.ADDIW, where this corresponds to `sext.w rd`. C.ADDIW is only valid when `rd` $\neq$ `x0`; the code points with `rd=x0` are reserved.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.82. c.addw

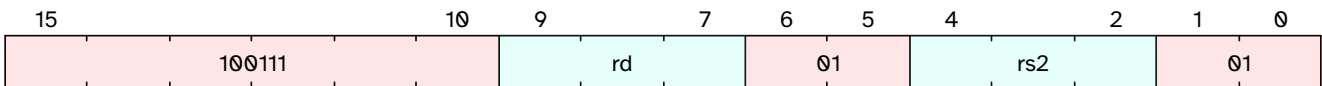
Synopsis

Add word

Mnemonic

rd, rs2

Encoding



Description

Add the 32-bit values in rs2 from rd, and store the result in rd. The rd and rs2 register indexes should be used as rd+8 and rs2+8 (registers x8-x15). C.ADDW expands into addw rd, rd, rs2.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.83. c.and

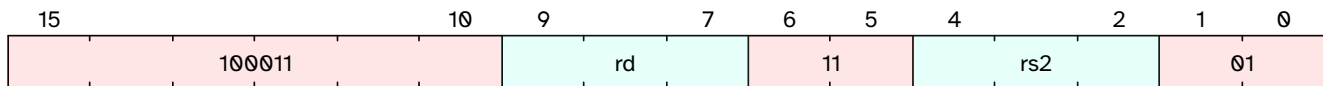
Synopsis

And

Mnemonic

rd, rs2

Encoding



Description

And rd with rs2, and store the result in rd The rd and rs2 register indexes should be used as rd+8 and rs2+8 (registers x8-x15). C.AND expands into and rd, rd, rs2.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.84. c.andi

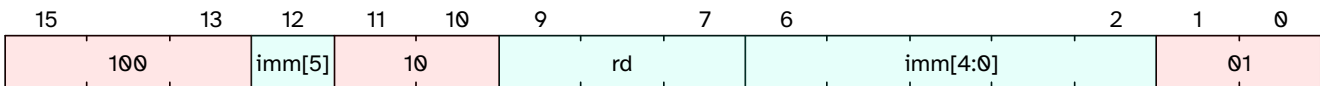
Synopsis

And immediate

Mnemonic

```
rd, imm
```

Encoding



Description

And an immediate to the value in rd, and store the result in rd. The rd register index should be used as rd+8 (registers x8-x15). C.ANDI expands into andi rd, rd, imm.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.85. c.beqz

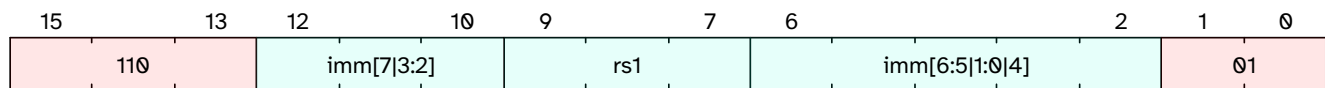
Synopsis

Branch if Equal Zero

Mnemonic

rs1, imm

Encoding



Description

C.BEQZ performs conditional control transfers. The offset is sign-extended and added to the pc to form the branch target address. It can therefore target a ± 256 B range. C.BEQZ takes the branch if the value in register rs1' is zero. It expands to `beq rs1, x0, offset`.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.86. c.bnez

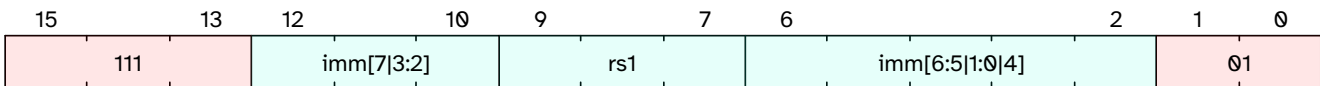
Synopsis

Branch if NOT Equal Zero

Mnemonic

rs1, imm

Encoding



Description

C.BEQZ performs conditional control transfers. The offset is sign-extended and added to the pc to form the branch target address. It can therefore target a ± 256 B range. C.BEQZ takes the branch if the value in register rs1' is NOT zero. It expands to `beq rs1, x0, offset`.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

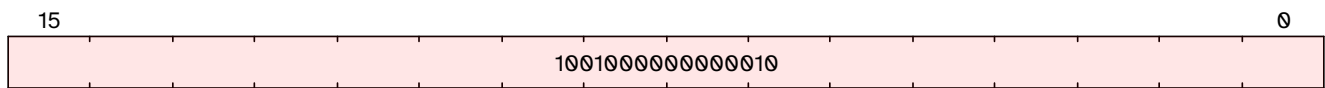
41.87. c.ebreak

Synopsis

Breakpoint exception.

Mnemonic

Encoding



Description

The C.EBREAK instruction is used by debuggers to cause control to be transferred back to a debugging environment. Unless overridden by an external debug environment, C.EBREAK raises a breakpoint exception and performs no other operation.



As described in the [C Standard Extension for Compressed Instructions](#), the [c.ebreak](#) instruction performs the same operation as the EBREAK instruction.

EBREAK causes the receiving privilege mode’s epc register to be set to the address of the EBREAK instruction itself, not the address of the following instruction. As EBREAK causes a synchronous exception, it is not considered to retire, and should not increment the [minstret](#) CSR.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.88. c.fld

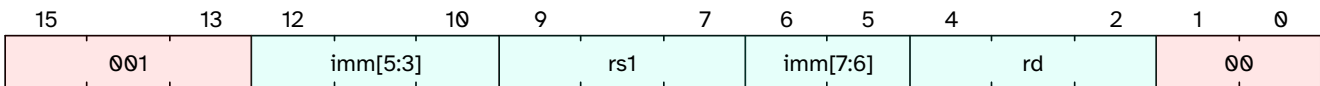
Synopsis

Load double-precision

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Loads a double precision floating-point value from memory into register rd. It computes an effective address by adding the zero-extended offset, scaled by 8, to the base address in register rs1. It expands to `fld rd, offset(rs1)`.

Included in

Extension	Version
C	~> 2.0.0
D	~> 2.2.0

41.89. c.fldsp

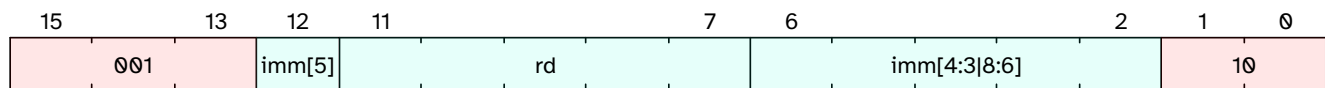
Synopsis

Load doubleword into floating-point register from stack

Mnemonic

```
fd, imm(sp)
```

Encoding



Description

Loads a double-precision floating-point value from memory into floating-point register rd. It computes its effective address by adding the zero-extended offset, scaled by 8, to the stack pointer, x2. It expands to `fld rd, offset(x2)`.

Included in

Extension	Version
C	~> 2.0.0
D	~> 2.2.0

41.90. c.flw

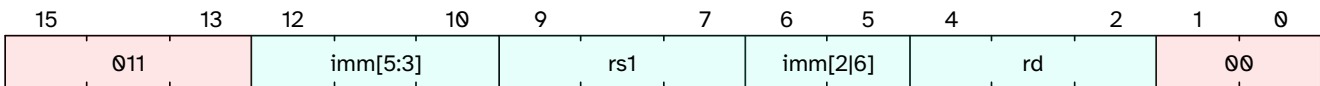
Synopsis

Load single-precision

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Loads a single precision floating-point value from memory into register rd. It computes an effective address by adding the zero-extended offset, scaled by 4, to the base address in register rs1. It expands to `flw rd, offset(rs1)`.

Included in

Extension	Version
C	~> 2.0.0
F	~> 2.2.0

41.91. c.flwsp

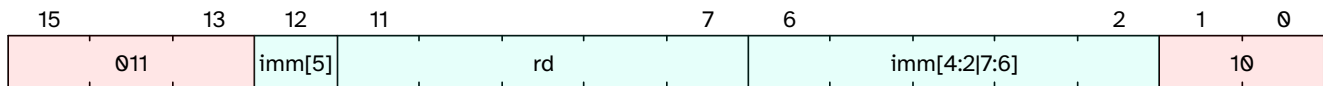
Synopsis

Load word into floating-point register from stack

Mnemonic

```
fd, imm(sp)
```

Encoding



Description

Loads a single-precision floating-point value from memory into floating-point register `rd`. It computes its effective address by adding the zero-extended offset, scaled by 4, to the stack pointer, `x2`. It expands to `flw rd, offset(x2)`.

Included in

Extension	Version
C	~> 2.0.0
F	~> 2.2.0

41.92. c.fsd

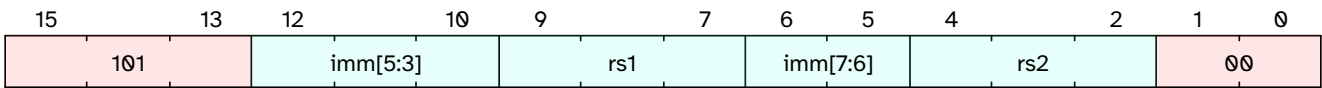
Synopsis

Store double-precision

Mnemonic

```
rs2, imm(rs1)
```

Encoding



Description

Stores a double precision floating-point value in register rs2 to memory. It computes an effective address by adding the zero-extended offset, scaled by 8, to the base address in register rs1. It expands to `fsd rs2, offset(rs1)`.

Included in

Extension	Version
C	~> 2.0.0
D	~> 2.2.0

41.93. c.fsdsp

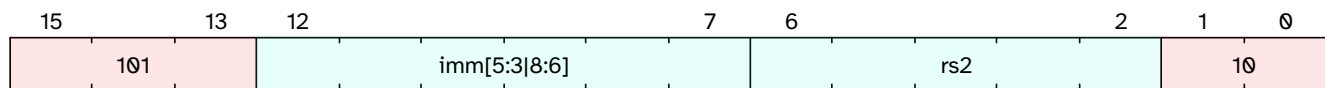
Synopsis

Store double-precision value to stack

Mnemonic

```
fs2, imm(sp)
```

Encoding



Description

Stores a double-precision floating-point value in floating-point register rs2 to memory. It computes an effective address by adding the zero-extended offset, scaled by 8, to the stack pointer, x2. It expands to `fsd rs2, offset(x2)`.

Included in

Extension	Version
C	~> 2.0.0
D	~> 2.2.0

41.94. c.fsw

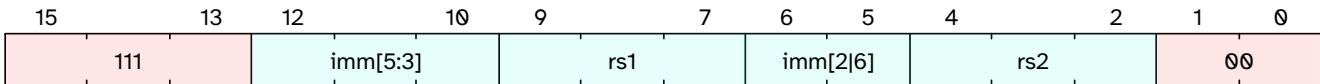
Synopsis

Store single-precision

Mnemonic

```
rs2, imm(rs1)
```

Encoding



Description

Stores a single precision floating-point value in register rs2 to memory. It computes an effective address by adding the zero-extended offset, scaled by 4, to the base address in register rs1. It expands to [fsw](#) rs2, offset(rs1).

Included in

Extension	Version
C	~> 2.0.0
F	~> 2.2.0

41.95. c.fswsp

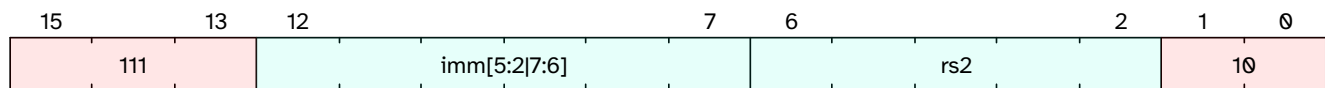
Synopsis

Store single-precision value to stack

Mnemonic

fs2, imm(sp)

Encoding



Description

Stores a single-precision floating-point value in floating-point register rs2 to memory. It computes an effective address by adding the zero-extended offset, scaled by 4, to the stack pointer, x2. It expands to [fsw](#) rs2, offset(x2).

Included in

Extension	Version
C	~> 2.0.0
F	~> 2.2.0

41.96. c.j

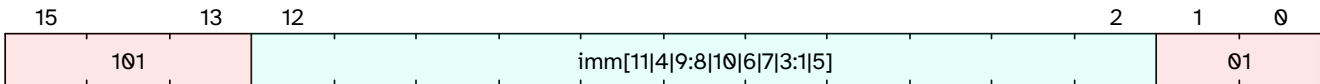
Synopsis

Jump

Mnemonic

imm

Encoding



Description

C.J performs an unconditional control transfer. The offset is sign-extended and added to the pc to form the jump target address. C.J can therefore target a ±2 KiB range. It expands to [jal](#) x0, offset.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.98. c.jalr

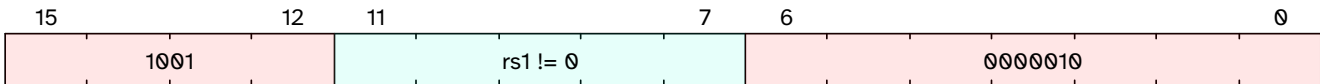
Synopsis

Jump and Link Register.

Mnemonic

rs1

Encoding



Description

C.JALR (jump and link register) performs the same operation as C.JR, but additionally writes the address of the instruction following the jump (pc+2) to the link register, x1. C.JALR expands to jalr x1, 0(rs1).

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.99. c.jr

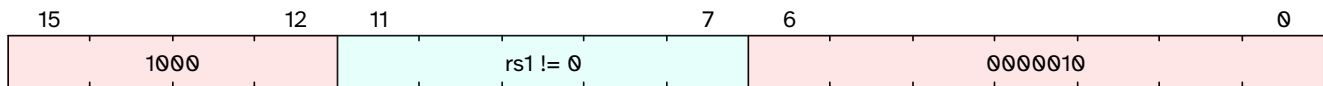
Synopsis

Jump Register

Mnemonic

rs1

Encoding



Description

C.JR (jump register) performs an unconditional control transfer to the address in register rs1. C.JR expands to jalr x0, 0(rs1).

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.100. c.lbu

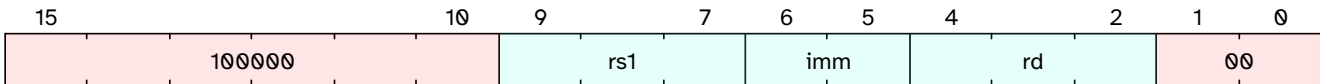
Synopsis

Load unsigned byte, 16-bit encoding

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Loads a 8-bit value from memory into register rd. It computes an effective address by adding the zero-extended offset, to the base address in register rs1. It expands to `lbu rd, offset(rs1)`.

Included in

Extension	Version
Zcb	~> 1.0.0
Zce	~> 1.0.0

41.101. c.ld

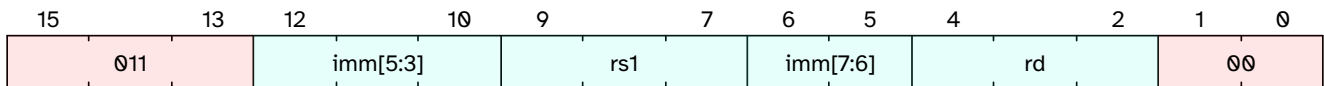
Synopsis

Load double

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Loads a 64-bit value from memory into register rd. It computes an effective address by adding the zero-extended offset, scaled by 8, to the base address in register rs1. It expands to `ld rd, offset(rs1)`.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.102. c.ldsp

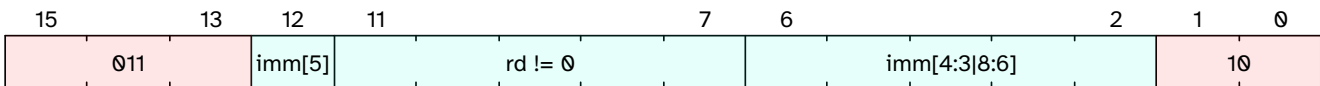
Synopsis

Load doubleword from stack pointer

Mnemonic

```
rd, imm(sp)
```

Encoding



Description

C.LDSP is an RV64C/RV128C-only instruction that loads a 64-bit value from memory into register rd. It computes its effective address by adding the zero-extended offset, scaled by 8, to the stack pointer, x2. It expands to `ld rd, offset(x2)`. C.LDSP is only valid when rd ≠ x0 the code points with rd=x0 are reserved.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.103. c.lh

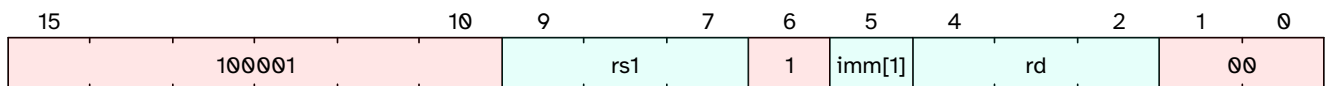
Synopsis

Load signed halfword, 16-bit encoding

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Loads a 16-bit value from memory into register rd. It computes an effective address by adding the zero-extended offset, to the base address in register rs1. It expands to `lh rd, offset(rs1)`.

Included in

Extension	Version
Zcb	~> 1.0.0
Zce	~> 1.0.0

41.104. c.lhu

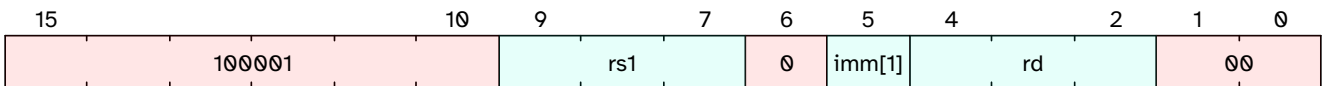
Synopsis

Load unsigned halfword, 16-bit encoding

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Loads a 16-bit value from memory into register rd. It computes an effective address by adding the zero-extended offset, to the base address in register rs1. It expands to `lhu rd, offset(rs1)`.

Included in

Extension	Version
Zcb	~> 1.0.0
Zce	~> 1.0.0

41.105. c.li

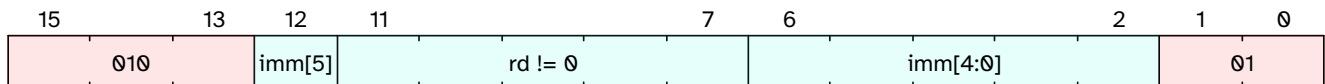
Synopsis

Load the sign-extended 6-bit immediate

Mnemonic

`rd, imm`

Encoding



Description

C.LI loads the sign-extended 6-bit immediate, `imm`, into register `rd`. C.LI expands into `addi rd, x0, imm`. C.LI is only valid when `rd` $\neq$ `x0`; the code points with `rd=x0` encode HINTs.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.106. c.lui

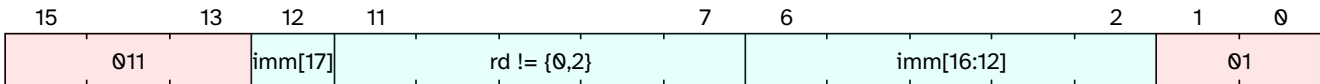
Synopsis

Load the non-zero 6-bit immediate field into bits 17-12 of the destination register

Mnemonic

```
rd, imm
```

Encoding



Description

C.LUI loads the non-zero 6-bit immediate field into bits 17-12 of the destination register, clears the bottom 12 bits, and sign-extends bit 17 into all higher bits of the destination. C.LUI expands into `lui rd, imm`. C.LUI is only valid when `rd != x0` and `rd != x2`, and when the immediate is not equal to zero. The code points with `imm=0` are reserved; the remaining code points with `rd=x0` are HINTs; and the remaining code points with `rd=x2` correspond to the C.ADDI16SP instruction

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.107. c.lw

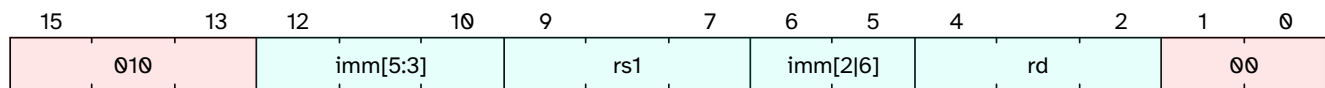
Synopsis

Load word

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Loads a 32-bit value from memory into register rd. It computes an effective address by adding the zero-extended offset, scaled by 4, to the base address in register rs1. It expands to `lw rd, offset(rs1)`.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.108. c.lwsp

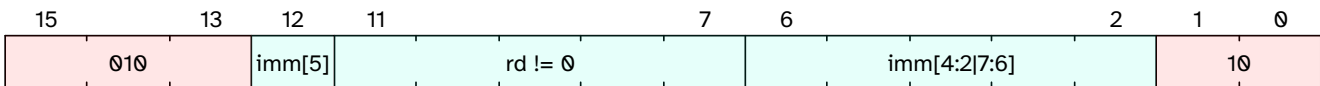
Synopsis

Load word from stack pointer

Mnemonic

```
rd, imm(sp)
```

Encoding



Description

Loads a 32-bit value from memory into register rd. It computes an effective address by adding the zero-extended offset, scaled by 4, to the stack pointer, x2. It expands to `lw rd, offset(x2)`. C.LWSP is only valid when rd ≠ x0. The code points with rd=x0 are reserved.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.109. c.mul

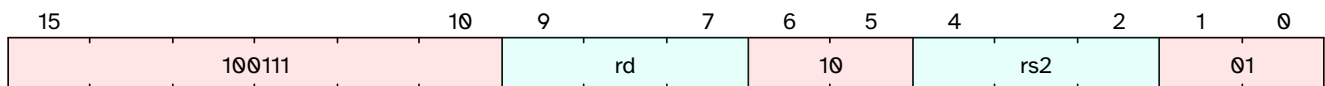
Synopsis

Multiply, 16-bit encoding

Mnemonic

rd, rs2

Encoding



Description

This instruction multiplies XLEN bits of the source operands from rsd' and rs2' and writes the lowest XLEN bits of the result to rsd'.

Included in

Extension	Version
Zcb	~> 1.0.0
Zmmul	~> 1.0.0

41.110. c.mv

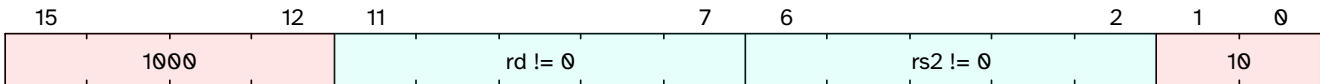
Synopsis

Move Register

Mnemonic

rd, rs2

Encoding



Description

C.MV (move register) performs copy of the data in register rs2 to register rd C.MV expands to addi rd, x0, rs2.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.111. c.nop

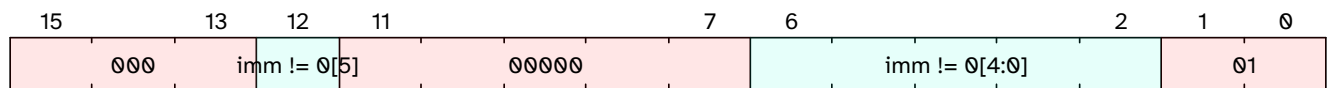
Synopsis

Non-operation

Mnemonic

imm

Encoding



Description

C.NOP expands into `addi x0, x0, imm`.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.112. c.not

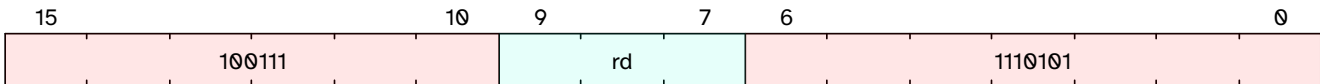
Synopsis

Bitwise not, 16-bit encoding

Mnemonic

rd

Encoding



Description

This instruction takes a single source/destination operand. This instruction takes the one’s complement of rd'/rs1' and writes the result to the same register.

Included in

Extension	Version
Zcb	~> 1.0.0
Zce	~> 1.0.0

41.113. c.or

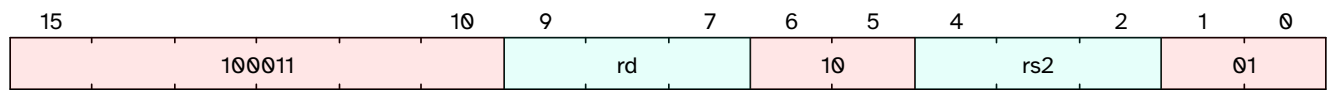
Synopsis

Or

Mnemonic

rd, rs2

Encoding



Description

Or rd with rs2, and store the result in rd The rd and rs2 register indexes should be used as rd+8 and rs2+8 (registers x8-x15). C.OR expands into or rd, rd, rs2.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.114. c.sb

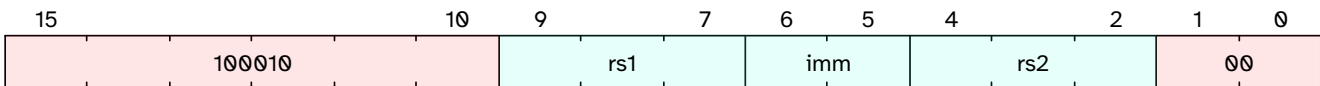
Synopsis

Store unsigned byte, 16-bit encoding

Mnemonic

rs2, imm(rs1)

Encoding



Description

Stores a 8-bit value from register rs2 into memory. It computes an effective address by adding the zero-extended offset, to the base address in register rs1. It expands to `sb rs2, offset(rs1)`.

Included in

Extension	Version
Zcb	~> 1.0.0
Zce	~> 1.0.0

41.115. c.sd

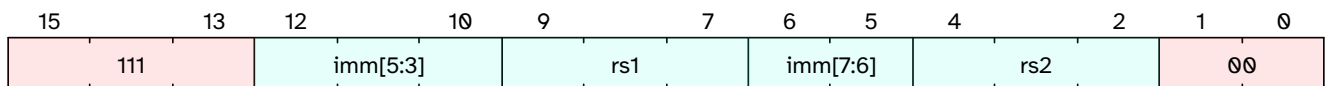
Synopsis

Store double

Mnemonic

rs2, imm(rs1)

Encoding



Description

Stores a 64-bit value in register rs2 to memory. It computes an effective address by adding the zero-extended offset, scaled by 8, to the base address in register rs1. It expands to [sd](#) rs2, offset(rs1).

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.116. c.sdsp

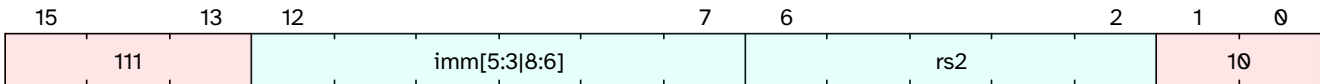
Synopsis

Store doubleword to stack

Mnemonic

rs2, imm(sp)

Encoding



Description

Stores a 64-bit value in register rs2 to memory. It computes an effective address by adding the zero-extended offset, scaled by 8, to the stack pointer, x2. It expands to `sd rs2, offset(x2)`.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.117. c.sext.b

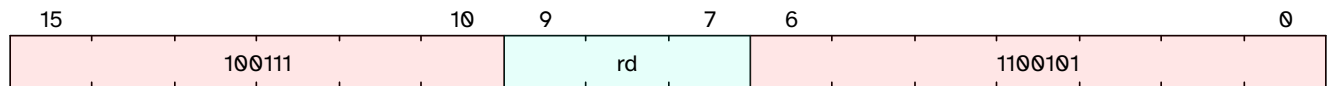
Synopsis

Sign-extend byte, 16-bit encoding

Mnemonic

rd

Encoding



Description

This instruction takes a single source/destination operand. This instruction sign-extends the least-significant byte of the source to XLEN by copying the most-significant bit in the byte (i.e., bit 7) to all of the more-significant bits.

Included in

Extension	Version
Zcb	~> 1.0.0
Zbb	~> 1.0.0

41.118. c.sext.h

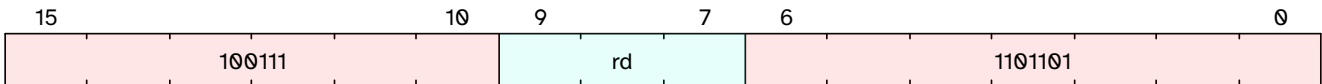
Synopsis

Sign-extend halfword, 16-bit encoding

Mnemonic

rd

Encoding



Description

This instruction takes a single source/destination operand. This instruction sign-extends the least-significant halfword of the source to XLEN by copying the most-significant bit in the halfword (i.e., bit 15) to all of the more-significant bits.

Included in

Extension	Version
Zcb	~> 1.0.0
Zbb	~> 1.0.0

41.119. c.sh

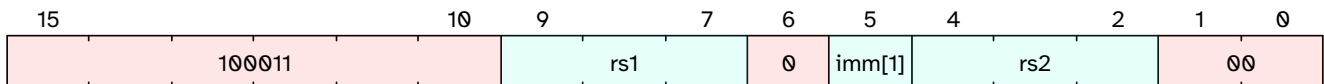
Synopsis

Store unsigned halfword, 16-bit encoding

Mnemonic

```
rs2, imm(rs1)
```

Encoding



Description

Stores a 16-bit value from register `rs2` into memory. It computes an effective address by adding the zero-extended offset, to the base address in register `rs1`. It expands to `sh rs2, offset(rs1)`.

Included in

Extension	Version
Zcb	~> 1.0.0
Zce	~> 1.0.0

41.120. c.slli

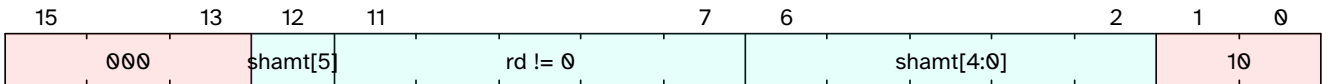
Synopsis

Shift left logical immediate

Mnemonic

rd, shamt

Encoding



Description

Shift the value in rd left by shamt, and store the result back in rd. C.SLLI expands into slli rd, rd, shamt.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.121. c.srai

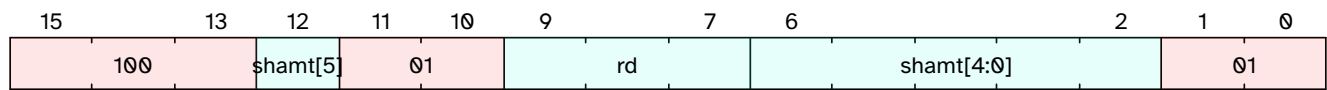
Synopsis

Shift right arithmetical immediate

Mnemonic

rd, shamt

Encoding



Description

Arithmetic shift (the original sign bit is copied into the vacated upper bits) the value in rd right by shamt, and store the result in rd. The rd register index should be used as rd+8 (registers x8-x15). C.SRAI expands into srai rd, rd, shamt.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.122. c.srli

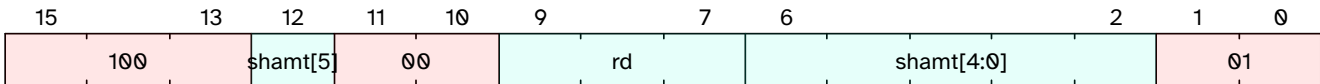
Synopsis

Shift right logical immediate

Mnemonic

```
rd, shamt
```

Encoding



Description

Shift the value in rd right by shamt, and store the result back in rd. The rd register index should be used as rd+8 (registers x8-x15). C.SRLI expands into srli rd, rd, shamt.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.123. c.sub

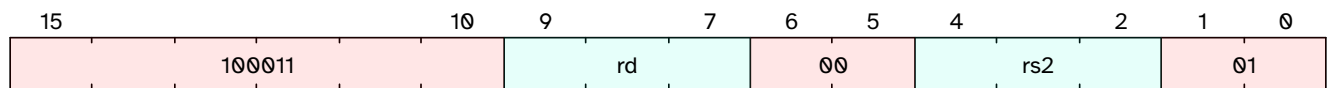
Synopsis

Subtract

Mnemonic

rd, rs2

Encoding



Description

Subtract the value in rs2 from rd, and store the result in rd. The rd and rs2 register indexes should be used as rd+8 and rs2+8 (registers x8-x15). C.SUB expands into sub rd, rd, rs2.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.124. c.subw

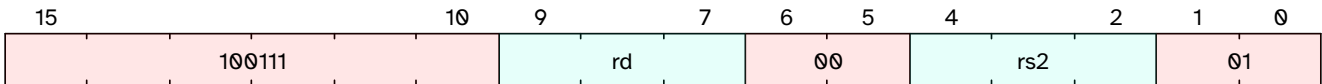
Synopsis

Subtract word

Mnemonic

rd, rs2

Encoding



Description

Subtract the 32-bit values in rs2 from rd, and store the result in rd. The rd and rs2 register indexes should be used as rd+8 and rs2+8 (registers x8-x15). C.SUBW expands into subw rd, rd, rs2.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.125. c.sw

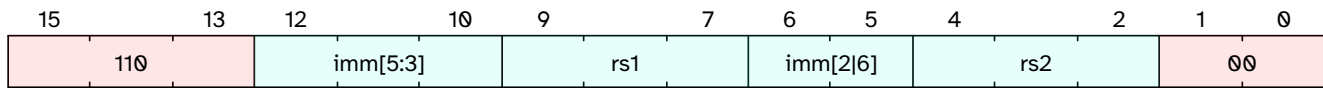
Synopsis

Store word

Mnemonic

```
rs2, imm(rs1)
```

Encoding



Description

Stores a 32-bit value in register rs2 to memory. It computes an effective address by adding the zero-extended offset, scaled by 4, to the base address in register rs1. It expands to [sw](#) rs2, offset(rs1).

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.126. c.swsp

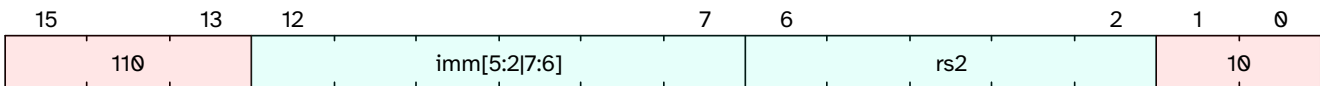
Synopsis

Store word to stack

Mnemonic

rs2, imm(sp)

Encoding



Description

Stores a 32-bit value in register rs2 to memory. It computes an effective address by adding the zero-extended offset, scaled by 4, to the stack pointer, x2. It expands to `sw rs2, offset(x2)`.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.127. c.xor

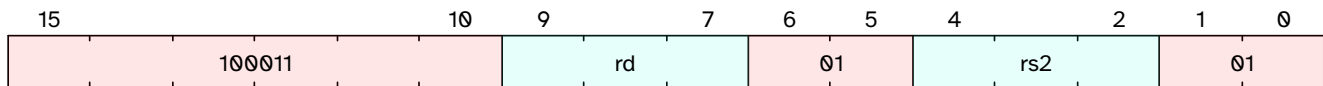
Synopsis

Exclusive Or

Mnemonic

rd, rs2

Encoding



Description

Exclusive or rd with rs2, and store the result in rd The rd and rs2 register indexes should be used as rd+8 and rs2+8 (registers x8-x15). C.XOR expands into xor rd, rd, rs2.

Included in

Extension	Version
C	~> 2.0.0
Zca	~> 1.0.0

41.128. c.zext.b

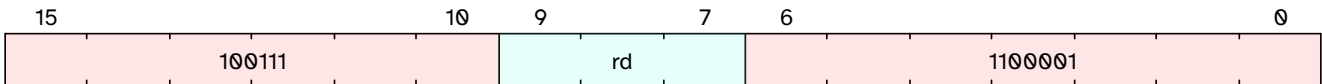
Synopsis

Zero-extend byte, 16-bit encoding

Mnemonic

rd

Encoding



Description

This instruction takes a single source/destination operand. This instruction zero-extends the least-significant byte of the source to XLEN by inserting 0’s into all of the bits more significant than 7.

Included in

Extension	Version
Zcb	~> 1.0.0
Zbb	~> 1.0.0

41.129. c.zext.h

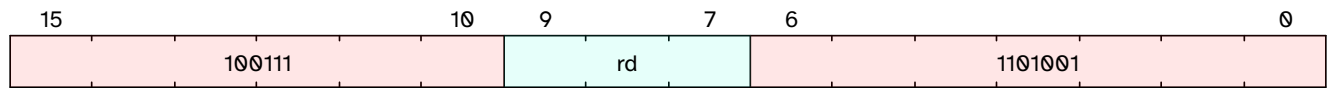
Synopsis

Zero-extend halfword, 16-bit encoding

Mnemonic

rd

Encoding



Description

This instruction takes a single source/destination operand. This instruction zero-extends the least-significant halfword of the source to XLEN by inserting 0’s into all of the bits more significant than 15.

Included in

Extension	Version
Zcb	~> 1.0.0
Zbb	~> 1.0.0

41.130. c.zext.w

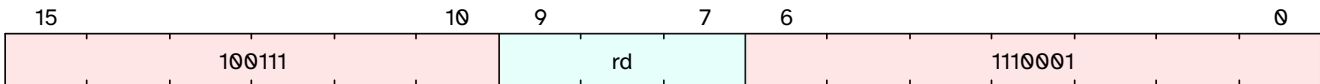
Synopsis

Zero-extend word, 16-bit encoding

Mnemonic

rd

Encoding



Description

This instruction takes a single source/destination operand. It zero-extends the least-significant word of the operand to XLEN bits by inserting zeros into all of the bits more significant than 31.

Included in

Extension	Version
Zcb	~> 1.0.0
Zbb	~> 1.0.0

41.131. cbo.clean

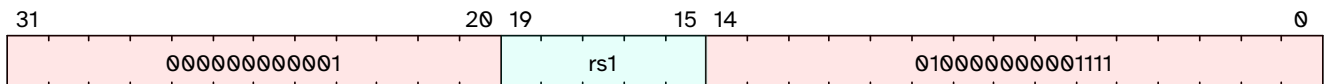
Synopsis

Cache Block Clean

Mnemonic

TODO

Encoding



Description

Cleans an entire cache block globally throughout the system.

Exactly what happens is coherence protocol-dependent, but in general it is expected that after this operation():

- The cache block will be in the clean (not dirty) state in any coherent cache holding a valid copy of the line.
- The data will be cleaned to a point such that an incoherent load can observe the cleaned data.

`cbo.clean` is ordered by FENCE instructions but not FENCE.I or SFENCE.VMA.

<%- if CACHE_BLOCK_SIZE.bit_length > [PMP_GRANULARITY, PMA_GRANULARITY].min -%> Both PMP and PMA access control must be the same for all bytes in the block; otherwise, `cbo.clean` has UNSPECIFIED behavior. <%- end -%>

Clean operations are treated as stores for page and access permissions. If permission checks fail, one of the following exceptions will occur:

```
<%- if ext?(:H) -%>
* `Store/AMO Guest-Page Fault` if virtual memory translation fails during G-
stage translation.
<%- end -%>
* `Store/AMO Page Fault` if virtual memory translation fails <% if ext?(:H)
%>when V=0 or during VS-stage translation<% end %>
* `Store/AMO Access Fault` if a PMP or PMA access check fails
```

<%- if CACHE_BLOCK_SIZE.bit_length ≤ [PMP_GRANULARITY, PMA_GRANULARITY].min -%> Because cache blocks are naturally aligned and always fit in a single PMP or PMA regions, the PMP and PMA access checks only need to check a single address in the line. <%- end -%>

CBO operations never raise a misaligned address fault.

Included in

Extension	Version
Zicbom	~> 1.0.0

DRAFT

41.132. cbo.flush

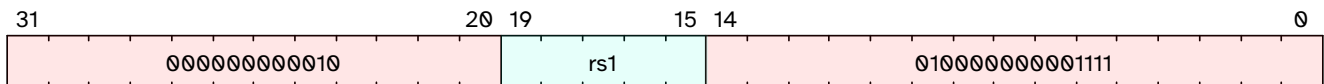
Synopsis

Cache Block Flush

Mnemonic

TODO

Encoding



Description

Flushes an entire cache block by cleaning it and then invalidating it in all caches.

`cbo.flush` is ordered by FENCE instructions but not FENCE.I or SFENCE.VMA.

<%- if CACHE_BLOCK_SIZE.bit_length > [PMP_GRANULARITY, PMA_GRANULARITY].min -%> Both PMP and PMA access control must be the same for all bytes in the block; otherwise, `cbo.flush` has UNSPECIFIED behavior. <%- end -%>

Flush operations are treated as stores for page and access permissions. If permission checks fail, one of the following exceptions will occur:

```

<%- if ext?(:H) -%>
* `Store/AMO Guest-Page Fault` if virtual memory translation fails during G-
stage translation.
<%- end -%>
* `Store/AMO Page Fault` if virtual memory translation fails <% if ext?(:H)
%>when V=0 or during VS-stage translation<% end %>
* `Store/AMO Access Fault` if a PMP or PMA access check fails.

```

<%- if CACHE_BLOCK_SIZE.bit_length ≤ [PMP_GRANULARITY, PMA_GRANULARITY].min -%> Because cache blocks are naturally aligned and always fit in a single PMP or PMA regions, the PMP and PMA access checks only need to check a single address in the line. <%- end -%>

CBO operations never raise a misaligned address fault.

Included in

Extension	Version
Zicbom	~> 1.0.0

41.133. cbo.inval

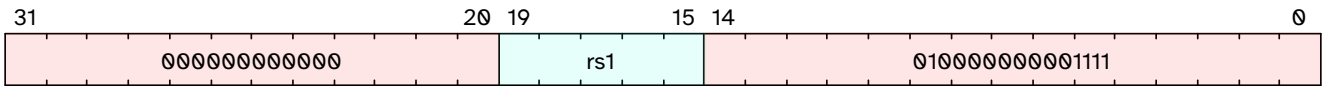
Synopsis

Cache Block Invalidate

Mnemonic

TODO

Encoding



Description

Either invalidates or flushes (clean + invalidate) a cache block, depending on the current mode and value of [menvcfg.CBIE](#), [senvcfg.CBIE](#), and/or [henvcfg.CBIE](#).

The instruction is an invalidate (without a clean) when:

- In M-mode
- In (H)S-mode and [menvcfg.CBIE](#) == 11
- In U-mode and [menvcfg.CBIE](#) == 11 and [senvcfg.CBIE](#) == 11
- In VS-mode and [menvcfg.CBIE](#) == 11 and [henvcfg.CBIE](#) == 11
- In VU-mode and [menvcfg.CBIE](#) == 11 and [henvcfg.CBIE](#) == 11 and [senvcfg.CBIE](#) == 11

Otherwise, if the instruction does not trap (see Access section), the operation is a flush. The table below summarizes the options.

menvcfg.CBIE	senvcfg.CBIE	henvcfg.CBIE	cbo.inval Operation				
			M-mode	S-mode	U-mode	VS-mode	VU-mode
00	-	-	Invalidate	Illegal Instruction	Illegal Instruction	Virtual Instruction	Virtual Instruction
01	00	00	Invalidate	Flush	Illegal Instruction	Virtual Instruction	Virtual Instruction
01	00	01	Invalidate	Flush	Illegal Instruction	Flush	Virtual Instruction
01	00	11	Invalidate	Flush	Illegal Instruction	Flush	Virtual Instruction
01	01	00	Invalidate	Flush	Flush	Virtual Instruction	Virtual Instruction
01	01	01	Invalidate	Flush	Flush	Flush	Flush
01	01	11	Invalidate	Flush	Flush	Flush	Flush

01	11	00	Invalida te	Flush	Flush	Virtual Instruction	Virtual Instruction
01	11	01	Invalida te	Flush	Flush	Flush	Flush
01	11	11	Invalida te	Flush	Flush	Flush	Flush
11	00	00	Invalida te	Invalidate	Illegal Instruction	Virtual Instruction	Virtual Instruction
11	00	01	Invalida te	Invalidate	Illegal Instruction	Flush	Virtual Instruction
11	00	11	Invalida te	Invalidate	Illegal Instruction	Invalidate	Virtual Instruction
11	01	00	Invalida te	Invalidate	Flush	Virtual Instruction	Virtual Instruction
11	01	01	Invalida te	Invalidate	Flush	Flush	Flush
11	01	11	Invalida te	Invalidate	Flush	Invalidate	Flush
11	11	00	Invalida te	Invalidate	Invalidate	Virtual Instruction	Virtual Instruction
11	11	01	Invalida te	Invalidate	Invalidate	Flush	Flush
11	11	11	Invalida te	Invalidate	Invalidate	Invalidate	Invalidate

`cbo.inval` is ordered by FENCE instructions but not FENCE.I or SFENCE.VMA.

<%- if CACHE_BLOCK_SIZE.bit_length > [PMP_GRANULARITY, PMA_GRANULARITY].min -%> Both PMP and PMA access control must be the same for all bytes in the block; otherwise, `cbo.zero` has UNSPECIFIED behavior. <%- end -%>

Invalidate operations are treated as stores for page and access permissions. If permission checks fail, one of the following exceptions will occur:

```

<%- if ext?(:H) -%>
* `Store/AMO Guest-Page Fault` if virtual memory translation fails during G-
stage translation.
<%- end -%>
* `Store/AMO Page Fault` if virtual memory translation fails <% if ext?(:H)
%>when V=0 or during VS-stage translation<% end %>
* `Store/AMO Access Fault` if a PMP or PMA access check fails.

```

<%- if CACHE_BLOCK_SIZE.bit_length ≤ [PMP_GRANULARITY, PMA_GRANULARITY].min -%> Because cache blocks are naturally aligned and always fit in a single PMP or PMA regions, the PMP and PMA access checks only need to check a single address in the line. <%- end -%>

CBO operations never raise a misaligned address fault.

Included in

Extension	Version
Zicbom	~> 1.0.0

DRAFT

41.134. cbo.zero

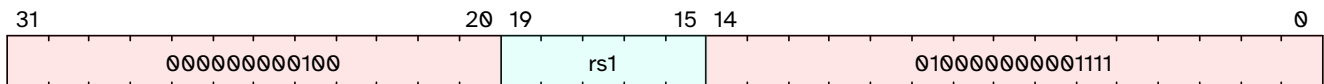
Synopsis

Cache Block Zero

Mnemonic

TODO

Encoding



Description

Zeros an entire cache block

The block zeroing does not need to be atomic.

[cbo.zero](#) is ordered by FENCE instructions but not FENCE.I or SFENCE.VMA.

<%- if CACHE_BLOCK_SIZE.bit_length > [PMP_GRANULARITY, PMA_GRANULARITY].min -%> Both PMP and PMA access control must be the same for all bytes in the block; otherwise, [cbo.zero](#) has UNSPECIFIED behavior. <%- end -%>

Clean operations are treated as stores for page and access permissions. If permission checks fail, one of the following exceptions will occur:

```
<%- if ext?(:H) -%>
* `Store/AMO Guest-Page Fault` if virtual memory translation fails during G-
stage translation.
<%- end -%>
* `Store/AMO Page Fault` if virtual memory translation fails <% if ext?(:H)
%>when V=0 or during VS-stage translation<% end %>
* `Store/AMO Access Fault` if a PMP or PMA access check fails.
```

<%- if CACHE_BLOCK_SIZE.bit_length ≤ [PMP_GRANULARITY, PMA_GRANULARITY].min -%> Because cache blocks are naturally aligned and always fit in a single PMP or PMA regions, the PMP and PMA access checks only need to check a single address in the line. <%- end -%>

CBO operations never raise a misaligned address fault.

Included in

Extension	Version
Zicboz	~> 1.0.0

41.135. ctmul

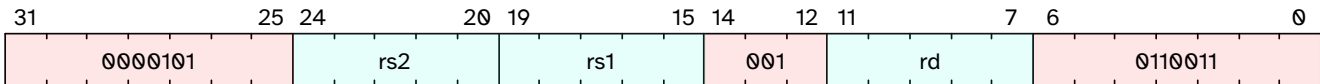
Synopsis

Carry-less multiply (low-part)

Mnemonic

rd, rs1, rs2

Encoding



Description

ctmul produces the lower half of the 2*XLEN carry-less product

Included in

Extension	Version
B	~> 1.0.0
Zbc	~> 1.0.0
Zbkc	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.136. ctmulh

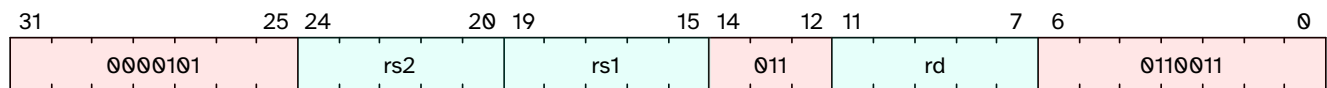
Synopsis

Carry-less multiply (high-part)

Mnemonic

rd, rs1, rs2

Encoding



Description

`ctmulh` produces the upper half of the $2 \times \text{XLEN}$ carry-less product

Included in

Extension	Version
B	~> 1.0.0
Zbc	~> 1.0.0
Zbkc	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.137. clmulr

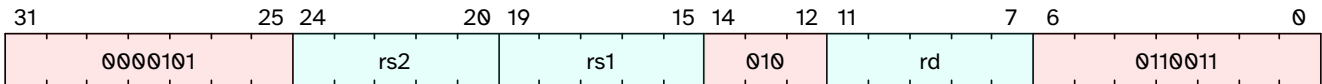
Synopsis

Carry-less multiply (reversed)

Mnemonic

rd, rs1, rs2

Encoding



Description

clmulr produces bits 2*XLEN-2:XLEN-1 of the 2*XLEN carry-less product

Included in

Extension	Version
B	~> 1.0.0
Zbc	~> 1.0.0

41.138. clz

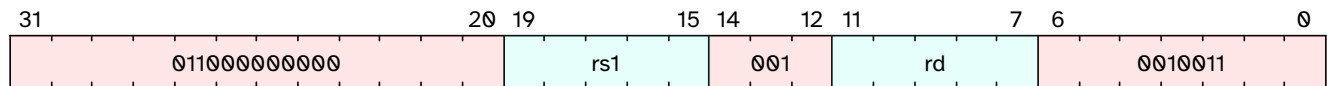
Synopsis

Count leading zero bits

Mnemonic

rd, rs1

Encoding



Description

This instruction counts the number of 0’s before the first 1, starting at the most-significant bit (i.e., XLEN-1) and progressing to bit 0. Accordingly, if the input is 0, the output is XLEN, and if the most-significant bit of the input is a 1, the output is 0.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.139. clzw

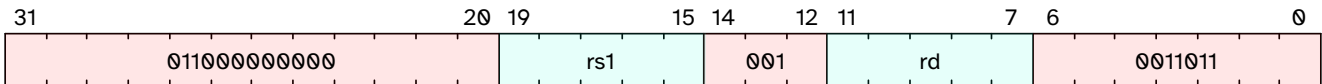
Synopsis

Count leading zero bits in word

Mnemonic

rd, rs1

Encoding



Description

This instruction counts the number of 0’s before the first 1 starting at bit 31 and progressing to bit 0. Accordingly, if the least-significant word is 0, the output is 32, and if the most-significant bit of the word (*i.e.*, bit 31) is a 1, the output is 0.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.140. cm.mva01s

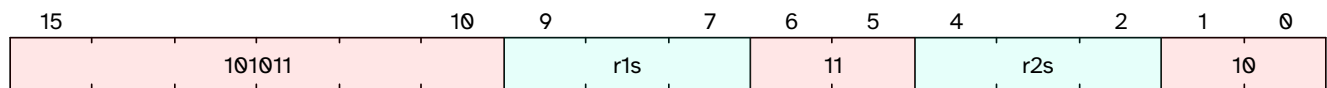
Synopsis

Move two s0-s7 registers into a0-a1

Mnemonic

r1s, r2s

Encoding



Description

This instruction moves r1s' into a0 and r2s' into a1. The execution is atomic, so it is not possible to observe state where only one of a0 or a1 have been updated. The encoding uses sreg number specifiers instead of xreg number specifiers to save encoding space. The mapping between them is specified in the pseudo-code below.

Included in

Extension	Version
Zcmp	~> 1.0.0

41.141. cm.mvsa01

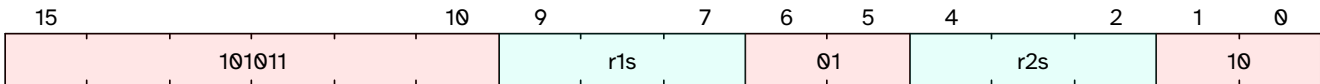
Synopsis

Move a0-a1 into two registers of s0-s7

Mnemonic

r1s, r2s

Encoding



Description

This instruction moves a0 into r1s' and a1 into r2s'. r1s' and r2s' must be different. The execution is atomic, so it is not possible to observe state where only one of r1s' or r2s' has been updated. The encoding uses sreg number specifiers instead of xreg number specifiers to save encoding space. The mapping between them is specified in the pseudo-code below.

Included in

Extension	Version
Zcmp	~> 1.0.0

41.142. cm.pop

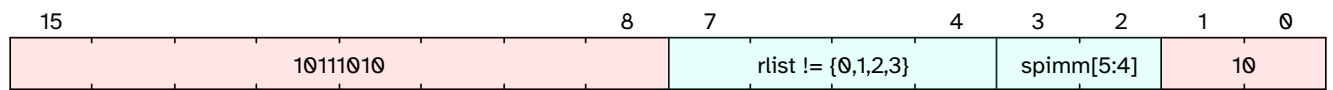
Synopsis

Destroy function call stack frame

Mnemonic

reg_list, stack_adj

Encoding



Description

Destroy stack frame: load ra and 0 to 12 saved registers from the stack frame, deallocate the stack frame. This instruction pops (loads) the registers in reg_list from stack memory, and then adjusts the stack pointer by stack_adj.

Restrictions on stack_adj:

- it must be enough to store all of the listed registers
- it must be a multiple of 16 (bytes):
 - for RV32 the allowed values are: 16, 32, 48, 64, 80, 96, 112
 - for RV64 the allowed values are: 16, 32, 48, 64, 80, 96, 112, 128, 144, 160

Included in

Extension	Version
Zcmp	~> 1.0.0

41.143. cm.popret

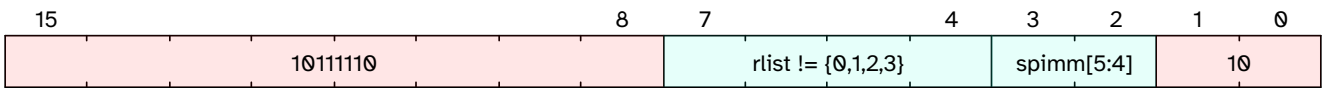
Synopsis

Destroy function call stack frame and return to ra.

Mnemonic

```
reg_list, stack_adj
```

Encoding



Description

Destroy stack frame: load ra and 0 to 12 saved registers from the stack frame, deallocate the stack frame, return to ra. This instruction pops (loads) the registers in reg_list from stack memory, and then adjusts the stack pointer by stack_adj and then return to ra.

Restrictions on stack_adj:

- it must be enough to store all of the listed registers
- it must be a multiple of 16 (bytes):
 - for RV32 the allowed values are: 16, 32, 48, 64, 80, 96, 112
 - for RV64 the allowed values are: 16, 32, 48, 64, 80, 96, 112, 128, 144, 160

Included in

Extension	Version
Zcmp	~> 1.0.0

41.144. cm.popretz

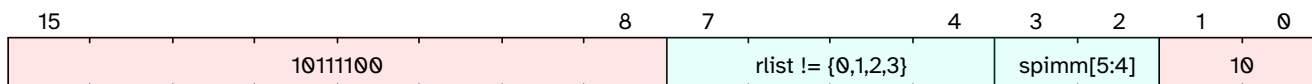
Synopsis

Destroy function call stack frame, move zero to a0 and return to ra.

Mnemonic

```
reg_list, stack_adj
```

Encoding



Description

Destroy stack frame: load ra and 0 to 12 saved registers from the stack frame, deallocate the stack frame, move zero to a0, return to ra. This instruction pops (loads) the registers in reg_list from stack memory, and then adjusts the stack pointer by stack_adj, move zero to a0 and then return to ra.

Restrictions on stack_adj:

- it must be enough to store all of the listed registers
- it must be a multiple of 16 (bytes):
 - for RV32 the allowed values are: 16, 32, 48, 64, 80, 96, 112
 - for RV64 the allowed values are: 16, 32, 48, 64, 80, 96, 112, 128, 144, 160

Included in

Extension	Version
Zcmp	~> 1.0.0

41.145. cm.push

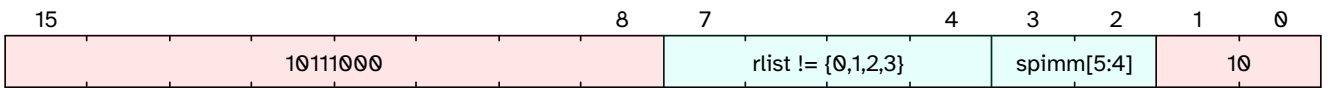
Synopsis

Create function call stack frame

Mnemonic

reg_list, -stack_adj

Encoding



Description

Create stack frame: store ra and 0 to 12 saved registers to the stack frame, optionally allocate additional stack space. This instruction pushes (stores) the registers in reg_list to the memory below the stack pointer, and then creates the stack frame by decrementing the stack pointer by stack_adj.

Restrictions on stack_adj:

- it must be enough to store all of the listed registers
- it must be a multiple of 16 (bytes):
 - for RV32 the allowed values are: 16, 32, 48, 64, 80, 96, 112
 - for RV64 the allowed values are: 16, 32, 48, 64, 80, 96, 112, 128, 144, 160

Included in

Extension	Version
Zcmp	~> 1.0.0

41.146. cpop

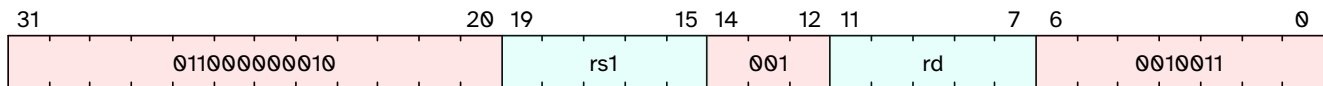
Synopsis

Count set bits

Mnemonic

```
rd, rs1
```

Encoding



Description

This instructions counts the number of 1’s (i.e., set bits) in the source register.

This operations is known as population count, popcount, sideways sum, bit summation, or Hamming weight.

The GCC builtin function `__builtin_popcount (unsigned int x)` is implemented by cpop on RV32 and by cpopw on RV64. The GCC builtin function `__builtin_popcountl (unsigned long x)` for LP64 is implemented by cpop on RV64.

Listing 26. Software Hint

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.147. cpop

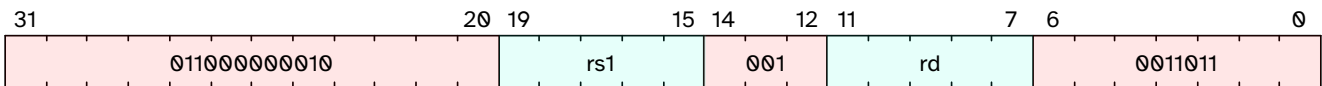
Synopsis

Count set bits in word

Mnemonic

```
rd, rs1
```

Encoding



Description

This instructions counts the number of 1’s (i.e., set bits) in the least-significant word of the source register.

This operations is known as population count, popcount, sideways sum, bit summation, or Hamming weight.

The GCC builtin function `__builtin_popcount (unsigned int x)` is implemented by cpop on RV32 and by cpop on RV64. The GCC builtin function `__builtin_popcountl (unsigned long x)` for LP64 is implemented by cpop on RV64.

Listing 27. Software Hint

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.148. csrrc

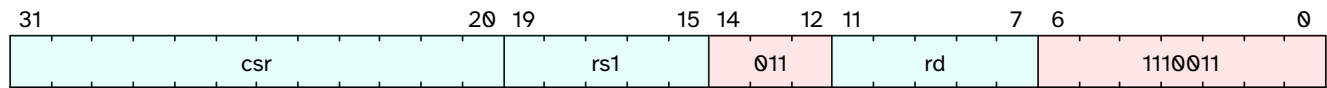
Synopsis

No synopsis available.

Mnemonic

rd, rs1, csr

Encoding



Description

No description available.

Included in

Extension	Version
Zicsr	~> 2.0.0

41.149. csrri

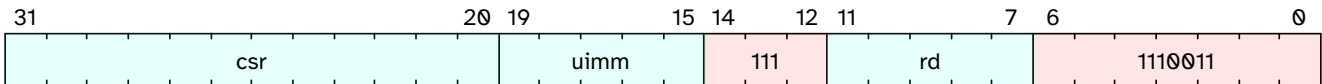
Synopsis

No synopsis available.

Mnemonic

rd, csr, imm

Encoding



Description

No description available.

Included in

Extension	Version
Zicsr	~> 2.0.0

41.150. csrrs

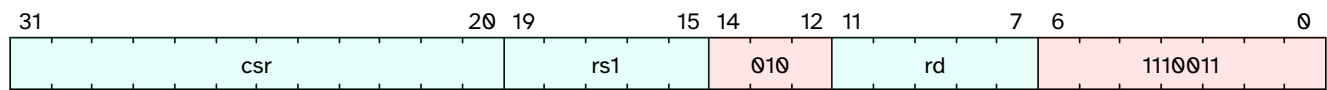
Synopsis

Atomic Read and Set Bits in CSR

Mnemonic

rd, rs1, csr

Encoding



Description

Atomically read and set bits in a CSR.

Reads the value of the CSR, zero-extends the value to XLEN bits, and writes it to integer register rd. The initial value in integer register rs1 is treated as a bit mask that specifies bit positions to be set in the CSR. Any bit that is high in rs1 will cause the corresponding bit to be set in the CSR, if that CSR bit is writable. Other bits in the CSR are not explicitly written.

Included in

Extension	Version
Zicsr	~> 2.0.0

41.151. csrrsi

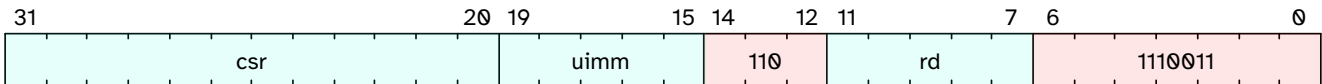
Synopsis

No synopsis available.

Mnemonic

rd, csr, imm

Encoding



Description

No description available.

Included in

Extension	Version
Zicsr	~> 2.0.0

41.152. csrrw

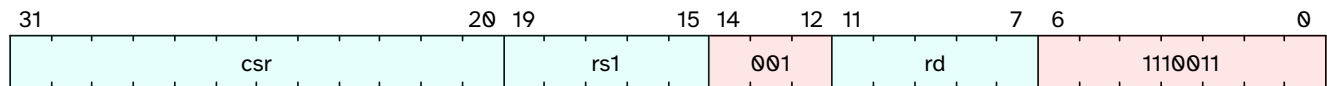
Synopsis

Atomic Read/Write CSR

Mnemonic

rd, rs1, csr

Encoding



Description

Atomically swap values in the CSRs and integer registers.

Read the old value of the CSR, zero-extends the value to xLEN bits, and then write it to integer register rd. The initial value in rs1 is written to the CSR. If rd=x0, then the instruction shall not read the CSR and shall not cause any of the side effects that might occur on a CSR read.

Included in

Extension	Version
Zicsr	~> 2.0.0

41.153. csrwi

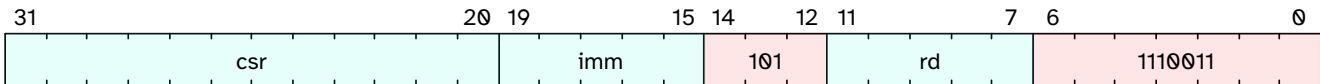
Synopsis

Atomic Read/Write CSR Immediate

Mnemonic

rd, zimm, csr

Encoding



Description

Atomically write CSR using a 5-bit immediate, and load the previous value into 'rd'.

Read the old value of the CSR, zero-extends the value to xLEN bits, and then write it to integer register rd. The 5-bit uimm field is zero-extended and written to the CSR. If rd=x0, then the instruction shall not read the CSR and shall not cause any of the side effects that might occur on a CSR read.

Included in

Extension	Version
Zicsr	~> 2.0.0

41.154. ctz

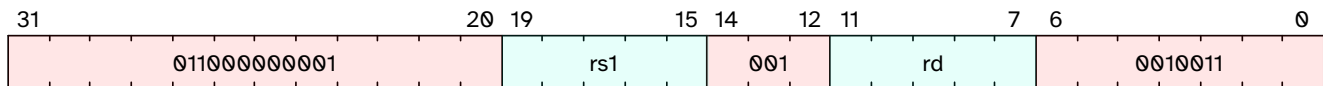
Synopsis

Count trailing zero bits

Mnemonic

rd, rs1

Encoding



Description

This instruction counts the number of 0’s before the first 1, starting at the least-significant bit (i.e., 0) and progressing to the most-significant bit (i.e., XLEN-1). Accordingly, if the input is 0, the output is XLEN, and if the least-significant bit of the input is a 1, the output is 0.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.155. ctzw

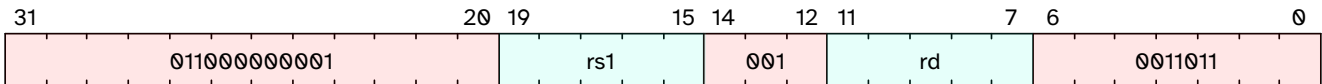
Synopsis

Count trailing zero bits in word

Mnemonic

rd, rs1

Encoding



Description

This instruction counts the number of 0’s before the first 1, starting at the least-significant bit (i.e., 0) and progressing to the most-significant bit of the least-significant word (i.e., 31). Accordingly, if the least-significant word is 0, the output is 32, and if the least-significant bit of the input is a 1, the output is 0.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.156. czero.eqz

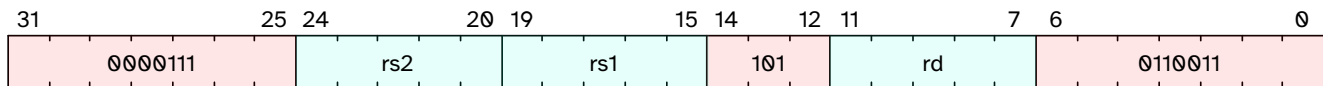
Synopsis

Conditional zero, if condition is equal to zero.

Mnemonic

```
rd, rs1, rs2
```

Encoding



Description

If rs2 contains the value zero, this instruction writes the value zero to rd. Otherwise, this instruction copies the contents of rs1 to rd. This instruction carries a syntactic dependency from both rs1 and rs2 to rd. Furthermore, if the Zkt extension is implemented, this instruction’s timing is independent of the data values in rs1 and rs2.

Included in

Extension	Version
Zicond	~> 1.0.0

41.157. czero.nez

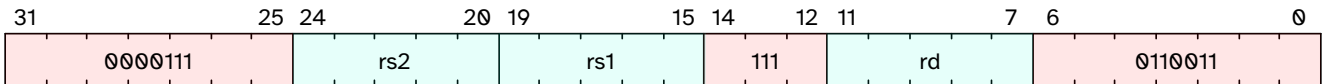
Synopsis

Conditional zero, if condition is nonzero.

Mnemonic

rd, rs1, rs2

Encoding



Description

If rs2 contains a nonzero value, this instruction writes the value zero to rd. Otherwise, this instruction copies the contents of rs1 to rd. This instruction carries a syntactic dependency from both rs1 and rs2 to rd. Furthermore, if the Zkt extension is implemented, this instruction’s timing is independent of the data values in rs1 and rs2.

Included in

Extension	Version
Zicond	~> 1.0.0

41.158. div

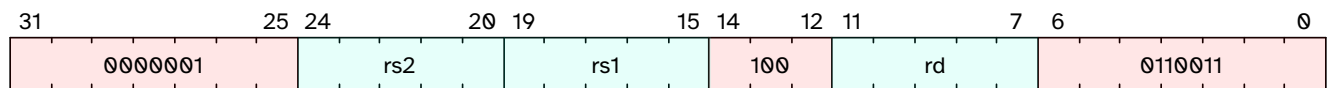
Synopsis

Signed division

Mnemonic

rd, rs1, rs2

Encoding



Description

Divide rs1 by rs2, and store the result in rd. The remainder is discarded.

Division by zero will put -1 into rd.

Division resulting in signed overflow (when most negative number is divided by -1) will put the most negative number into rd;

Included in

Extension	Version
M	~> 2.0.0

41.159. divu

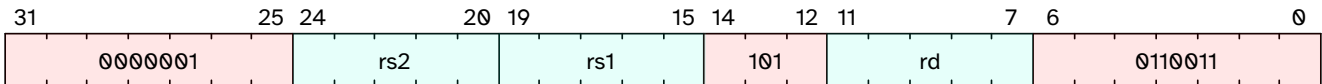
Synopsis

Unsigned division

Mnemonic

rd, rs1, rs2

Encoding



Description

Divide unsigned values in rs1 by rs2, and store the result in rd.

The remainder is discarded.

If the value in rs2 is zero, rd gets the largest unsigned value.

Included in

Extension	Version
M	~> 2.0.0

41.160. divuw

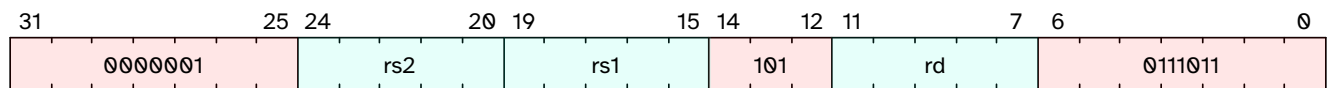
Synopsis

Unsigned 32-bit division

Mnemonic

rd, rs1, rs2

Encoding



Description

Divide the unsigned 32-bit values in rs1 and rs2, and store the sign-extended result in rd.

The remainder is discarded.

If the value in rs2 is zero, rd is written with all 1s.

Included in

Extension	Version
M	~> 2.0.0

41.161. divw

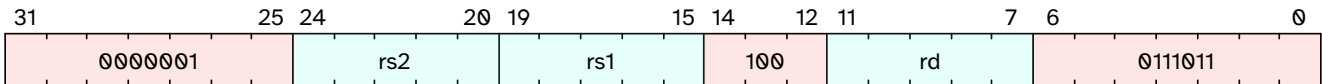
Synopsis

Signed 32-bit division

Mnemonic

rd, rs1, rs2

Encoding



Description

Divide the lower 32-bits of register rs1 by the lower 32-bits of register rs2, and store the sign-extended result in rd.

The remainder is discarded.

Division by zero will put -1 into rd.

Division resulting in signed overflow (when most negative number is divided by -1) will put the most negative number into rd;

Included in

Extension	Version
M	~> 2.0.0

41.162. dret

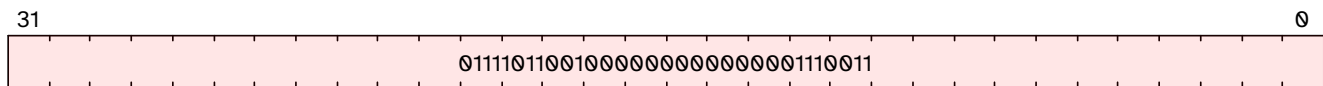
Synopsis

No synopsis available.

Mnemonic

dret

Encoding



Description

No description available.

Included in

Extension	Version
Sdext	~> 1.0.0

DRAFT

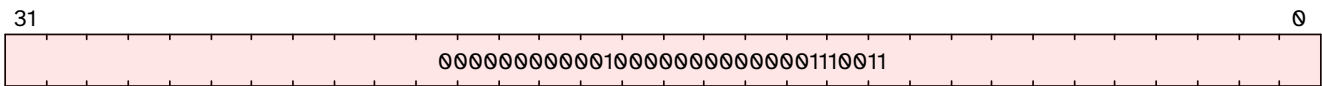
41.163. ebreak

Synopsis

Breakpoint exception

Mnemonic

Encoding



Description

The EBREAK instruction is used by debuggers to cause control to be transferred back to a debugging environment. Unless overridden by an external debug environment, EBREAK raises a breakpoint exception and performs no other operation.



As described in the [C Standard Extension for Compressed Instructions](#), the [c.ebreak](#) instruction performs the same operation as the EBREAK instruction.

EBREAK causes the receiving privilege mode’s epc register to be set to the address of the EBREAK instruction itself, not the address of the following instruction. As EBREAK causes a synchronous exception, it is not considered to retire, and should not increment the [minstret](#) CSR.

Included in

Extension	Version
I	~> 2.1.0

Environment call

[illegible]

ECALL causes the receiving privilege mode's epc register to be set to the address of the ECALL instruction itself, not the address of the following instruction. As ECALL causes a synchronous exception, it is not considered to retire, and should not increment the `minstret` CSR.

Extension	Version
I	~> 2.1.0

41.165. fadd.d

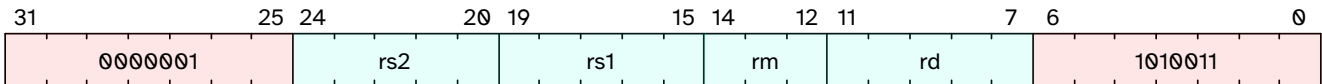
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.166. fadd.h

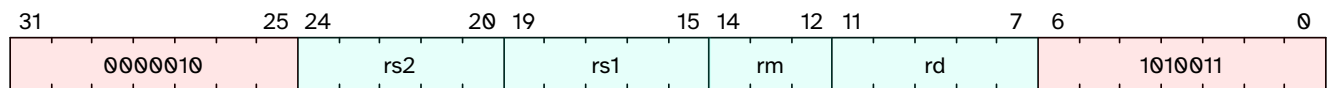
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.167. fadd.q

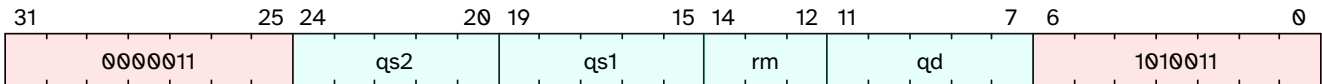
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.168. fadd.s

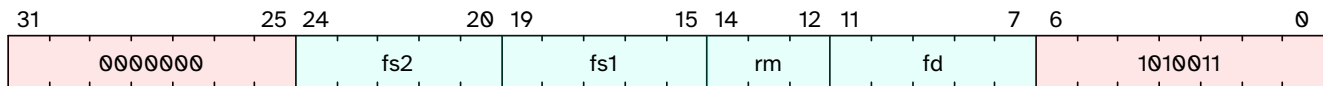
Synopsis

Single-precision floating-point addition

Mnemonic

fd, fs1, fs2, rm

Encoding



Description

Do the single-precision floating-point addition of fs1 and fs2 and store the result in fd. rm is the dynamic Rounding Mode.

Included in

Extension	Version
F	~> 2.2.0

41.169. fclass.d

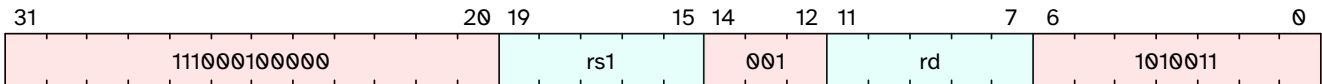
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.170. fclass.h

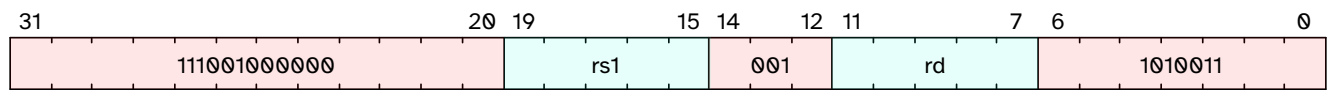
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.171. fclass.q

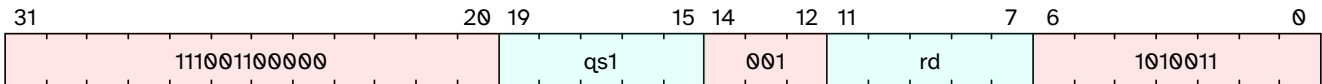
Synopsis

No synopsis available.

Mnemonic

rd, qs1

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.172. fclass.s

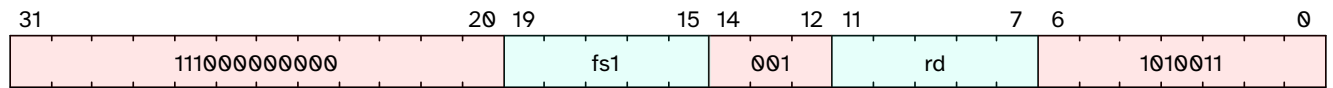
Synopsis

Single-precision floating-point classify.

Mnemonic

```
rd, fs1
```

Encoding



Description

The `fclass.s` instruction examines the value in floating-point register `fs1` and writes to integer register `rd` a 10-bit mask that indicates the class of the floating-point number. The format of the mask is described in the table below. The corresponding bit in `rd` will be set if the property is true and clear otherwise. All other bits in `rd` are cleared. Note that exactly one bit in `rd` will be set. `fclass.s` does not set the floating-point exception flags.

Table 100. Format of result of `fclass` instruction.

rd bit	Meaning
0	<i>rs1</i> is $-\infty$.
1	<i>rs1</i> is a negative normal number.
2	<i>rs1</i> is a negative subnormal number.
3	<i>rs1</i> is -0 .
4	<i>rs1</i> is $+0$.
5	<i>rs1</i> is a positive subnormal number.
6	<i>rs1</i> is a positive normal number.
7	<i>rs1</i> is $+\infty$.
8	<i>rs1</i> is a signaling NaN.
9	<i>rs1</i> is a quiet NaN.

Included in

Extension	Version
F	~> 2.2.0

41.173. fcvf.bf16.s

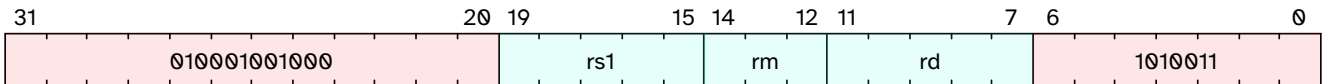
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfbfmin	~> 1.0.0

41.174. fcvt.d.h

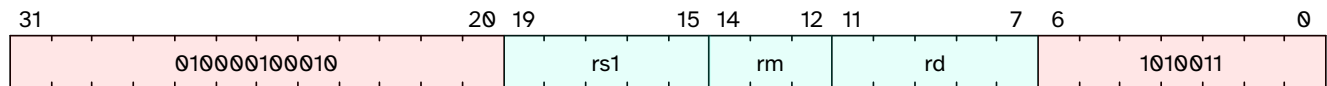
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfh	~> 1.0.0

41.175. fcvt.d.l

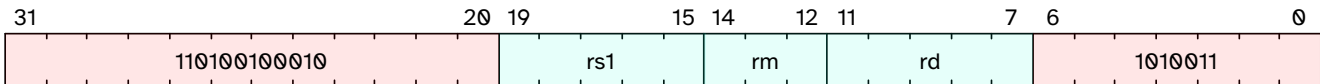
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.176. fcvt.d.lu

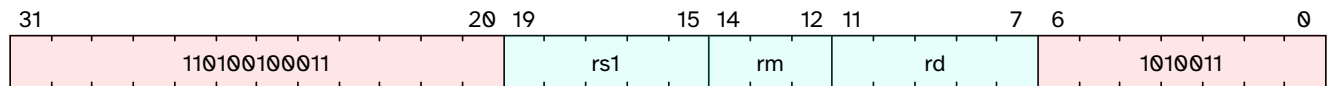
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.177. fcvt.d.q

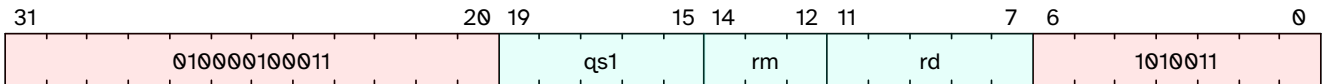
Synopsis

No synopsis available.

Mnemonic

rd, qs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.178. fcvt.d.s

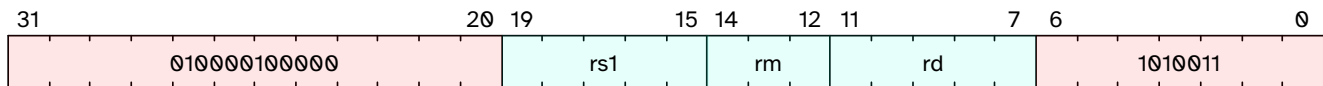
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

DRAFT

41.179. fcvt.d.w

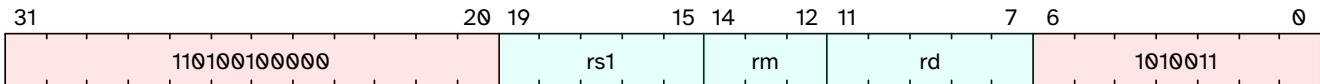
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.180. fcvt.d.wu

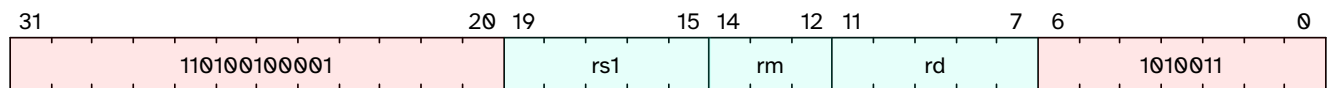
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.181. fcvt.h.d

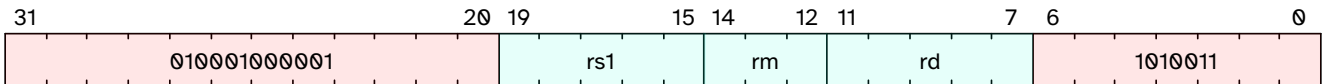
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfh	~> 1.0.0

41.182. fcvt.h.l

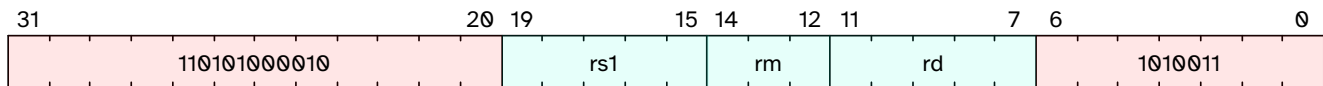
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

DRAFT

41.183. fcvt.h.lu

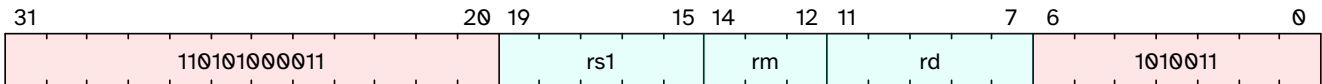
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.184. fcvt.h.q

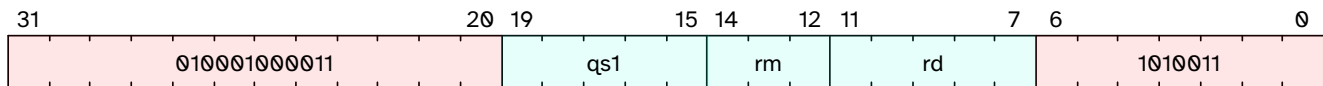
Synopsis

No synopsis available.

Mnemonic

rd, qs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0
Zfh	~> 1.0.0

41.185. fcvt.h.s

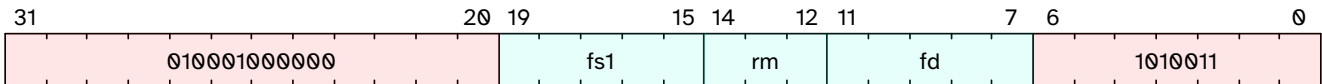
Synopsis

Convert half-precision float to a single-precision float

Mnemonic

fd, rs1

Encoding



Description

Converts a half-precision number in floating-point register *fs1* into a single-precision floating-point number in floating-point register *fd*.

[fcvt.h.s](#) rounds according to the *rm* field.

All floating-point conversion instructions set the Inexact exception flag if the rounded result differs from the operand value and the Invalid exception flag is not set.

Included in

Extension	Version
Zfh	~> 1.0.0
Zfhmin	~> 1.0.0

41.186. fcvt.h.w

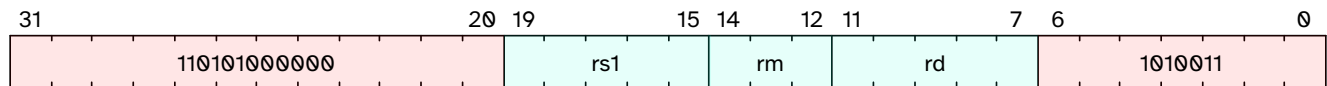
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.187. fcv.t.h.wu

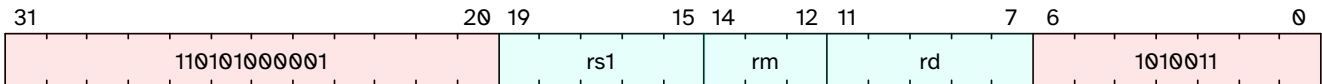
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.188. fcvt.Ld

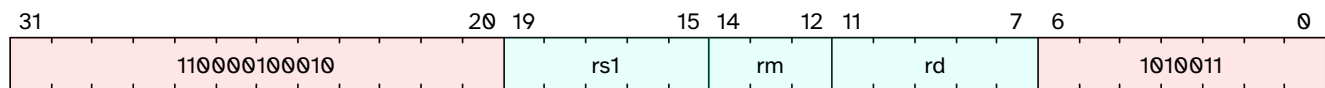
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.189. fcvt.Lh

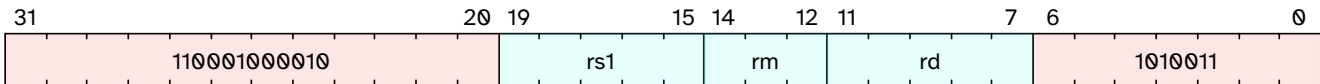
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.190. fcvt.lq

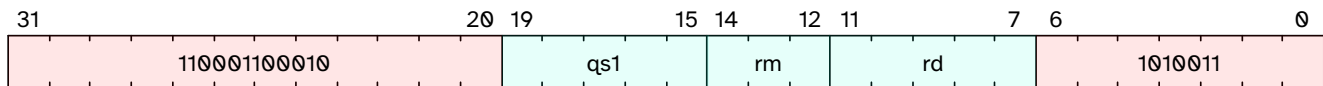
Synopsis

No synopsis available.

Mnemonic

rd, qs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

DRAFT

41.191. fcvt.l.s

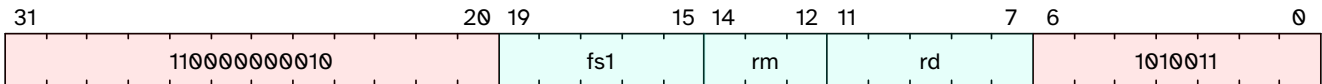
Synopsis

No synopsis available.

Mnemonic

rd, fs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

41.192. fcvt.lu.d

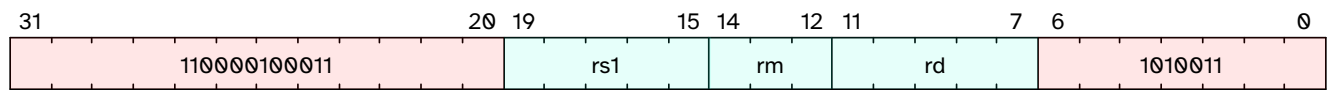
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.193. fcvt.lu.h

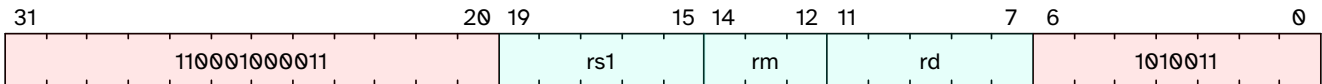
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.194. fcvt.lu.q

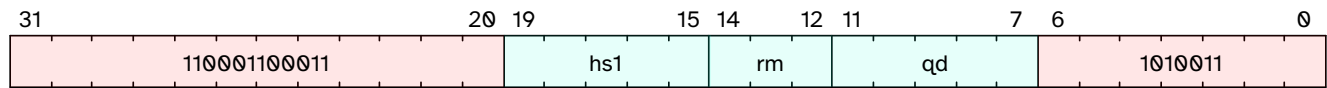
Synopsis

No synopsis available.

Mnemonic

qd, hs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.195. fcvt.lu.s

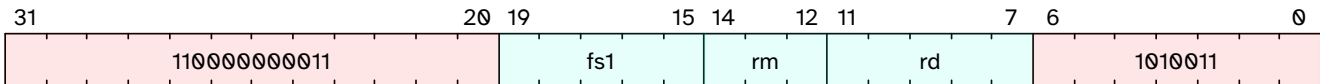
Synopsis

No synopsis available.

Mnemonic

rd, fs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

41.196. fcvt.q.d

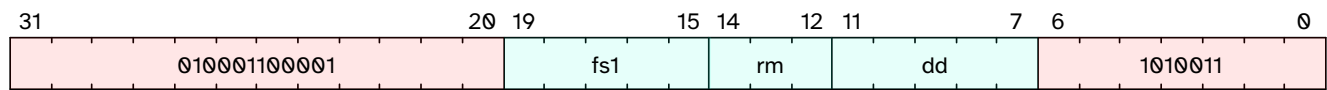
Synopsis

No synopsis available.

Mnemonic

dd, fs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.197. fcvt.q.h

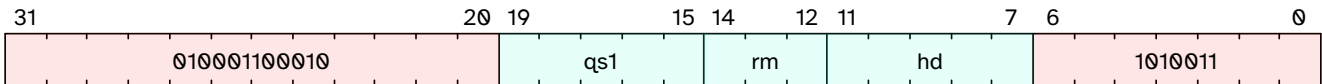
Synopsis

No synopsis available.

Mnemonic

hd, qs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0
Zfh	~> 1.0.0

41.198. fcvt.q.l

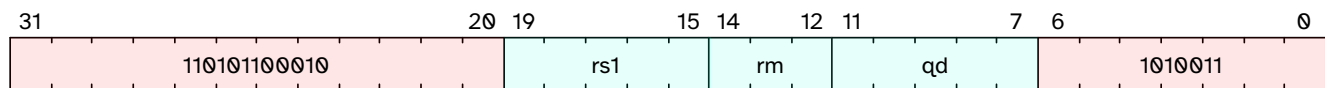
Synopsis

No synopsis available.

Mnemonic

qd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.199. fcvt.q.lu

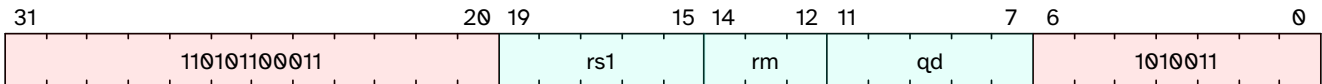
Synopsis

No synopsis available.

Mnemonic

qd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.200. fcvt.q.s

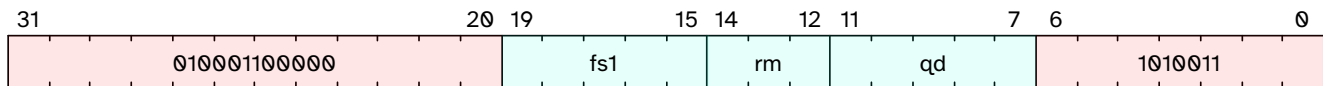
Synopsis

No synopsis available.

Mnemonic

qd, fs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.201. fcvt.q.w

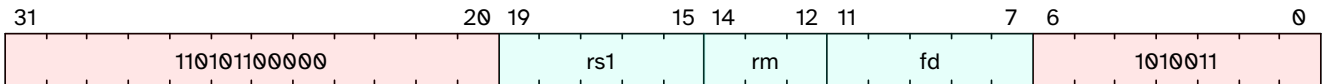
Synopsis

No synopsis available.

Mnemonic

fd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.202. fcvt.q.wu

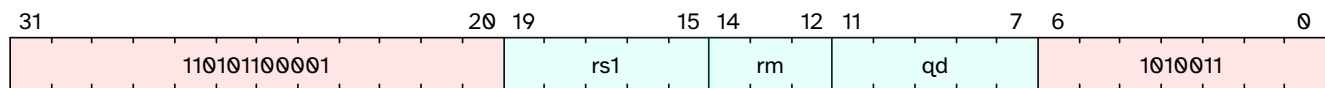
Synopsis

No synopsis available.

Mnemonic

qd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.203. fcvt.s.bf16

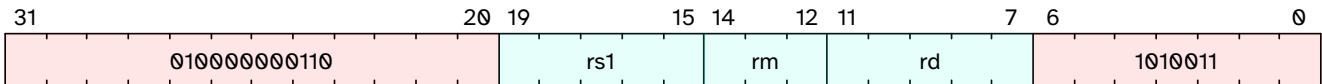
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfbfmin	~> 1.0.0

41.204. fcvt.s.d

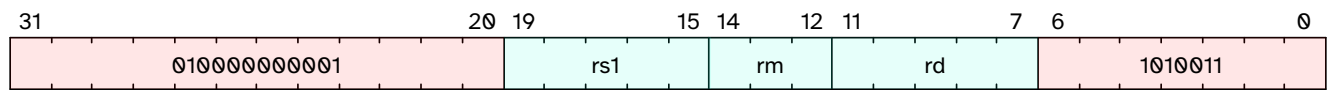
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.205. fcvt.s.h

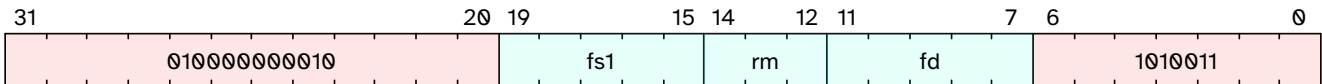
Synopsis

Convert single-precision float to a half-precision float

Mnemonic

fd, rs1

Encoding



Description

Converts a single-precision number in floating-point register *fs1* into a half-precision floating-point number in floating-point register *fd*.

[fcvt.s.h](#) will never round, and so the 'rm' field is effectively ignored.

Included in

Extension	Version
Zfh	~> 1.0.0
Zfhmin	~> 1.0.0

41.206. fcvt.s.l

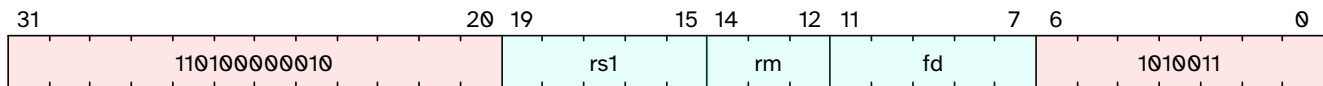
Synopsis

No synopsis available.

Mnemonic

fd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

41.207. fcvt.s.lu

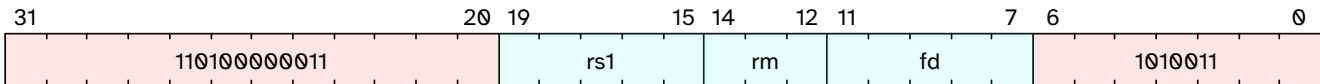
Synopsis

No synopsis available.

Mnemonic

fd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

41.208. fcvt.s.q

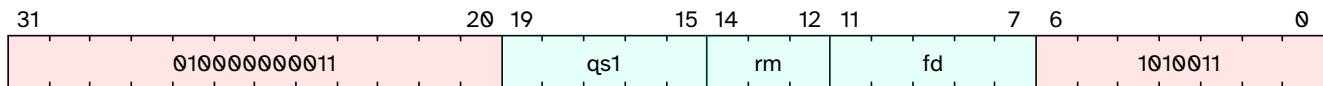
Synopsis

No synopsis available.

Mnemonic

fd, qs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

DRAFT

41.209. fcvt.s.w

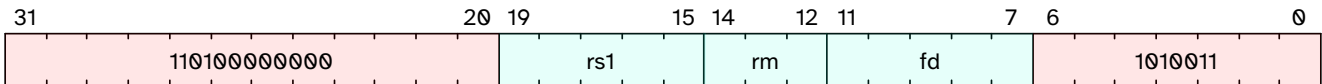
Synopsis

Convert signed 32-bit integer to single-precision float

Mnemonic

fd, rs1

Encoding



Description

Converts a 32-bit signed integer in integer register *rs1* into a floating-point number in floating-point register *fd*.

All floating-point to integer and integer to floating-point conversion instructions round according to the *rm* field. A floating-point register can be initialized to floating-point positive zero using `fcvt.s.w rd, x0`, which will never set any exception flags.

All floating-point conversion instructions set the Inexact exception flag if the rounded result differs from the operand value and the Invalid exception flag is not set.

Included in

Extension	Version
F	~> 2.2.0

41.210. fcvt.s.wu

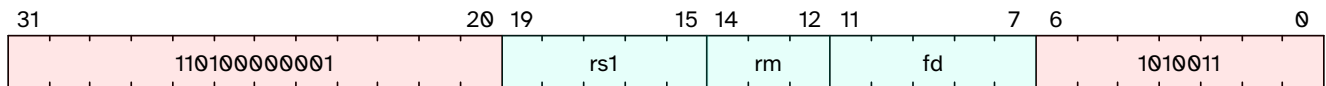
Synopsis

No synopsis available.

Mnemonic

fd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

41.211. fcvt.w.d

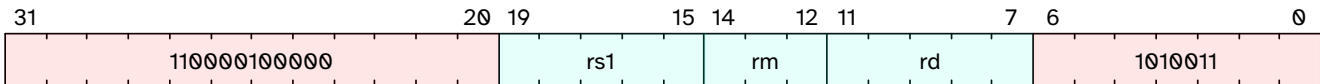
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.212. fcvt.w.h

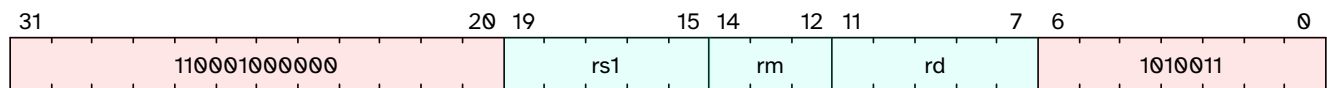
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.213. fcvt.w.q

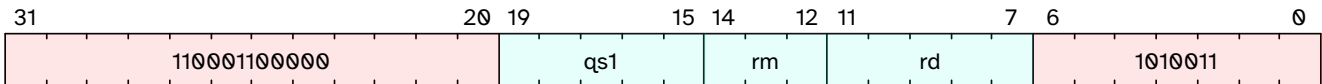
Synopsis

No synopsis available.

Mnemonic

rd, qs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.214. fcvt.w.s

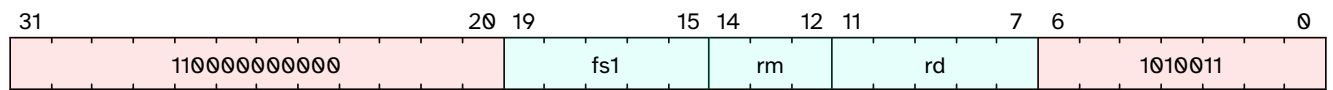
Synopsis

Convert single-precision float to integer word to signed 32-bit integer.

Mnemonic

```
rd, fs1
```

Encoding



Description

Converts a floating-point number in floating-point register *fs1* to a signed 32-bit integer indicates integer register *rd*.

For XLEN >32, [fcvt.w.s](#) sign-extends the 32-bit result to the destination register width.

If the rounded result is not representable as a 32-bit signed integer, it is clipped to the nearest value and the invalid flag is set.

The range of valid inputs and behavior for invalid inputs are:

	Value
Minimum valid input (after rounding)	-2^{31}
Maximum valid input (after rounding)	$2^{31} - 1$
Output for out-of-range negative input	-2^{31}
Output for $-\infty$	-2^{31}
Output for out-of-range positive input	$2^{31} - 1$
Output for $+\infty$ for NaN	$2^{31} - 1$

All floating-point to integer and integer to floating-point conversion instructions round according to the *rm* field. A floating-point register can be initialized to floating-point positive zero using `fcvt.s.w rd, x0`, which will never set any exception flags.

All floating-point conversion instructions set the Inexact exception flag if the rounded result differs from the operand value and the Invalid exception flag is not set.

Included in

Extension	Version
F	~> 2.2.0

41.215. fcvt.wu.d

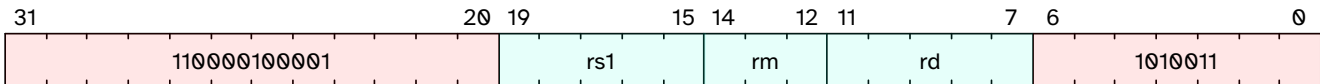
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.216. fcvt.wu.h

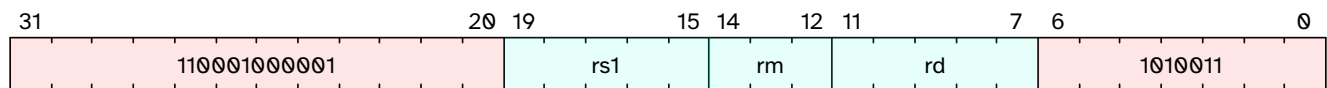
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.217. fcvt.wu.q

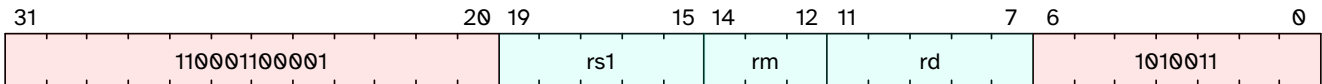
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.218. fcvt.wu.s

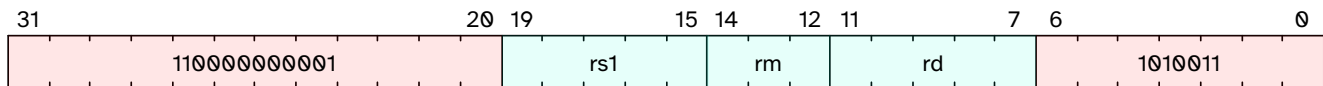
Synopsis

No synopsis available.

Mnemonic

rd, fs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

DRAFT

41.219. fcvtnmod.w.d

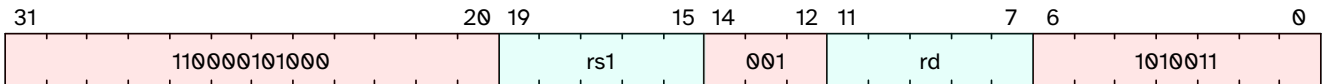
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfa	~> 1.0.0

41.220. fdiv.d

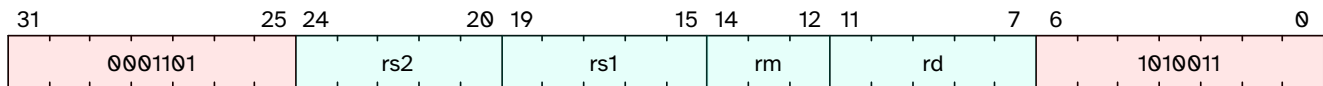
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.221. fdiv.h

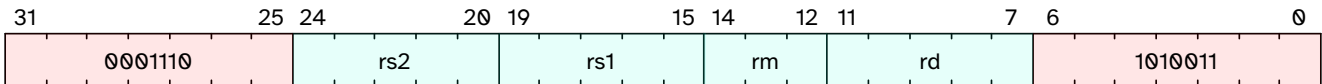
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.222. fdiv.q

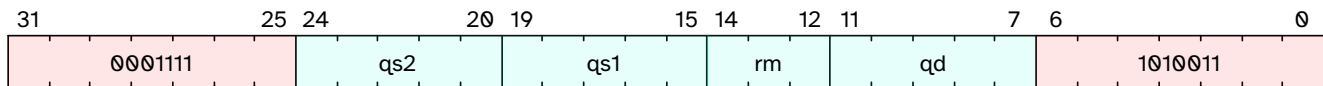
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.223. fdiv.s

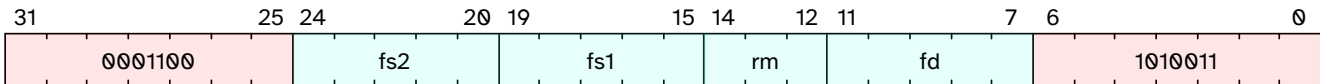
Synopsis

No synopsis available.

Mnemonic

fd, fs1, fs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

41.224. fence

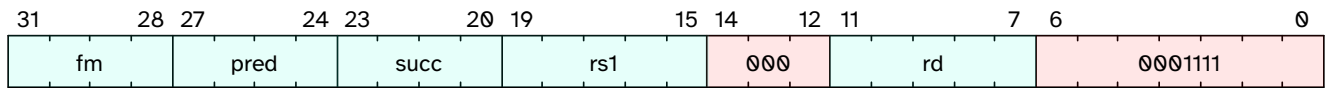
Synopsis

Memory ordering fence

Mnemonic

TODO

Encoding



Description

Orders memory operations.

The `fence` instruction is used to order device I/O and memory accesses as viewed by other RISC-V harts and external devices or coprocessors. Any combination of device input (I), device output (O), memory reads (R), and memory writes (W) may be ordered with respect to any combination of the same. Informally, no other RISC-V hart or external device can observe any operation in the *successor* set following a `fence` before any operation in the *predecessor* set preceding the `fence`.

The predecessor and successor fields have the same format to specify operation types:

pred				succ			
27	26	25	24	23	22	21	20
PI	PO	PR	PW	SI	SO	SR	SW

Table 101. Fence mode encoding

fm field	Mnemonic	Meaning
0000	none	Normal Fence
1000	TSO	With FENCE RW,RW: exclude write-to-read ordering; otherwise: Reserved for future use.
other		Reserved for future use.

When the mode field `fm` is 0001 and both the predecessor and successor sets are 'RW', then the instruction acts as a special-case `fence.tso`. `fence.tso` orders all load operations in its predecessor set before all memory operations in its successor set, and all store operations in its predecessor set before all store operations in its successor set. This leaves non-AMO store operations in the `fence.tso`'s predecessor set unordered with non-AMO loads in its successor set.

When mode field `fm` is not 0001, or when mode field `fm` is 0001 but the `pred` and `succ` fields are not both 'RW' (0x3), then the fence acts as a baseline fence (e.g., `fm` is effectively 0000). This is unaffected by the FIOM bits, described below (implicit promotion does not change how `fence.tso` is decoded).

The `rs1` and `rd` fields are unused and ignored.

In modes other than M-mode, `fence` is further affected by `menvcfg.FIOM`, `senvcfg.FIOM`<% if ext?:(H)

%>, and/or [henvcfg.FIOM](#)<% end %> as follows:

Table 102. Effective PR/PW/SR/SW in (H)S-mode

Table 103. Effective PR/PW/SR/SW in U-mode

<%- if ext?:(H) -%> .Effective PR/PW/SR/SW in VS-mode and VU-mode

<%- end -%>

Included in

Extension	Version
I	~> 2.1.0

DRAFT

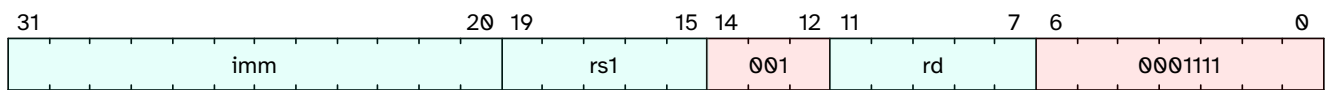
41.225. fence.i

Synopsis

Instruction fence

Mnemonic

Encoding



Description

The FENCE.I instruction is used to synchronize the instruction and data streams. RISC-V does not guarantee that stores to instruction memory will be made visible to instruction fetches on a RISC-V hart until that hart executes a FENCE.I instruction. A FENCE.I instruction ensures that a subsequent instruction fetch on a RISC-V hart will see any previous data stores already visible to the same RISC-V hart. FENCE.I does *not* ensure that other RISC-V harts' instruction fetches will observe the local hart's stores in a multiprocessor system. To make a store to instruction memory visible to all RISC-V harts, the writing hart also has to execute a data FENCE before requesting that all remote RISC-V harts execute a FENCE.I.

The unused fields in the FENCE.I instruction, *imm*[11:0], *rs1*, and *rd*, are reserved for finer-grain fences in future extensions. For forward compatibility, base implementations shall ignore these fields, and standard software shall zero these fields.



Because FENCE.I only orders stores with a hart's own instruction fetches, application code should only rely upon FENCE.I if the application thread will not be migrated to a different hart. The EEI can provide mechanisms for efficient multiprocessor instruction-stream synchronization.

Included in

Extension	Version
Zifencei	~> 2.0.0

41.226. feq.d

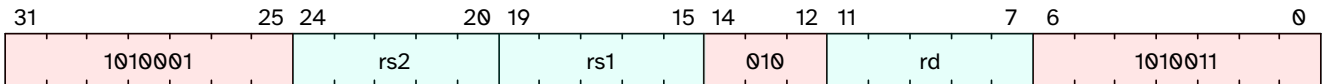
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.227. feq.h

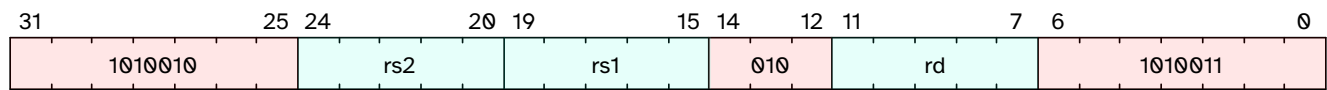
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.228. feq.q

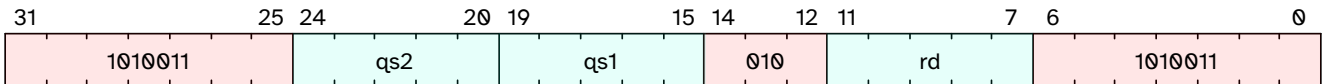
Synopsis

No synopsis available.

Mnemonic

rd, qs1, qs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.229. feq.s

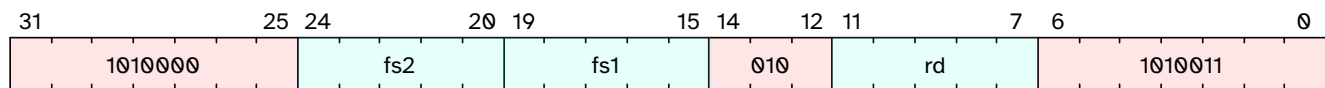
Synopsis

Single-precision floating-point equal

Mnemonic

rd, fs1, fs2

Encoding



Description

Writes 1 to *rd* if *fs1* and *fs2* are equal, and 0 otherwise.

If either operand is NaN, the result is 0 (not equal). If either operand is a signaling NaN, the invalid flag is set.

Positive zero is considered equal to negative zero.

Included in

Extension	Version
F	~> 2.2.0

41.230. fld

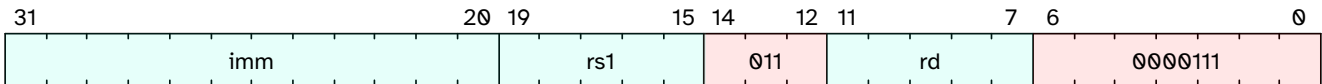
Synopsis

No synopsis available.

Mnemonic

rd, rs1, imm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.231. fle.d

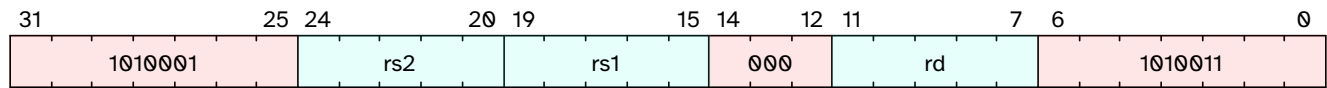
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

DRAFT

41.232. fle.h

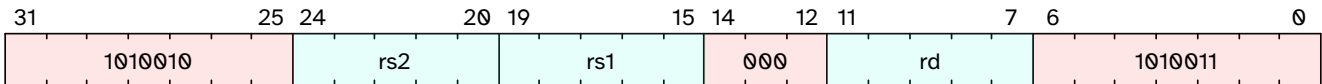
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.233. fle.q

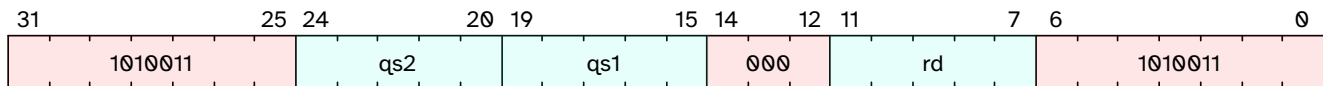
Synopsis

No synopsis available.

Mnemonic

rd, qs1, qs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.234. fle.s

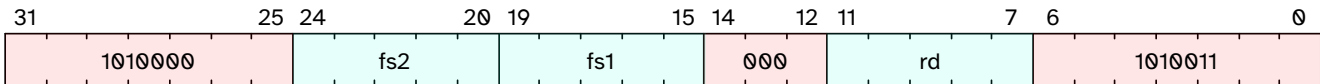
Synopsis

Single-precision floating-point less than or equal

Mnemonic

rd, fs1, fs2

Encoding



Description

Writes 1 to *rd* if *fs1* is less than or equal to *fs2*, and 0 otherwise.

If either operand is NaN, the result is 0 (not equal). If either operand is a NaN (signaling or quiet), the invalid flag is set.

Positive zero and negative zero are considered equal.

Included in

Extension	Version
F	~> 2.2.0

41.235. fleq.d

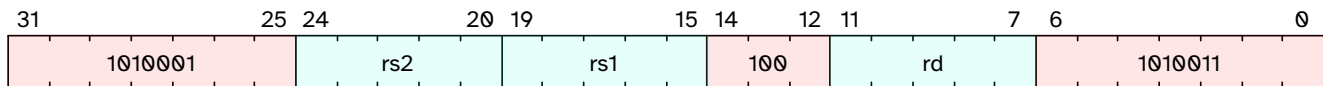
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfa	~> 1.0.0

41.236. fleq.h

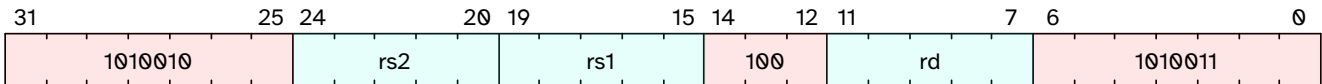
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0
Zfh	~> 1.0.0

41.237. fleq.q

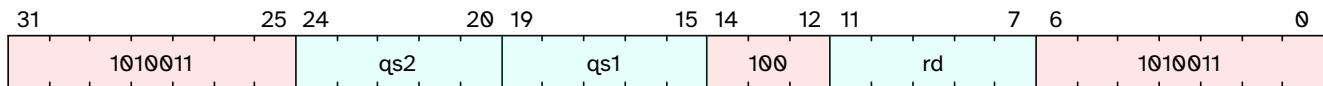
Synopsis

No synopsis available.

Mnemonic

rd, qs1, qs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0
Zfa	~> 1.0.0

41.238. fleq.s

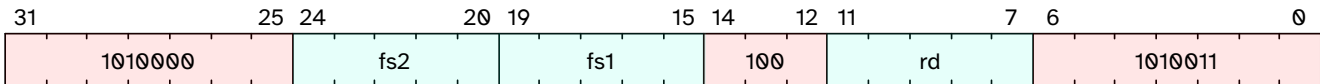
Synopsis

No synopsis available.

Mnemonic

rd, fs1, fs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0

41.239. flh

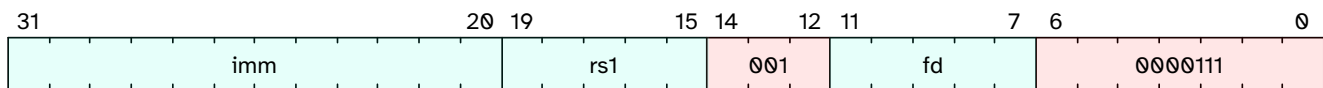
Synopsis

Half-precision floating-point load

Mnemonic

```
fd, imm(rs1)
```

Encoding



Description

The `flh` instruction loads a single-precision floating-point value from memory at address $rs1 + imm$ into floating-point register rd .

`flh` does not modify the bits being transferred; in particular, the payloads of non-canonical NaNs are preserved.

`flh` is only guaranteed to execute atomically if the effective address is naturally aligned.

Included in

Extension	Version
Zfh	~> 1.0.0
Zfhmin	~> 1.0.0
Zhinx	~> 1.0.0

41.240. fli.d

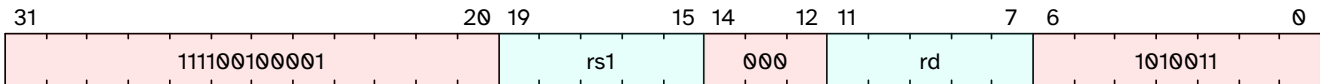
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfa	~> 1.0.0

41.241. fli.h

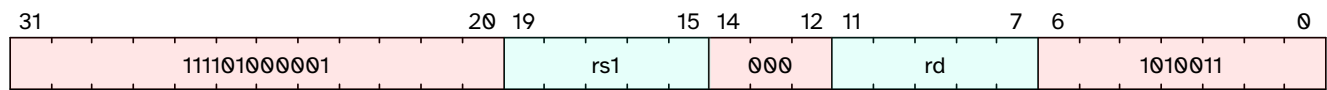
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0
Zfh	~> 1.0.0

41.242. fli.q

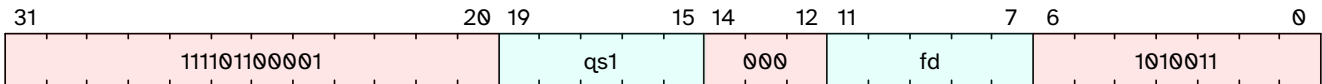
Synopsis

No synopsis available.

Mnemonic

fd, qs1

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0
Zfa	~> 1.0.0

41.243. fli.s

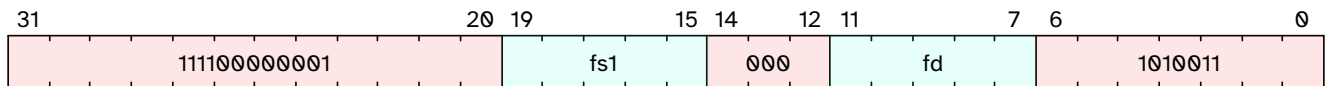
Synopsis

No synopsis available.

Mnemonic

fd, fs1

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0

41.244. flq

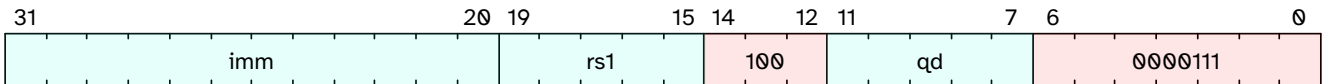
Synopsis

No synopsis available.

Mnemonic

qd, rs1, imm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.245. ftt.d

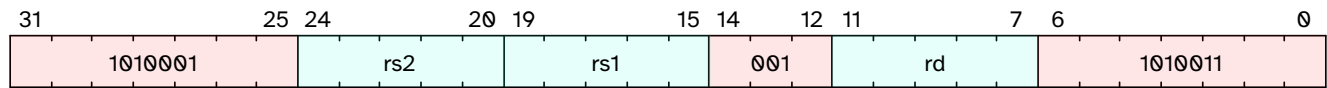
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.246. flt.h

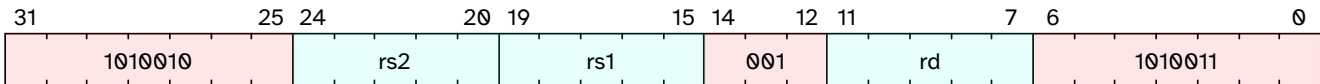
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.247. flt.q

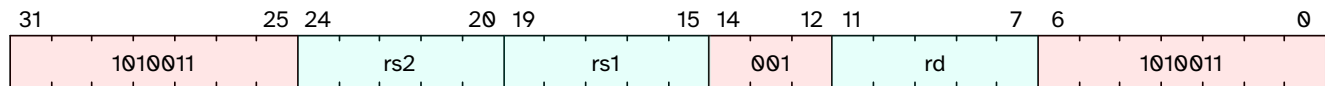
Synopsis

No synopsis available.

Mnemonic

rd, qs1, qs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

DRAFT

41.248. ftt.s

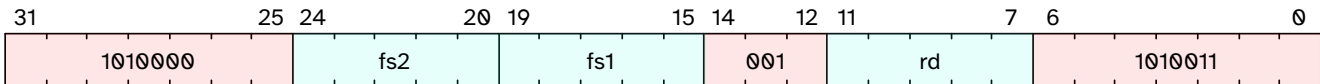
Synopsis

Single-precision floating-point less than

Mnemonic

rd, fs1, fs2

Encoding



Description

Writes 1 to *rd* if *fs1* is less than *fs2*, and 0 otherwise.

If either operand is NaN, the result is 0 (not equal). If either operand is a NaN (signaling or quiet), the invalid flag is set.

Included in

Extension	Version
F	~> 2.2.0

41.249. fltq.d

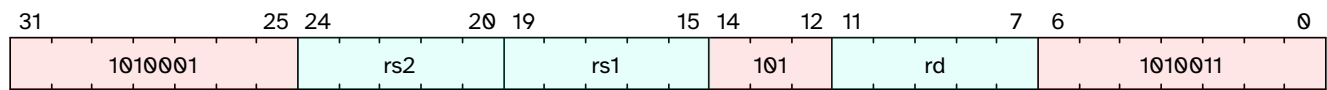
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfa	~> 1.0.0

41.250. fltq.h

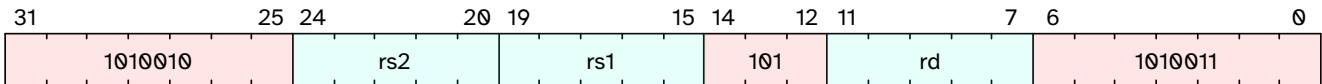
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0
Zfh	~> 1.0.0

41.251. fltq.q

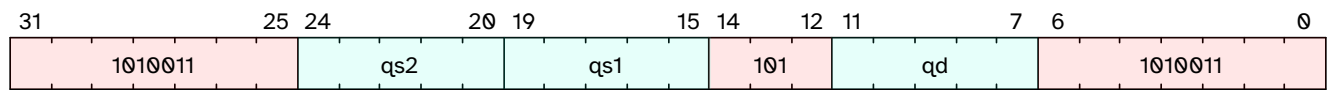
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0
Zfa	~> 1.0.0

41.252. fltq.s

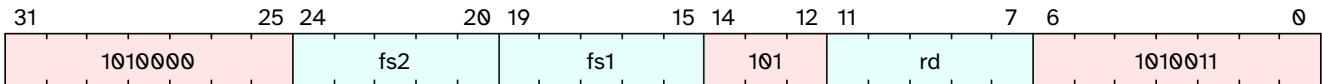
Synopsis

No synopsis available.

Mnemonic

rd, fs1, fs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0

41.253. flw

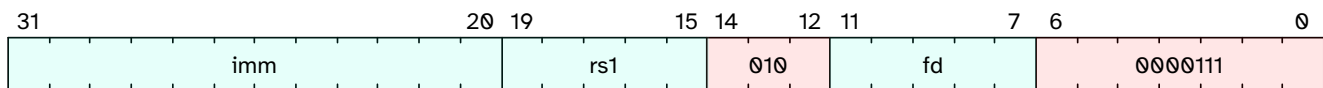
Synopsis

Single-precision floating-point load

Mnemonic

```
fd, rs1, imm
```

Encoding



Description

The `flw` instruction loads a single-precision floating-point value from memory at address $rs1 + imm$ into floating-point register fd .

`flw` does not modify the bits being transferred; in particular, the payloads of non-canonical NaNs are preserved.

Included in

Extension	Version
F	~> 2.2.0

41.254. fmadd.d

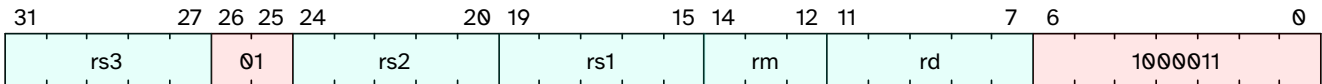
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.255. fmadd.h

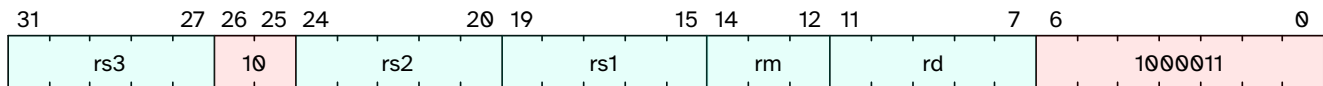
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

DRAFT

41.256. fmadd.q

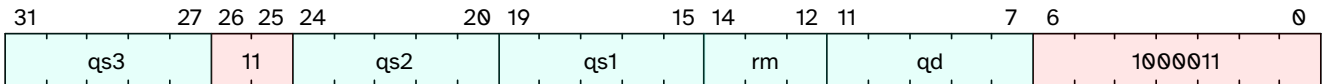
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2, qs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.257. fmadd.s

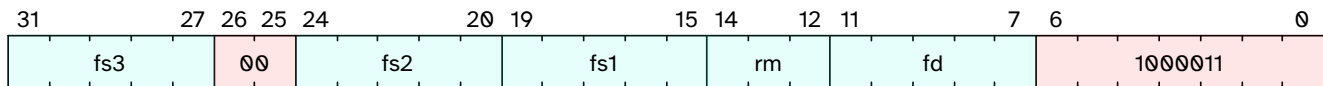
Synopsis

No synopsis available.

Mnemonic

fd, fs1, fs2, fs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

DRAFT

41.258. fmax.d

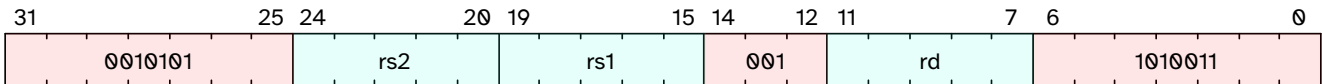
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.259. fmax.h

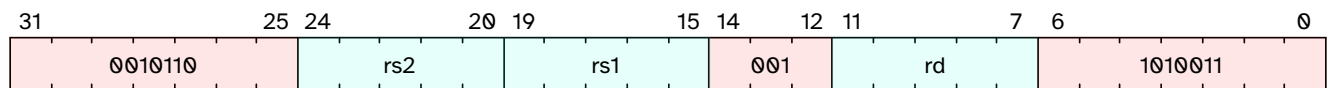
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.260. fmax.q

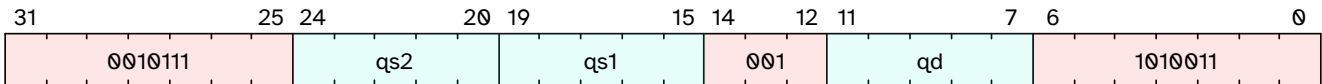
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.261. fmax.s

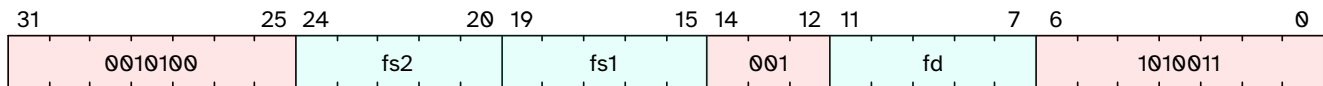
Synopsis

No synopsis available.

Mnemonic

fd, fs1, fs2

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

DRAFT

41.262. fmaxm.d

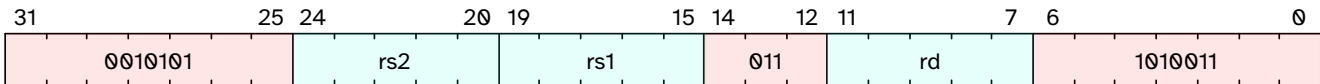
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfa	~> 1.0.0

41.263. fmaxm.h

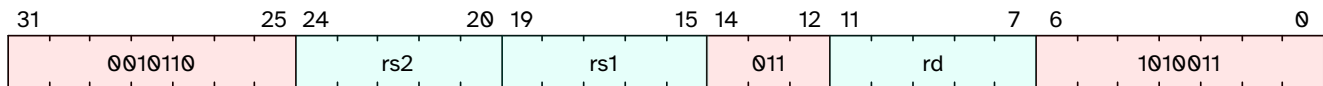
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0
Zfh	~> 1.0.0

41.264. fmaxm.q

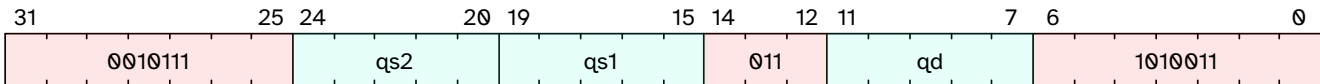
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0
Zfa	~> 1.0.0

41.265. fmaxm.s

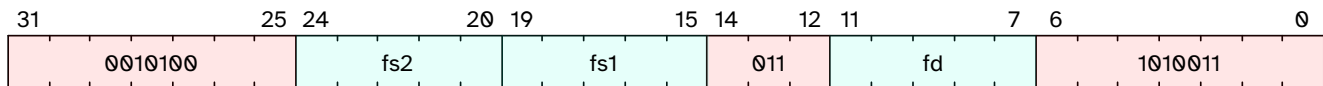
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0

41.266. fmin.d

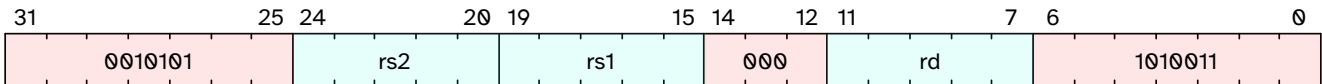
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.267. fmin.h

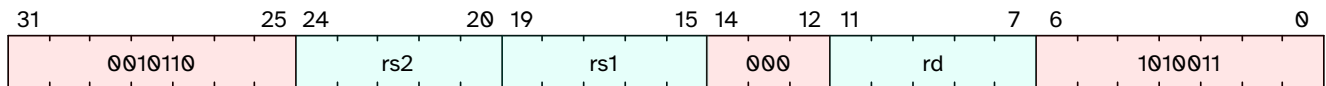
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.268. fmin.q

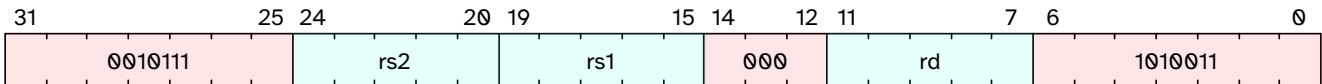
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.269. fmin.s

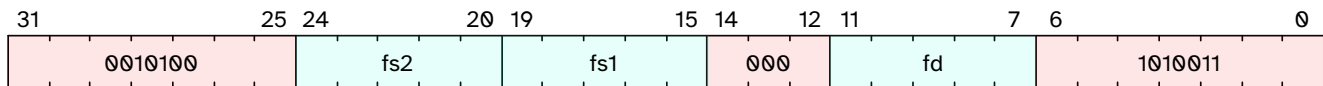
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

41.270. fminm.d

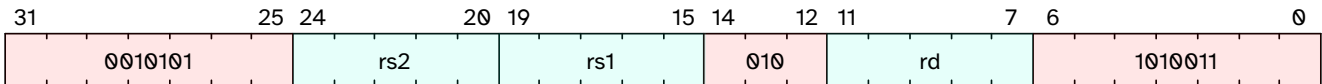
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfa	~> 1.0.0

41.271. fminm.h

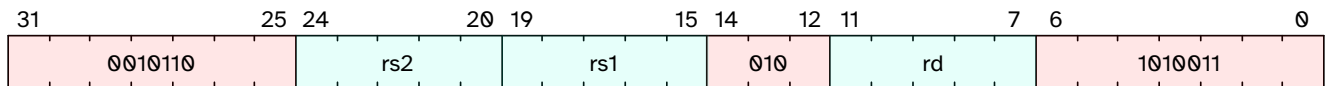
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0
Zfh	~> 1.0.0

41.272. fminm.q

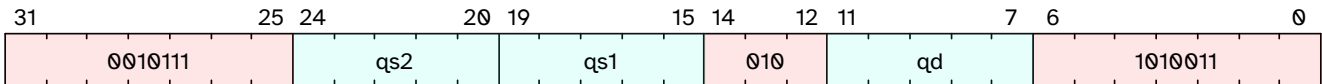
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0
Zfa	~> 1.0.0

41.273. fminm.s

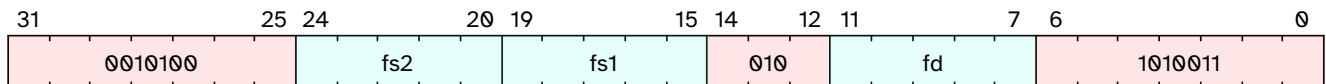
Synopsis

No synopsis available.

Mnemonic

fd, fs1, fs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0

41.274. fmsub.d

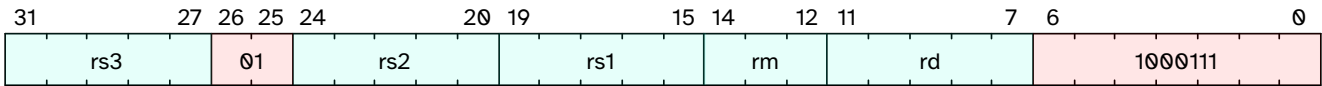
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.275. fmsub.h

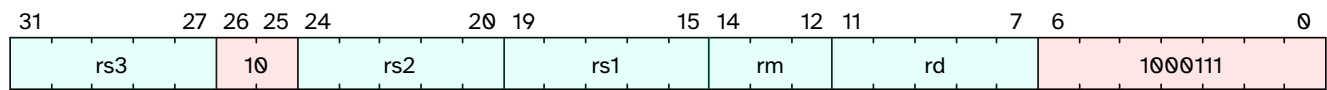
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.276. fmsub.q

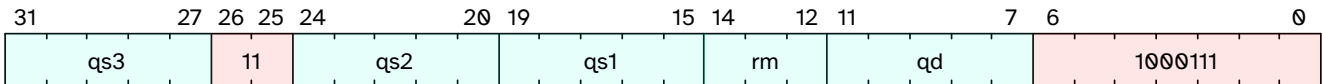
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2, qs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.277. fmsub.s

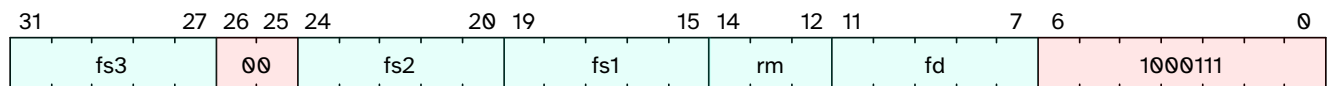
Synopsis

No synopsis available.

Mnemonic

fd, fs1, fs2, fs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

41.278. fmul.d

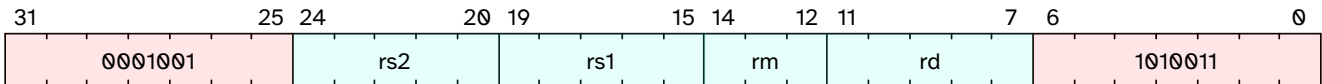
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.279. fmul.h

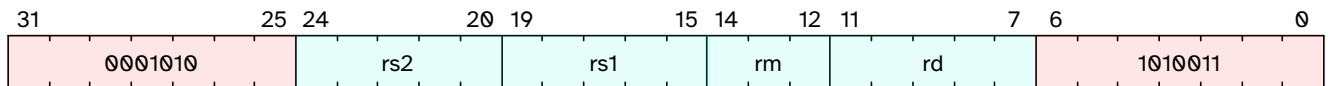
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.280. fmul.q

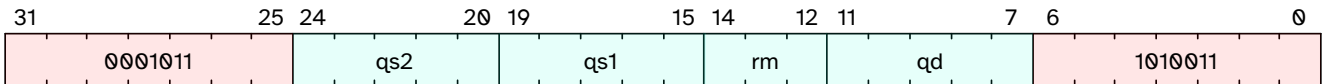
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.281. fmul.s

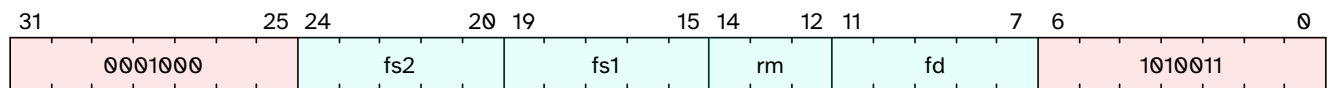
Synopsis

No synopsis available.

Mnemonic

fd, fs1, fs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

41.282. fmv.d.x

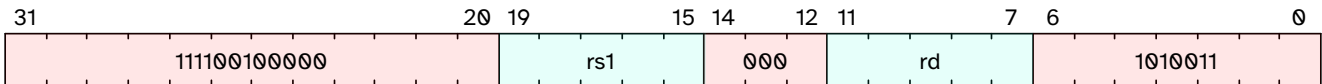
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.283. fmv.h.x

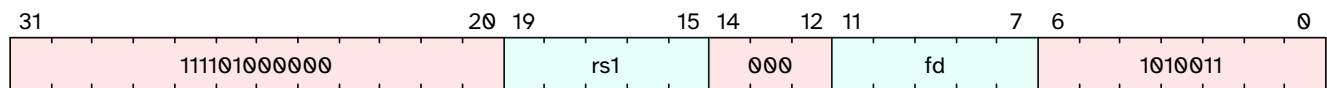
Synopsis

Half-precision floating-point move from integer

Mnemonic

fd, rs1

Encoding



Description

Moves the half-precision value encoded in IEEE 754-2008 standard encoding from the lower 16 bits of integer register `rs1` to the floating-point register `fd`. The bits are not modified in the transfer, and in particular, the payloads of non-canonical NaNs are preserved.

Included in

Extension	Version
F	~> 2.2.0

41.284. fmv.w.x

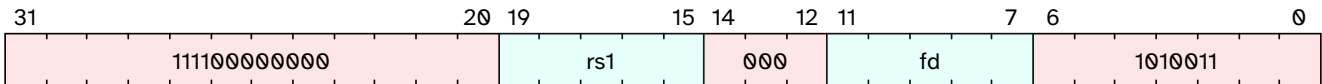
Synopsis

Single-precision floating-point move from integer

Mnemonic

fd, rs1

Encoding



Description

Moves the single-precision value encoded in IEEE 754-2008 standard encoding from the lower 32 bits of integer register rs1 to the floating-point register fd. The bits are not modified in the transfer, and in particular, the payloads of non-canonical NaNs are preserved.

Included in

Extension	Version
F	~> 2.2.0

41.285. fmv.x.d

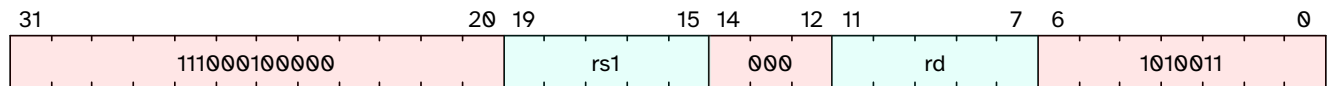
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

DRAFT

41.286. fmv.x.h

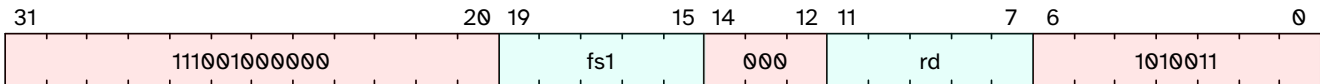
Synopsis

Move half-precision value from floating-point to integer register

Mnemonic

```
rd, fs1
```

Encoding



Description

Moves the half-precision value in floating-point register rs1 represented in IEEE 754-2008 encoding to the lower 16 bits of integer register rd.

The bits are not modified in the transfer, and in particular, the payloads of non-canonical NaNs are preserved.

The highest XLEN-16 bits of the destination register are filled with copies of the floating-point number’s sign bit.

Included in

Extension	Version
Zfh	~> 1.0.0
Zfhmin	~> 1.0.0
Zhinx	~> 1.0.0

41.287. fmv.x.w

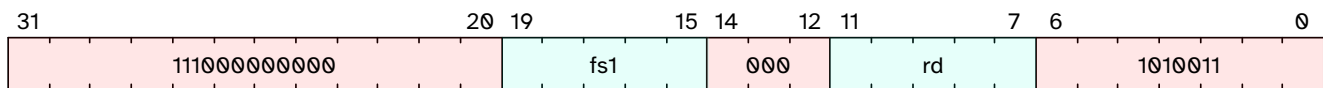
Synopsis

Move single-precision value from floating-point to integer register

Mnemonic

rd, fs1

Encoding



Description

Moves the single-precision value in floating-point register rs1 represented in IEEE 754-2008 encoding to the lower 32 bits of integer register rd. The bits are not modified in the transfer, and in particular, the payloads of non-canonical NaNs are preserved. For RV64, the higher 32 bits of the destination register are filled with copies of the floating-point number’s sign bit.

Included in

Extension	Version
F	~> 2.2.0

41.288. fmvh.x.d

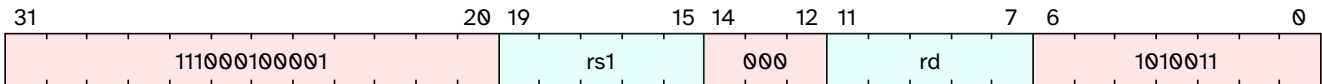
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfa	~> 1.0.0

41.289. fmvh.x.q

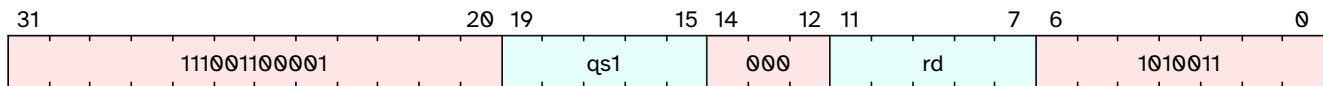
Synopsis

No synopsis available.

Mnemonic

rd, qs1

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0
Zfa	~> 1.0.0

41.290. fmv.d.x

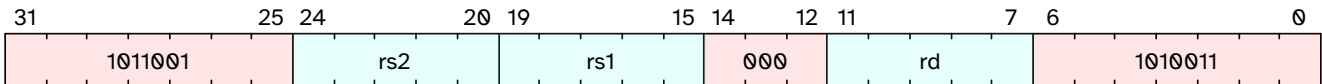
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfa	~> 1.0.0

41.291. fmvq.q.x

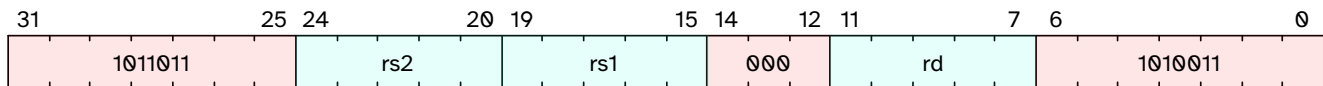
Synopsis

No synopsis available.

Mnemonic

qd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0
Zfa	~> 1.0.0

41.292. fnmadd.d

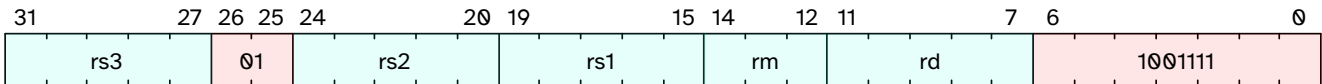
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.293. fnmadd.h

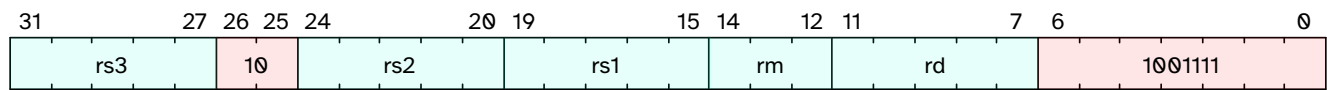
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.294. fnmadd.q

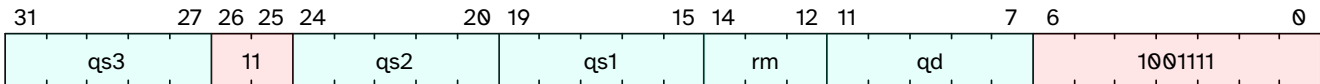
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2, qs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.295. fnmadd.s

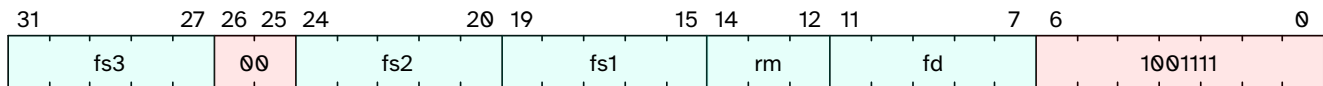
Synopsis

No synopsis available.

Mnemonic

fd, fs1, fs2, fs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

DRAFT

41.296. fnmsub.d

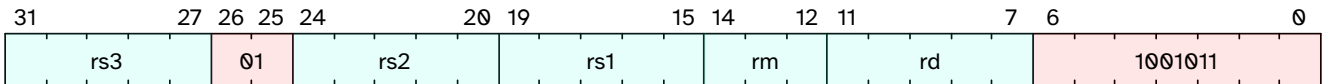
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.297. fnmsub.h

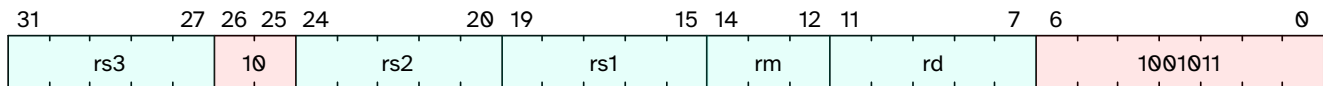
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.298. fnmsub.q

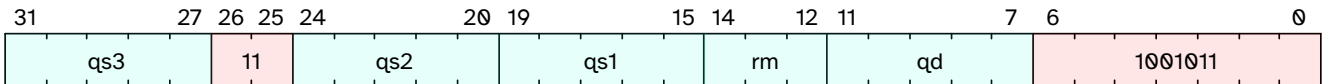
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2, qs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.299. fnmsub.s

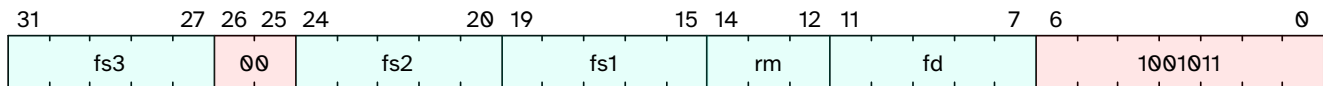
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rs3, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

DRAFT

41.300. fround.d

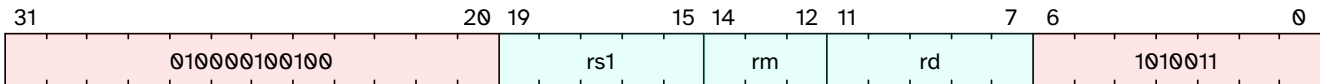
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfa	~> 1.0.0

41.301. fround.h

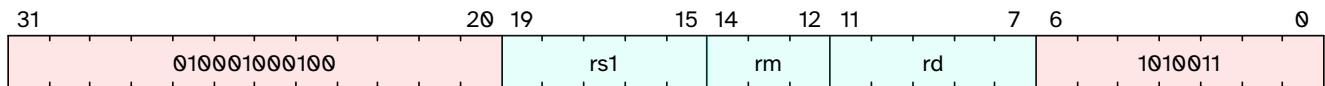
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0
Zfh	~> 1.0.0

41.302. fround.q

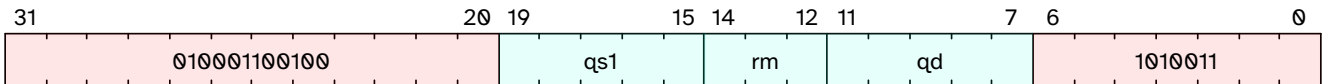
Synopsis

No synopsis available.

Mnemonic

qd, qs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0
Zfa	~> 1.0.0

41.303. fround.s

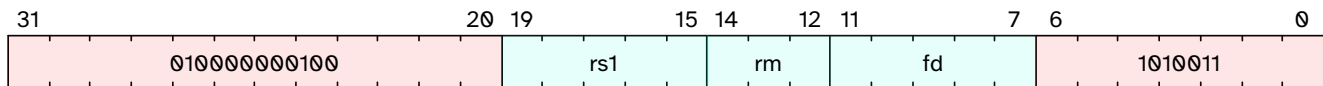
Synopsis

No synopsis available.

Mnemonic

fd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0

41.304. froundnx.d

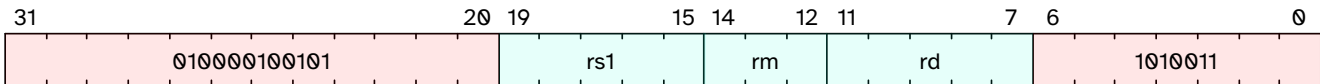
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0
Zfa	~> 1.0.0

41.305. froundnx.h

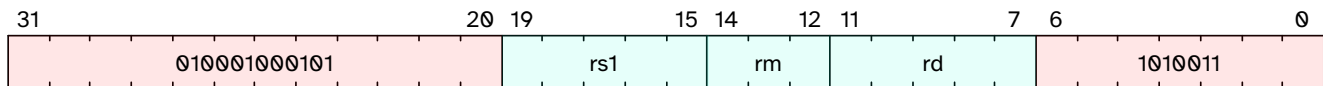
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0
Zfh	~> 1.0.0

41.306. froundnx.q

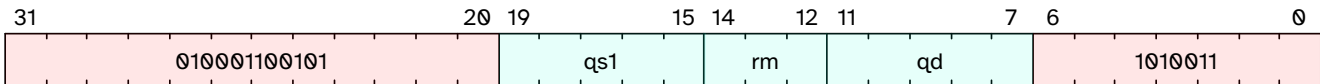
Synopsis

No synopsis available.

Mnemonic

qd, qs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0
Zfa	~> 1.0.0

41.307. froundnx.s

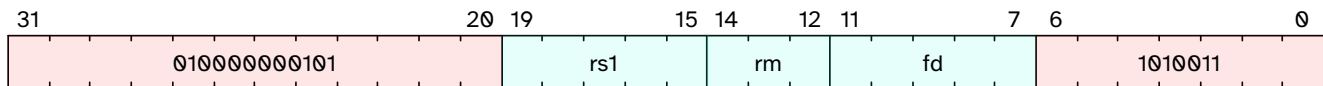
Synopsis

No synopsis available.

Mnemonic

fd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfa	~> 1.0.0

41.308. fsd

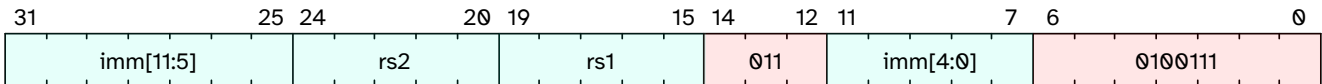
Synopsis

No synopsis available.

Mnemonic

rs1, rs2, imm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.309. fsgnj.d

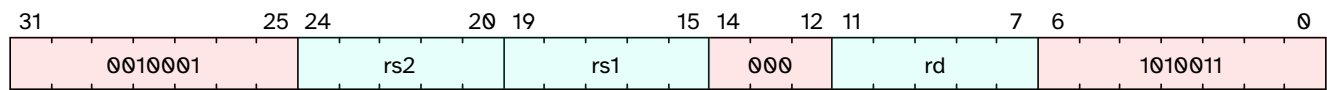
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

41.310. fsgnj.h

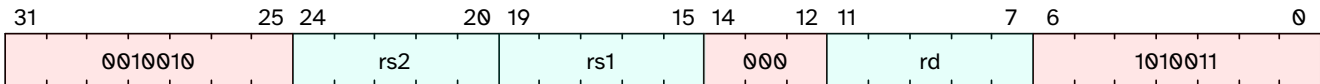
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.311. fsgnj.q

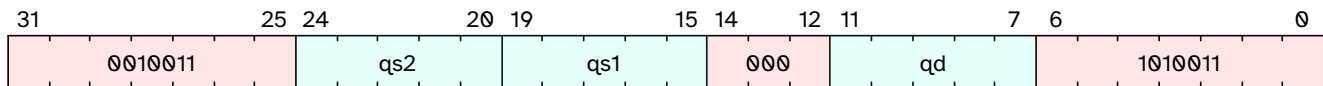
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

DRAFT

41.312. fsgnj.s

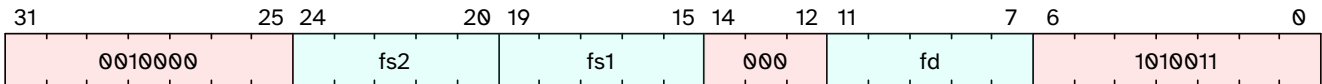
Synopsis

Single-precision sign inject

Mnemonic

fd, fs1, fs2

Encoding



Description

Writes *fd* with sign bit of *fs2* and the exponent and mantissa of *fs1*.

Sign-injection instructions do not set floating-point exception flags, nor do they canonicalize NaNs.

Included in

Extension	Version
F	~> 2.2.0

41.313. fsgnjn.d

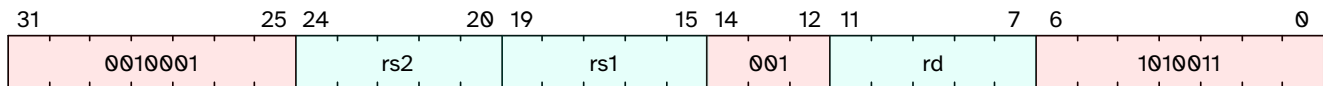
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

DRAFT

41.314. fsgnjn.h

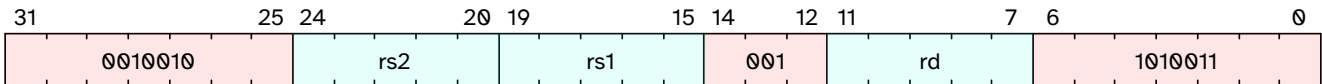
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.315. fsgnjn.q

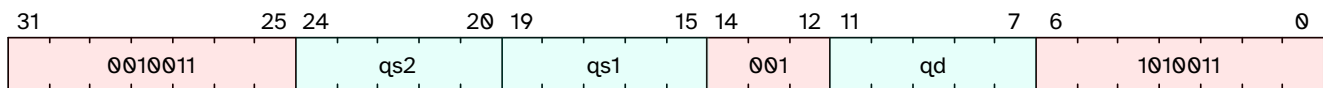
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

DRAFT

41.316. fsgnfn.s

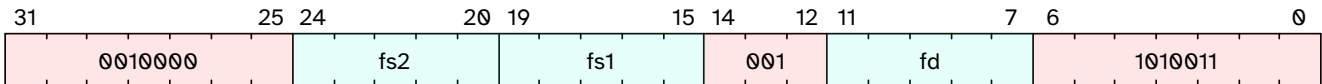
Synopsis

Single-precision sign inject negate

Mnemonic

fd, fs1, fs2

Encoding



Description

Writes *fd* with the opposite of the sign bit of *fs2* and the exponent and mantissa of *fs1*.

Sign-injection instructions do not set floating-point exception flags, nor do they canonicalize NaNs.

Included in

Extension	Version
F	~> 2.2.0

41.317. fsgnjx.d

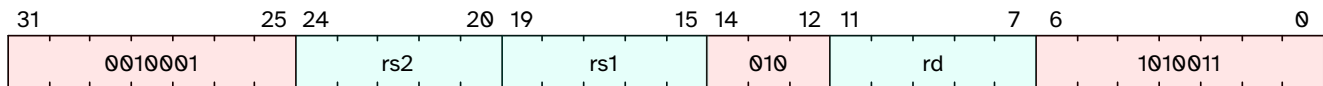
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

DRAFT

41.318. fsgnjx.h

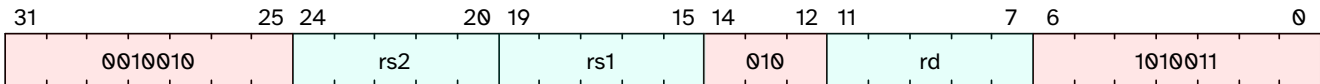
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.319. fsgnjx.q

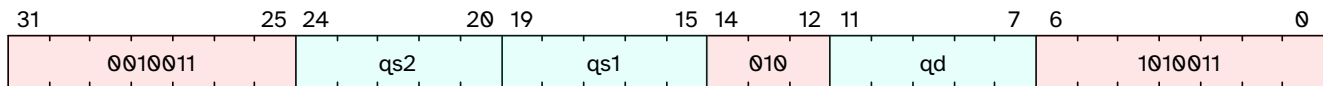
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

DRAFT

41.320. fsgnjx.s

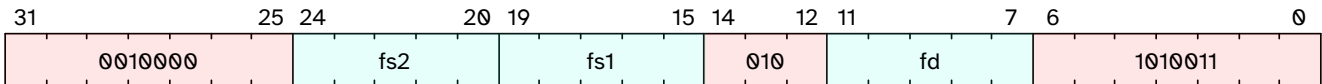
Synopsis

Single-precision sign inject exclusive or

Mnemonic

fd, fs1, fs2

Encoding



Description

Writes *fd* with the xor of the sign bits of *fs2* and *fs1* and the exponent and mantissa of *fs1*.

Sign-injection instructions do not set floating-point exception flags, nor do they canonicalize NaNs.

Included in

Extension	Version
F	~> 2.2.0

41.321. fsh

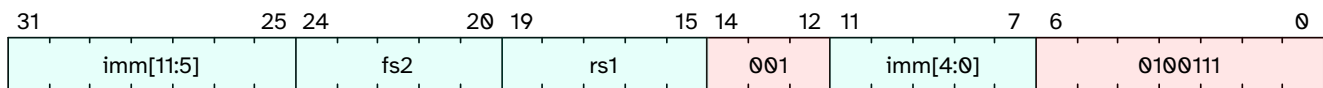
Synopsis

Half-precision floating-point store

Mnemonic

```
fs2, imm(rs1)
```

Encoding



Description

The `fsh` instruction stores a half-precision floating-point value from register *rd* to memory at address *rs1* + *imm*.

`fsh` does not modify the bits being transferred; in particular, the payloads of non-canonical NaNs are preserved.

`fsh` ignores all but the lower 16 bits in *rs2*.

`fsh` is only guaranteed to execute atomically if the effective address is naturally aligned.

Included in

Extension	Version
Zfh	~> 1.0.0
Zfhmin	~> 1.0.0
Zhinx	~> 1.0.0

41.322. fsq

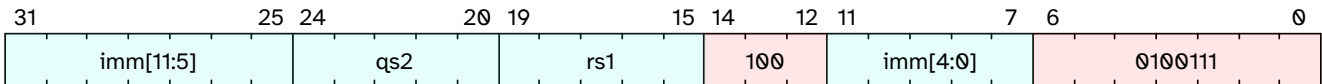
Synopsis

No synopsis available.

Mnemonic

rs1, qs2, imm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.323. fsqrt.d

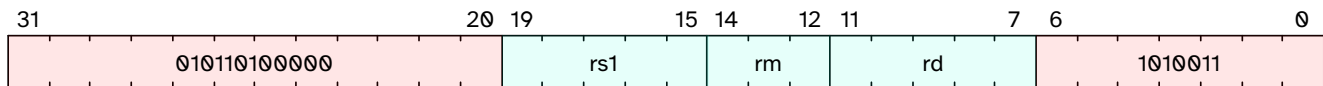
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

DRAFT

41.324. fsqrt.h

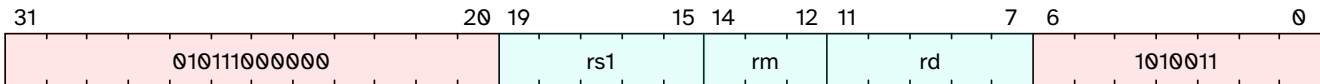
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.325. fsqrt.q

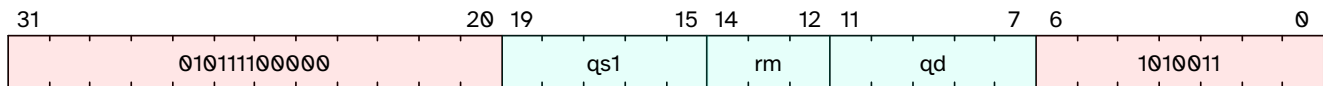
Synopsis

No synopsis available.

Mnemonic

qd, qs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.326. fsqrt.s

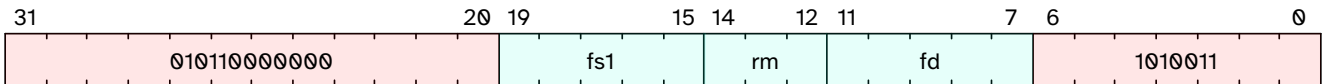
Synopsis

No synopsis available.

Mnemonic

fd, fs1, rm

Encoding



Description

No description available.

Included in

Extension	Version
F	~> 2.2.0

41.327. fsub.d

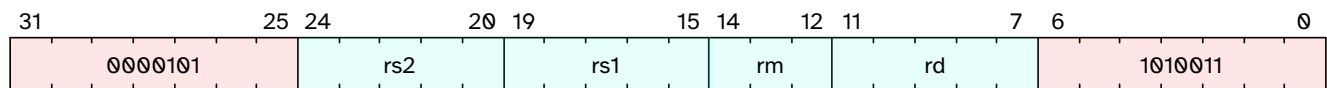
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
D	~> 2.2.0

DRAFT

41.328. fsub.h

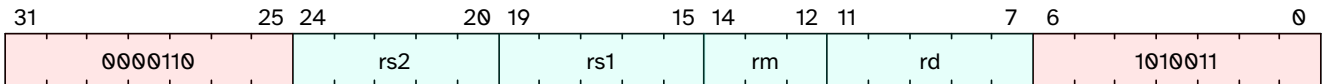
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
Zfh	~> 1.0.0

41.329. fsub.q

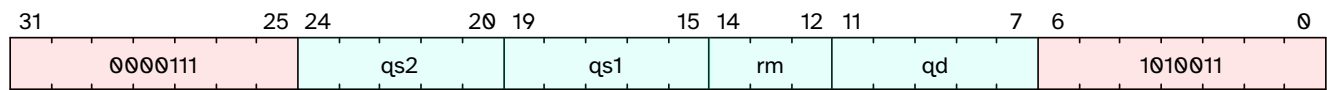
Synopsis

No synopsis available.

Mnemonic

qd, qs1, qs2, rm

Encoding



Description

No description available.

Included in

Extension	Version
Q	~> 1.0.0

41.330. fsub.s

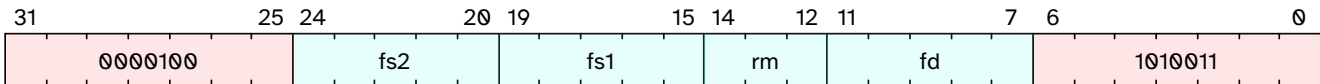
Synopsis

Single-precision floating-point subtraction

Mnemonic

fd, fs1, fs2, rm

Encoding



Description

Do the single-precision floating-point subtraction of fs2 from fs1 and store the result in fd. rm is the dynamic Rounding Mode.

Included in

Extension	Version
F	~> 2.2.0

41.331. fsw

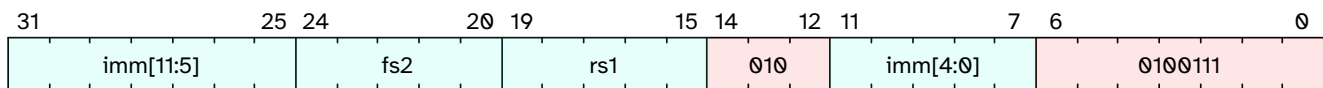
Synopsis

Single-precision floating-point store

Mnemonic

```
fs2, rs1, imm
```

Encoding



Description

The **fsw** instruction stores a single-precision floating-point value in *fs2* to memory at address *rs1* + *imm*.

fsw does not modify the bits being transferred; in particular, the payloads of non-canonical NaNs are preserved.

Included in

Extension	Version
F	~> 2.2.0

41.332. hfence.gvma

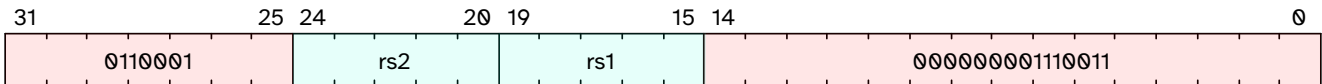
Synopsis

No synopsis available.

Mnemonic

rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

41.333. hfence.vvma

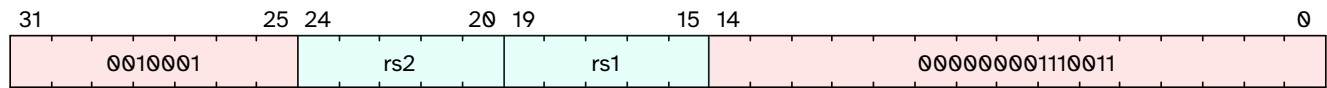
Synopsis

No synopsis available.

Mnemonic

rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

41.334. hinval.gvma

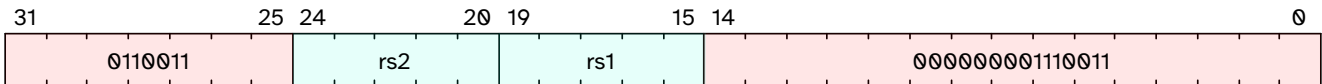
Synopsis

Invalidate cached address translations

Mnemonic

rs1, rs2

Encoding



Description

[hinval.gvma](#) has the same semantics as [sINVAL.vma](#) except that it combines with [sfence.w.inval](#) and [sfence.inval.ir](#) to replace [hfence.gvma](#) and uses VMID instead of ASID.

Included in

Extension	Version
Svinval	~> 1.0.0
H	~> 1.0.0

41.335. hINVAL.VVMA

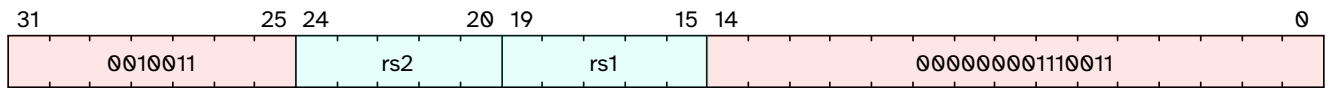
Synopsis

Invalidate cached address translations

Mnemonic

rs1, rs2

Encoding



Description

[hINVAL.VVMA](#) has the same semantics as [SINVAL.VMA](#) except that it combines with [SFENCE.W.INVALID](#) and [SFENCE.INVALID.IR](#) to replace [HFENCE.VVMA](#).

Included in

Extension	Version
SINVAL	~> 1.0.0
H	~> 1.0.0

41.336. hlv.b

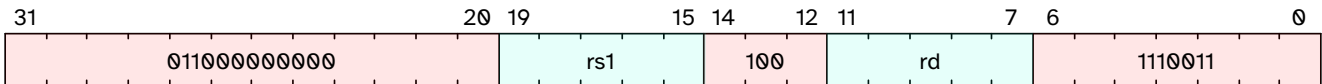
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

41.337. hlv.bu

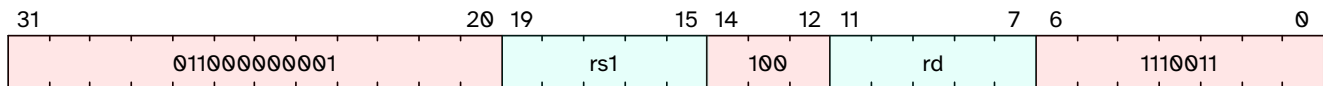
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

DRAFT

41.338. hlv.d

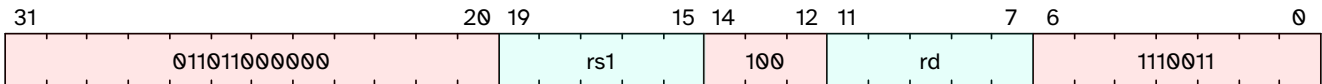
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

41.339. hlv.h

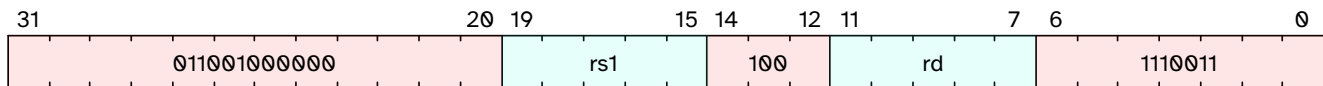
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

DRAFT

41.340. hlv.hu

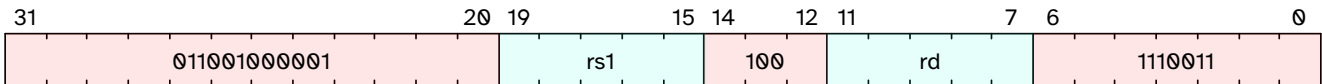
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

41.341. hlv.w

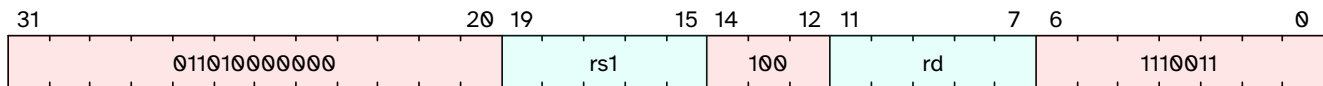
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

DRAFT

41.342. hlv.wu

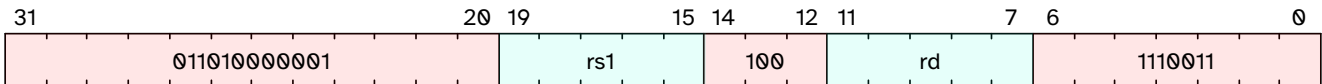
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

41.343. hlvx.hu

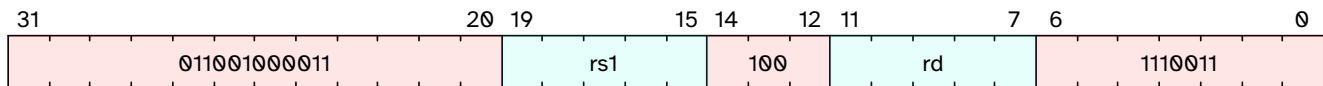
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

DRAFT

41.344. hlvx.wu

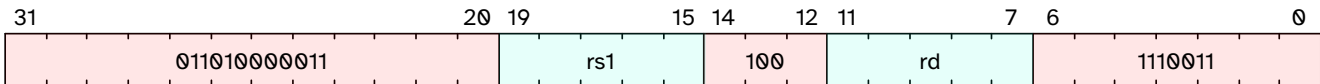
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

41.345. hsv.b

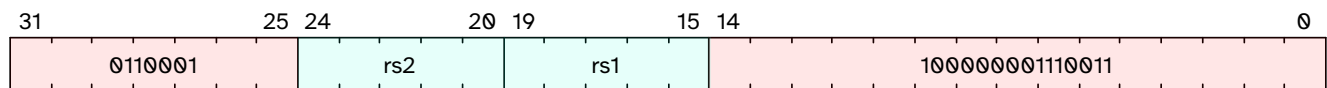
Synopsis

No synopsis available.

Mnemonic

rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

DRAFT

41.346. hsv.d

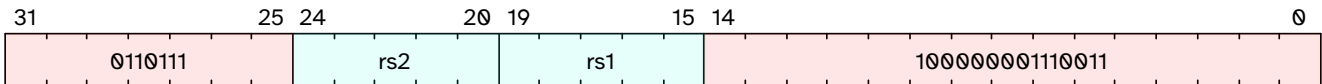
Synopsis

No synopsis available.

Mnemonic

rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

41.347. hsv.h

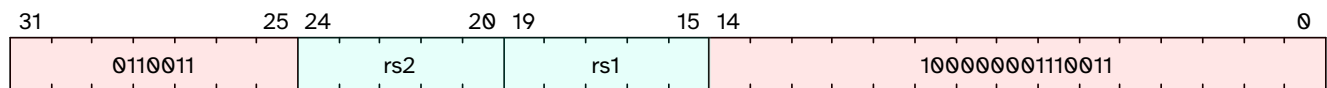
Synopsis

No synopsis available.

Mnemonic

rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

DRAFT

41.348. hsv.w

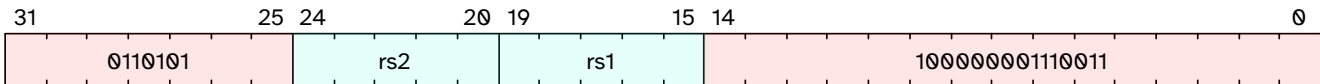
Synopsis

No synopsis available.

Mnemonic

rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
H	~> 1.0.0

41.349. jal

Synopsis

Jump and link

Mnemonic

rd, imm

Encoding



Description

Jump to a PC-relative offset and store the return address in rd.

Included in

Extension	Version
I	~> 2.1.0

41.350. jalr

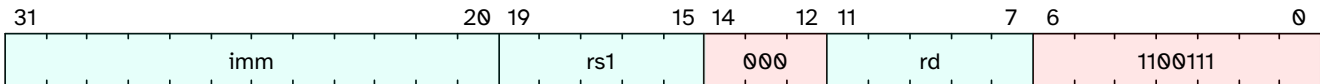
Synopsis

Jump and link register

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Jump to an address formed by adding rs1 to a signed offset then clearing the least significant bit, and store the return address in rd.

Included in

Extension	Version
I	~> 2.1.0

41.351. lb

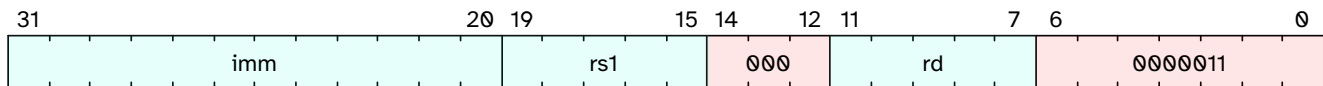
Synopsis

Load byte

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Load 8 bits of data into register *rd* from an address formed by adding *rs1* to a signed offset. Sign extend the result.

Included in

Extension	Version
I	~> 2.1.0

41.352. lb.aq

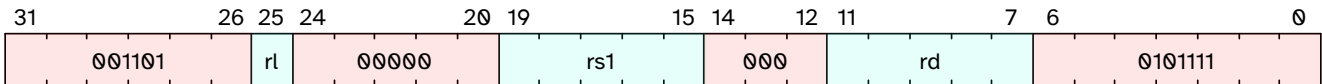
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zalasr	~> 0.3.5

41.353. lbu

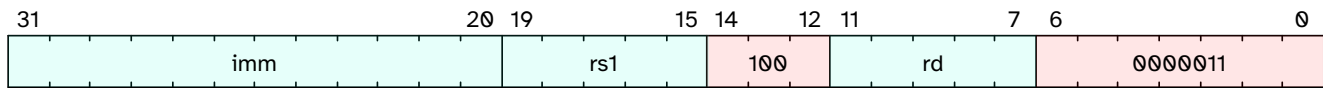
Synopsis

Load byte unsigned

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Load 8 bits of data into register `rd` from an address formed by adding `rs1` to a signed offset. Zero extend the result.

Included in

Extension	Version
I	~> 2.1.0

41.354. ld

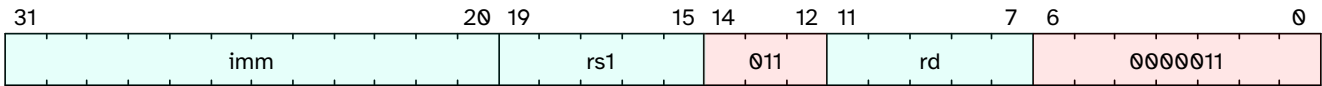
Synopsis

Load doubleword

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Load 64 bits of data into register `rd` from an address formed by adding `rs1` to a signed offset.

Included in

Extension	Version
I	~> 2.1.0

41.355. ld.aq

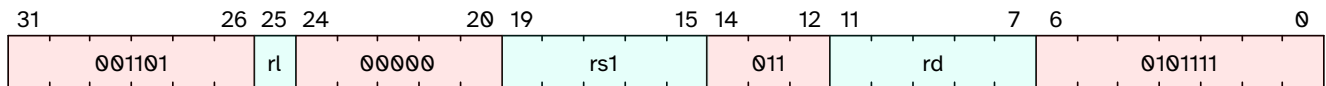
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zalasr	~> 0.3.5

41.356. lh

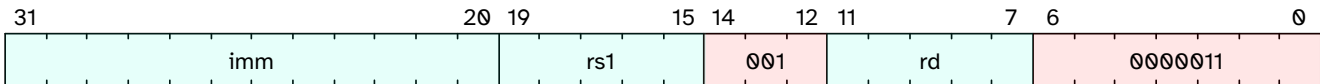
Synopsis

Load halfword

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Load 16 bits of data into register `rd` from an address formed by adding `rs1` to a signed offset. Sign extend the result.

Included in

Extension	Version
I	~> 2.1.0

41.357. lh.aq

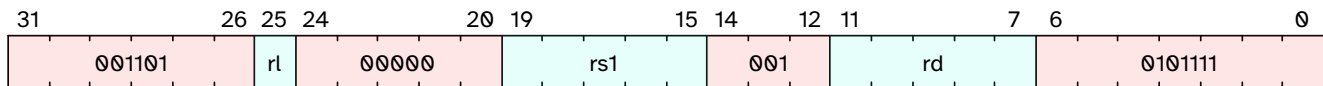
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zalasr	~> 0.3.5

DRAFT

41.358. lhu

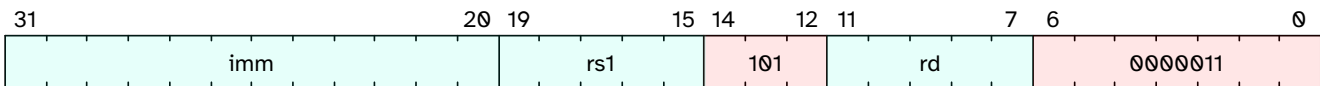
Synopsis

Load halfword unsigned

Mnemonic

```
rd, imm(rs1)
```

Encoding



Description

Load 16 bits of data into register `rd` from an address formed by adding `rs1` to a signed offset. Zero extend the result.

Included in

Extension	Version
I	~> 2.1.0

41.359. lpad

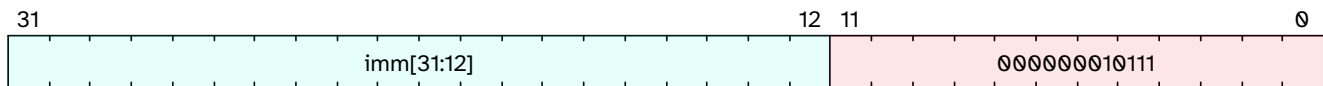
Synopsis

No synopsis available.

Mnemonic

imm

Encoding



Description

No description available.

Included in

Extension	Version
Zicfilp	~> 1.0.0

41.360. lr.d

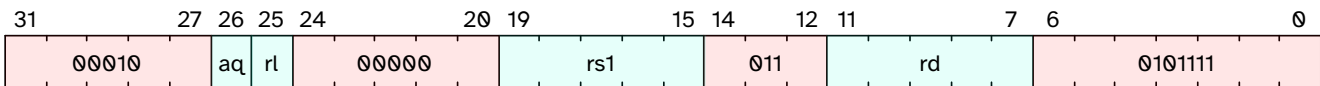
Synopsis

Load reserved doubleword

Mnemonic

```
rd, rs1
```

Encoding



Description

Loads a word from the address in rs1, places the value in rd, and registers a *reservation set* — a set of bytes that subsumes the bytes in the addressed word.

The address in rs1 must be 8-byte aligned.

If the address is not naturally aligned, a `LoadAddressMisaligned` exception or an `LoadAccessFault` exception will be generated. The access-fault exception can be generated for a memory access that would otherwise be able to complete except for the misalignment, if the misaligned access should not be emulated.

An implementation can register an arbitrarily large reservation set on each LR, provided the reservation set includes all bytes of the addressed data word or doubleword. An SC can only pair with the most recent LR in program order. An SC may succeed only if no store from another hart to the reservation set can be observed to have occurred between the LR and the SC, and if there is no other SC between the LR and itself in program order. An SC may succeed only if no write from a device other than a hart to the bytes accessed by the LR instruction can be observed to have occurred between the LR and SC. Note this LR might have had a different effective address and data size, but reserved the SC’s address as part of the reservation set.

Following this model, in systems with memory translation, an SC is allowed to succeed if the earlier LR reserved the same location using an alias with a different virtual address, but is also allowed to fail if the virtual address is different.

To accommodate legacy devices and buses, writes from devices other than RISC-V harts are only required to invalidate reservations when they overlap the bytes accessed by the LR. These writes are not required to invalidate the reservation when they access other bytes in the reservation set.

Software should not set the *rl* bit on an LR instruction unless the *aq* bit is also set. LR.*rl* and SC.*aq* instructions are not guaranteed to provide any stronger ordering than those with both bits clear, but may result in lower performance.

Included in

Extension	Version
A	~> 2.1.0
Zalrsc	~> 1.0.0

DRAFT

41.361. lr.w

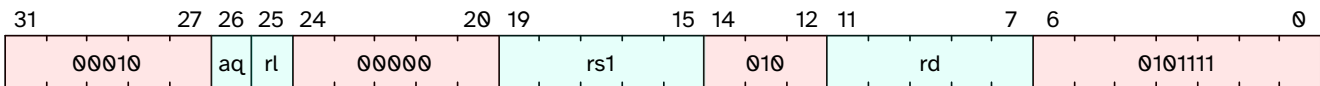
Synopsis

Load reserved word

Mnemonic

```
rd, rs1
```

Encoding



Description

Loads a word from the address in rs1, places the sign-extended value in rd, and registers a *reservation set* — a set of bytes that subsumes the bytes in the addressed word.

<%- if XLEN == 64 -%> The 32-bit load result is sign-extended to 64-bits. <%- end -%>

The address in rs1 must be naturally aligned to the size of the operand (i.e., eight-byte aligned for doublewords and four-byte aligned for words).

If the address is not naturally aligned, a LoadAddressMisaligned exception or an LoadAccessFault exception will be generated. The access-fault exception can be generated for a memory access that would otherwise be able to complete except for the misalignment, if the misaligned access should not be emulated.

An implementation can register an arbitrarily large reservation set on each LR, provided the reservation set includes all bytes of the addressed data word or doubleword. An SC can only pair with the most recent LR in program order. An SC may succeed only if no store from another hart to the reservation set can be observed to have occurred between the LR and the SC, and if there is no other SC between the LR and itself in program order. An SC may succeed only if no write from a device other than a hart to the bytes accessed by the LR instruction can be observed to have occurred between the LR and SC. Note this LR might have had a different effective address and data size, but reserved the SC’s address as part of the reservation set.

Following this model, in systems with memory translation, an SC is allowed to succeed if the earlier LR reserved the same location using an alias with a different virtual address, but is also allowed to fail if the virtual address is different.

To accommodate legacy devices and buses, writes from devices other than RISC-V harts are only required to invalidate reservations when they overlap the bytes accessed by the LR. These writes are not required to invalidate the reservation when they access other bytes in

the reservation set.

Software should not set the *rl* bit on an LR instruction unless the *aq* bit is also set. LR.*rl* and SC.*aq* instructions are not guaranteed to provide any stronger ordering than those with both bits clear, but may result in lower performance.

Included in

Extension	Version
A	~> 2.1.0
Zalrsc	~> 1.0.0

DRAFT

41.362. lui

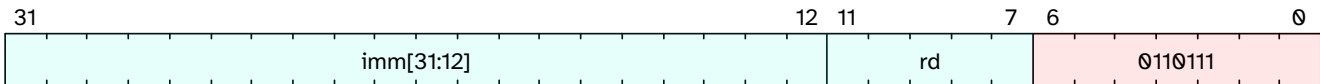
Synopsis

Load upper immediate

Mnemonic

```
rd, imm
```

Encoding



Description

Load the zero-extended imm into rd.

Included in

Extension	Version
I	~> 2.1.0

41.363. lw

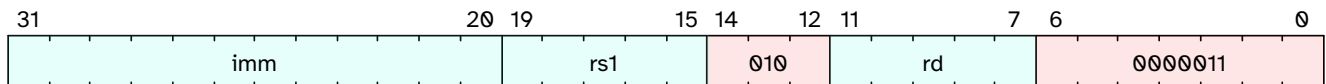
Synopsis

Load word

Mnemonic

`rd, imm(rs1)`

Encoding



Description

Load 32 bits of data into register `rd` from an address formed by adding `rs1` to a signed offset. Sign extend the result.

Included in

Extension	Version
I	~> 2.1.0

41.364. lw.aq

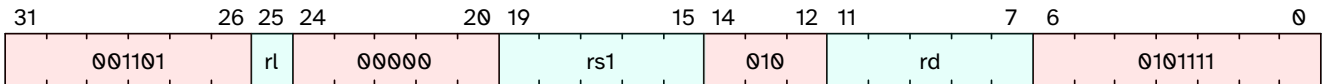
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zalasr	~> 0.3.5

41.365. lwu

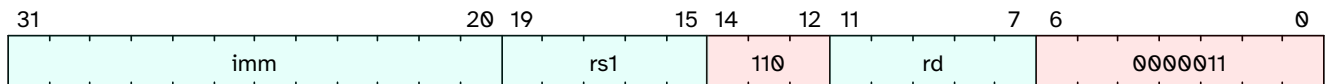
Synopsis

Load word unsigned

Mnemonic

`rd, imm(rs1)`

Encoding



Description

Load 64 bits of data into register `rd` from an address formed by adding `rs1` to a signed offset. Zero extend the result.

Included in

Extension	Version
I	~> 2.1.0

41.366. max

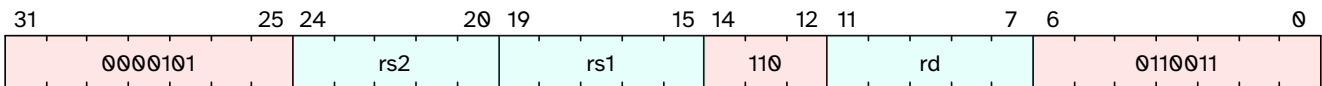
Synopsis

Maximum

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction returns the larger of two signed integers.



Software Hint

Calculating the absolute value of a signed integer can be performed using the following sequence: `neg rD,rS` followed by ``max rD,rS,rD`. When using this common sequence, it is suggested that they are scheduled with no intervening instructions so that implementations that are so optimized can fuse them together.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.367. maxu

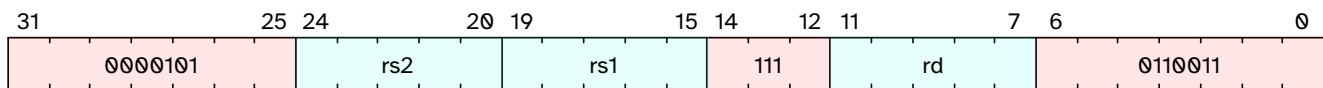
Synopsis

Unsigned maximum

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction returns the larger of two unsigned integers.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.368. min

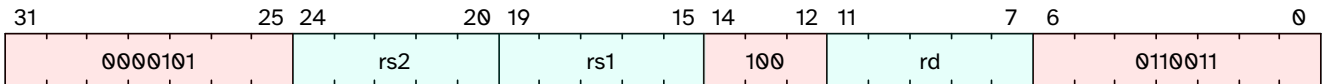
Synopsis

Minimum

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction returns the smaller of two signed integers.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.369. minu

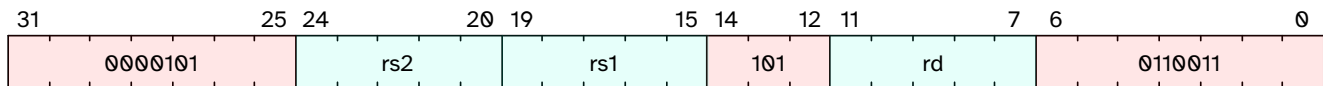
Synopsis

Unsigned minimum

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction returns the smaller of two unsigned integers.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.370. mnret

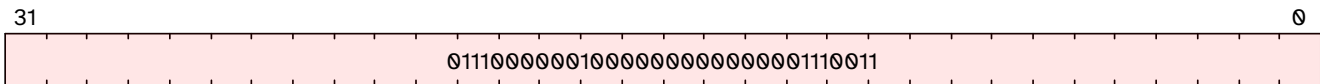
Synopsis

No synopsis available.

Mnemonic

mnret

Encoding



Description

No description available.

Included in

Extension	Version
Smrnmi	~> 1.0.0

41.371. mop.r.n

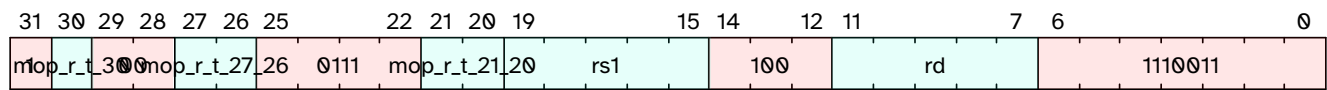
Synopsis

No synopsis available.

Mnemonic

mop_r_t_30, mop_r_t_27_26, mop_r_t_21_20, rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zimop	~> 1.0.0

41.372. mop.rr.n

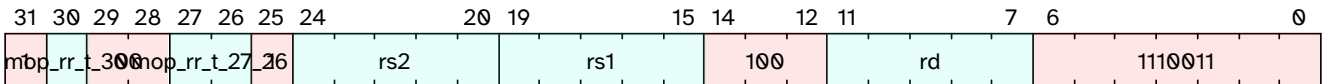
Synopsis

No synopsis available.

Mnemonic

mop_rr_t_30, mop_rr_t_27_26, rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zimop	~> 1.0.0

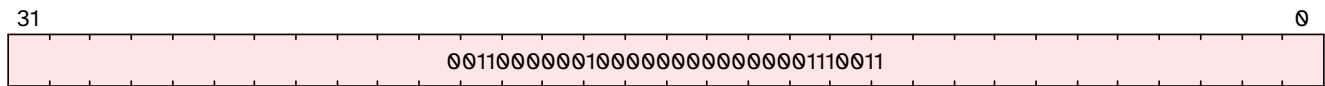
41.373. mret

Synopsis

Machine Exception Return

Mnemonic

Encoding



Description

Returns from an exception in M-mode.

Included in

Extension	Version
Sm	~> 1.11.0

41.374. mul

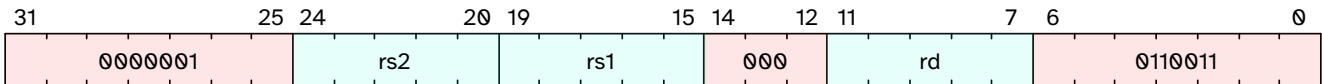
Synopsis

Signed multiply

Mnemonic

```
rd, rs1, rs2
```

Encoding



Description

MUL performs an XLEN-bitxXLEN-bit multiplication of rs1 by rs2 and places the lower XLEN bits in the destination register. Any overflow is thrown away.



If both the high and low bits of the same product are required, then the recommended code sequence is: MULH[[S]U] rdh, rs1, rs2; MUL rdl, rs1, rs2 (source register specifiers must be in same order and rdh cannot be the same as rs1 or rs2). Microarchitectures can then fuse these into a single multiply operation instead of performing two separate multiplies.

Included in

Extension	Version
M	~> 2.0.0
Zmmul	~> 1.0.0

41.375. mulh

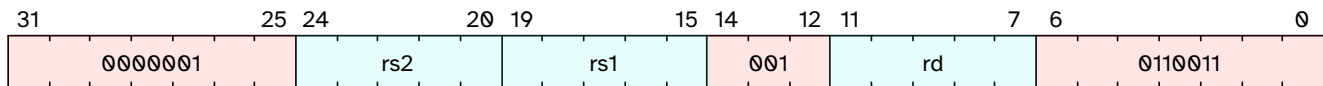
Synopsis

Signed multiply high

Mnemonic

rd, rs1, rs2

Encoding



Description

Multiply the signed values in rs1 to rs2, and store the upper half of the result in rd. The lower half is thrown away.

If both the upper and lower halves are needed, it suggested to use the sequence:

```
mulh rdh, rs1, rs2
mul  rdl, rs1, rs2
---
```

Microarchitectures may look for that sequence and fuse the operations.

Included in

Extension	Version
M	~> 2.0.0
Zmmul	~> 1.0.0

41.376. mulhsu

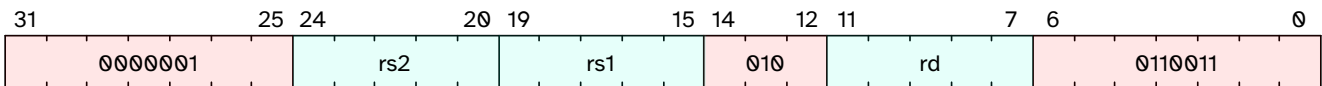
Synopsis

Signed/unsigned multiply high

Mnemonic

rd, rs1, rs2

Encoding



Description

Multiply the signed value in rs1 by the unsigned value in rs2, and store the upper half of the result in rd. The lower half is thrown away.

If both the upper and lower halves are needed, it suggested to use the sequence:

```
mulhsu rdh, rs1, rs2
mul    rdl, rs1, rs2
---
```

Microarchitectures may look for that sequence and fuse the operations.

Included in

Extension	Version
M	~> 2.0.0
Zmmul	~> 1.0.0

41.377. mulhu

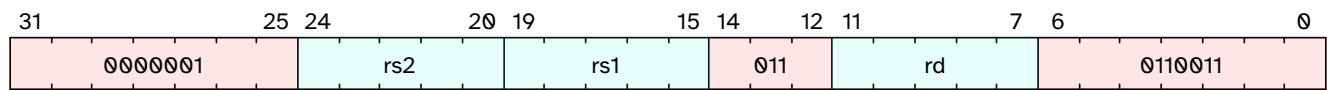
Synopsis

Unsigned multiply high

Mnemonic

rd, rs1, rs2

Encoding



Description

Multiply the unsigned values in rs1 to rs2, and store the upper half of the result in rd. The lower half is thrown away.

If both the upper and lower halves are needed, it suggested to use the sequence:

```
mulhu rdh, rs1, rs2
mul    rdl, rs1, rs2
---
```

Microarchitectures may look for that sequence and fuse the operations.

Included in

Extension	Version
M	~> 2.0.0
Zmmul	~> 1.0.0

41.378. mulw

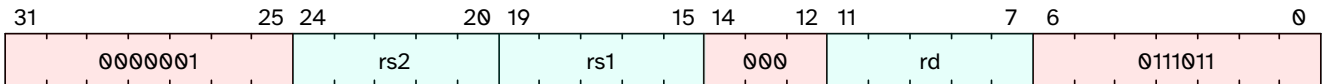
Synopsis

Signed 32-bit multiply

Mnemonic

```
rd, rs1, rs2
```

Encoding



Description

Multiplies the lower 32 bits of the source registers, placing the sign-extension of the lower 32 bits of the result into the destination register.

Any overflow is thrown away.



In RV64, MUL can be used to obtain the upper 32 bits of the 64-bit product, but signed arguments must be proper 32-bit signed values, whereas unsigned arguments must have their upper 32 bits clear. If the arguments are not known to be sign- or zero-extended, an alternative is to shift both arguments left by 32 bits, then use MULH[[S]U].

Included in

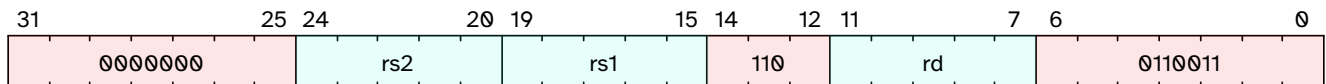
Extension	Version
M	~> 2.0.0
Zmmul	~> 1.0.0

41.379. or**Synopsis**

Or

Mnemonic

rd, rs1, rs2

Encoding**Description**

Or rs1 with rs2, and store the result in rd

Included in

Extension	Version
I	~> 2.1.0

41.380. orc.b

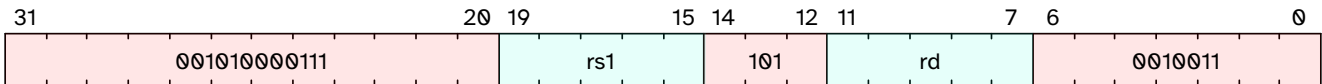
Synopsis

Bitware OR-combine, byte granule

Mnemonic

rd, rs1, rs2

Encoding



Description

Combines the bits within each byte using bitwise logical OR. This sets the bits of each byte in the result rd to all zeros if no bit within the respective byte of rs is set, or to all ones if any bit within the respective byte of rs is set.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.381. ori

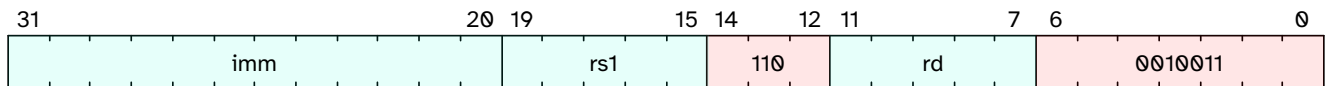
Synopsis

Or immediate

Mnemonic

rd, rs1, imm

Encoding



Description

Or an immediate to the value in rs1, and store the result in rd

Included in

Extension	Version
I	~> 2.1.0

41.382. orn

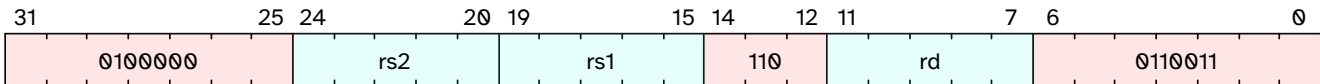
Synopsis

OR with inverted operand

Mnemonic

```
rd, rs1, rs2
```

Encoding



Description

This instruction performs the bitwise logical OR operation between rs1 and the bitwise inversion of rs2.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0
Zbkb	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.383. pack

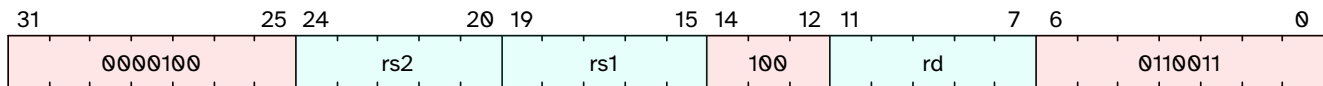
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zbkb	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.384. packh

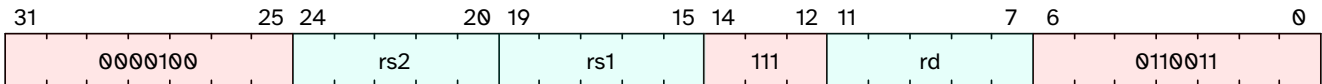
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zbkb	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.385. packw

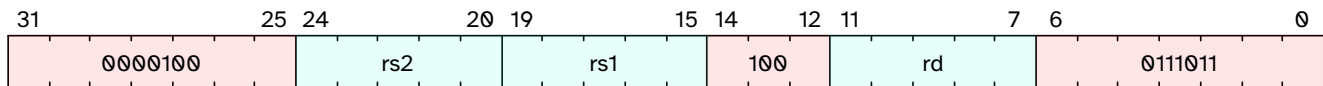
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zbkb	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.386. rem

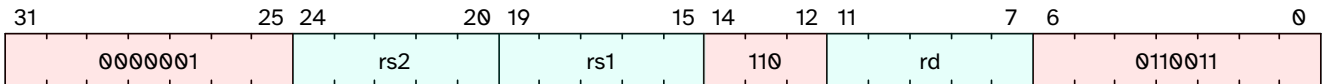
Synopsis

Signed remainder

Mnemonic

rd, rs1, rs2

Encoding



Description

Calculate the remainder of signed division of rs1 by rs2, and store the result in rd.

If the value in register rs2 is zero, write the value in rs1 into rd;

If the result of the division overflows, write zero into rd;

Included in

Extension	Version
M	~> 2.0.0

41.387. remu

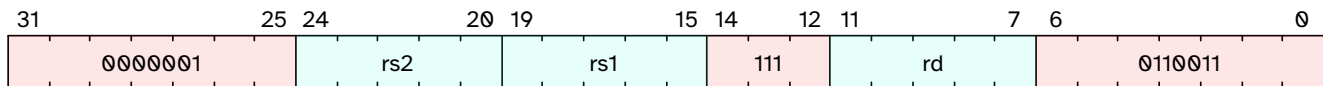
Synopsis

Unsigned remainder

Mnemonic

rd, rs1, rs2

Encoding



Description

Calculate the remainder of unsigned division of rs1 by rs2, and store the result in rd.

Included in

Extension	Version
M	~> 2.0.0

41.388. remuw

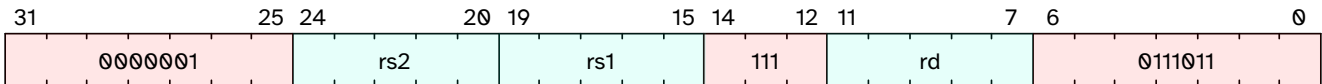
Synopsis

Unsigned 32-bit remainder

Mnemonic

rd, rs1, rs2

Encoding



Description

Calculate the remainder of unsigned division of the 32-bit values in rs1 by rs2, and store the sign-extended result in rd.

If the value in rs2 is zero, rd gets the sign-extended value in rs1.

Included in

Extension	Version
M	~> 2.0.0

41.389. remw

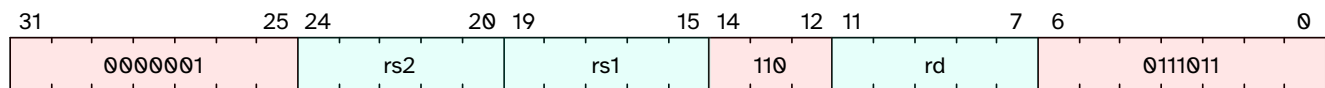
Synopsis

Signed 32-bit remainder

Mnemonic

rd, rs1, rs2

Encoding



Description

Calculate the remainder of signed division of the 32-bit values rs1 by rs2, and store the sign-extended result in rd.

If the value in register rs2 is zero, write the sign-extended 32-bit value in rs1 into rd;

If the result of the division overflows, write zero into rd;

Included in

Extension	Version
M	~> 2.0.0

41.390. rev8

Synopsis

Byte-reverse register (RV64 encoding)

Mnemonic

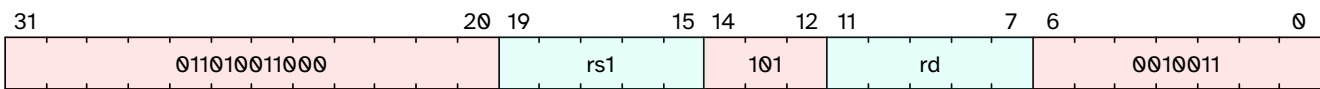
rd, rs1

Encoding

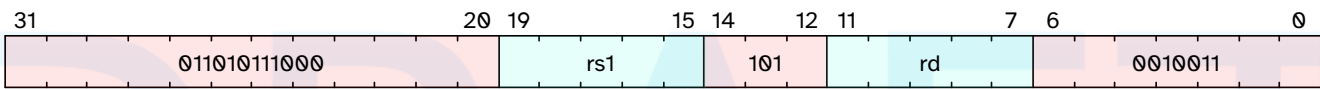


This instruction has different encodings in RV32 and RV64

RV32



RV64



Description

This instruction reverses the order of the bytes in rs1.



The rev8 mnemonic corresponds to different instruction encodings in RV32 and RV64.



The byte-reverse operation is only available for the full register width. To emulate word-sized and halfword-sized byte-reversal, perform a rev8 rd,rs followed by a srar rd,rd,K, where K is XLEN-32 and XLEN-16, respectively.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0
Zbkb	~> 1.0.0

41.391. rol

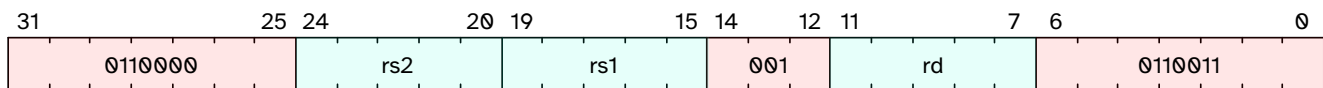
Synopsis

Rotate left (Register)

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction performs a rotate left of rs1 by the amount in least-significant $\log_2(\text{XLEN})$ bits of rs2.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0
Zbkb	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.392. rolw

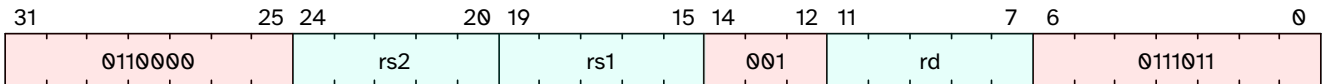
Synopsis

Rotate left word (Register)

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction performs a rotate left of the least-significant word of rs1 by the amount in least-significant 5 bits of rs2. The resulting word value is sign-extended by copying bit 31 to all of the more-significant bits.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0
Zbkb	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.393. ror

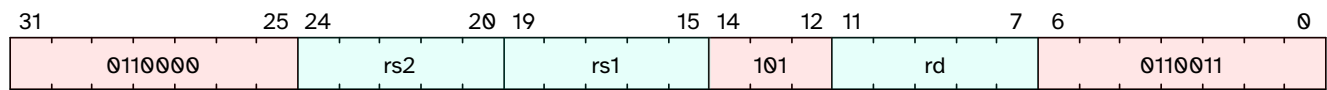
Synopsis

Rotate right (Register)

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction performs a rotate right of rs1 by the amount in least-significant $\log_2(\text{XLEN})$ bits of rs2.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0
Zbkb	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.394. rori

Synopsis

Rotate right (Immediate)

Mnemonic

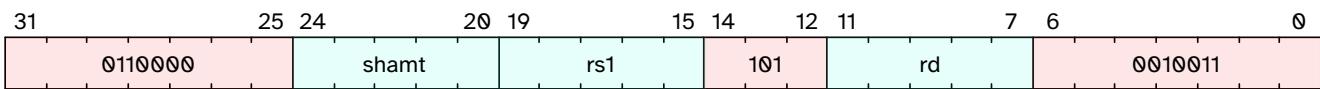
rd, rs1, shamt

Encoding

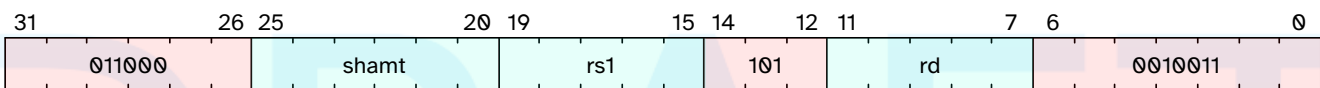


This instruction has different encodings in RV32 and RV64

RV32



RV64



Description

This instruction performs a rotate right of rs1 by the amount in the least-significant log2(XLEN) bits of shamt. For RV32, the encodings corresponding to shamt[5]=1 are reserved.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0
Zbkb	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.395. roriw

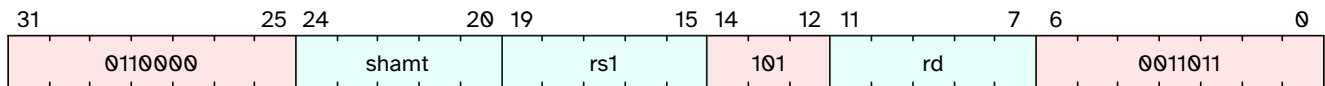
Synopsis

Rotate right word (Immediate)

Mnemonic

rd, rs1, shamt

Encoding



Description

This instruction performs a rotate right on the least-significant word of rs1 by the amount in the least-significant log2(XLEN) bits of shamt. The resulting word value is sign-extended by copying bit 31 to all of the more-significant bits.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0
Zbkb	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.396. row

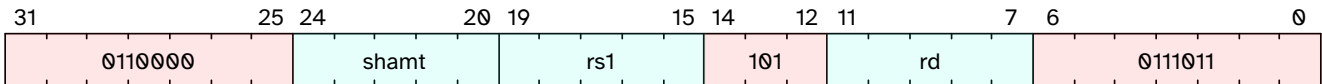
Synopsis

Rotate right word (Register)

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction performs a rotate right on the least-significant word of rs1 by the amount in least-significant 5 bits of rs2. The resultant word is sign-extended by copying bit 31 to all of the more-significant bits.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0
Zbkb	~> 1.0.0
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zks	~> 1.0.0

41.397. sb

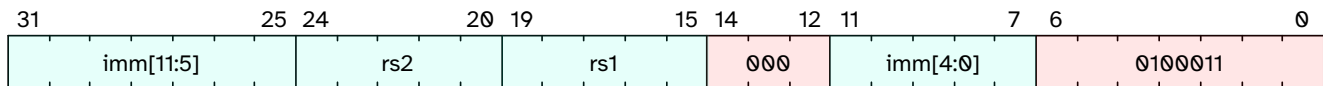
Synopsis

Store byte

Mnemonic

```
rs2, imm(rs1)
```

Encoding



Description

Store 8 bits of data from register rs2 to an address formed by adding rs1 to a signed offset.

Included in

Extension	Version
I	~> 2.1.0

41.398. sb.rl

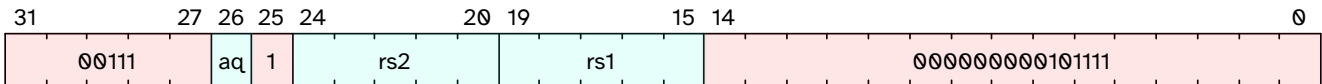
Synopsis

No synopsis available.

Mnemonic

rs1, rs2, aq

Encoding



Description

No description available.

Included in

Extension	Version
Zalasr	~> 0.3.5

41.399. sc.d

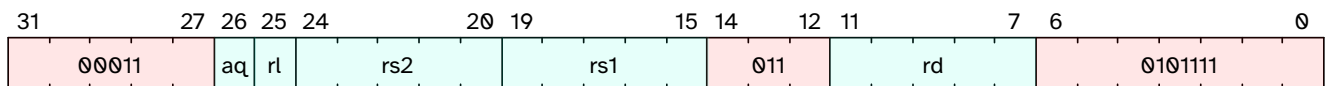
Synopsis

Store conditional doubleword

Mnemonic

rd, rs2, rs1

Encoding



Description

`sc.d` conditionally writes a doubleword in `rs2` to the address in `rs1`: the `sc.d` succeeds only if the reservation is still valid and the reservation set contains the bytes being written. If the `sc.d` succeeds, the instruction writes the doubleword in `rs2` to memory, and it writes zero to `rd`. If the `sc.d` fails, the instruction does not write to memory, and it writes a nonzero value to `rd`. For the purposes of memory protection, a failed `sc.d` may be treated like a store. Regardless of success or failure, executing an `sc.d` instruction invalidates any reservation held by this hart.

The failure code with value 1 encodes an unspecified failure. Other failure codes are reserved at this time. Portable software should only assume the failure code will be non-zero.

The address held in `rs1` must be naturally aligned to the size of the operand (*i.e.*, eight-byte aligned). If the address is not naturally aligned, an address-misaligned exception or an access-fault exception will be generated. The access-fault exception can be generated for a memory access that would otherwise be able to complete except for the misalignment, if the misaligned access should not be emulated.



Emulating misaligned LR/SC sequences is impractical in most systems.

Misaligned LR/SC sequences also raise the possibility of accessing multiple reservation sets at once, which present definitions do not provide for.

An implementation can register an arbitrarily large reservation set on each LR, provided the reservation set includes all bytes of the addressed data word or doubleword. An SC can only pair with the most recent LR in program order. An SC may succeed only if no store from another hart to the reservation set can be observed to have occurred between the LR and the SC, and if there is no other SC between the LR and itself in program order. An SC may succeed only if no write from a device other than a hart to the bytes accessed by the LR instruction can be observed to have occurred between the LR and SC. Note this LR might have had a different effective address and data size, but reserved the SC's address as part of the reservation set.

Following this model, in systems with memory translation, an SC is allowed to succeed if the earlier LR reserved the same location using an alias with a different virtual address, but is also allowed to fail if the virtual address is different.

To accommodate legacy devices and buses, writes from devices other than RISC-V harts are only required to invalidate reservations when they overlap the bytes accessed by the LR. These writes are not required to invalidate the reservation when they access other bytes in the reservation set.

The SC must fail if the address is not within the reservation set of the most recent LR in program order. The SC must fail if a store to the reservation set from another hart can be observed to occur between the LR and SC. The SC must fail if a write from some other device to the bytes accessed by the LR can be observed to occur between the LR and SC. (If such a device writes the reservation set but does not write the bytes accessed by the LR, the SC may or may not fail.) An SC must fail if there is another SC (to any address) between the LR and the SC in program order. The precise statement of the atomicity requirements for successful LR/SC sequences is defined by the Atomicity Axiom of the memory model.

The platform should provide a means to determine the size and shape of the reservation set.

A platform specification may constrain the size and shape of the reservation set.

A store-conditional instruction to a scratch word of memory should be used to forcibly invalidate any existing load reservation:

- *during a preemptive context switch, and*
- *if necessary when changing virtual to physical address mappings, such as when migrating pages that might contain an active reservation.*

The invalidation of a hart's reservation when it executes an LR or SC imply that a hart can only hold one reservation at a time, and that an SC can only pair with the most recent LR, and LR with the next following SC, in program order. This is a restriction to the Atomicity Axiom in Section 18.1 that ensures software runs correctly on expected common implementations that operate in this manner.

An SC instruction can never be observed by another RISC-V hart before the LR instruction that established the reservation.

The LR/SC sequence can be given acquire semantics by setting the `aq` bit on the LR instruction. The LR/SC sequence can be given release semantics by by setting the `rl` bit on the SC instruction. Assuming suitable mappings for other atomic operations, setting the `aq` bit on the LR instruction, and setting the `rl` bit on the SC instruction makes the LR/SC sequence sequentially consistent in the C memory_order_seq_cst sense. Such a sequence does not act as a fence for ordering ordinary load and store instructions before and after the sequence. Specific instruction mappings for other C atomic operations, or stronger notions of "sequential consistency", may require both bits to be set on either or both of the LR or SC instruction.

If neither bit is set on either LR or SC, the LR/SC sequence can be observed to occur before or after surrounding memory operations from the same RISC-V hart. This can be appropriate when the LR/SC sequence is used to implement a parallel reduction

operation.

Software should not set the *rl* bit on an LR instruction unless the *aq* bit is also set. LR.*rl* and SC.*aq* instructions are not guaranteed to provide any stronger ordering than those with both bits clear, but may result in lower performance.

Included in

Extension	Version
A	~> 2.1.0
Zalrsc	~> 1.0.0

DRAFT

41.400. sc.w

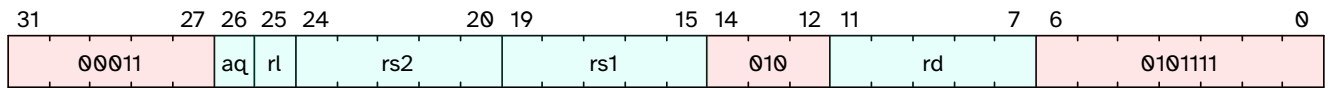
Synopsis

Store conditional word

Mnemonic

rd, rs2, rs1

Encoding



Description

sc.w conditionally writes a word in rs2 to the address in rs1: the sc.w succeeds only if the reservation is still valid and the reservation set contains the bytes being written. If the sc.w succeeds, the instruction writes the word in rs2 to memory, and it writes zero to rd. If the sc.w fails, the instruction does not write to memory, and it writes a nonzero value to rd. For the purposes of memory protection, a failed sc.w may be treated like a store. Regardless of success or failure, executing an sc.w instruction invalidates any reservation held by this hart.

<%- if XLEN == 64 -%>

If a value other than 0 or 1 is defined as a result for sc.w, the value will be sign-extended into rd. <%- end -%>

The failure code with value 1 encodes an unspecified failure. Other failure codes are reserved at this time. Portable software should only assume the failure code will be non-zero.

The address held in rs1 must be naturally aligned to the size of the operand (i.e., eight-byte aligned for doublewords and four-byte aligned for words). If the address is not naturally aligned, an address-misaligned exception or an access-fault exception will be generated. The access-fault exception can be generated for a memory access that would otherwise be able to complete except for the misalignment, if the misaligned access should not be emulated.

Emulating misaligned LR/SC sequences is impractical in most systems.
Misaligned LR/SC sequences also raise the possibility of accessing multiple reservation sets at once, which present definitions do not provide for.

An implementation can register an arbitrarily large reservation set on each LR, provided the reservation set includes all bytes of the addressed data word or doubleword. An SC can only pair with the most recent LR in program order. An SC may succeed only if no store from another hart to the reservation set can be observed to have occurred between the LR and the SC, and if there is no other SC between the LR and itself in program order. An SC may succeed only if no write from a device other than a hart to the bytes accessed by the LR instruction can be observed to have occurred between the LR and SC. Note this LR might have had a different effective address and data size, but reserved the SC's address as part of the reservation set.

Following this model, in systems with memory translation, an SC is allowed to succeed if the earlier LR reserved the same location using an alias with a different virtual address, but is also allowed to fail if the virtual address is different.

To accommodate legacy devices and buses, writes from devices other than RISC-V harts are only required to invalidate reservations when they overlap the bytes accessed by the LR.

These writes are not required to invalidate the reservation when they access other bytes in the reservation set.

The SC must fail if the address is not within the reservation set of the most recent LR in program order. The SC must fail if a store to the reservation set from another hart can be observed to occur between the LR and SC. The SC must fail if a write from some other device to the bytes accessed by the LR can be observed to occur between the LR and SC. (If such a device writes the reservation set but does not write the bytes accessed by the LR, the SC may or may not fail.) An SC must fail if there is another SC (to any address) between the LR and the SC in program order. The precise statement of the atomicity requirements for successful LR/SC sequences is defined by the Atomicity Axiom of the memory model.

The platform should provide a means to determine the size and shape of the reservation set.

A platform specification may constrain the size and shape of the reservation set.

A store-conditional instruction to a scratch word of memory should be used to forcibly invalidate any existing load reservation:



- *during a preemptive context switch, and*
- *if necessary when changing virtual to physical address mappings, such as when migrating pages that might contain an active reservation.*

The invalidation of a hart's reservation when it executes an LR or SC imply that a hart can only hold one reservation at a time, and that an SC can only pair with the most recent LR, and LR with the next following SC, in program order. This is a restriction to the Atomicity Axiom in Section 18.1 that ensures software runs correctly on expected common implementations that operate in this manner.

An SC instruction can never be observed by another RISC-V hart before the LR instruction that established the reservation.



The LR/SC sequence can be given acquire semantics by setting the `aq` bit on the LR instruction. The LR/SC sequence can be given release semantics by by setting the `rl` bit on the SC instruction. Assuming suitable mappings for other atomic operations, setting the `aq` bit on the LR instruction, and setting the `rl` bit on the SC instruction makes the LR/SC sequence sequentially consistent in the C memory_order_seq_cst sense. Such a sequence does not act as a fence for ordering ordinary load and store

instructions before and after the sequence. Specific instruction mappings for other C atomic operations, or stronger notions of "sequential consistency", may require both bits to be set on either or both of the LR or SC instruction.

If neither bit is set on either LR or SC, the LR/SC sequence can be observed to occur before or after surrounding memory operations from the same RISC-V hart. This can be appropriate when the LR/SC sequence is used to implement a parallel reduction operation.

Software should not set the *rl* bit on an LR instruction unless the *aq* bit is also set. LR.*rl* and SC.*aq* instructions are not guaranteed to provide any stronger ordering than those with both bits clear, but may result in lower performance.

Included in

Extension	Version
A	~> 2.1.0
Zalrsc	~> 1.0.0

DRAFT

41.401. sctrclr

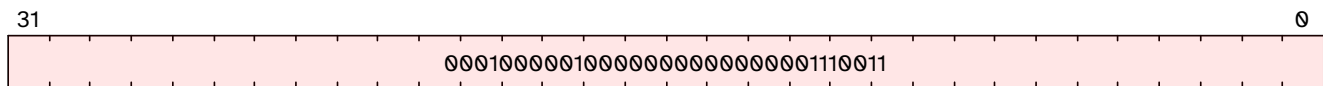
Synopsis

No synopsis available.

Mnemonic

sctrclr

Encoding



Description

No description available.

Included in

Extension	Version
Smdbltrp	~> 1.0.0

DRAFT

41.402. sd

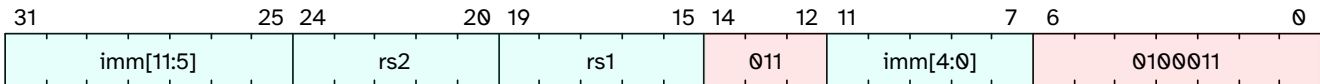
Synopsis

Store doubleword

Mnemonic

```
rs2, imm(rs1)
```

Encoding



Description

Store 64 bits of data from register rs2 to an address formed by adding rs1 to a signed offset.

Included in

Extension	Version
I	~> 2.1.0

41.403. sd.rl

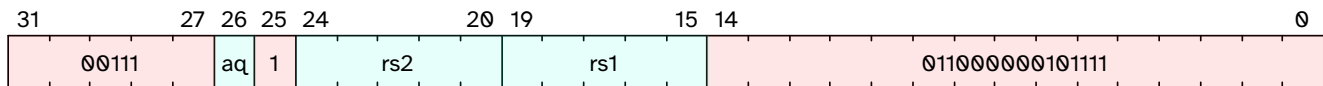
Synopsis

No synopsis available.

Mnemonic

rs1, rs2, aq

Encoding



Description

No description available.

Included in

Extension	Version
Zalasr	~> 0.3.5

DRAFT

41.404. sext.b

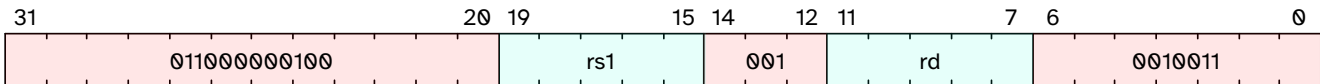
Synopsis

Sign-extend byte

Mnemonic

rd, rs1

Encoding



Description

This instruction sign-extends the least-significant byte in the source to XLEN by copying the most-significant bit in the byte (i.e., bit 7) to all of the more-significant bits.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

41.405. sext.h

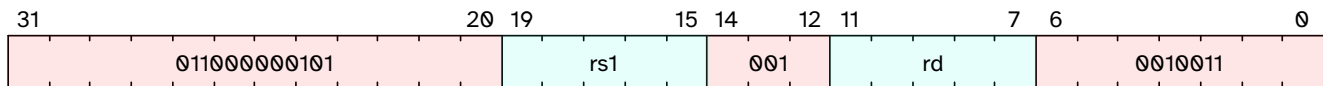
Synopsis

Sign-extend halfword

Mnemonic

rd, rs1

Encoding



Description

This instruction sign-extends the least-significant halfword in the source to XLEN by copying the most-significant bit in the halfword (i.e., bit 15) to all of the more-significant bits.

Included in

Extension	Version
B	~> 1.0.0
Zbb	~> 1.0.0

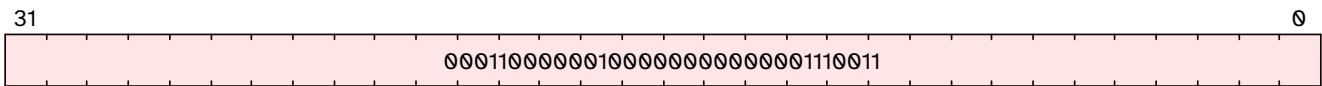
41.406. sfence.inval.ir

Synopsis

Order implicit page table reads after invalidation

Mnemonic

Encoding



Description

The [sfence.inval.ir](#) instruction guarantees that any previous [sINVAL.VMA](#) instructions executed by the current hart are ordered before subsequent implicit references by that hart to the memory-management data structures.

Included in

Extension	Version
SINVAL	~> 1.0.0

41.407. sfence.vma

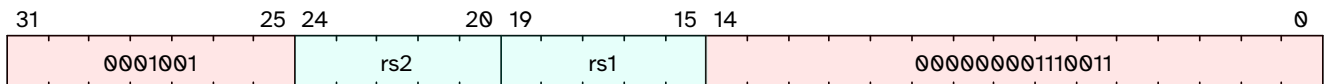
Synopsis

Supervisor memory-management fence

Mnemonic

rs1, rs2

Encoding



Description

The supervisor memory-management fence instruction `SFENCE.VMA` is used to synchronize updates to in-memory memory-management data structures with current execution. Instruction execution causes implicit reads and writes to these data structures; however, these implicit references are ordinarily not ordered with respect to explicit loads and stores. Executing an `SFENCE.VMA` instruction guarantees that any previous stores already visible to the current RISC-V hart are ordered before certain implicit references by subsequent instructions in that hart to the memory-management data structures. The specific set of operations ordered by `SFENCE.VMA` is determined by `rs1` and `rs2`, as described below. `SFENCE.VMA` is also used to invalidate entries in the address-translation cache associated with a hart (see [\[sv32algorithm\]](#)). Further details on the behavior of this instruction are described in [\[virt-control\]](#) and [\[pmp-vmem\]](#).



The `SFENCE.VMA` is used to flush any local hardware caches related to address translation. It is specified as a fence rather than a TLB flush to provide cleaner semantics with respect to which instructions are affected by the flush operation and to support a wider variety of dynamic caching structures and memory-management schemes. `SFENCE.VMA` is also used by higher privilege levels to synchronize page table writes and the address translation hardware.

`SFENCE.VMA` orders only the local hart's implicit references to the memory-management data structures.



Consequently, other harts must be notified separately when the memory-management data structures have been modified. One approach is to use 1) a local data fence to ensure local writes are visible globally, then 2) an interprocessor interrupt to the other thread, then 3) a local `SFENCE.VMA` in the interrupt handler of the remote thread, and finally 4) signal back to originating thread that operation is complete. This is, of course, the RISC-V analog to a TLB shutdown.

For the common case that the translation data structures have only been modified for a single address mapping (i.e., one page or superpage), `rs1` can specify a virtual address within that mapping to effect a translation fence for that mapping only. Furthermore, for the common case that the translation data structures have only been modified for a single address-space identifier, `rs2` can specify the address space. The behavior of `SFENCE.VMA` depends on `rs1` and `rs2` as follows:

- If `rs1=x0` and `rs2=x0`, the fence orders all reads and writes made to any level of the page tables, for

all address spaces. The fence also invalidates all address-translation cache entries, for all address spaces.

- If $rs1=x0$ and $rs2\neq x0$, the fence orders all reads and writes made to any level of the page tables, but only for the address space identified by integer register $rs2$. Accesses to *global* mappings (see [\[translation\]](#)) are not ordered. The fence also invalidates all address-translation cache entries matching the address space identified by integer register $rs2$, except for entries containing global mappings.
- If $rs1\neq x0$ and $rs2=x0$, the fence orders only reads and writes made to leaf page table entries corresponding to the virtual address in $rs1$, for all address spaces. The fence also invalidates all address-translation cache entries that contain leaf page table entries corresponding to the virtual address in $rs1$, for all address spaces.
- If $rs1\neq x0$ and $rs2\neq x0$, the fence orders only reads and writes made to leaf page table entries corresponding to the virtual address in $rs1$, for the address space identified by integer register $rs2$. Accesses to global mappings are not ordered. The fence also invalidates all address-translation cache entries that contain leaf page table entries corresponding to the virtual address in $rs1$ and that match the address space identified by integer register $rs2$, except for entries containing global mappings.

If the value held in $rs1$ is not a valid virtual address, then the SFENCE.VMA instruction has no effect. No exception is raised in this case.

When $rs2\neq x0$, bits SXLEN-1:ASIDMAX of the value held in $rs2$ are reserved for future standard use. Until their use is defined by a standard extension, they should be zeroed by software and ignored by current implementations. Furthermore, if ASIDLLEN<ASIDMAX, the implementation shall ignore bits ASIDMAX-1:ASIDLLEN of the value held in $rs2$.



It is always legal to over-fence, e.g., by fencing only based on a subset of the bits in $rs1$ and/or $rs2$, and/or by simply treating all SFENCE.VMA instructions as having $rs1=x0$ and/or $rs2=x0$. For example, simpler implementations can ignore the virtual address in $rs1$ and the ASID value in $rs2$ and always perform a global fence. The choice not to raise an exception when an invalid virtual address is held in $rs1$ facilitates this type of simplification.

An implicit read of the memory-management data structures may return any translation for an address that was valid at any time since the most recent SFENCE.VMA that subsumes that address. The ordering implied by SFENCE.VMA does not place implicit reads and writes to the memory-management data structures into the global memory order in a way that interacts cleanly with the standard RVWMO ordering rules. In particular, even though an SFENCE.VMA orders prior explicit accesses before subsequent implicit accesses, and those implicit accesses are ordered before their associated explicit accesses, SFENCE.VMA does not necessarily place prior explicit accesses before subsequent explicit accesses in the global memory order. These implicit loads also need not otherwise obey normal program order semantics with respect to prior loads or stores to the same address.



A consequence of this specification is that an implementation may use any translation for an address that was valid at any time since the most recent SFENCE.VMA that subsumes that address. In particular, if a leaf PTE is modified but a subsuming SFENCE.VMA is not executed, either the old translation or the new translation will be used, but the choice is unpredictable. The behavior is otherwise well-defined.

In a conventional TLB design, it is possible for multiple entries to match a single

address if, for example, a page is upgraded to a superpage without first clearing the original non-leaf PTE's valid bit and executing an `SFENCE.VMA` with `rs1=x0`. In this case, a similar remark applies: it is unpredictable whether the old non-leaf PTE or the new leaf PTE is used, but the behavior is otherwise well defined.

Another consequence of this specification is that it is generally unsafe to update a PTE using a set of stores of a width less than the width of the PTE, as it is legal for the implementation to read the PTE at any time, including when only some of the partial stores have taken effect.

This specification permits the caching of PTEs whose V (Valid) bit is clear. Operating systems must be written to cope with this possibility, but implementers are reminded that eagerly caching invalid PTEs will reduce performance by causing additional page faults.

Implementations must only perform implicit reads of the translation data structures pointed to by the current contents of the `satp` register or a subsequent valid (`V=1`) translation data structure entry, and must only raise exceptions for implicit accesses that are generated as a result of instruction execution, not those that are performed speculatively.

Changes to the `sstatus` fields `SUM` and `MXR` take effect immediately, without the need to execute an `SFENCE.VMA` instruction. Changing `satp.MODE` from Bare to other modes and vice versa also takes effect immediately, without the need to execute an `SFENCE.VMA` instruction. Likewise, changes to `satp.ASID` take effect immediately.

The following common situations typically require executing an `SFENCE.VMA` instruction:

- When software recycles an ASID (i.e., reassociates it with a different page table), it should first change `satp` to point to the new page table using the recycled ASID, then execute `SFENCE.VMA` with `rs1=x0` and `rs2` set to the recycled ASID. Alternatively, software can execute the same `SFENCE.VMA` instruction while a different ASID is loaded into `satp`, provided the next time `satp` is loaded with the recycled ASID, it is simultaneously loaded with the new page table.
- If the implementation does not provide ASIDs, or software chooses to always use ASID 0, then after every `satp` write, software should execute `SFENCE.VMA` with `rs1=x0`. In the common case that no global translations have been modified, `rs2` should be set to a register other than `x0` but which contains the value zero, so that global translations are not flushed.
- If software modifies a non-leaf PTE, it should execute `SFENCE.VMA` with `rs1=x0`. If any PTE along the traversal path had its `G` bit set, `rs2` must be `x0`; otherwise, `rs2` should be set to the ASID for which the translation is being modified.
- If software modifies a leaf PTE, it should execute `SFENCE.VMA` with `rs1` set to a virtual address within the page. If any PTE along the traversal path had its `G` bit set, `rs2` must be `x0`; otherwise, `rs2` should be set to the ASID for which the translation is being modified.
- For the special cases of increasing the permissions on a leaf PTE and changing an invalid PTE to a valid leaf, software may choose to execute the `SFENCE.VMA` lazily. After modifying the PTE but before executing `SFENCE.VMA`, either the new



or old permissions will be used. In the latter case, a page-fault exception might occur, at which point software should execute SFENCE.VMA in accordance with the previous bullet point.

If a hart employs an address-translation cache, that cache must appear to be private to that hart. In particular, the meaning of an ASID is local to a hart; software may choose to use the same ASID to refer to different address spaces on different harts.



A future extension could redefine ASIDs to be global across the SEE, enabling such options as shared translation caches and hardware support for broadcast TLB shutdown. However, as OSes have evolved to significantly reduce the scope of TLB shutdowns using novel ASID-management techniques, we expect the local-ASID scheme to remain attractive for its simplicity and possibly better scalability.

For implementations that make `satp.MODE` read-only zero (always Bare), attempts to execute an SFENCE.VMA instruction might raise an illegal-instruction exception.

Included in

Extension	Version
S	~> 1.11.0

DRAFT

41.409. sh

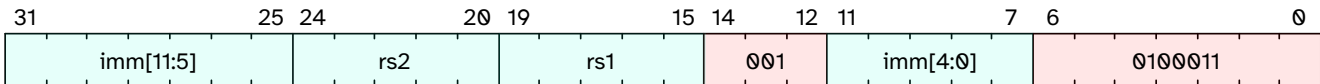
Synopsis

Store halfword

Mnemonic

rs2, imm(rs1)

Encoding



Description

Store 16 bits of data from register rs2 to an address formed by adding rs1 to a signed offset.

Included in

Extension	Version
I	~> 2.1.0

41.410. sh.rl

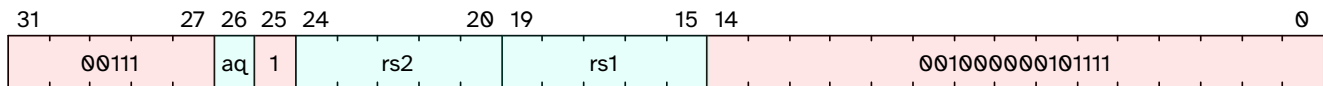
Synopsis

No synopsis available.

Mnemonic

rs1, rs2, aq

Encoding



Description

No description available.

Included in

Extension	Version
Zalasr	~> 0.3.5

DRAFT

41.411. sh1add

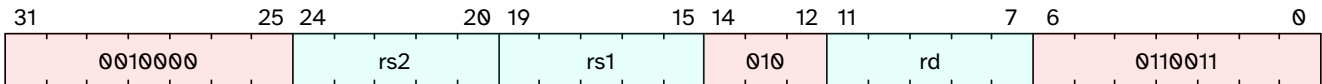
Synopsis

Shift left by 1 and add

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction shifts rs1 to the left by 1 bit and adds it to rs2.

Included in

Extension	Version
B	~> 1.0.0
Zba	~> 1.0.0

41.412. sh1add.uw

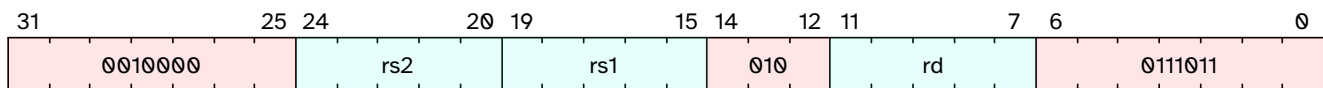
Synopsis

Shift unsigned word left by 1 and add

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction performs an XLEN-wide addition of two addends. The first addend is rs2. The second addend is the unsigned value formed by extracting the least-significant word of rs1 and shifting it left by 1 place.

Included in

Extension	Version
B	~> 1.0.0
Zba	~> 1.0.0

41.413. sh2add

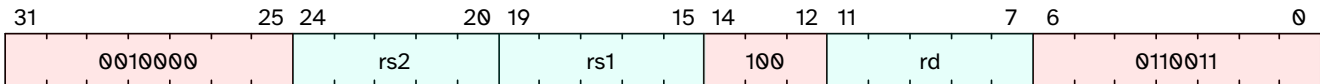
Synopsis

Shift left by 2 and add

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction shifts rs1 to the left by 2 places and adds it to rs2.

Included in

Extension	Version
B	~> 1.0.0
Zba	~> 1.0.0

41.414. sh2add.uw

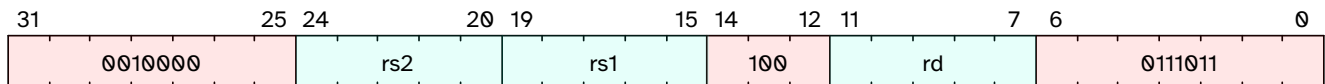
Synopsis

Shift unsigned word left by 2 and add

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction performs an XLEN-wide addition of two addends. The first addend is rs2. The second addend is the unsigned value formed by extracting the least-significant word of rs1 and shifting it left by 2 places.

Included in

Extension	Version
B	~> 1.0.0
Zba	~> 1.0.0

41.415. sh3add

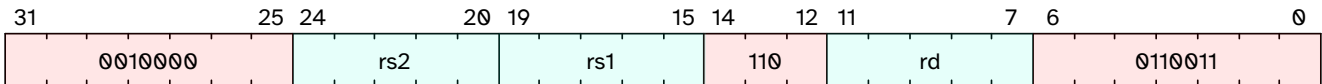
Synopsis

Shift left by 3 and add

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction shifts `rs1` to the left by 3 places and adds it to `rs2`.

Included in

Extension	Version
B	~> 1.0.0
Zba	~> 1.0.0

41.416. sh3add.uw

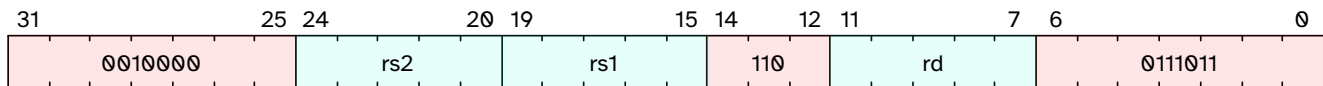
Synopsis

Shift unsigned word left by 3 and add

Mnemonic

rd, rs1, rs2

Encoding



Description

This instruction performs an XLEN-wide addition of two addends. The first addend is rs2. The second addend is the unsigned value formed by extracting the least-significant word of rs1 and shifting it left by 3 places.

Included in

Extension	Version
B	~> 1.0.0
Zba	~> 1.0.0

41.417. sha256sig0

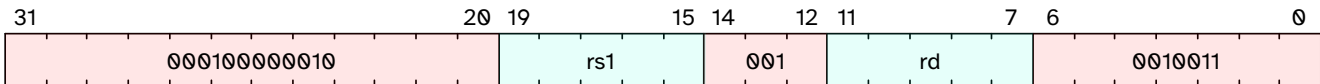
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.418. sha256sig1

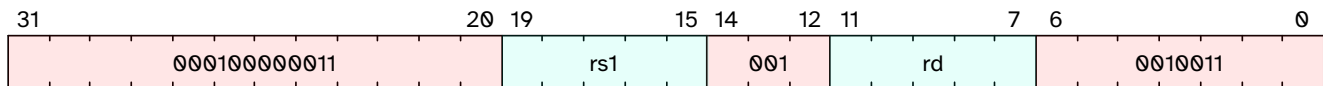
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.419. sha256sum0

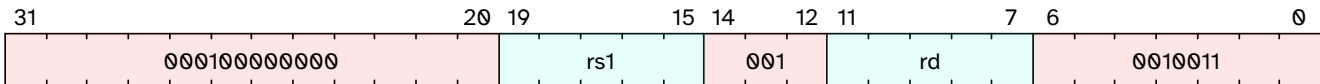
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.420. sha256sum1

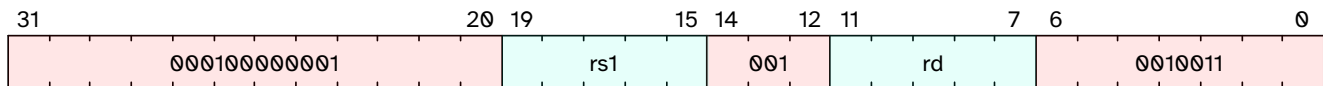
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.421. sha512sig0

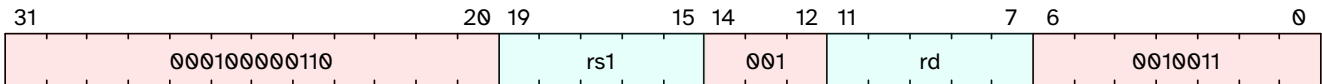
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.422. sha512sig0h

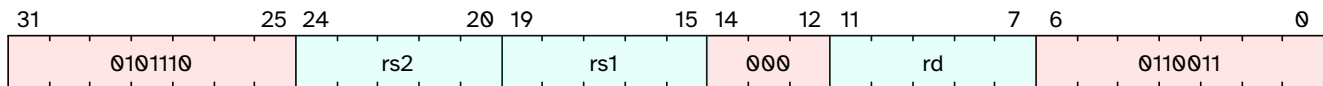
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.423. sha512sig0l

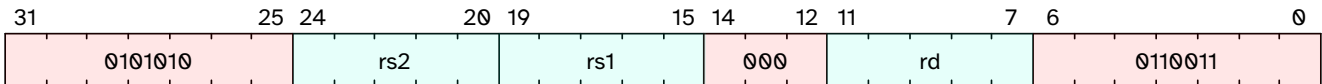
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.424. sha512sig1

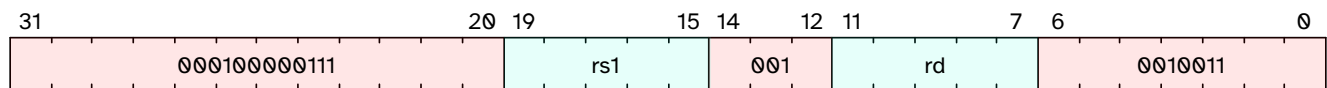
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.425. sha512sig1h

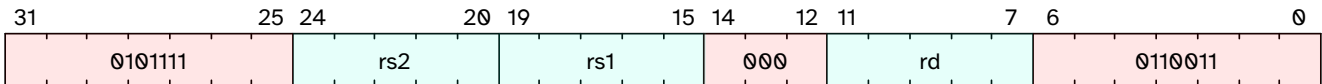
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.426. sha512sig1l

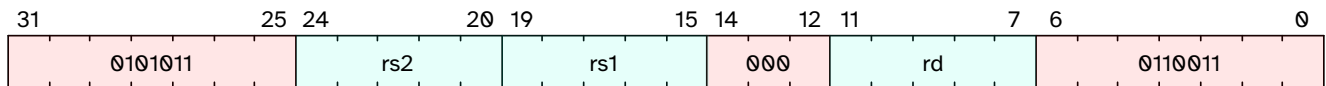
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.427. sha512sum0

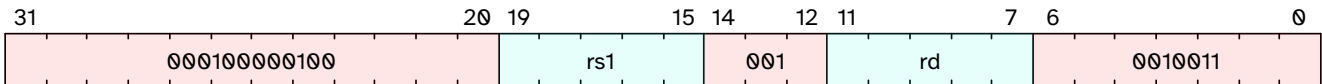
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.428. sha512sum0r

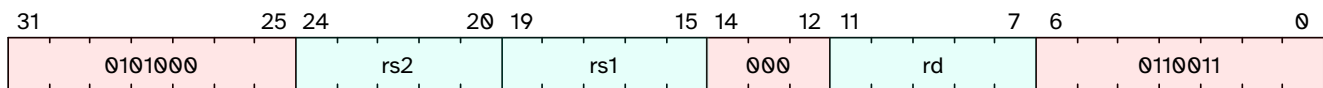
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.429. sha512sum1

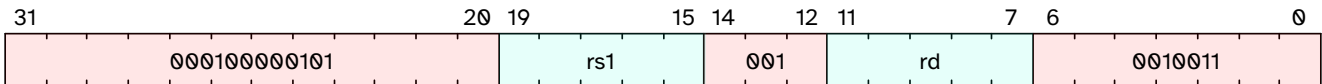
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.430. sha512sum1r

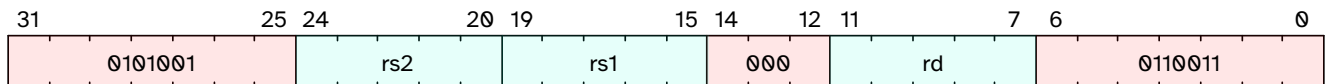
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2

Encoding



Description

No description available.

Included in

Extension	Version
Zk	~> 1.0.0
Zkn	~> 1.0.0
Zknh	~> 1.0.0

41.431. sinval.vma

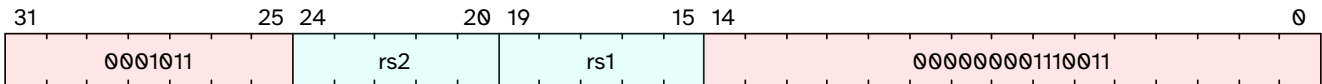
Synopsis

Invalidate cached address translations

Mnemonic

rs1, rs2

Encoding



Description

The [sinval.vma](#) instruction invalidates any address-translation cache entries that an [sfence.vma](#) instruction with the same values of rs1 and rs2 would invalidate. However, unlike [sfence.vma](#), [sinval.vma](#) instructions are only ordered with respect to [sfence.vma](#), [sfence.w.inval](#), and [sfence.inval.ir](#) instructions as defined below.

Included in

Extension	Version
Svinval	~> 1.0.0

41.432. sll

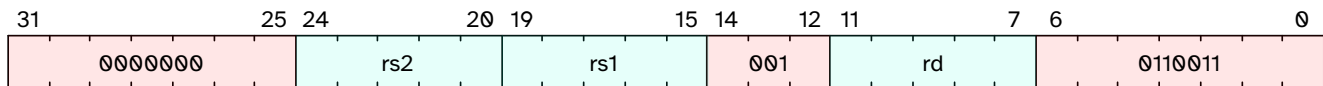
Synopsis

Shift left logical

Mnemonic

rd, rs1, rs2

Encoding



Description

Shift the value in rs1 left by the value in the lower 6 bits of rs2, and store the result in rd.

Included in

Extension	Version
I	~> 2.1.0

41.433. slli

Synopsis

Shift left logical immediate

Mnemonic

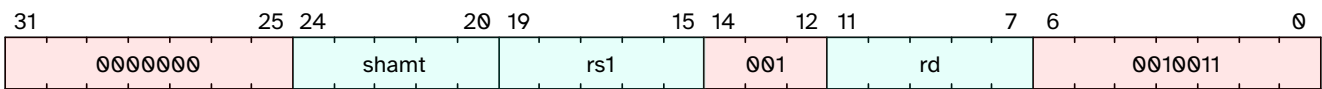
rd, rs1, shamt

Encoding

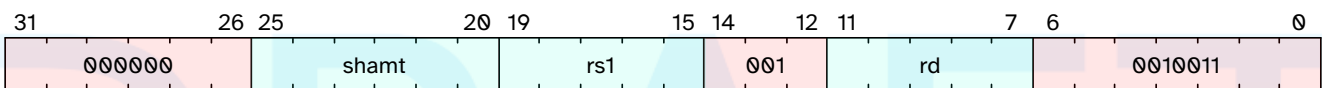


This instruction has different encodings in RV32 and RV64

RV32



RV64



Description

Shift the value in rs1 left by shamt, and store the result in rd

Included in

Extension	Version
I	~> 2.1.0

41.434. slli.uw

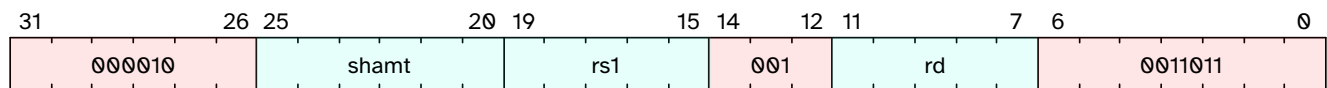
Synopsis

Shift left unsigned word (Immediate)

Mnemonic

rd, rs1, shamt

Encoding



Description

This instruction takes the least-significant word of rs1, zero-extends it, and shifts it left by the immediate.



This instruction is the same as [slli](#) with `zext.w` performed on rs1 before shifting.

Included in

Extension	Version
B	~> 1.0.0
Zba	~> 1.0.0

41.435. slliw

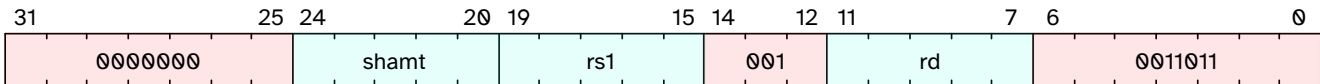
Synopsis

Shift left logical immediate word

Mnemonic

rd, rs1, shamt

Encoding



Description

Shift the 32-bit value in rs1 left by shamt, and store the sign-extended result in rd

Included in

Extension	Version
I	~> 2.1.0

41.436. sllw

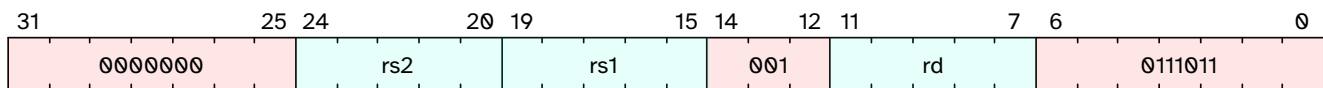
Synopsis

Shift left logical word

Mnemonic

rd, rs1, rs2

Encoding



Description

Shift the 32-bit value in rs1 left by the value in the lower 5 bits of rs2, and store the sign-extended result in rd.

Included in

Extension	Version
I	~> 2.1.0

41.437. slt

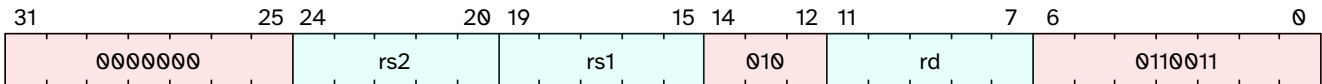
Synopsis

Set on less than

Mnemonic

rd, rs1, rs2

Encoding



Description

Places the value 1 in register rd if register rs1 is less than the value in register rs2, where both sources are treated as signed numbers, else 0 is written to rd.

Included in

Extension	Version
I	~> 2.1.0

41.438. slti

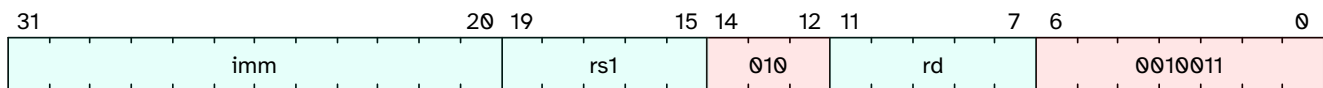
Synopsis

Set on less than immediate

Mnemonic

```
rd, rs1, imm
```

Encoding



Description

Places the value 1 in register `rd` if register `rs1` is less than the sign-extended immediate when both are treated as signed numbers, else 0 is written to `rd`.

Included in

Extension	Version
I	~> 2.1.0

41.439. sltiu

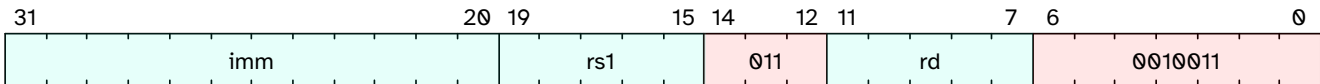
Synopsis

Set on less than immediate unsigned

Mnemonic

```
rd, rs1, imm
```

Encoding



Description

Places the value 1 in register `rd` if register `rs1` is less than the sign-extended immediate when both are treated as unsigned numbers (*i.e.*, the immediate is first sign-extended to XLEN bits then treated as an unsigned number), else 0 is written to `rd`.



sltiu rd, rs1, 1 sets rd to 1 if rs1 equals zero, otherwise sets rd to 0 (assembler pseudoinstruction SEQZ rd, rs).

Included in

Extension	Version
I	~> 2.1.0

41.440. sltu

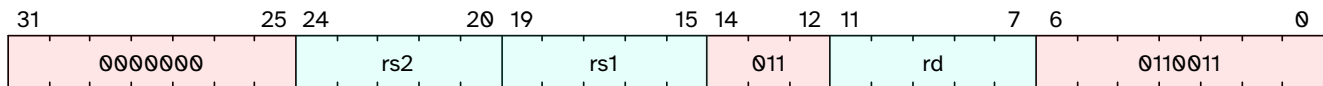
Synopsis

Set on less than unsigned

Mnemonic

rd, rs1, rs2

Encoding



Description

Places the value 1 in register rd if register rs1 is less than the value in register rs2, where both sources are treated as unsigned numbers, else 0 is written to rd.

Included in

Extension	Version
I	~> 2.1.0

41.441. sm3p0

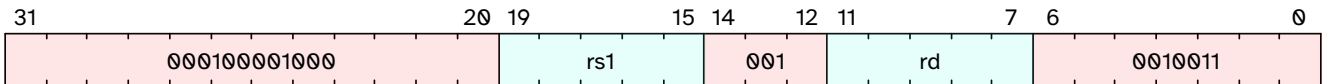
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zks	~> 1.0.0
Zksh	~> 1.0.0

41.442. sm3p1

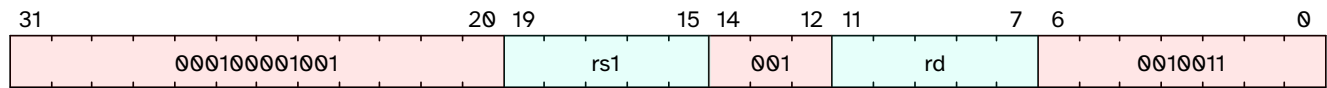
Synopsis

No synopsis available.

Mnemonic

rd, rs1

Encoding



Description

No description available.

Included in

Extension	Version
Zks	~> 1.0.0
Zksh	~> 1.0.0

41.443. sm4ed

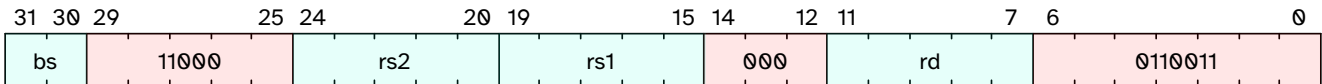
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, bs

Encoding



Description

No description available.

Included in

Extension	Version
Zks	~> 1.0.0
Zksed	~> 1.0.0

41.444. sm4ks

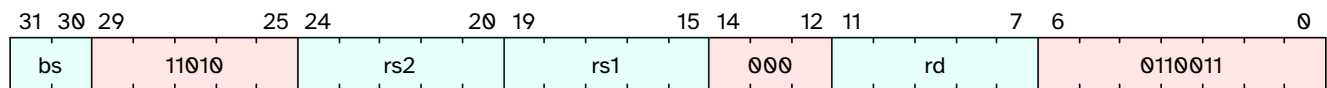
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, bs

Encoding



Description

No description available.

Included in

Extension	Version
Zks	~> 1.0.0
Zksed	~> 1.0.0

41.445. sra

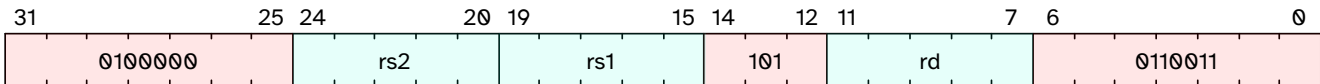
Synopsis

Shift right arithmetic

Mnemonic

rd, rs1, rs2

Encoding



Description

Arithmetic shift the value in rs1 right by the value in the lower 5 bits of rs2, and store the result in rd.

Included in

Extension	Version
I	~> 2.1.0

41.446. srai

Synopsis

Shift right arithmetic immediate

Mnemonic

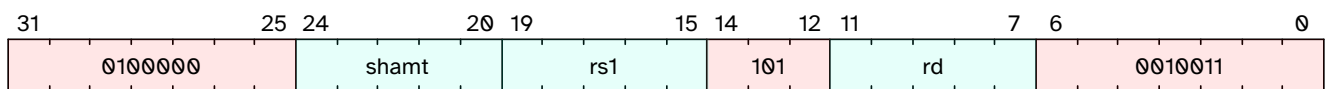
rd, rs1, shamt

Encoding

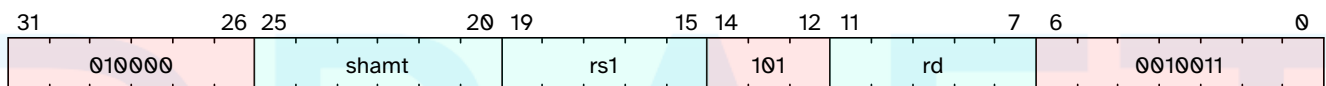


This instruction has different encodings in RV32 and RV64

RV32



RV64



Description

Arithmetic shift (the original sign bit is copied into the vacated upper bits) the value in rs1 right by shamt, and store the result in rd.

Included in

Extension	Version
I	~> 2.1.0

41.447. sraiw

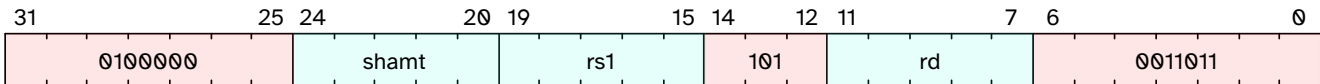
Synopsis

Shift right arithmetic immediate word

Mnemonic

rd, rs1, shamt

Encoding



Description

Arithmetic shift (the original sign bit is copied into the vacated upper bits) the 32-bit value in rs1 right by shamt, and store the sign-extended result in rd.

Included in

Extension	Version
I	~> 2.1.0

41.448. sraw

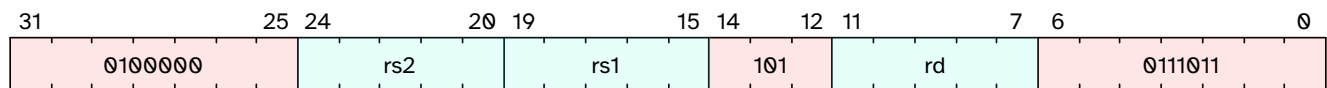
Synopsis

Shift right arithmetic word

Mnemonic

rd, rs1, rs2

Encoding



Description

Arithmetic shift the 32-bit value in rs1 right by the value in the lower 5 bits of rs2, and store the sign-extended result in rd.

Included in

Extension	Version
I	~> 2.1.0

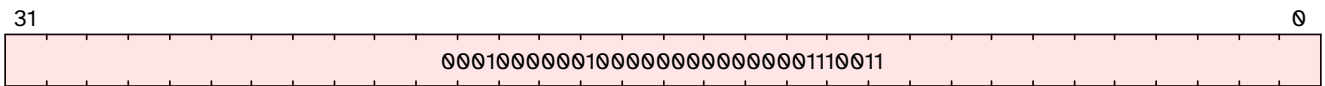
41.449. sret

Synopsis

Supervisor Exception Return

Mnemonic

Encoding



Description

Returns from an exception.

When `sret` is allowed to execute, its behavior depends on whether or not the current privilege mode is virtualized.

When the current privilege mode is (H)S-mode or M-mode

`sret` sets `hstatus` = 0, `mstatus.SPP` = 0, `mstatus.SIE` = `mstatus.SPIE`, and `mstatus.SPIE` = 1, changes the privilege mode according to the table below, and then jumps to the address in `sepc`.

Table 104. Next privilege mode following an `sret` in (H)S-mode or M-mode

mstatus.SPP	hstatus.SPVP	Mode after sret
0	0	U-mode
0	1	VU-mode
1	0	(H)S-mode
1	1	VS-mode

When the current privilege mode is VS-mode

`sret` sets `vsstatus.SPP` = 0, `vsstatus.SIE` = `vsstatus.SPIE`, and `vsstatus.SPIE` = 1, changes the privilege mode according to the table below, and then jumps to the address in `vsepc`.

Table 105. Next privilege mode following an `sret` in (H)S-mode or M-mode

vsstatus.SPP	Mode after sret
0	VU-mode
1	VS-mode

Included in

Extension	Version
S	~> 1.11.0

41.450. srl

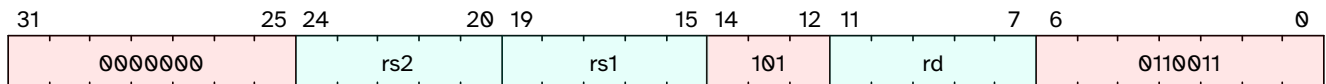
Synopsis

Shift right logical

Mnemonic

rd, rs1, rs2

Encoding



Description

Logical shift the value in rs1 right by the value in the lower bits of rs2, and store the result in rd.

Included in

Extension	Version
I	~> 2.1.0

41.451. srli

Synopsis

Shift right logical immediate

Mnemonic

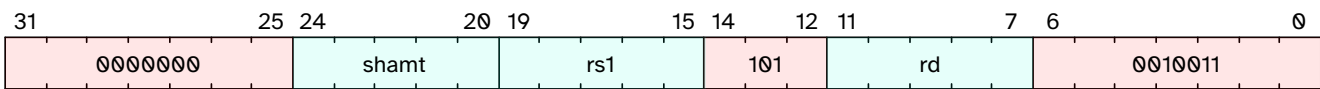
rd, rs1, shamt

Encoding

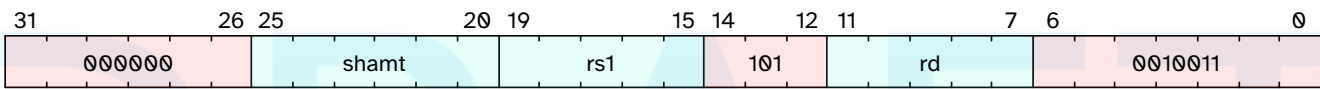


This instruction has different encodings in RV32 and RV64

RV32



RV64



Description

Shift the value in rs1 right by shamt, and store the result in rd

Included in

Extension	Version
I	~> 2.1.0

41.452. srlw

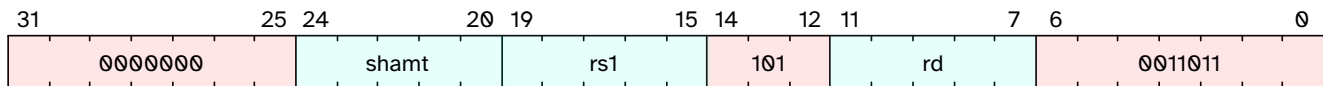
Synopsis

Shift right logical immediate word

Mnemonic

rd, rs1, shamt

Encoding



Description

Shift the 32-bit value in rs1 right by shamt, and store the sign-extended result in rd

Included in

Extension	Version
I	~> 2.1.0

41.453. srlw

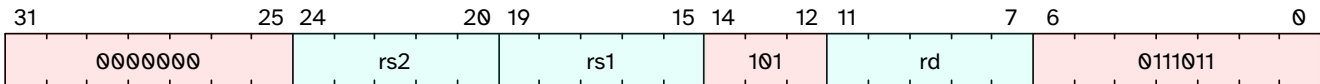
Synopsis

Shift right logical word

Mnemonic

rd, rs1, rs2

Encoding



Description

Logical shift the 32-bit value in rs1 right by the value in the lower 5 bits of rs2, and store the sign-extended result in rd.

Included in

Extension	Version
I	~> 2.1.0

41.454. ssamoswap.d

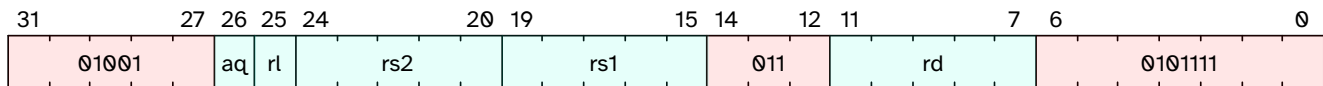
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zicfiss	~> 1.0.0

DRAFT

41.455. ssamoswap.w

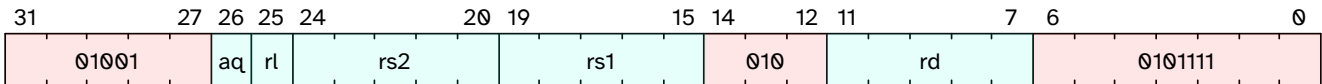
Synopsis

No synopsis available.

Mnemonic

rd, rs1, rs2, aq, rl

Encoding



Description

No description available.

Included in

Extension	Version
Zicfiss	~> 1.0.0

41.456. sspopchk.x1

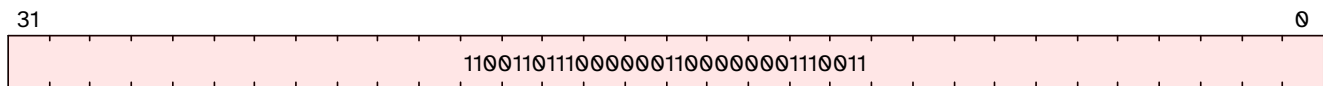
Synopsis

No synopsis available.

Mnemonic

sspopchk_r1

Encoding



Description

No description available.

Included in

Extension	Version
Zicfiss	~> 1.0.0

41.457. sspopchk.x5

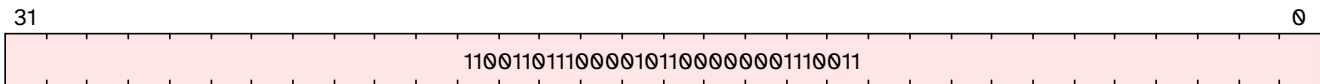
Synopsis

No synopsis available.

Mnemonic

sspopchk_r5

Encoding



Description

No description available.

Included in

Extension	Version
Zicfiss	~> 1.0.0

41.458. sspush.x1

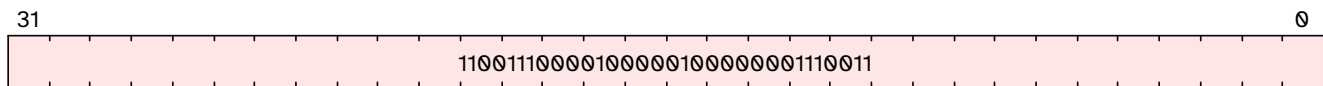
Synopsis

No synopsis available.

Mnemonic

sspush_r1

Encoding



Description

No description available.

Included in

Extension	Version
Zicfiss	~> 1.0.0

41.459. sspush.x5

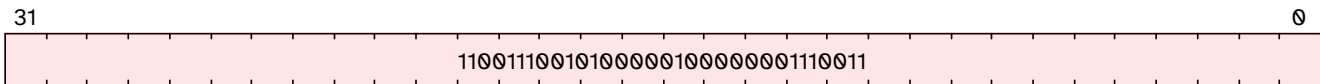
Synopsis

No synopsis available.

Mnemonic

sspush_r5

Encoding



Description

No description available.

Included in

Extension	Version
Zicfiss	~> 1.0.0

41.460. ssrdp

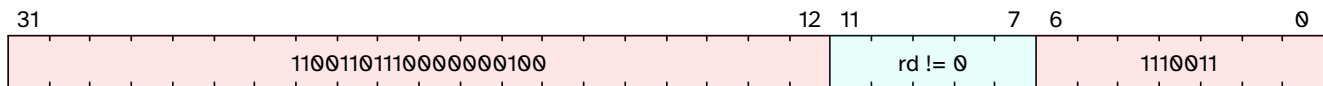
Synopsis

No synopsis available.

Mnemonic

rd

Encoding



Description

No description available.

Included in

Extension	Version
Zicfiss	~> 1.0.0

DRAFT

41.461. sub

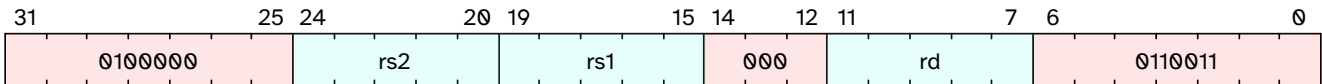
Synopsis

Subtract

Mnemonic

rd, rs1, rs2

Encoding



Description

Subtract the value in rs2 from rs1, and store the result in rd

Included in

Extension	Version
I	~> 2.1.0

41.462. subw

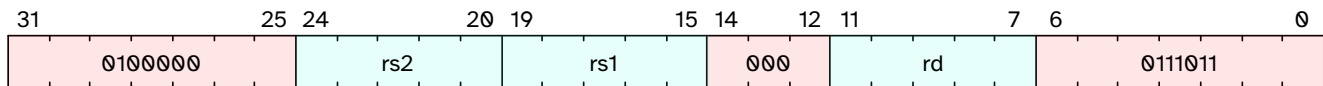
Synopsis

Subtract word

Mnemonic

rd, rs1, rs2

Encoding



Description

Subtract the 32-bit values in rs2 from rs1, and store the sign-extended result in rd

Included in

Extension	Version
I	~> 2.1.0

41.463. sw

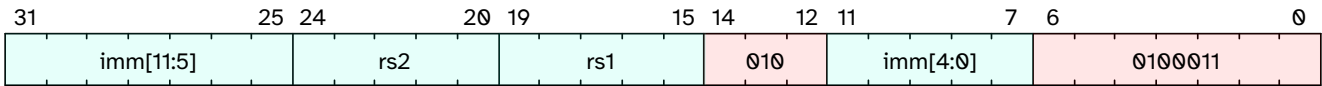
Synopsis

Store word

Mnemonic

```
rs2, imm(rs1)
```

Encoding



Description

Store 32 bits of data from register rs2 to an address formed by adding rs1 to a signed offset.

Included in

Extension	Version
I	~> 2.1.0

41.464. sw.rl

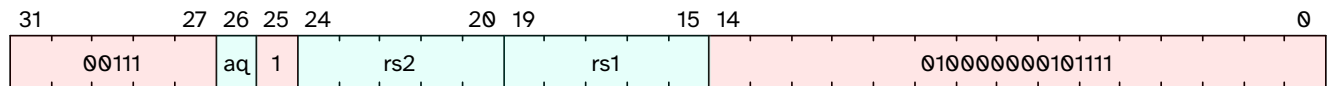
Synopsis

No synopsis available.

Mnemonic

rs1, rs2, aq

Encoding



Description

No description available.

Included in

Extension	Version
Zalasr	~> 0.3.5

DRAFT

41.465. unzip

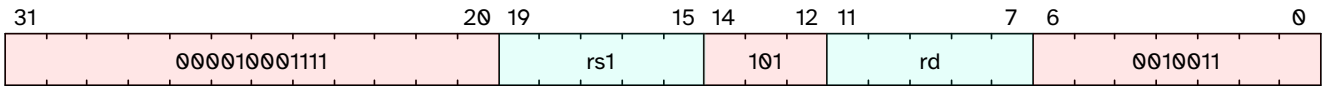
Synopsis

Bit deinterleave

Mnemonic

rd, rs1

Encoding



Description

This instruction gathers bits from the high and low halves of the source word into odd/even bit positions in the destination word. It is the inverse of the zip instruction. This instruction is available only on RV32.

Included in

Extension	Version
Zbkb	~> 1.0.0

41.466. vaadd.vv

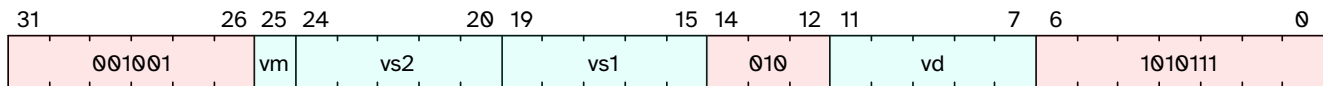
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.467. vaadd.vx

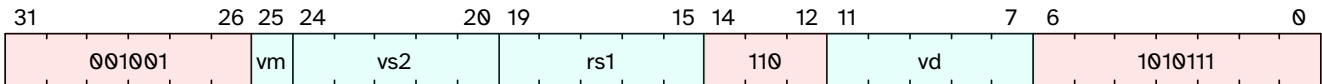
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.468. vaaddu.vv

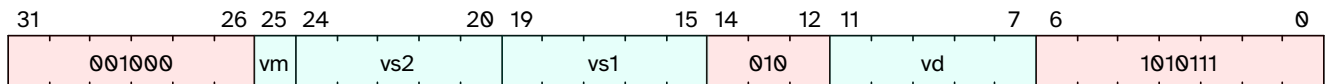
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.469. vaaddu.vx

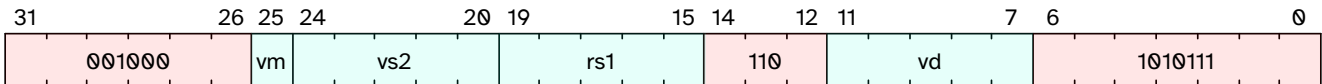
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.470. vadc.vim

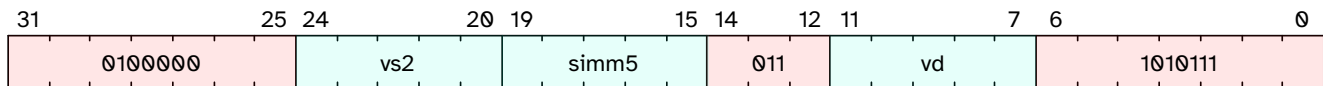
Synopsis

No synopsis available.

Mnemonic

vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.471. vadc.vvm

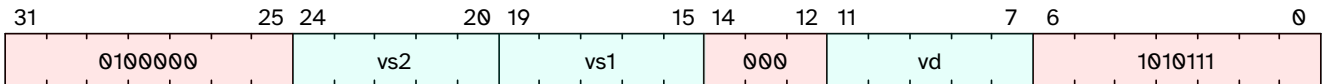
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.472. vadc.vxm

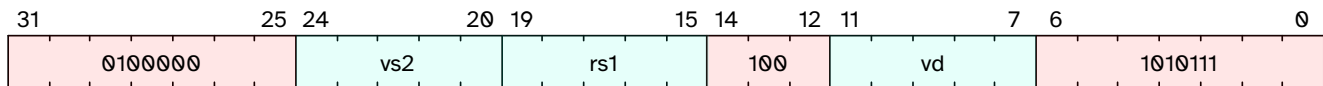
Synopsis

No synopsis available.

Mnemonic

vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.473. vadd.vi

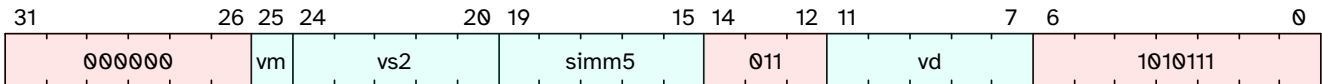
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.474. vadd.vw

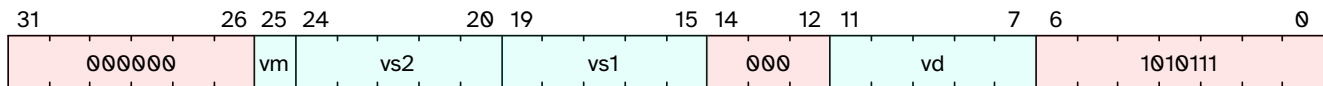
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.475. vadd.vx

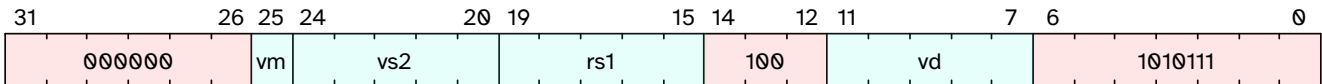
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.476. vaesdf.vs

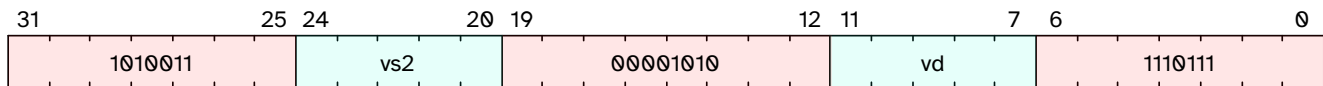
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvkned	~> 1.0.0

41.477. vaesdf.vv

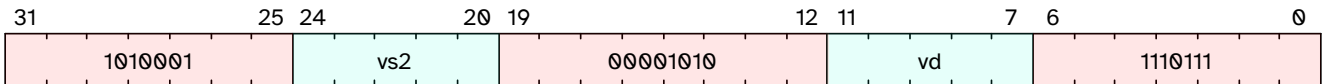
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvkned	~> 1.0.0

41.478. vaesdm.vs

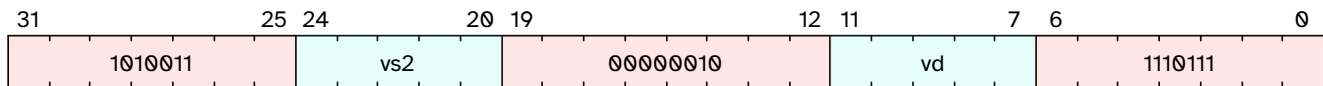
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvkned	~> 1.0.0

41.479. vaesdm.vv

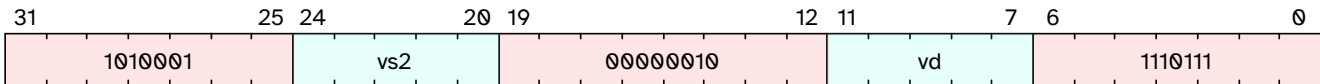
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvkned	~> 1.0.0

41.480. vaesef.vs

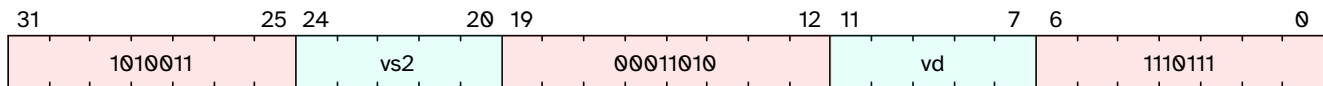
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvkned	~> 1.0.0

41.481. vaesef.vv

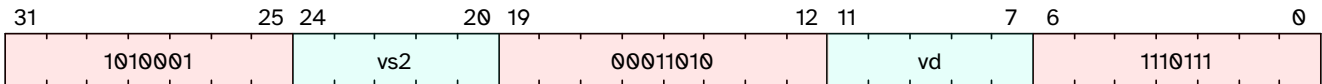
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvkned	~> 1.0.0

41.482. vaesem.vs

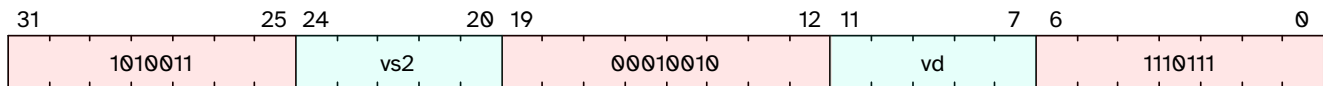
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvkned	~> 1.0.0

41.483. vaesem.vv

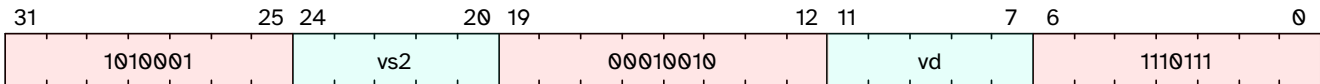
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvkned	~> 1.0.0

41.484. vaeskf1.vi

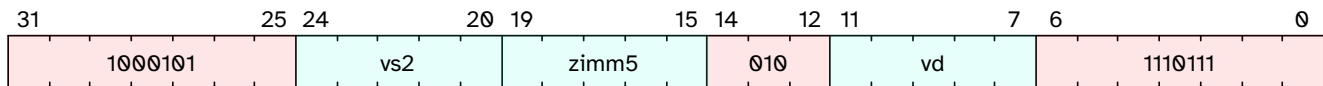
Synopsis

No synopsis available.

Mnemonic

vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvkned	~> 1.0.0

41.485. vaeskf2.vi

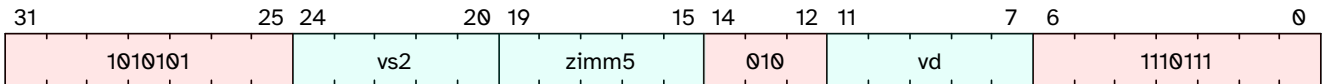
Synopsis

No synopsis available.

Mnemonic

vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvkned	~> 1.0.0

41.486. vaesz.vs

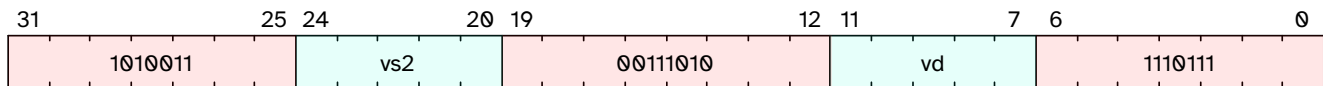
Synopsis

Vector AES round zero

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvkned	~> 1.0.0

41.487. vand.vi

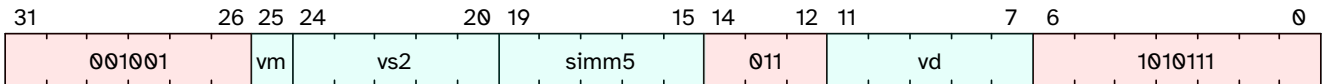
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.488. vand.vv

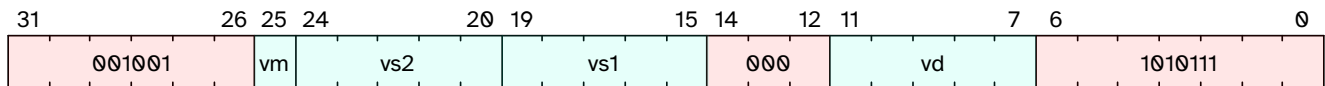
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.489. vand.vx

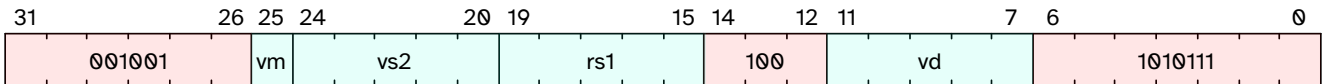
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.490. vandn.vv

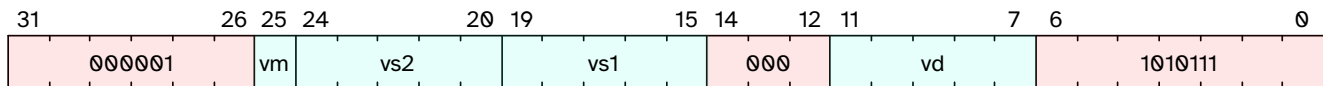
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.491. vandn.vx

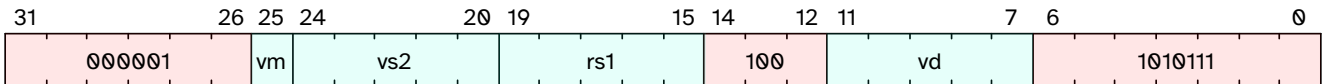
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.492. vasub.vv

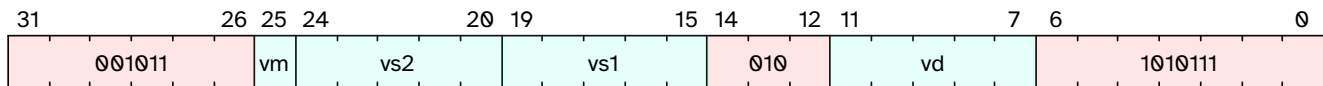
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.493. vasub.vx

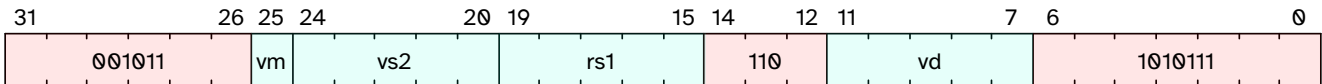
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.494. vasubu.vv

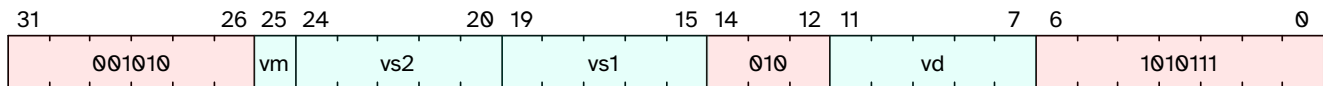
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.495. vasubu.vx

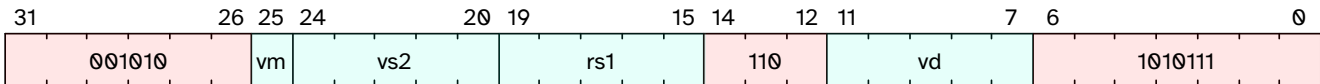
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.496. vbrev.v

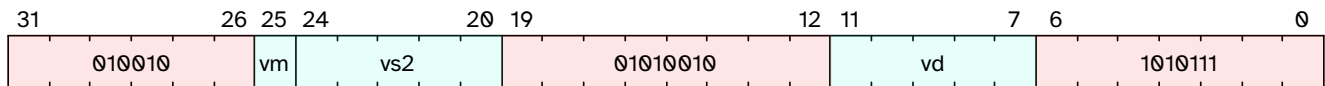
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.497. vbrev8.v

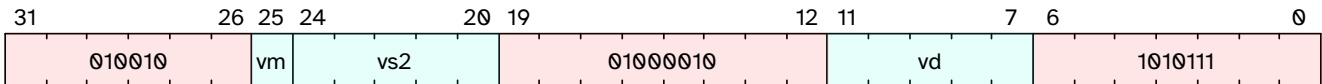
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.498. vclmul.vv

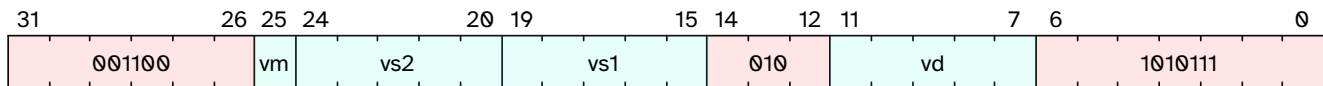
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbc	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.499. vclmul.vx

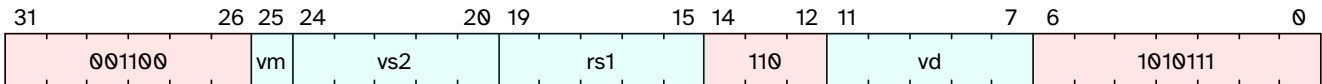
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbc	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.500. vclmulh.vv

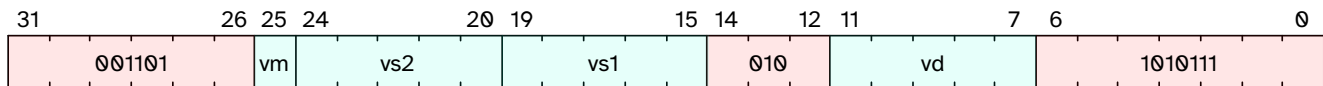
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbc	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.501. vclmulh.vx

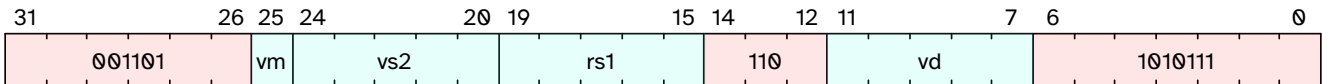
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbc	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.502. vclz.v

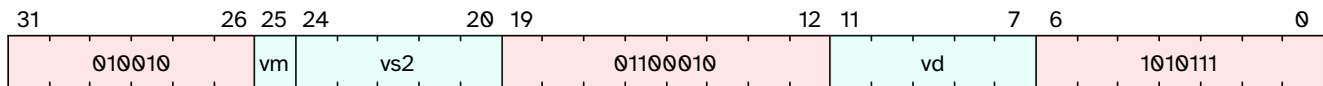
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.503. vcompress.vm

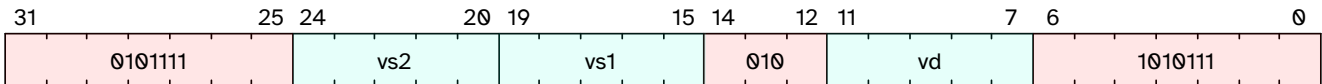
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.504. vcpop.m

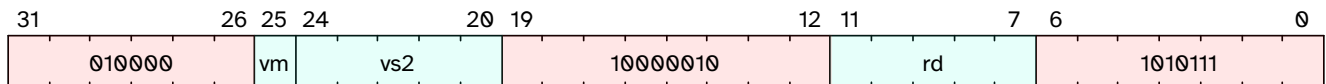
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.505. vcpop.v

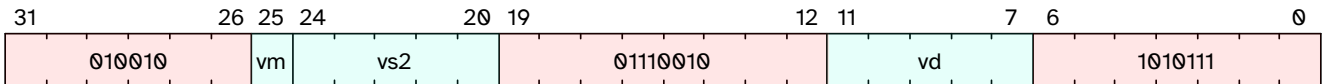
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.506. vctz.v

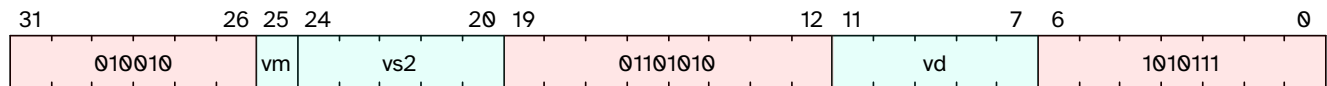
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.507. vdiv.vv

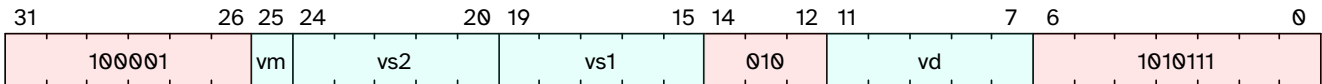
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.508. vdiv.vx

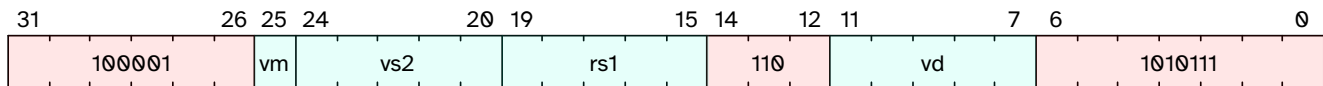
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.509. vdivu.vv

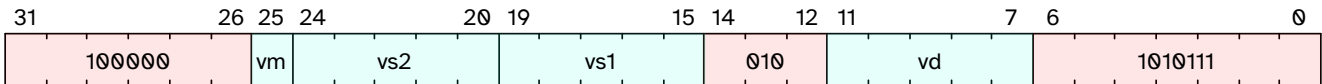
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.510. vdivu.vx

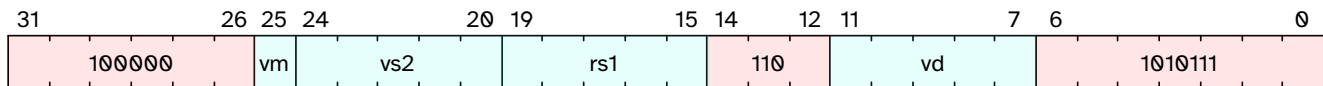
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.511. vfadd.vf

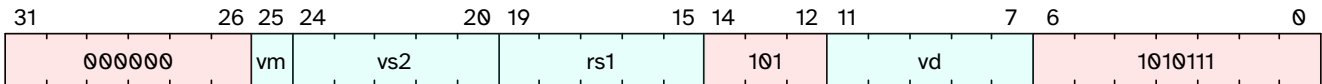
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.512. vfadd.wv

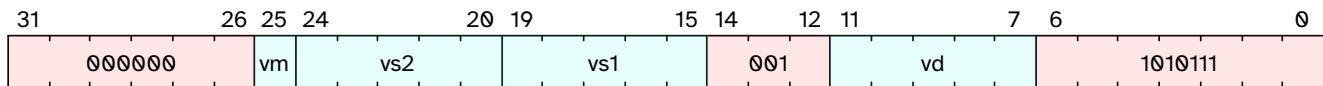
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.513. vfclass.v

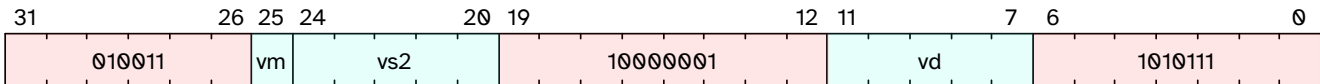
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.514. vfcvt.f.x.v

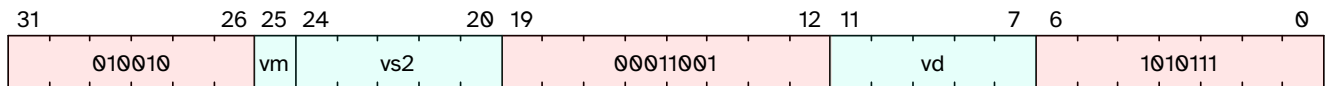
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.515. vfcvt.f.xu.v

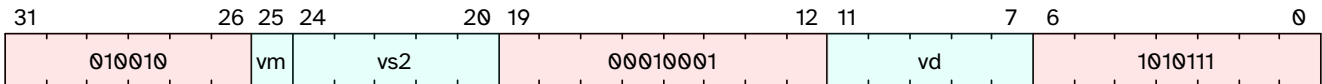
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.516. vfcvt.rtz.x.f.v

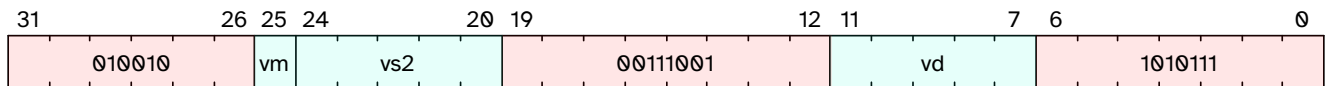
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.517. vfcvt.rtz.xu.f.v

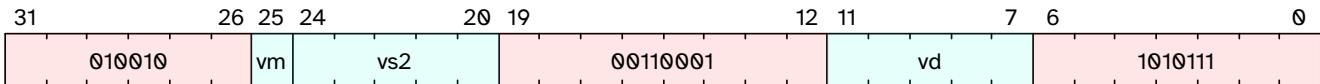
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.518. vfcvt.x.f.v

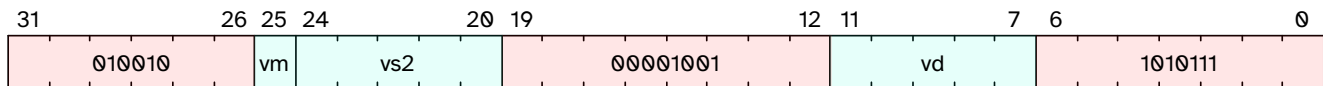
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.519. vfcvt.xu.f.v

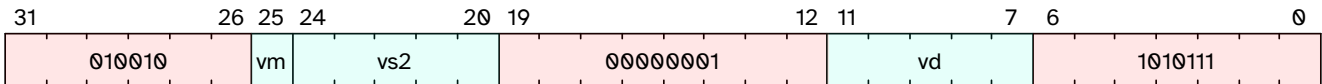
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.520. vfddiv.vf

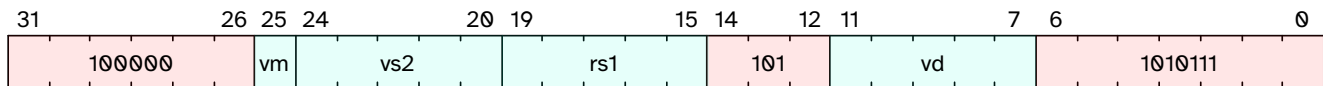
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.521. vfdiv.vv

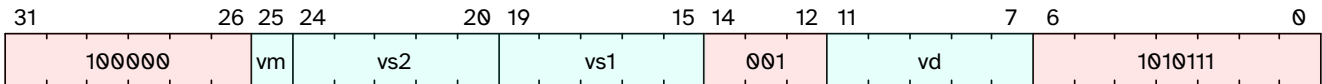
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.522. vfirst.m

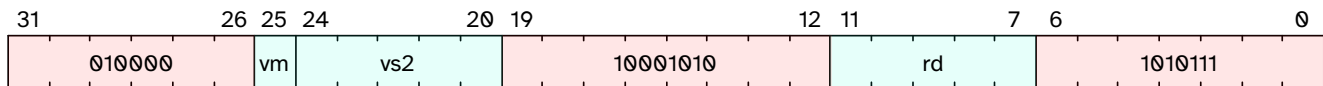
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.523. vfmacc.vf

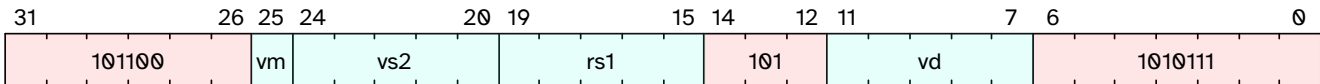
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.524. vfmacc.vv

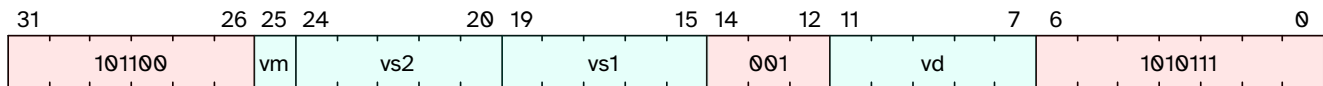
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.525. vfmadd.vf

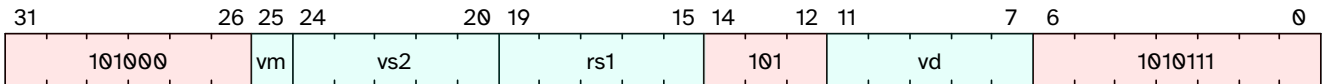
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.526. vfmadd.vv

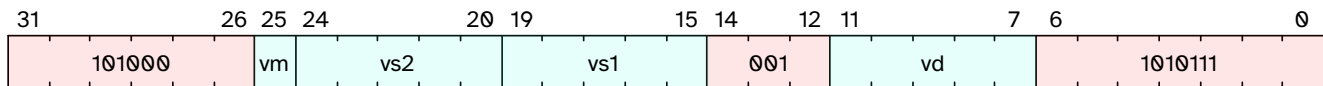
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.527. vfm_{max}.vf

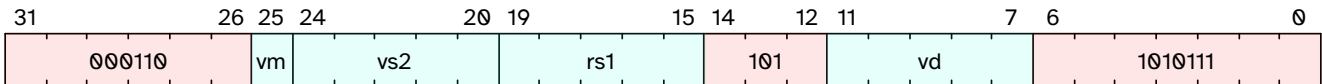
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.528. vfm_{max}.w

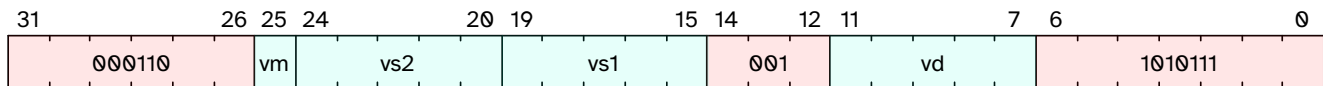
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.529. vfmerge.vfm

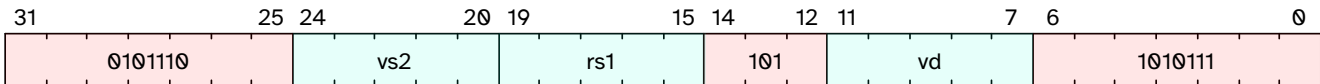
Synopsis

No synopsis available.

Mnemonic

vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.530. vfmin.vf

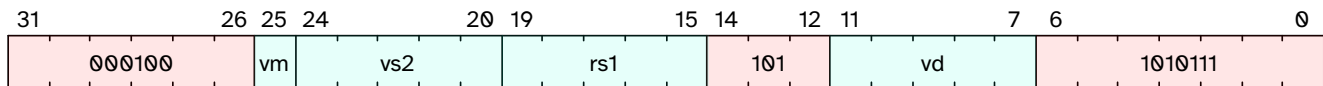
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.531. vfmin.wv

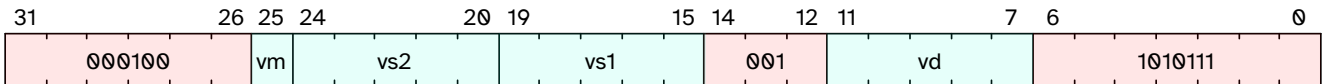
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.532. vfmsac.vf

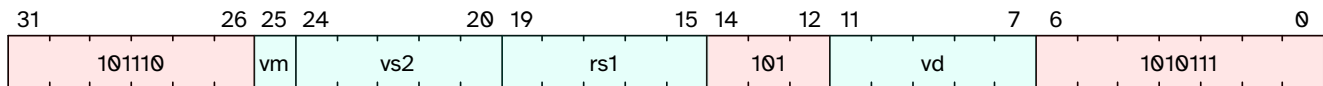
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.533. vfmsac.vv

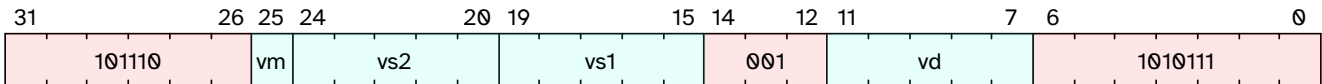
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.534. vfmsub.vf

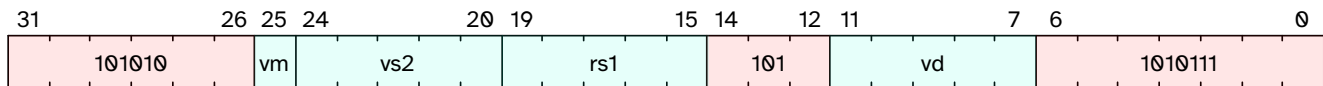
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.535. vfmsub.vv

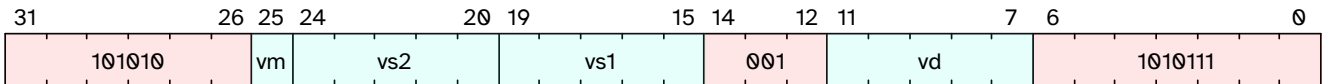
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.536. vfmul.vf

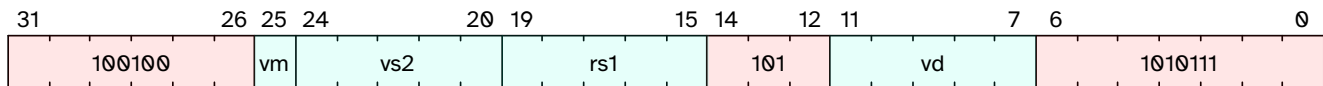
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.537. vfmul.vv

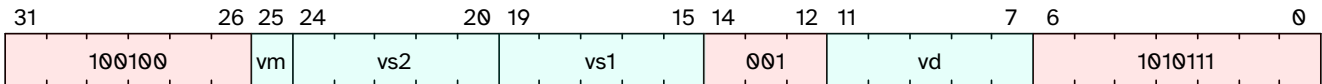
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.538. vfmv.f.s

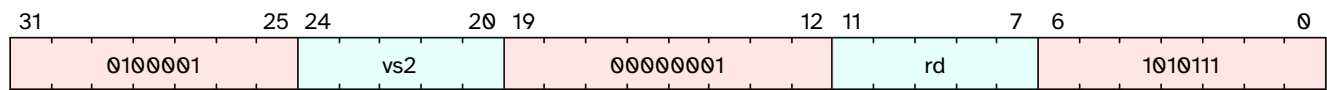
Synopsis

No synopsis available.

Mnemonic

vs2, rd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.539. vfmv.s.f

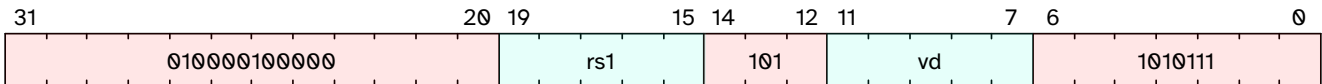
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.540. vfmv.v.f

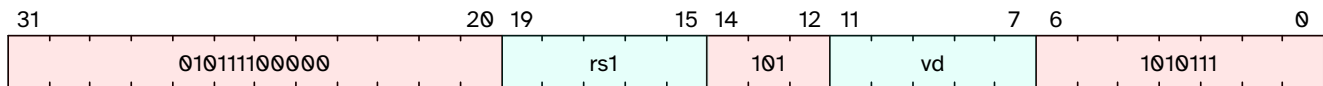
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.541. vfncvt.f.f.w

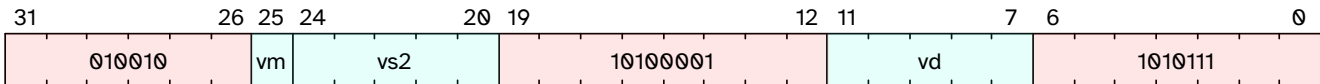
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.542. vfncvt.f.x.w

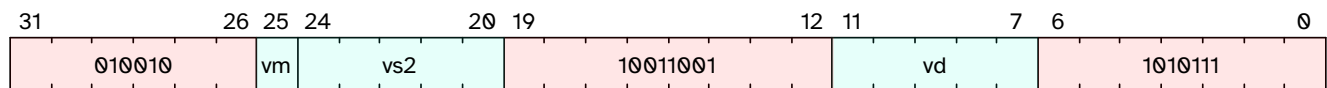
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.543. vfncvt.f.xu.w

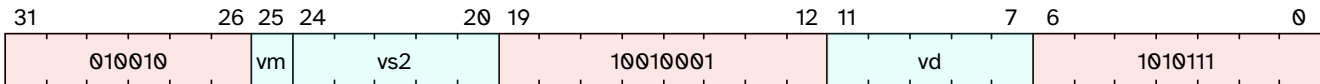
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.544. vfncvt.rod.f.f.w

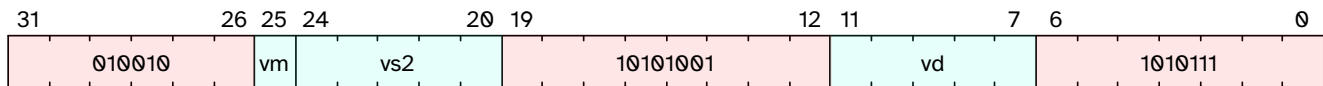
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.545. vfncvt.rtz.x.f.w

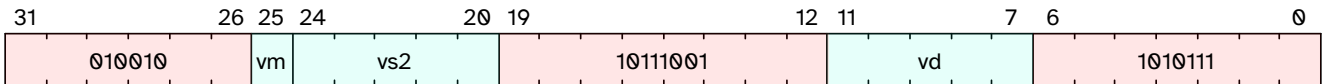
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.546. vfncvt.rtz.xu.f.w

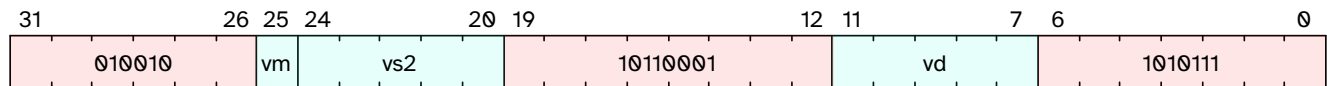
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.547. vfncvt.x.f.w

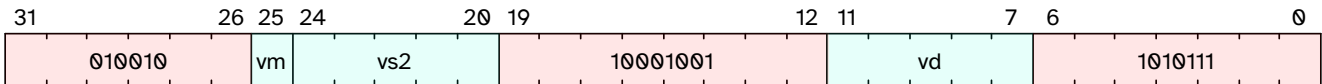
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.548. vfncvt.xu.f.w

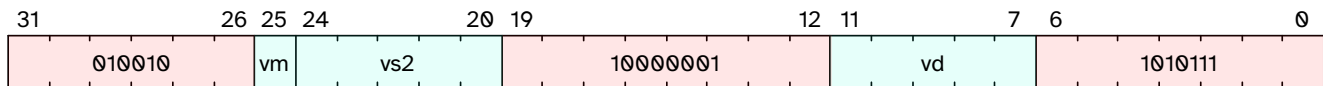
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.549. vfncvtbf16.f.f.w

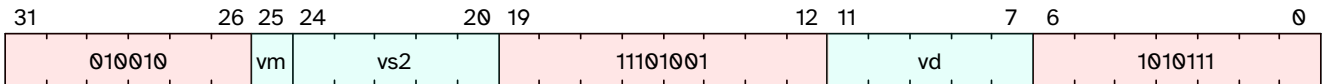
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvfbfmin	~> 1.0.0

41.550. vfnmacc.vf

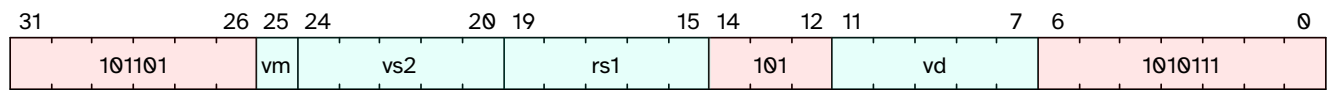
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.551. vfnmacc.vv

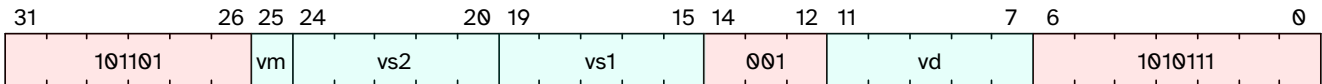
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.552. vfnmadd.vf

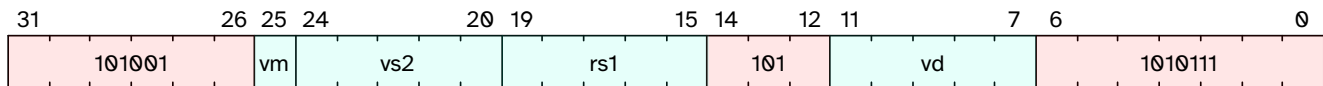
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.553. vfnmadd.vv

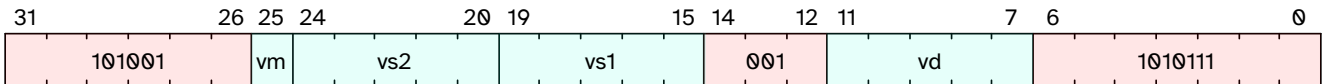
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.554. vfnmsac.vf

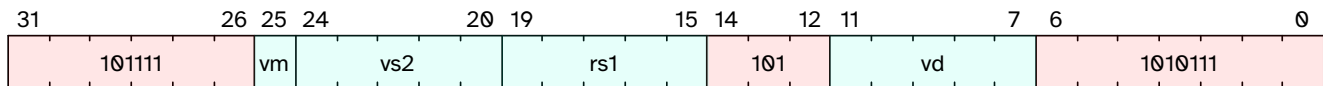
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.555. vfnmsac.vv

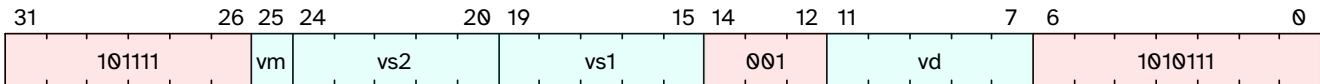
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.556. vfnmsub.vf

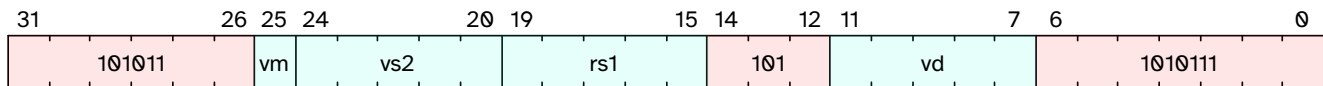
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.557. vfnmsub.vv

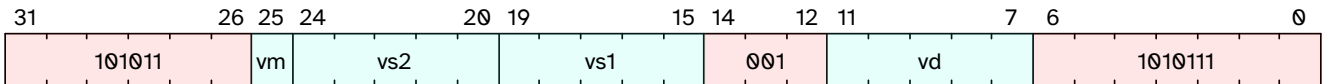
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.558. vfrdiv.vf

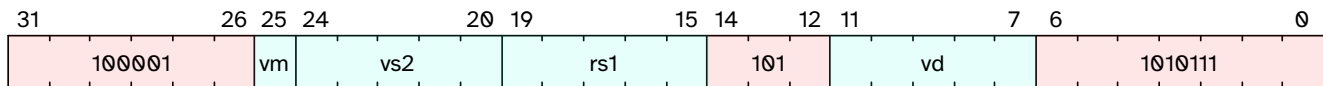
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.559. vfrec7.v

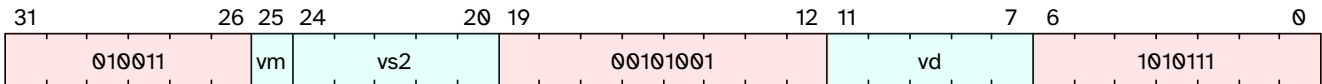
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.560. vfredmax.vs

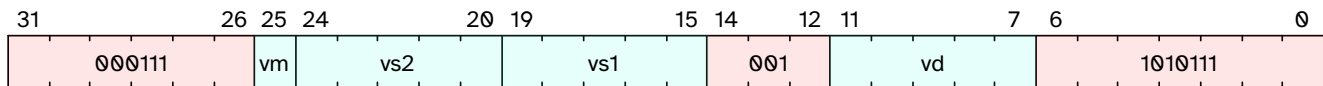
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.561. vfredmin.vs

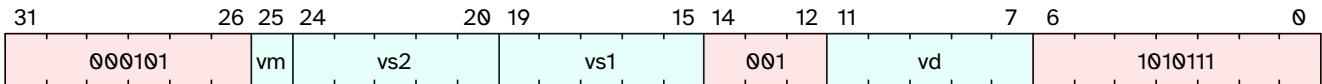
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.562. vfredosum.vs

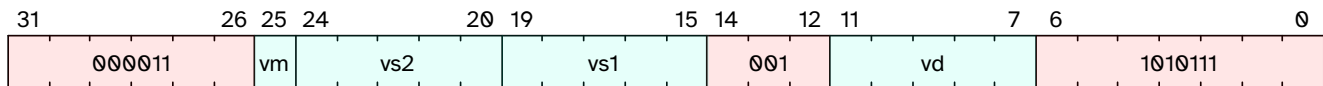
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.563. vfredusum.vs

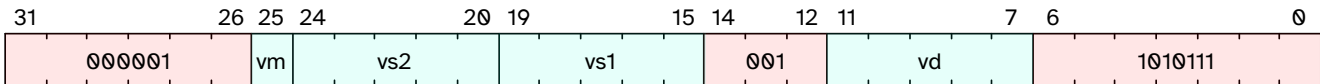
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.564. vfrsqr7.v

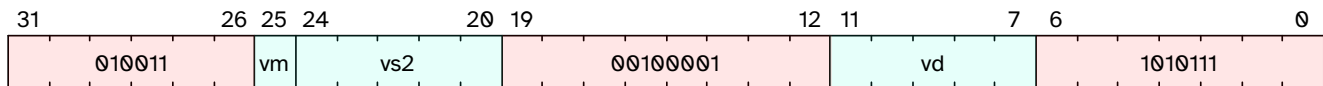
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.565. vfrsub.vf

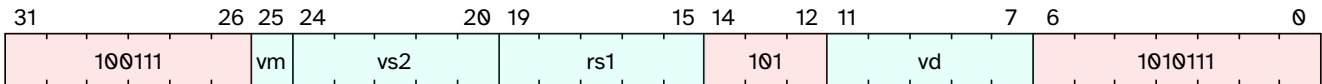
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.566. vfsgnj.vf

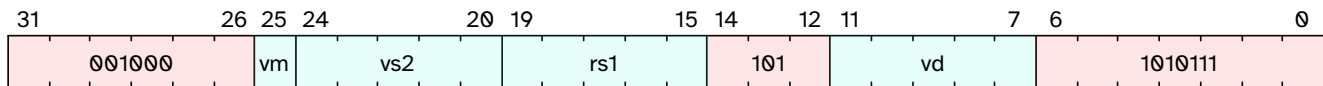
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.567. vfsgnj.vv

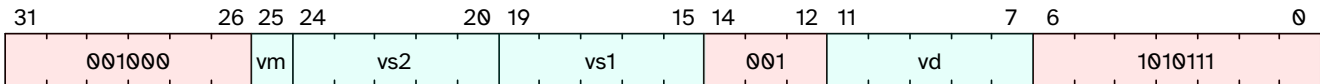
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.568. vfgsnjn.vf

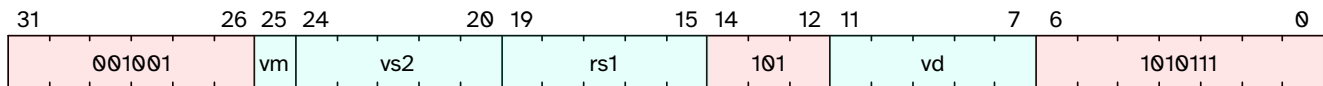
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.569. vfsgnjn.vv

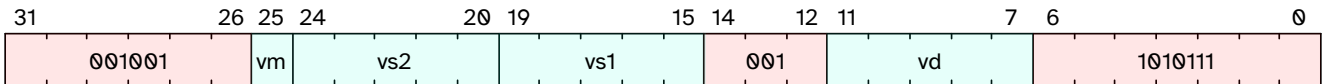
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.570. vfsgnjx.vf

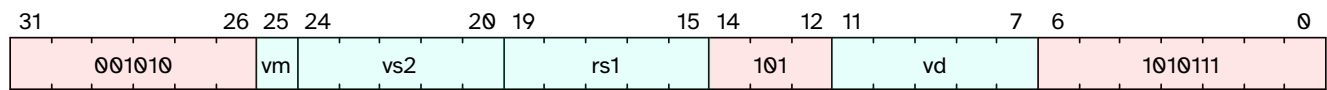
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.571. vfgsnjx.vv

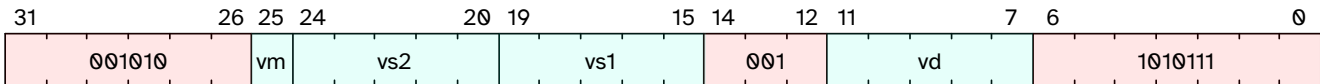
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.572. vfslide1down.vf

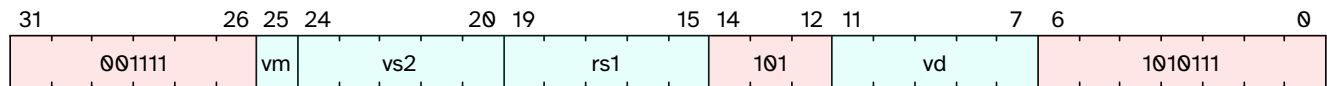
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.573. vslide1up.vf

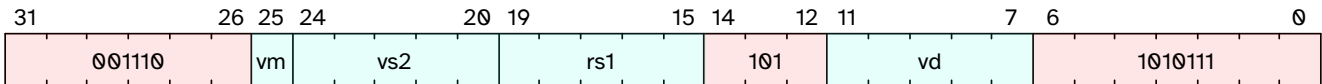
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.574. vfsqrt.v

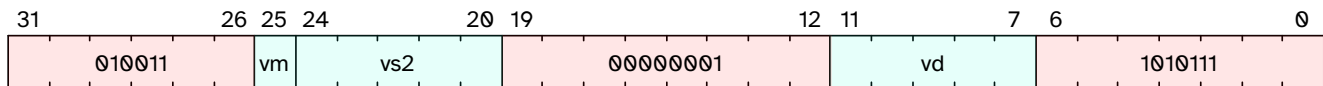
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.575. vsub.vf

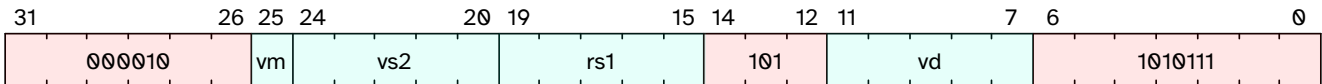
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.576. vsub.wv

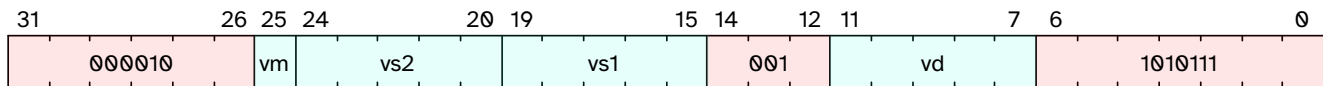
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.577. vfwadd.vf

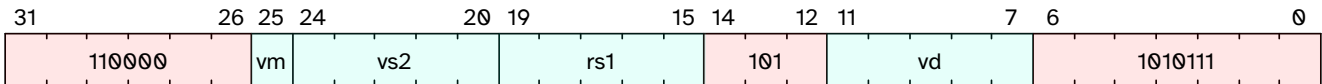
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.578. vfwadd.wv

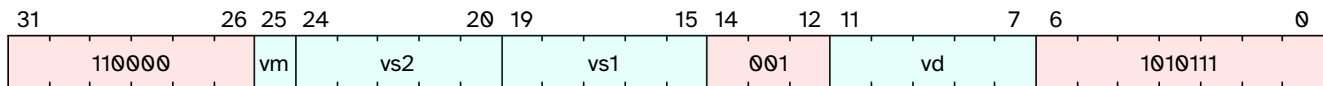
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.579. vfwadd.wf

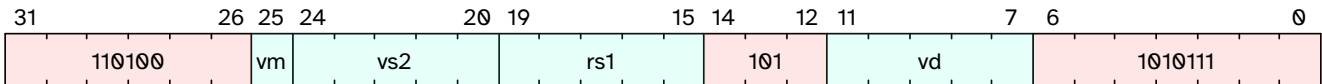
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.580. vfwadd.wv

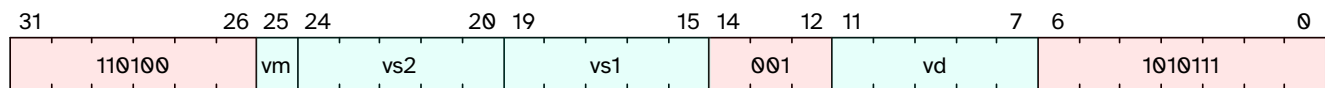
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.581. vfwcvt.f.f.v

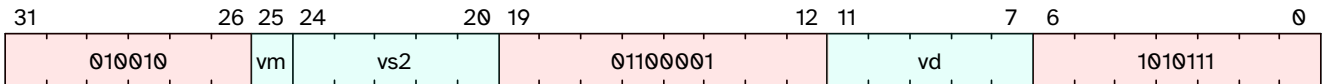
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.582. vfwcvt.f.x.v

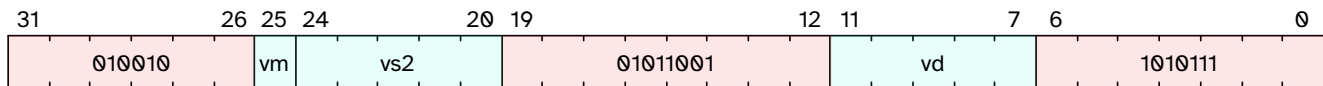
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.583. vfwcvt.f.xu.v

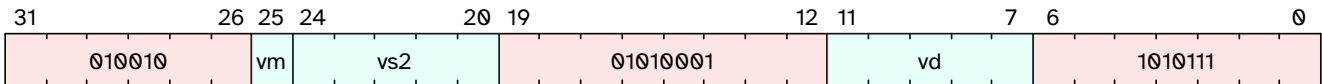
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.584. vfwcvt.rtz.x.f.v

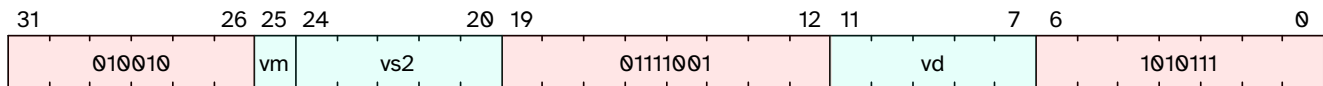
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.585. vfwcvt.rtz.xu.f.v

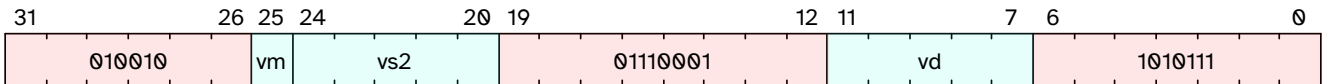
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.586. vfwcvt.x.f.v

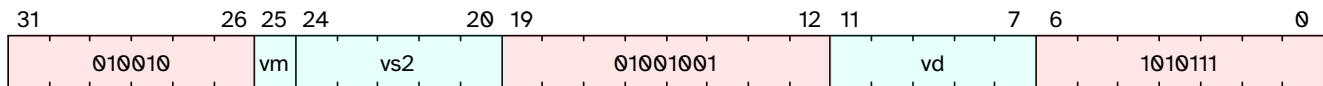
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.587. vfwcvt.xu.f.v

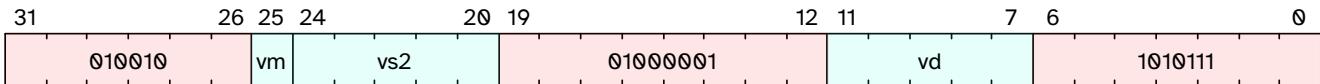
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.588. vfwcvtb16.f.f.v

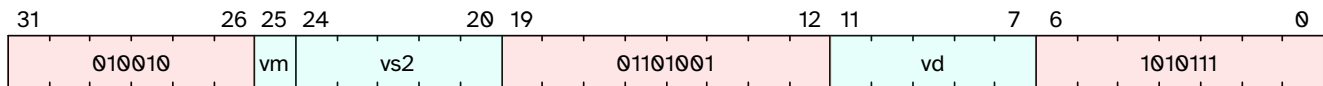
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvfbfmin	~> 1.0.0

DRAFT

41.589. vfwmaccc.vf

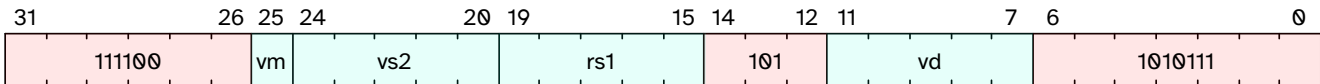
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.590. vfwmaccc.vv

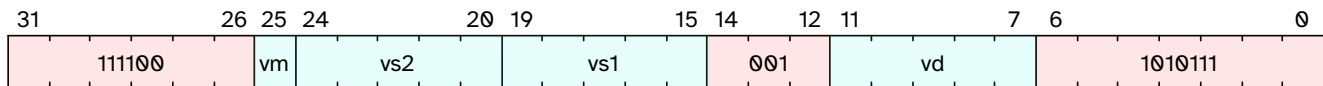
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.591. vfwmaccbf16.vf

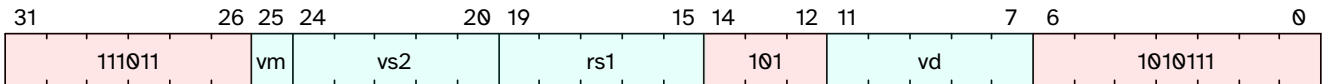
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvfbfwna	~> 1.0.0

41.592. vfwmaccbf16.vv

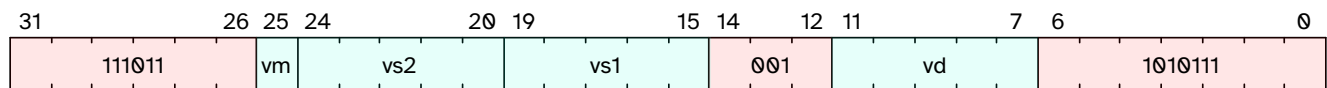
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvfbfwna	~> 1.0.0

DRAFT

41.593. vfwmsac.vf

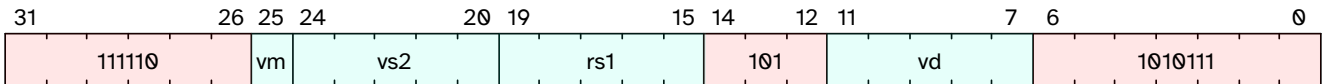
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.594. vfwmsac.vv

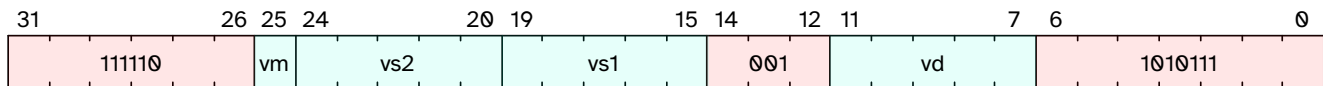
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.595. vfwmul.vf

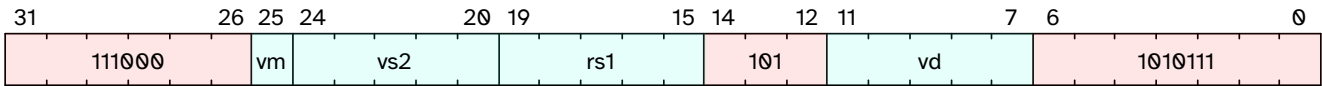
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.596. vfwmul.vv

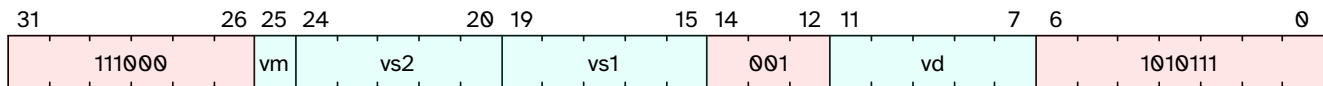
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.597. vfwnmacc.vf

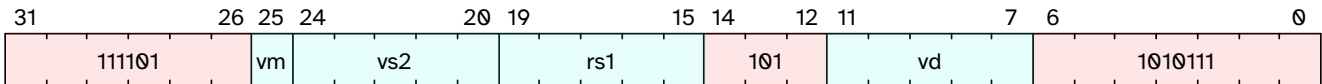
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.598. vfwnmacc.vv

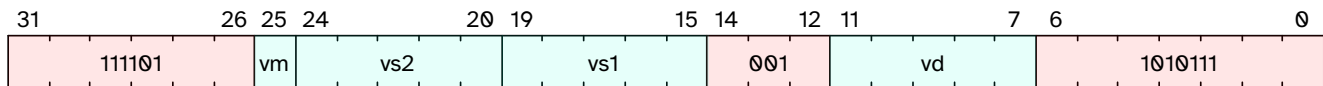
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.599. vfwnmsac.vf

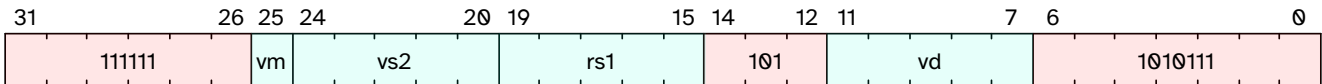
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.600. vfnmsac.vv

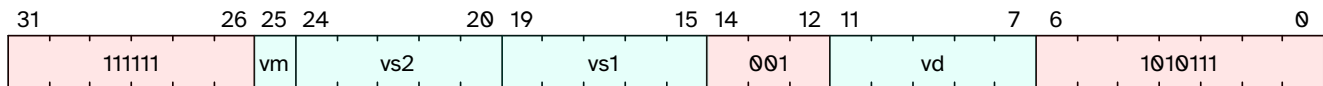
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.601. vfwredosum.vs

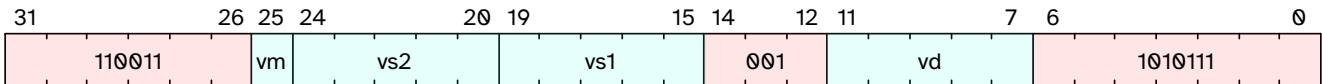
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.602. vfwredusum.vs

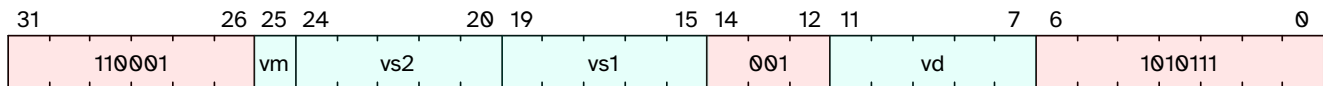
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.603. vfwsb.vf

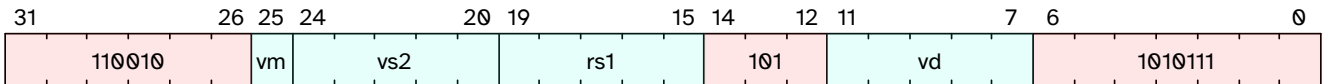
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.604. vfwsb.vv

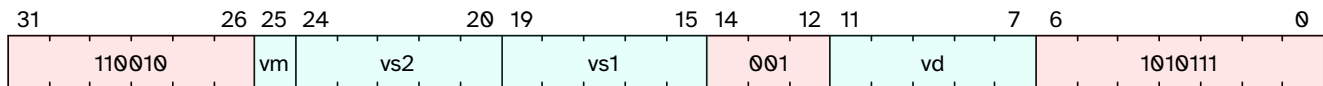
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.605. vfwsb.wf

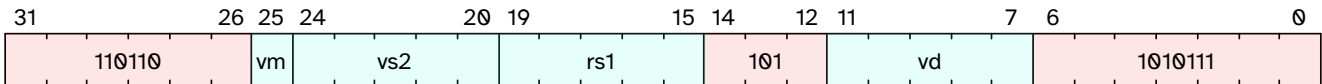
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.606. vfwsb.wv

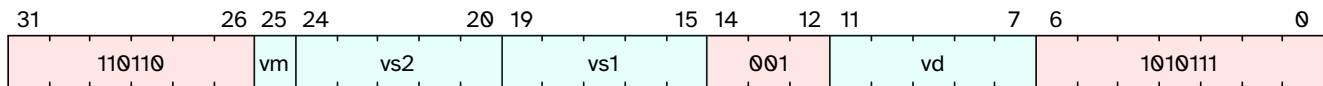
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.607. vghsh.vv

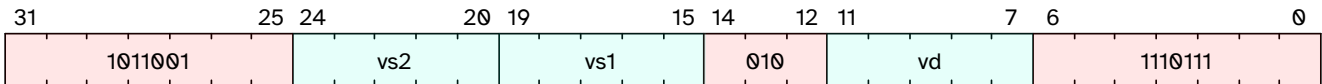
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkg	~> 1.0.0

41.608. vgmul.vv

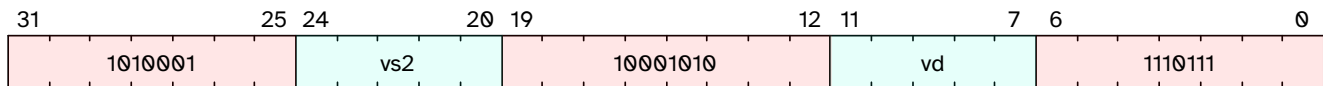
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkg	~> 1.0.0

41.609. vid.v

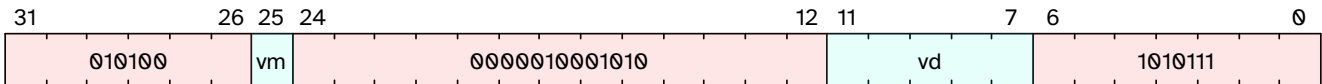
Synopsis

No synopsis available.

Mnemonic

vm, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.610. viota.m

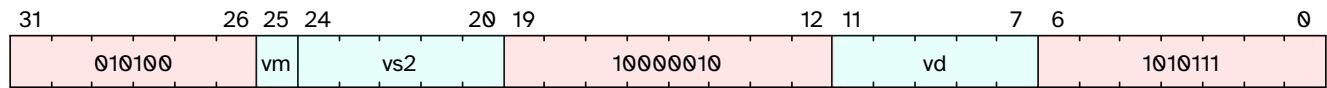
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.611. vl1re16.v

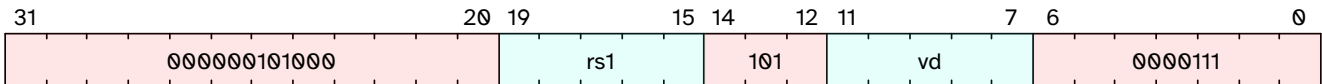
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.612. vl1re32.v

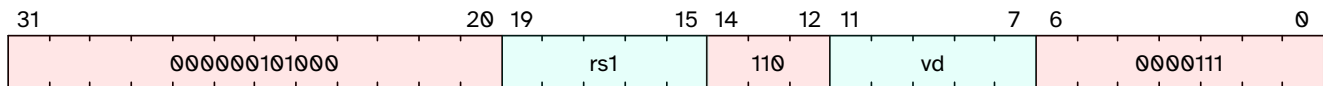
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.613. vl1re64.v

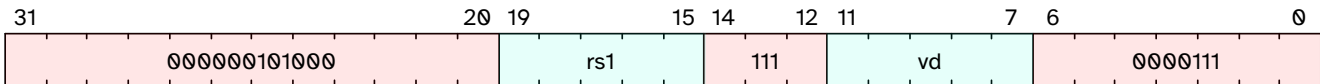
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.614. vl1re8.v

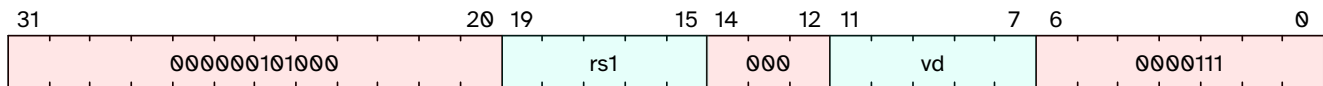
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.615. vl2re16.v

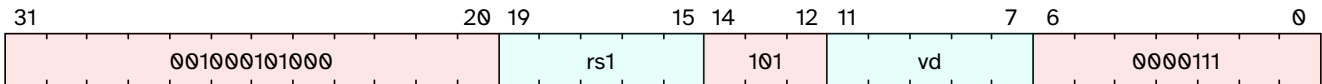
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.616. vl2re32.v

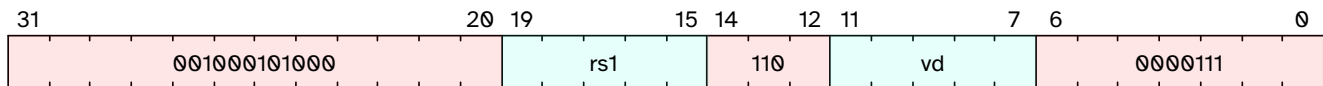
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.617. vl2re64.v

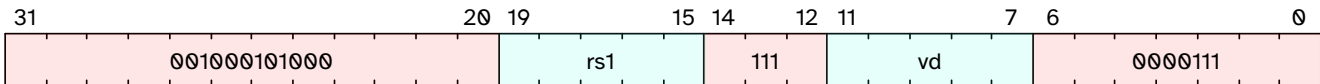
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.618. vl2re8.v

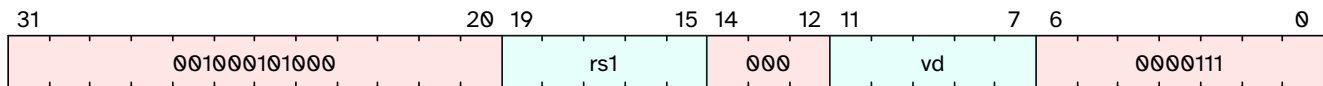
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.619. vl4re16.v

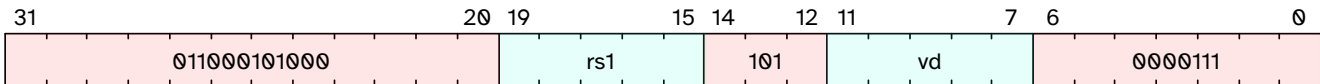
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.620. vl4re32.v

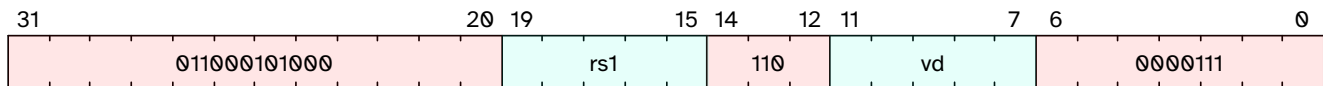
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.621. vl4re64.v

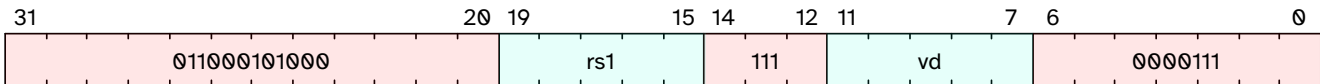
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.622. vl4re8.v

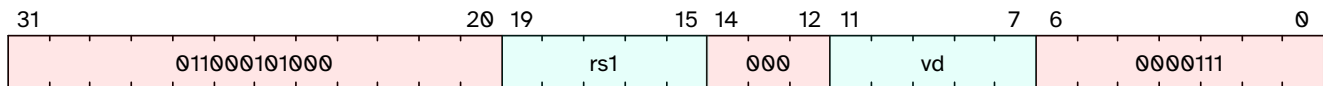
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.623. vl8re16.v

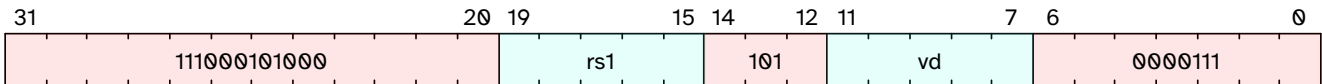
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.624. vl8re32.v

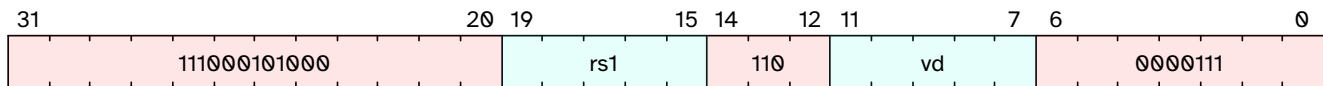
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.625. vl8re64.v

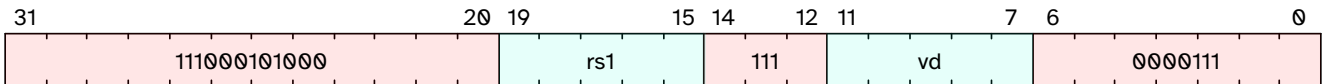
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.626. vl8re8.v

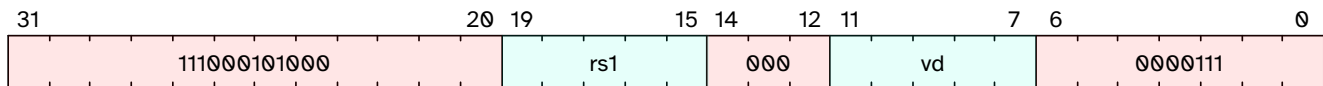
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.627. vle16.v

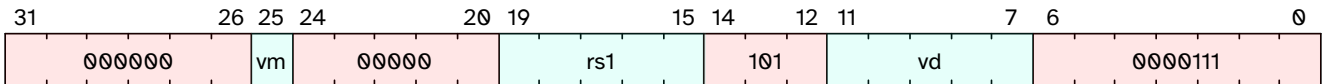
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.628. vle16ff.v

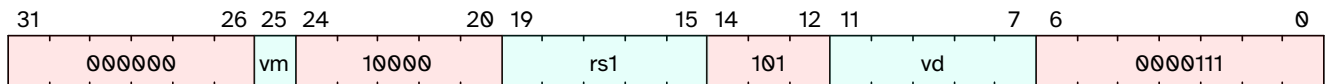
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.629. vle32.v

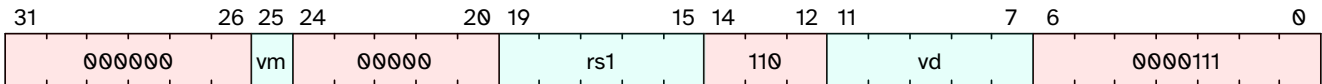
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.630. vle32ff.v

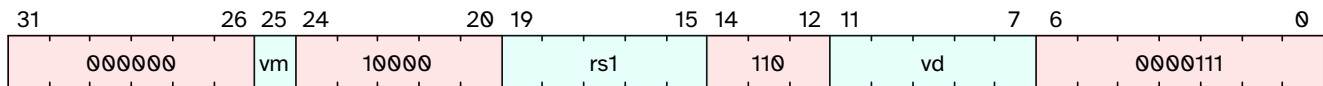
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.631. vle64.v

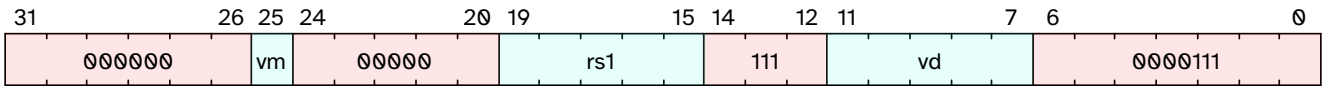
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.632. vle64ff.v

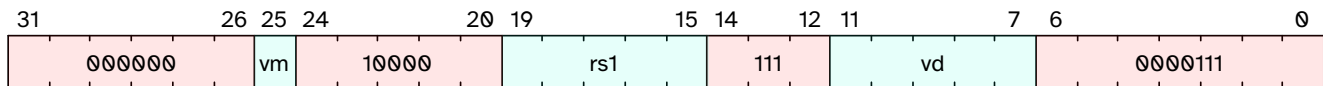
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.633. vle8.v

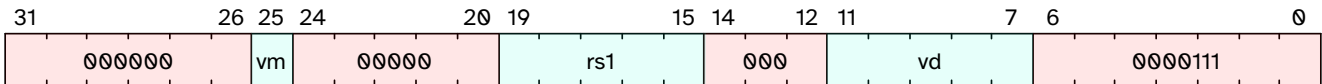
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.634. vle8ff.v

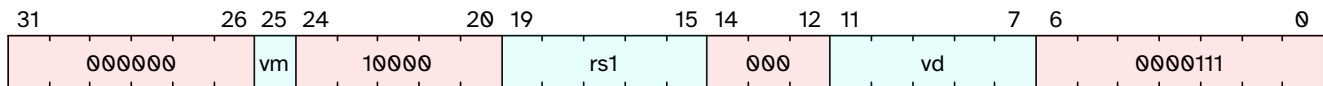
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.635. vlm.v

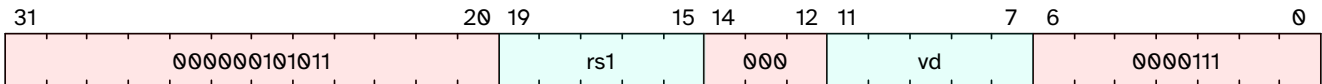
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.636. vloxei16.v

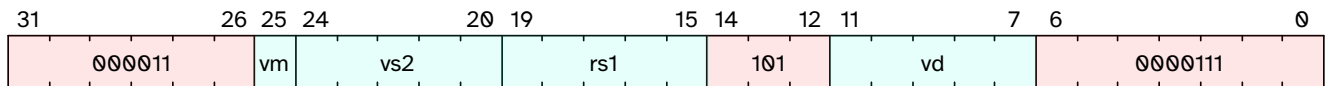
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.637. vloxei32.v

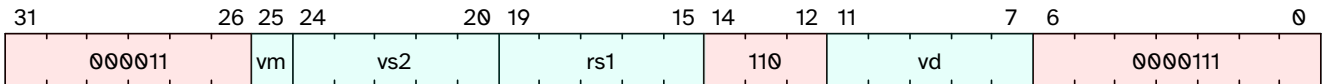
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.638. vloxei64.v

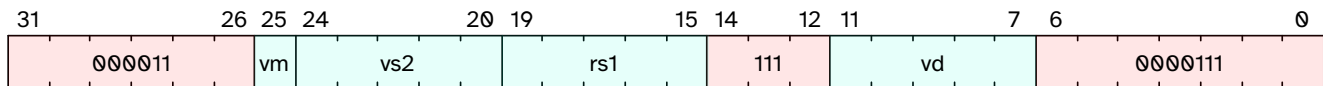
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.639. vloxei8.v

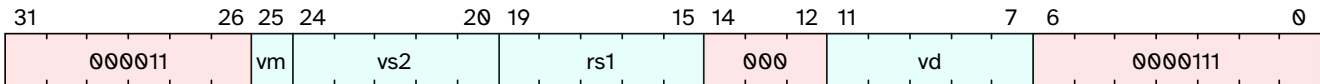
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.640. vloxseg2ei16.v

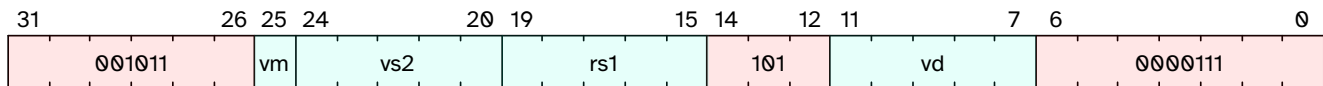
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.641. vloxseg2ei32.v

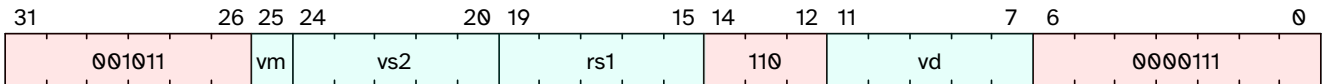
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.642. vloxseg2ei64.v

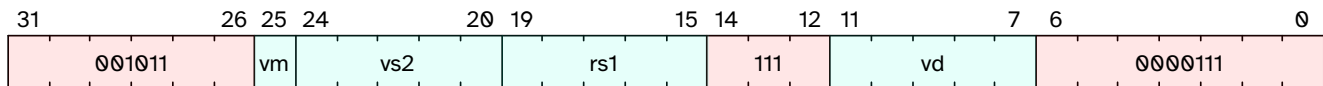
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.643. vloxseg2ei8.v

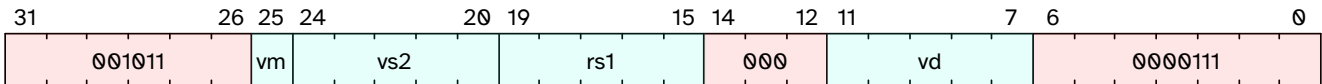
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.644. vloxseg3ei16.v

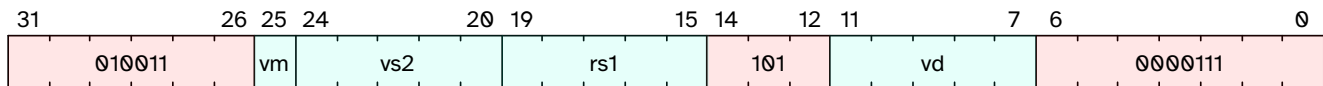
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.645. vloxseg3ei32.v

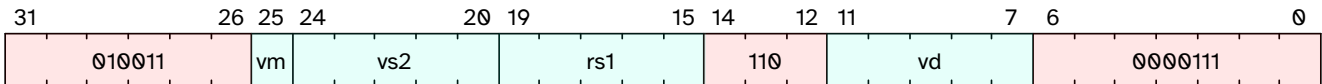
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.646. vloxseg3ei64.v

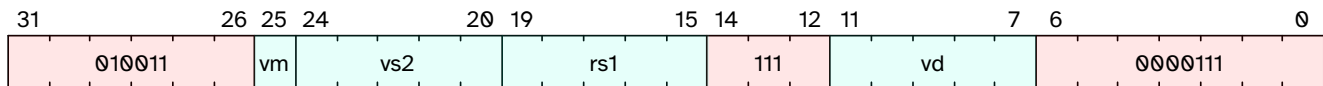
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.647. vloxseg3ei8.v

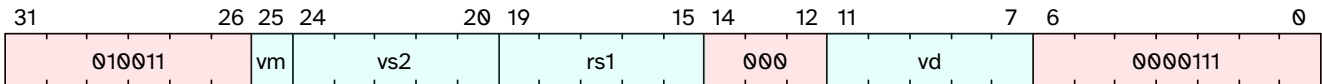
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.648. vloxseg4ei16.v

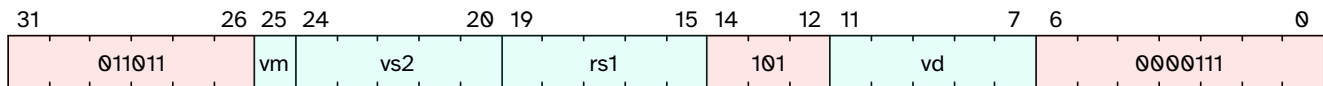
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.649. vloxseg4ei32.v

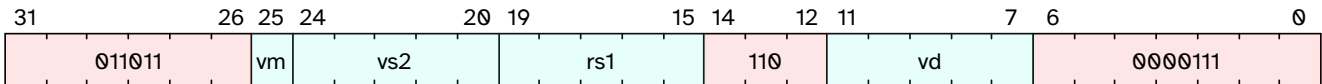
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.650. vloxseg4ei64.v

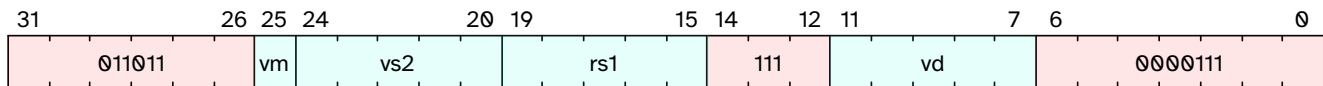
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.651. vloxseg4ei8.v

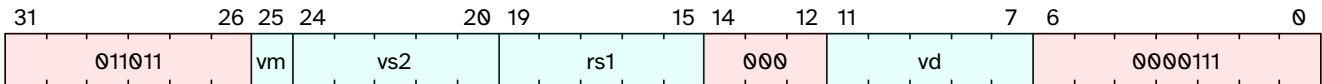
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.652. vloxseg5ei16.v

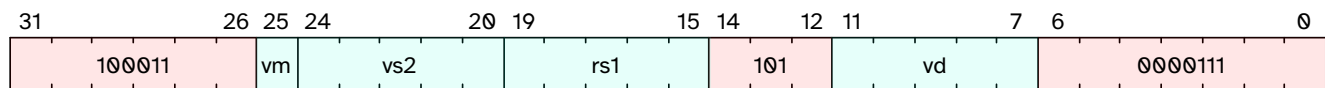
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.653. vloxseg5ei32.v

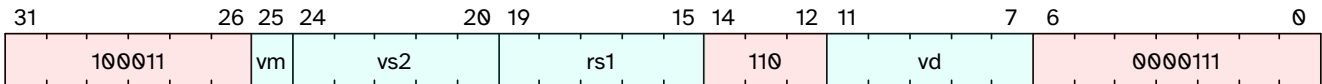
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.654. vloxseg5ei64.v

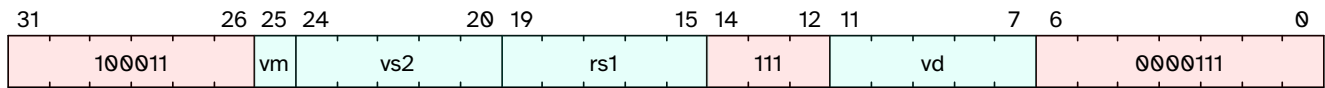
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.655. vloxseg5ei8.v

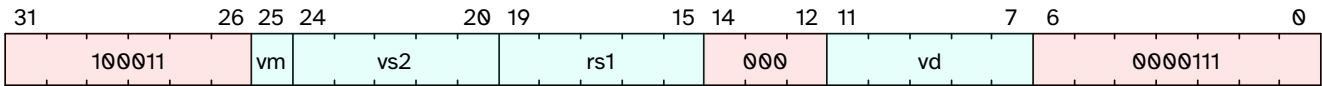
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.656. vloxseg6ei16.v

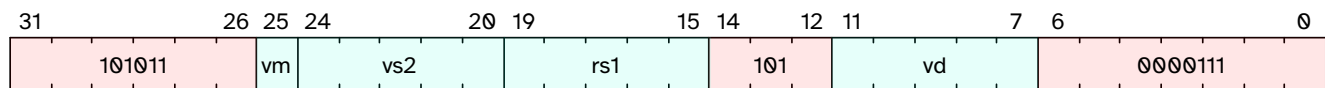
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.657. vloxseg6ei32.v

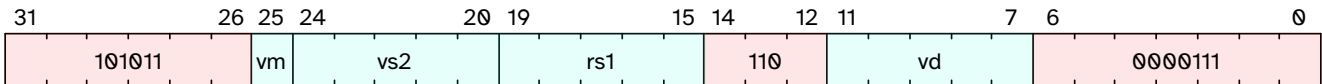
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.658. vloxseg6ei64.v

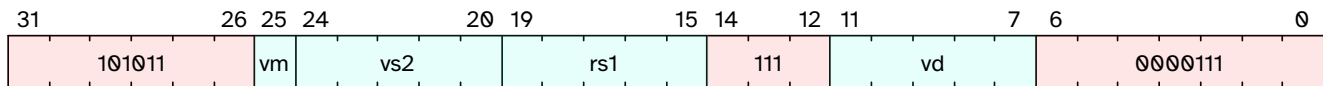
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.659. vloxseg6ei8.v

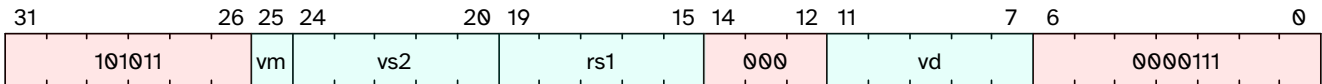
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.660. vloxseg7ei16.v

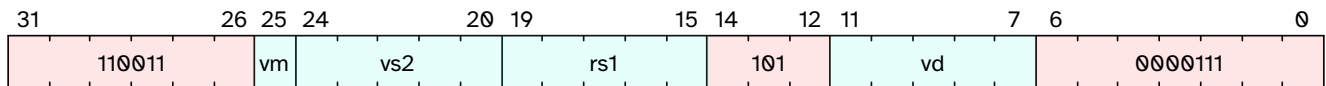
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.661. vloxseg7ei32.v

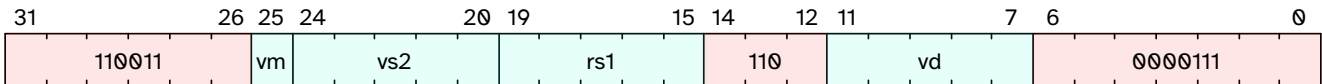
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.662. vloxseg7ei64.v

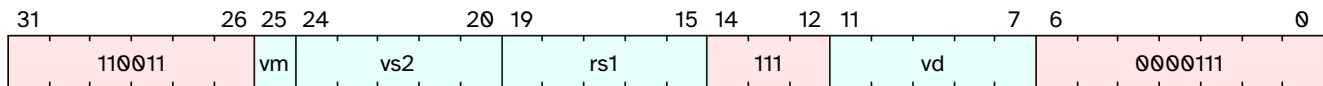
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.663. vloxseg7ei8.v

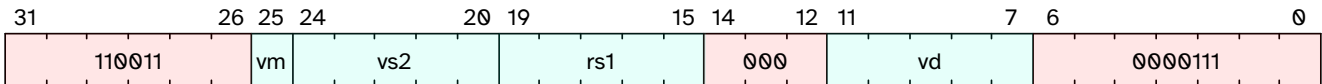
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.664. vloxseg8ei16.v

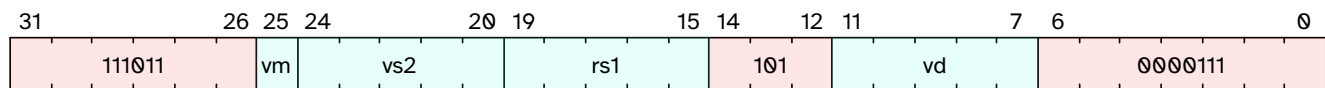
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.665. vloxseg8ei32.v

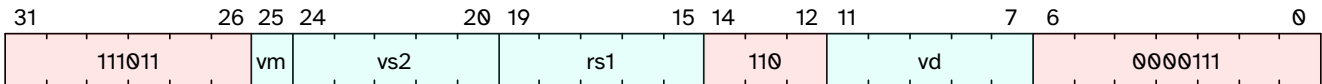
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.666. vloxseg8ei64.v

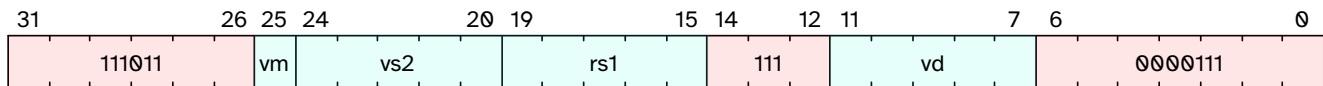
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.667. vloxseg8ei8.v

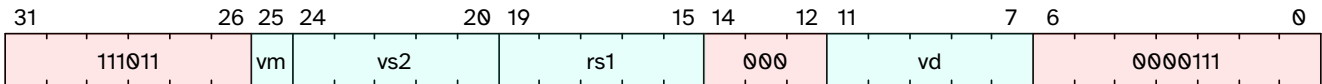
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.668. vlse16.v

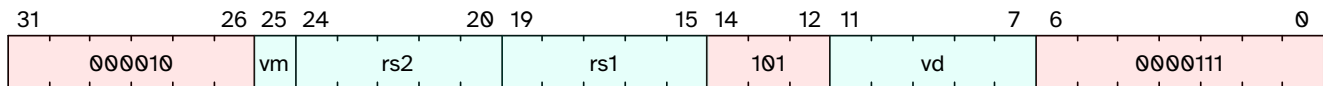
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.669. vlse32.v

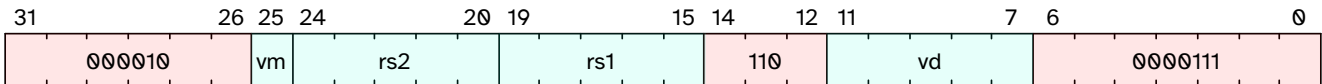
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.670. vlse64.v

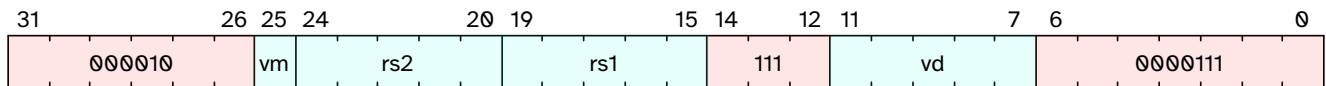
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.671. vlse8.v

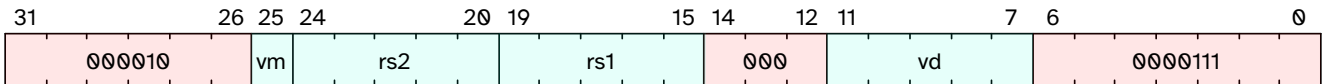
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.672. vlseg2e16.v

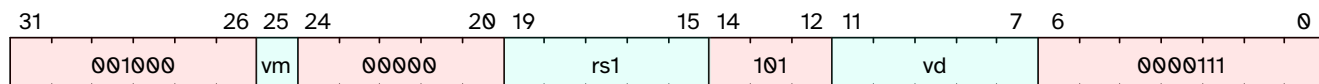
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.673. vlseg2e16ff.v

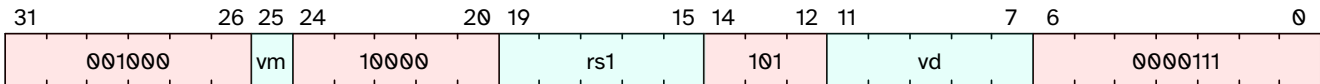
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.674. vlseg2e32.v

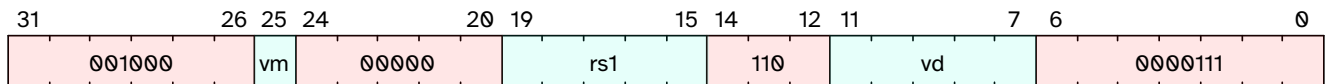
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.675. vlseg2e32ff.v

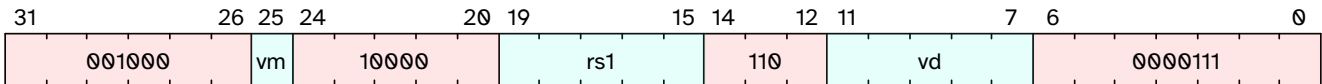
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.676. vlseg2e64.v

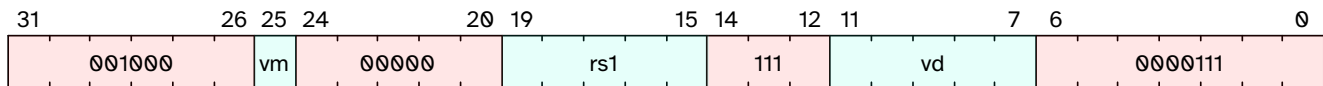
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.677. vlseg2e64ff.v

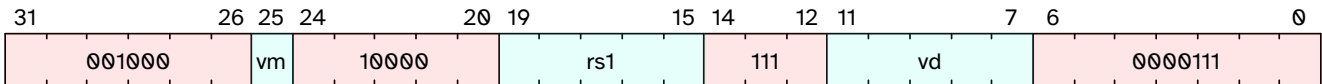
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.678. vlseg2e8.v

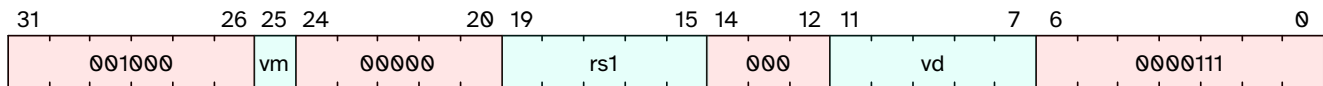
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.679. vlseg2e8ff.v

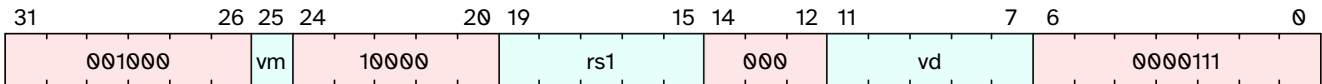
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.680. vlseg3e16.v

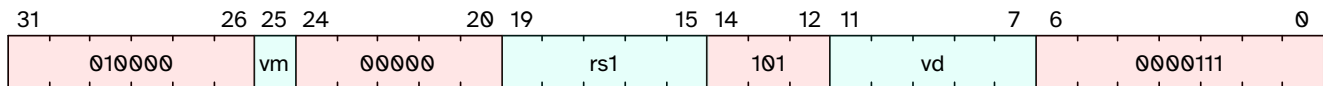
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.681. vlseg3e16ff.v

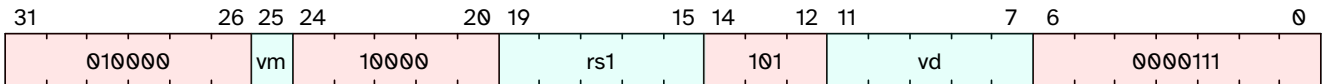
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.682. vlseg3e32.v

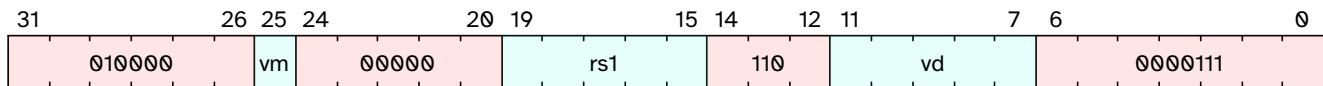
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.683. vlseg3e32ff.v

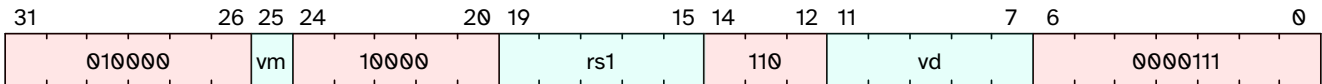
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.684. vlseg3e64.v

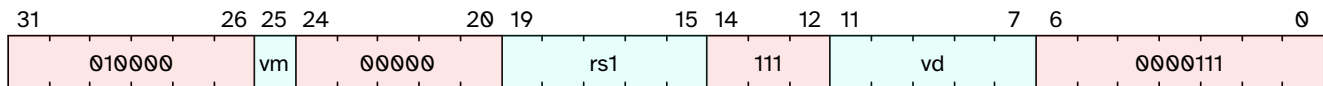
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.685. vlseg3e64ff.v

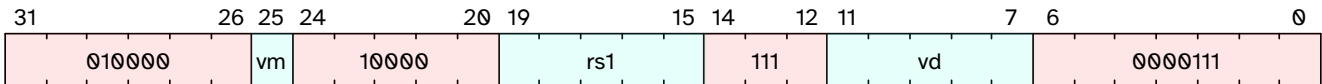
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.686. vlseg3e8.v

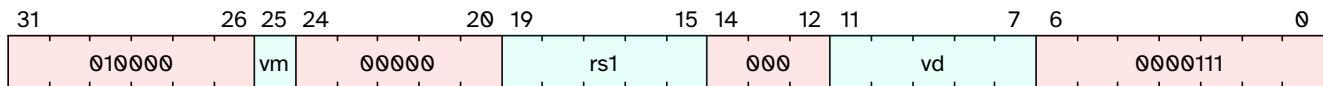
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.687. vlseg3e8ff.v

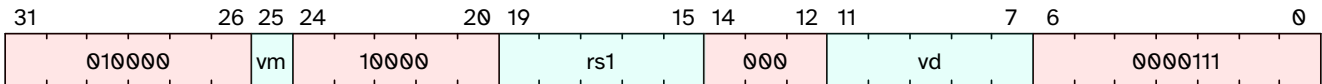
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.688. vlseg4e16.v

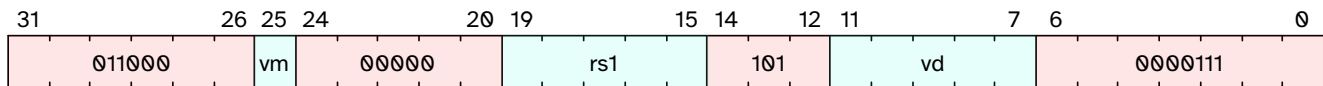
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.689. vlseg4e16ff.v

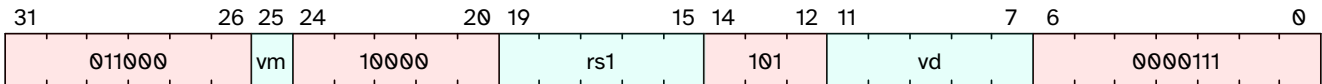
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.690. vlseg4e32.v

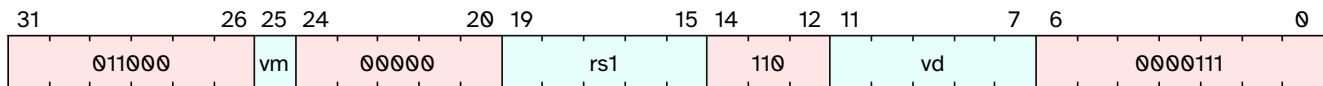
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.691. vlseg4e32ff.v

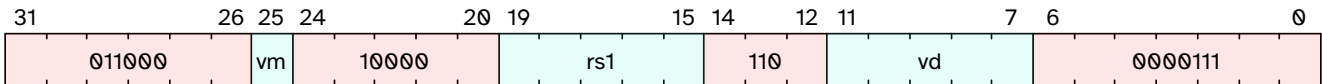
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.692. vlseg4e64.v

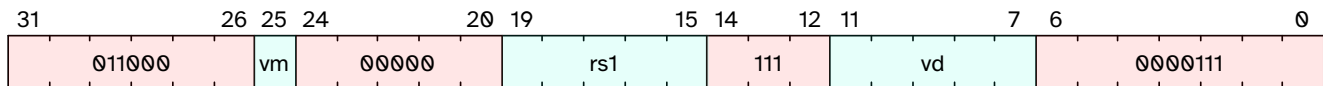
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.693. vlseg4e64ff.v

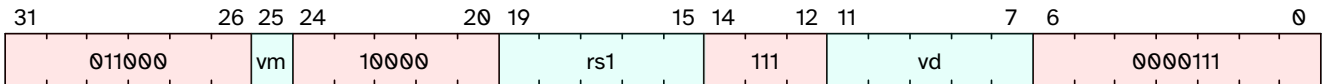
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.694. vlseg4e8.v

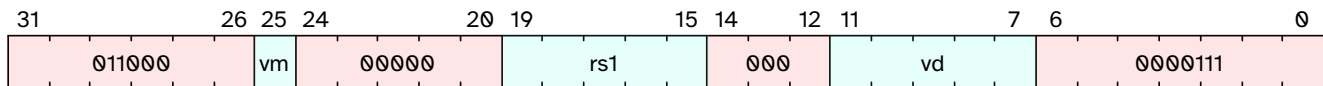
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.695. vlseg4e8ff.v

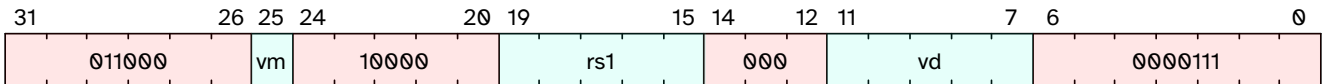
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.696. vlseg5e16.v

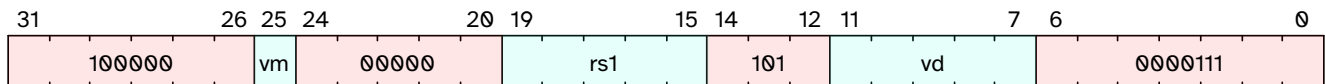
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.697. vlseg5e16ff.v

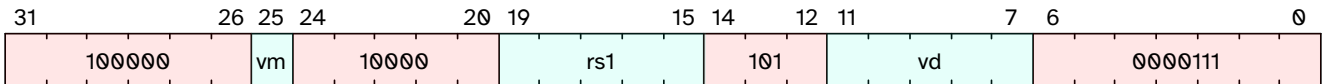
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.698. vlseg5e32.v

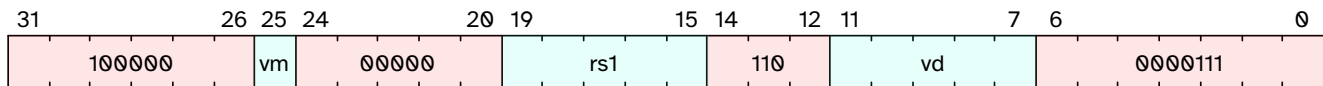
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.699. vlseg5e32ff.v

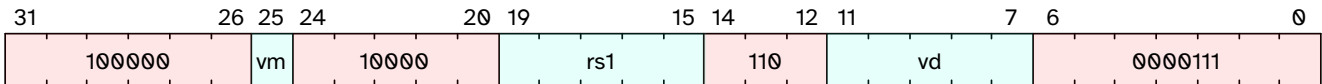
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.700. vlseg5e64.v

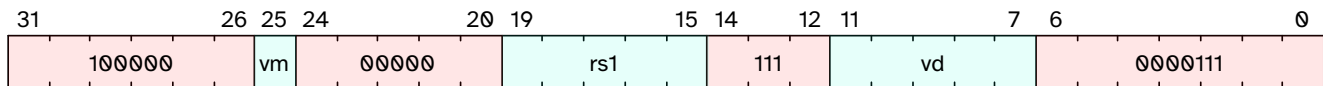
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.701. vlseg5e64ff.v

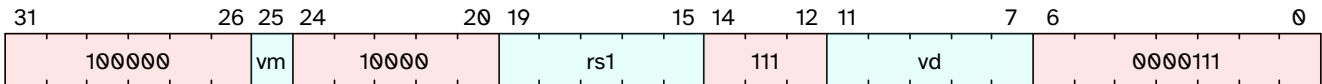
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.702. vlseg5e8.v

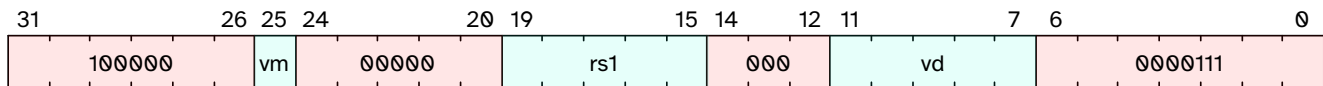
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.703. vlseg5e8ff.v

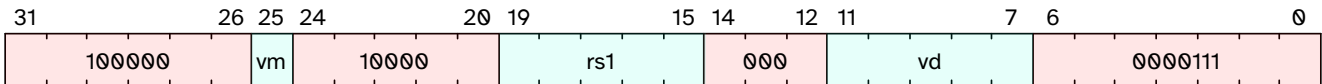
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.704. vlseg6e16.v

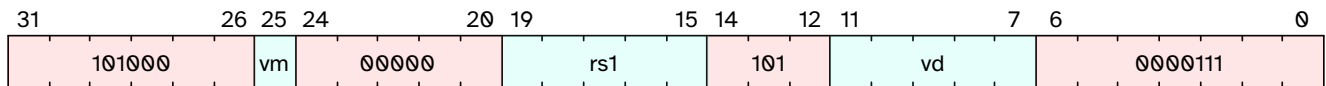
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.705. vlseg6e16ff.v

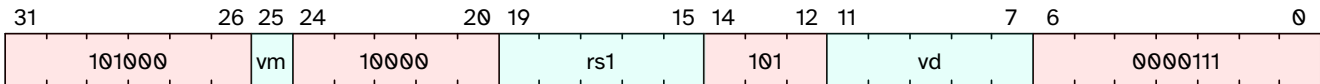
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.706. vlseg6e32.v

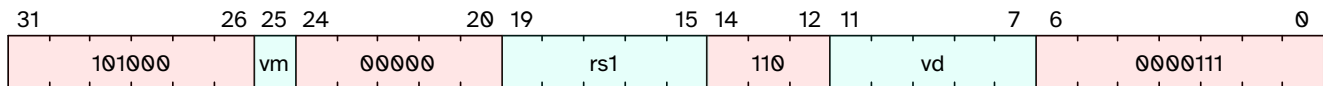
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.707. vlseg6e32ff.v

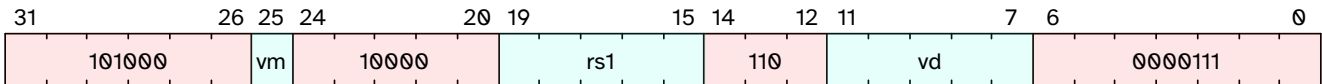
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.708. vlseg6e64.v

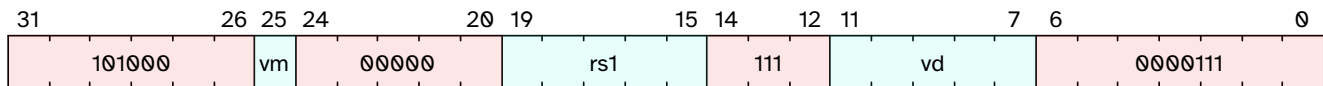
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.709. vlseg6e64ff.v

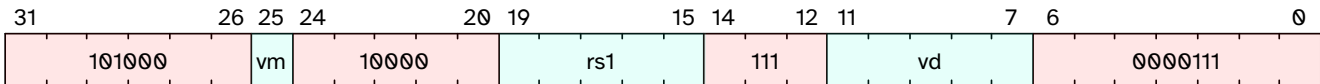
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.710. vlseg6e8.v

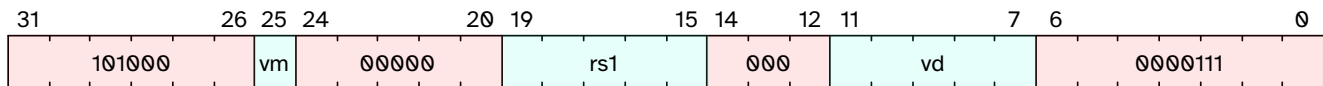
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.711. vlseg6e8ff.v

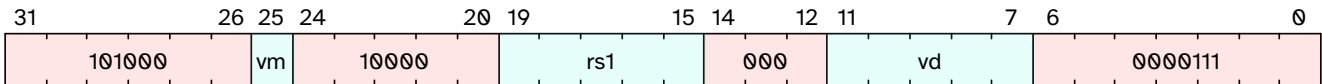
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.712. vlseg7e16.v

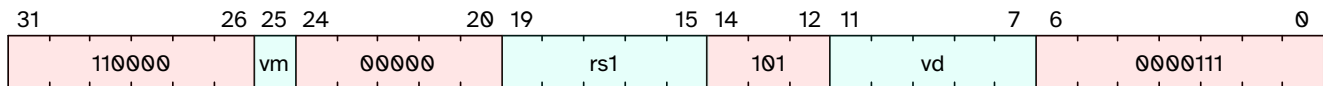
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.713. vlseg7e16ff.v

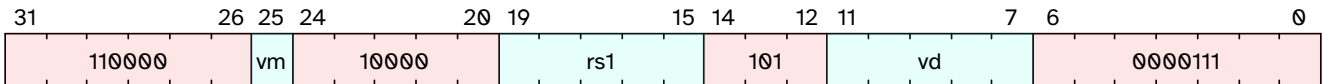
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.714. vlseg7e32.v

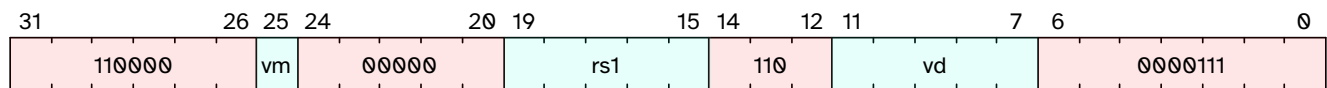
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.715. vlseg7e32ff.v

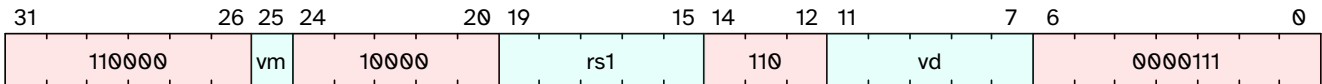
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.716. vlseg7e64.v

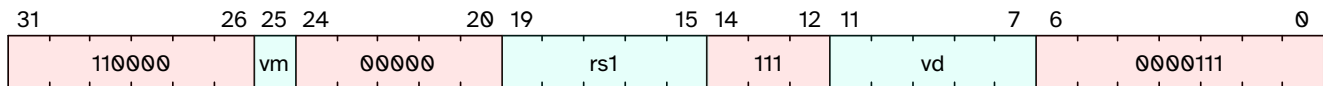
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.717. vlseg7e64ff.v

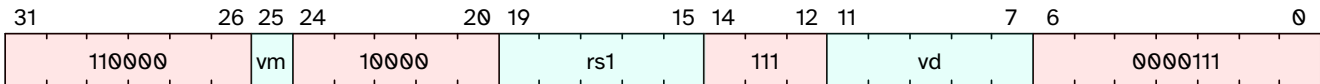
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.718. vlseg7e8.v

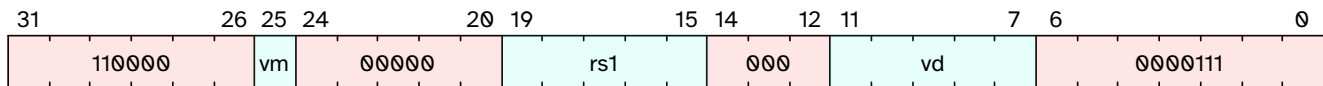
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.719. vlseg7e8ff.v

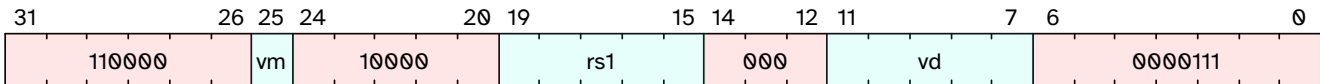
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.720. vlseg8e16.v

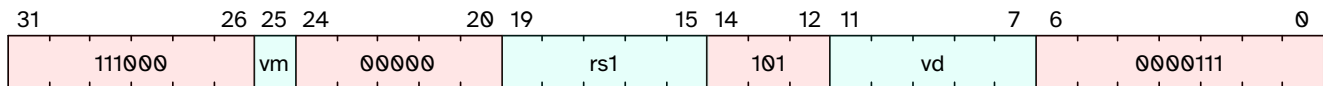
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.721. vlseg8e16ff.v

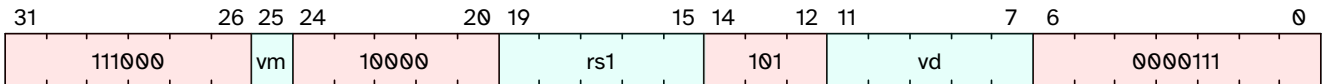
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.722. vlseg8e32.v

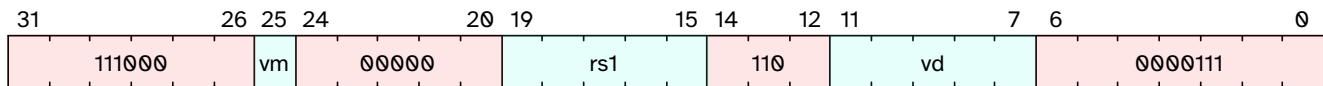
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.723. vlseg8e32ff.v

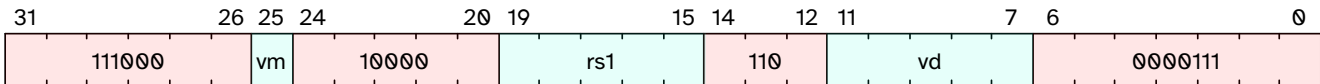
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.724. vlseg8e64.v

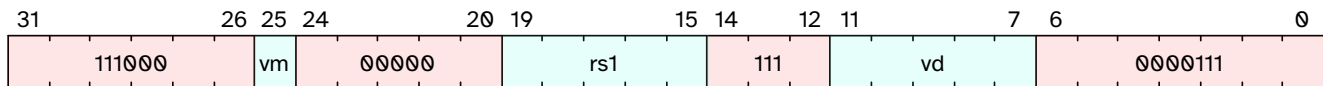
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.725. vlseg8e64ff.v

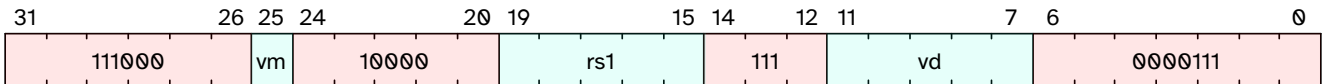
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.726. vlseg8e8.v

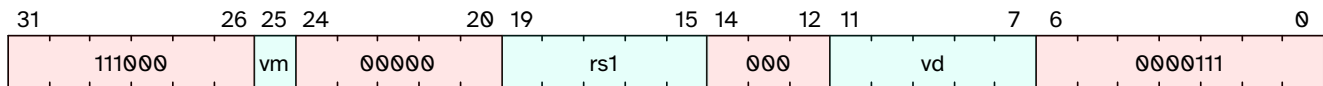
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.727. vlseg8e8ff.v

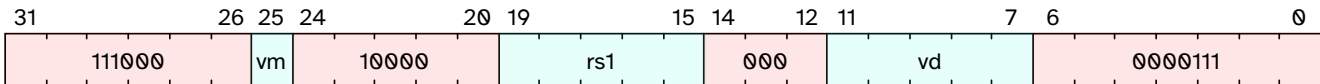
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.728. vlsseg2e16.v

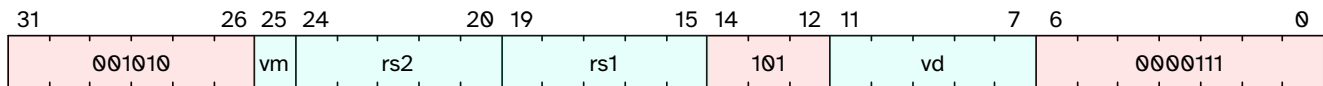
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.729. vlsseg2e32.v

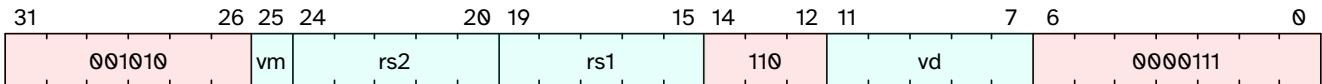
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.730. vlsseg2e64.v

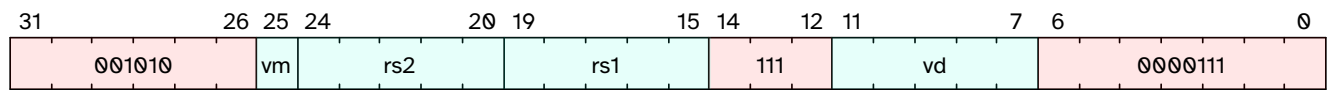
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.731. vlsseg2e8.v

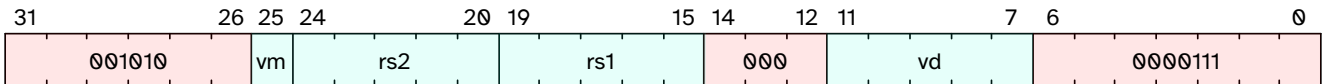
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.732. vlsseg3e16.v

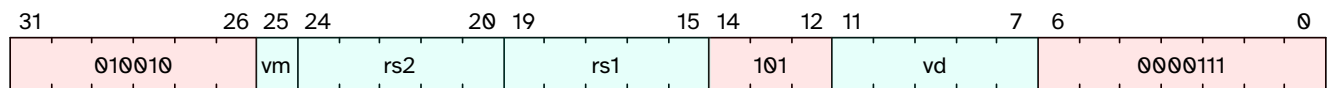
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.733. vlsseg3e32.v

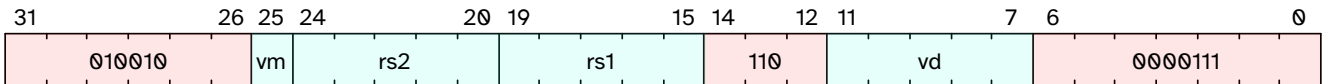
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.734. vlsseg3e64.v

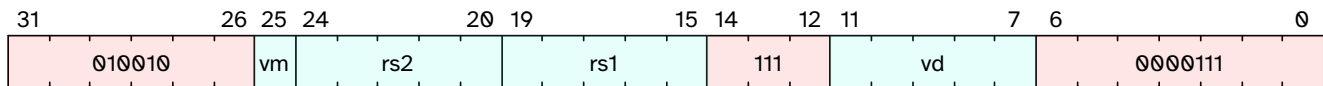
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.735. vlsseg3e8.v

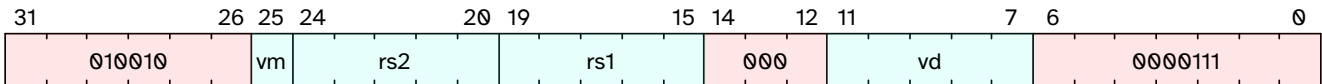
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.736. vlsseg4e16.v

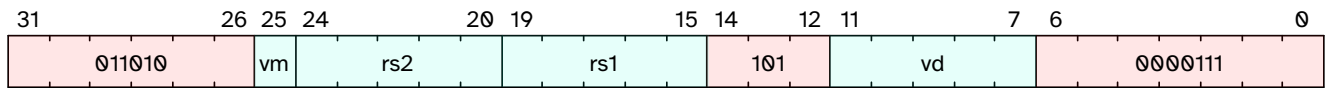
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.737. vlsseg4e32.v

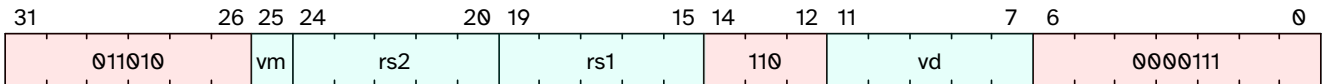
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.738. vlsseg4e64.v

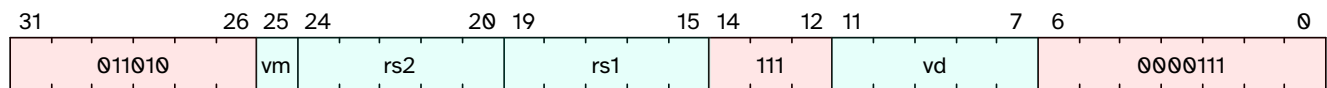
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.739. vlsseg4e8.v

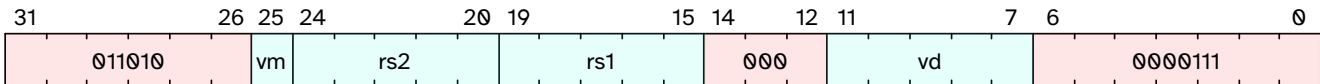
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.740. vlsseg5e16.v

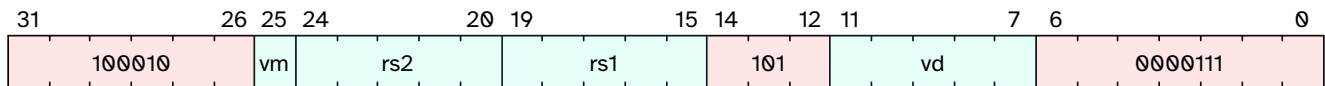
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.741. vlsseg5e32.v

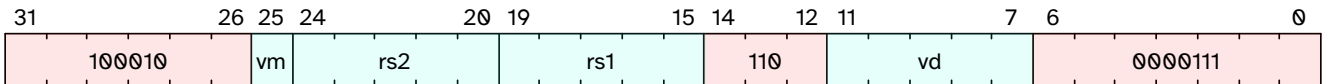
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.742. vlsseg5e64.v

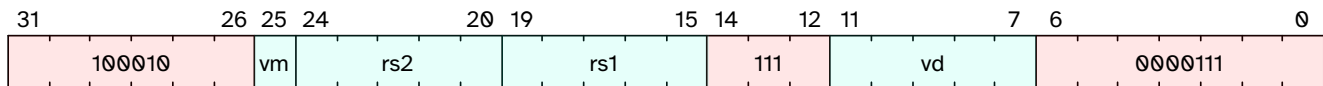
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.743. vlsseg5e8.v

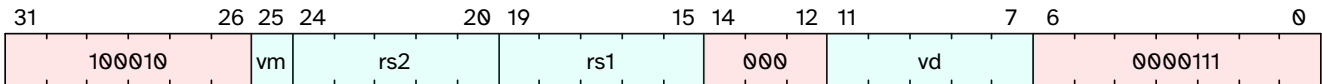
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.744. vlsseg6e16.v

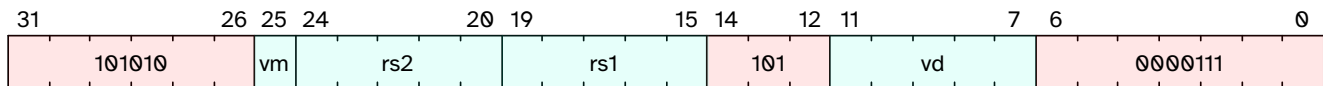
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.745. vlsseg6e32.v

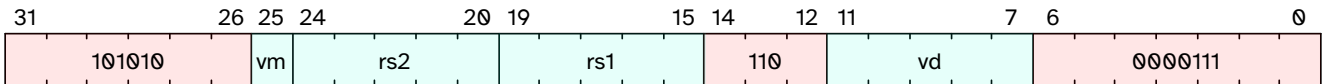
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.746. vlsseg6e64.v

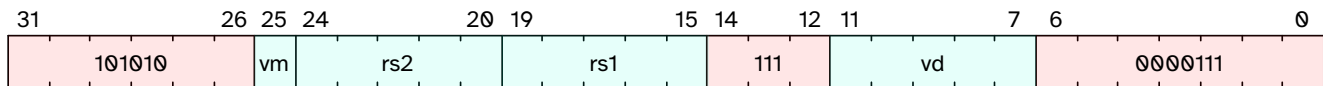
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.747. vlsseg6e8.v

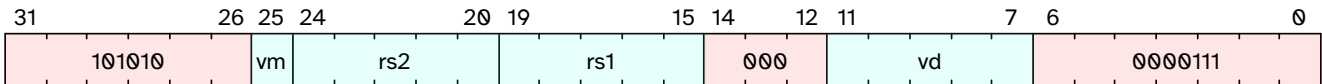
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.748. vlsseg7e16.v

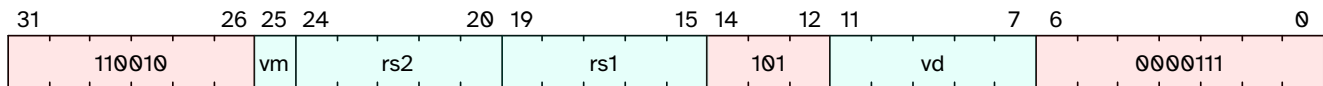
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.749. vlsseg7e32.v

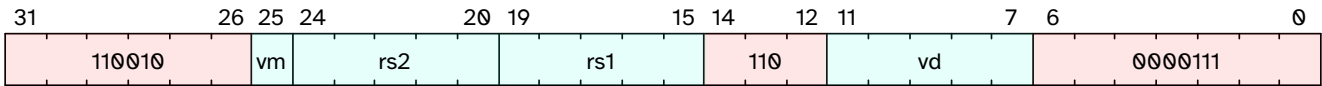
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.750. vlsseg7e64.v

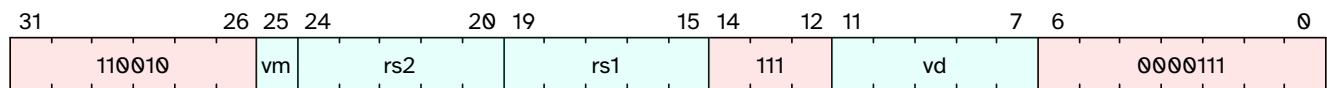
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.751. vlsseg7e8.v

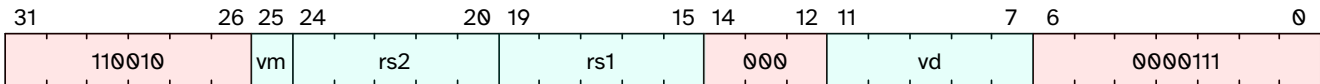
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.752. vlsseg8e16.v

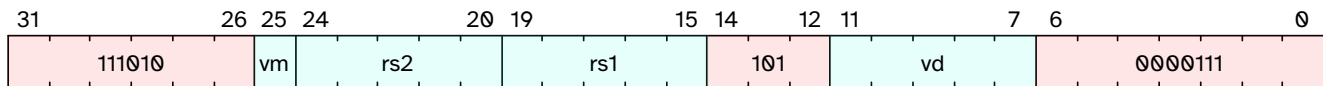
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.753. vlsseg8e32.v

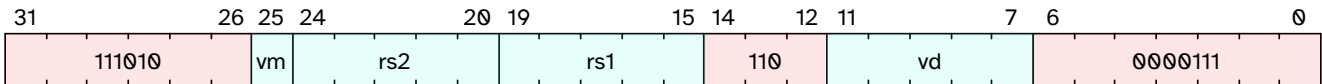
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.754. vlsseg8e64.v

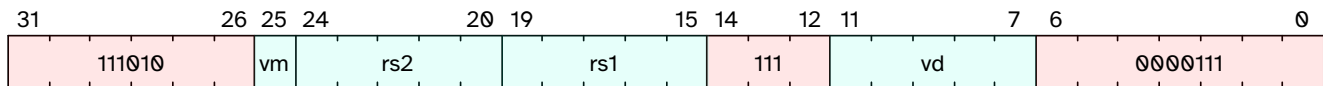
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.755. vlsseg8e8.v

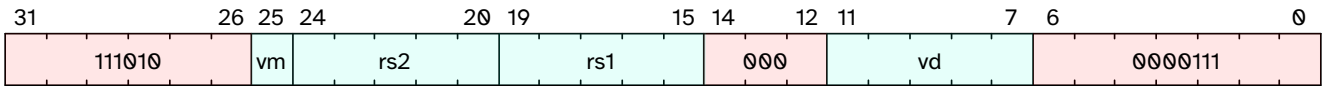
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.756. vluxei16.v

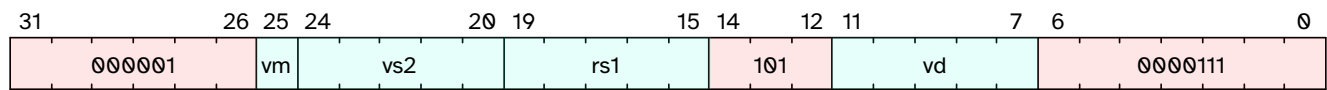
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.757. vluxe32.v

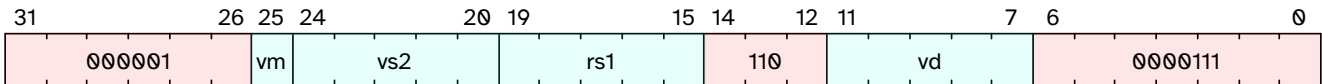
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.758. vluxe64.v

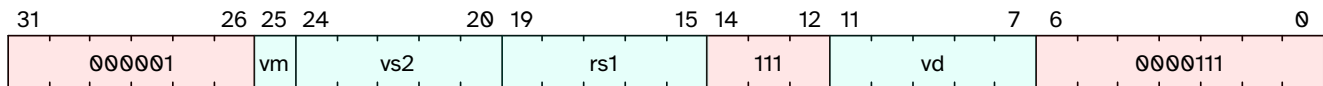
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.759. vluxeï8.v

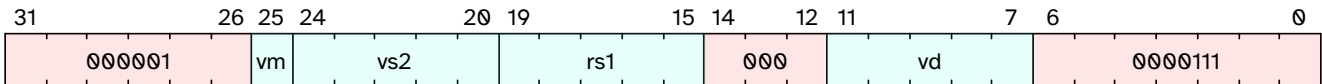
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.760. vluxseg2ei16.v

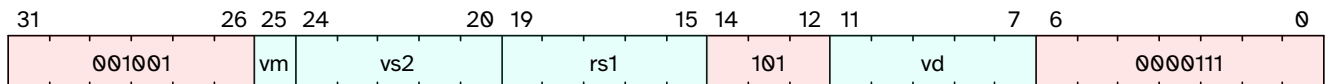
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.761. vluxseg2ei32.v

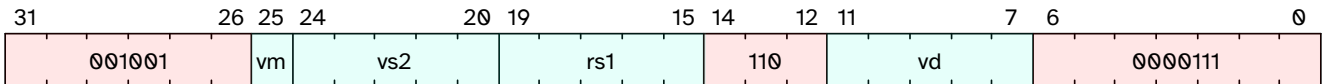
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.762. vluxseg2ei64.v

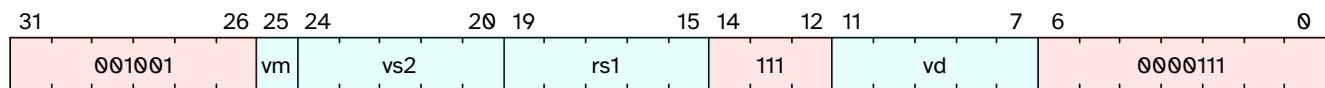
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.763. vluxseg2ei8.v

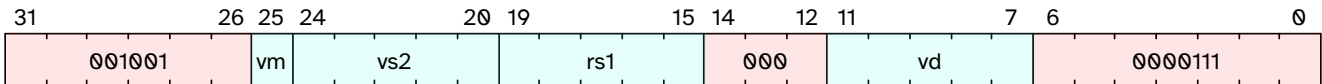
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.764. vluxseg3ei16.v

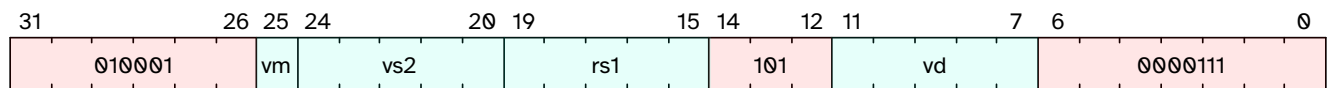
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.765. vluxseg3ei32.v

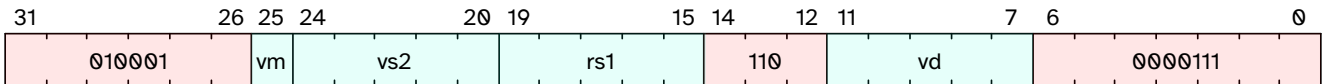
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.766. vluxseg3ei64.v

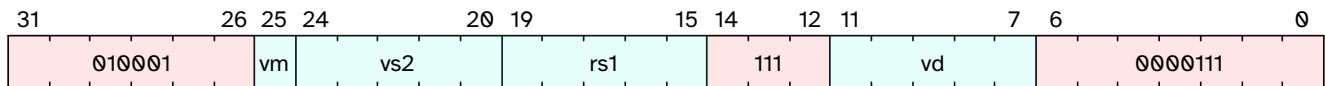
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.767. vluxseg3ei8.v

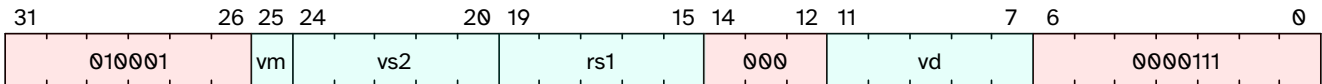
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.768. vluxseg4ei16.v

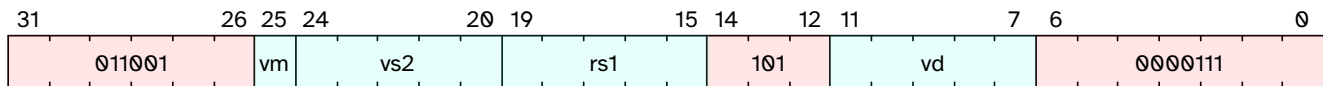
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.769. vluxseg4ei32.v

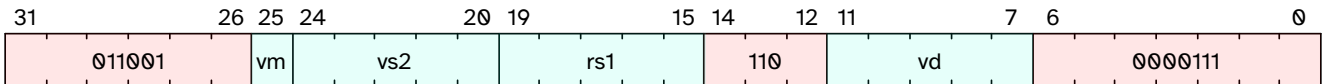
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.770. vluxseg4ei64.v

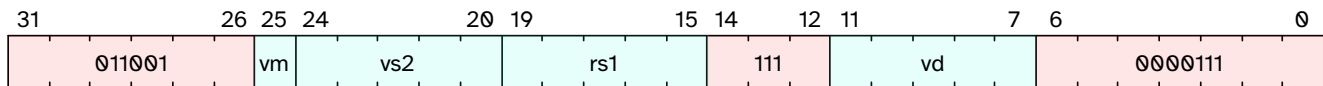
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.771. vluxseg4ei8.v

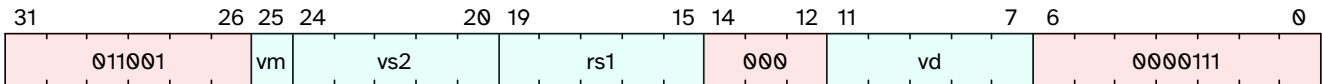
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.772. vluxseg5ei16.v

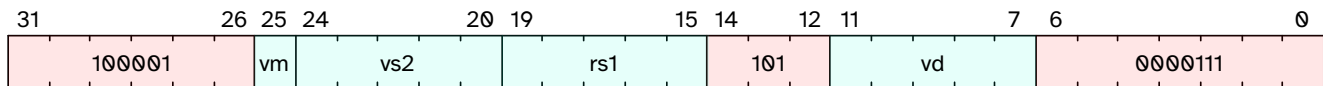
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.773. vluxseg5ei32.v

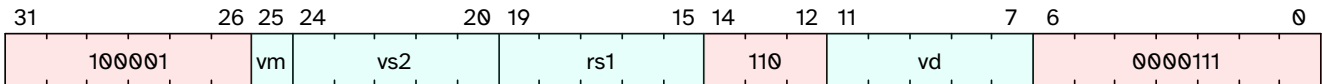
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.774. vluxseg5ei64.v

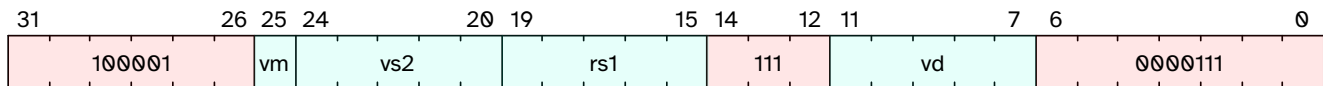
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.775. vluxseg5ei8.v

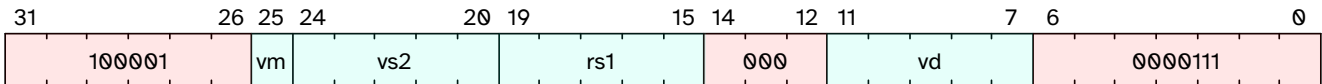
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.776. vluxseg6ei16.v

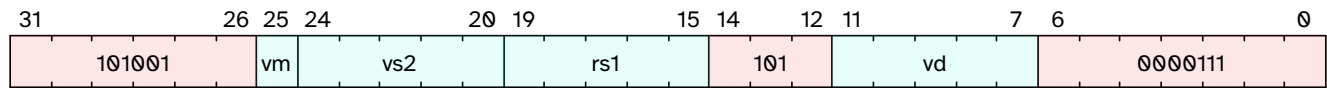
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.777. vluxseg6ei32.v

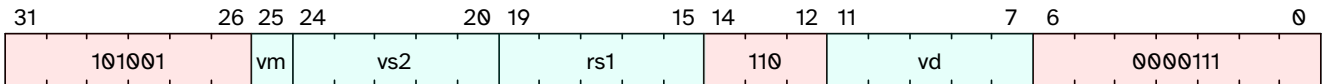
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.778. vluxseg6ei64.v

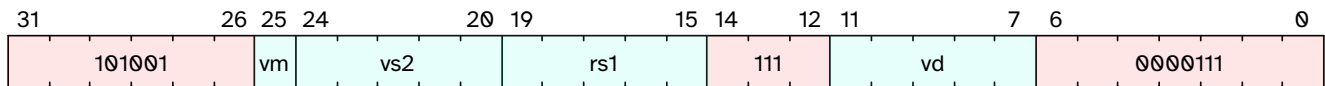
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.779. vluxseg6ei8.v

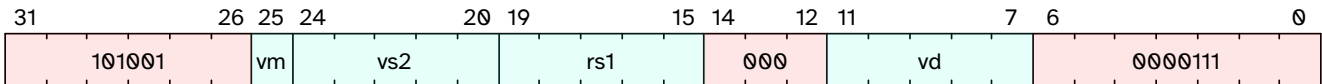
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.780. vluxseg7ei16.v

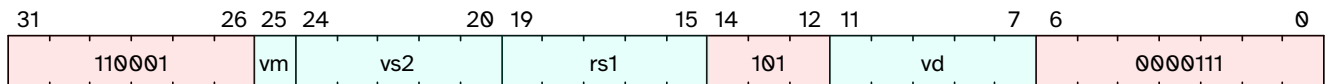
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.781. vluxseg7ei32.v

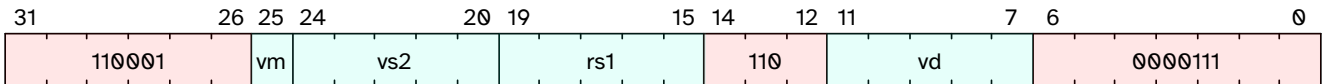
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.782. vluxseg7ei64.v

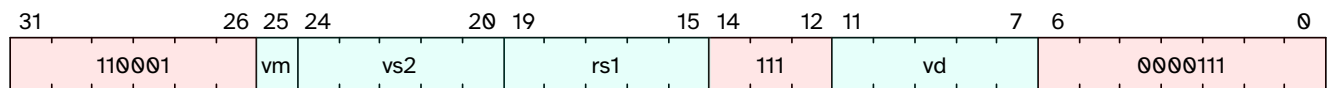
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.783. vluxseg7ei8.v

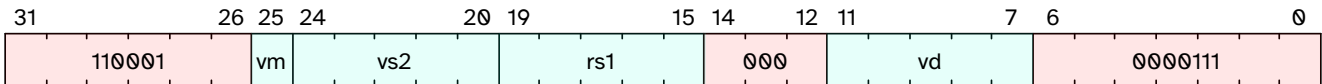
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.784. vluxseg8ei16.v

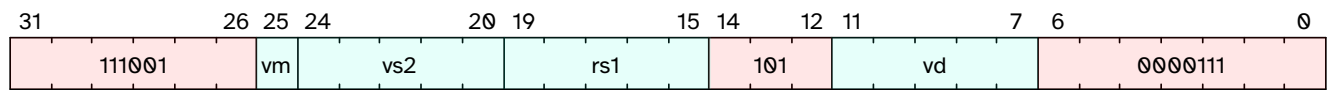
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.785. vluxseg8ei32.v

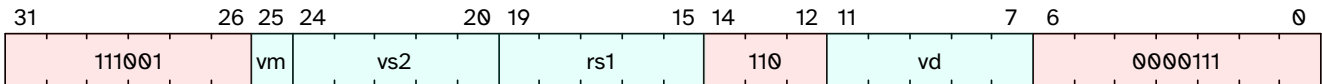
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.786. vluxseg8ei64.v

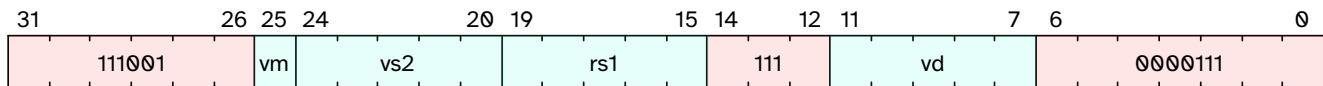
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.787. vluxseg8ei8.v

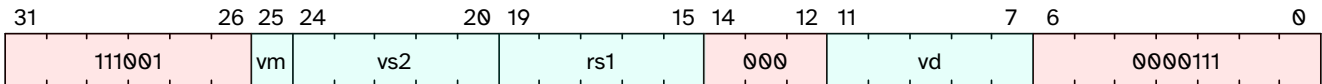
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.788. vmacc.vv

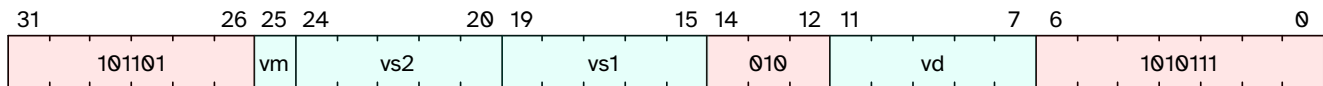
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.789. vmacc.vx

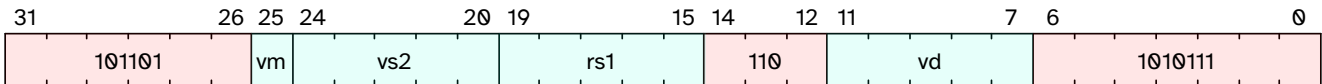
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.790. vmadc.vi

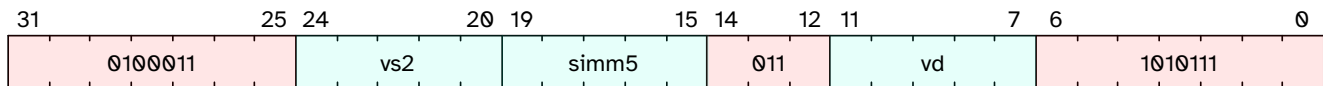
Synopsis

No synopsis available.

Mnemonic

vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.791. vmadc.vim

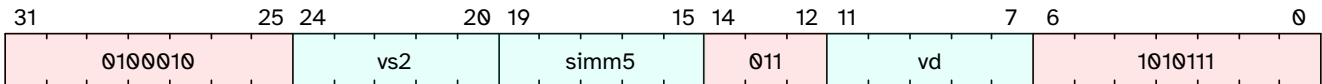
Synopsis

No synopsis available.

Mnemonic

vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.792. vmadc.vv

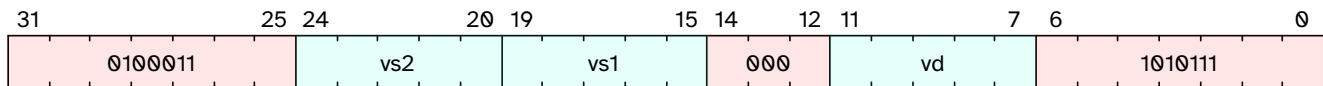
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.793. vmadc.vvm

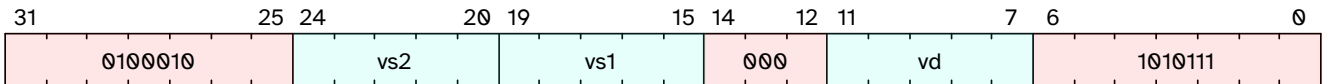
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.794. vmadc.vx

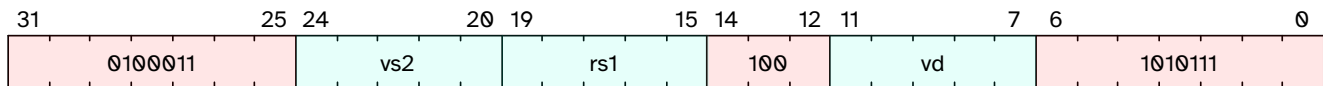
Synopsis

No synopsis available.

Mnemonic

vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.795. vmadc.vxm

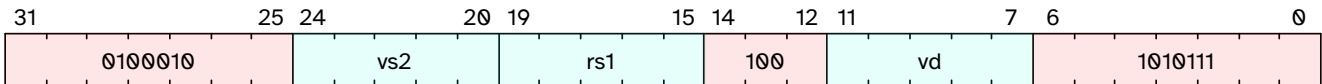
Synopsis

No synopsis available.

Mnemonic

vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.796. vmadd.vv

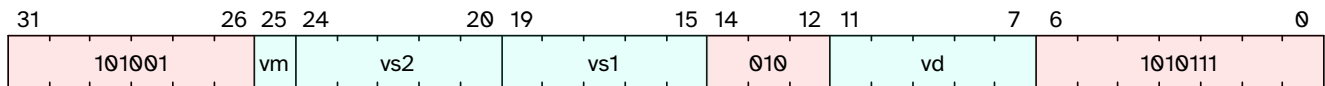
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.797. vmadd.vx

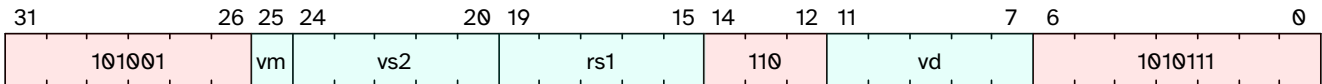
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.798. vmand.mm

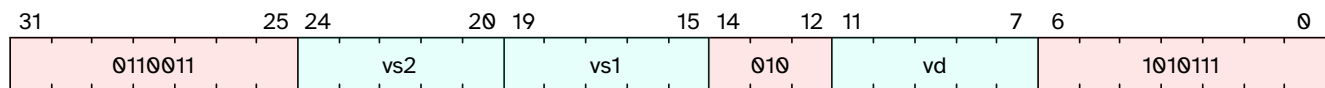
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.799. vmandn.mm

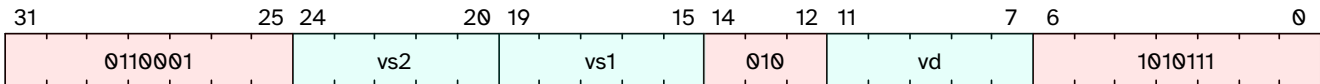
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.800. vmax.vv

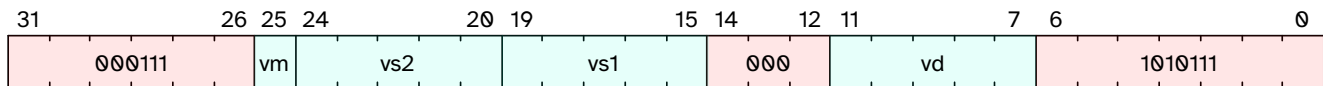
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.801. vmax.vx

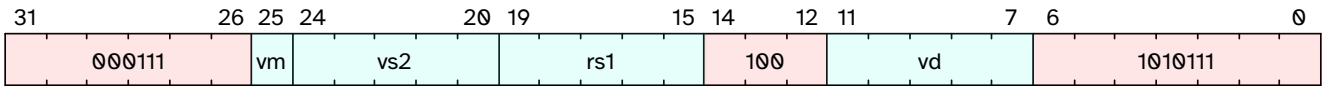
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.802. vmaxu.vv

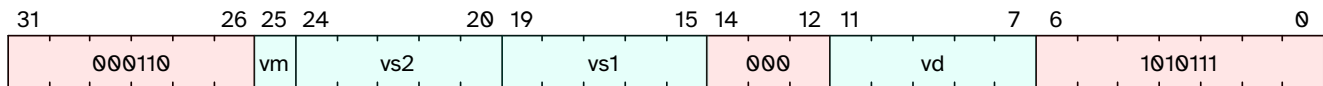
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.803. vmaxu.vx

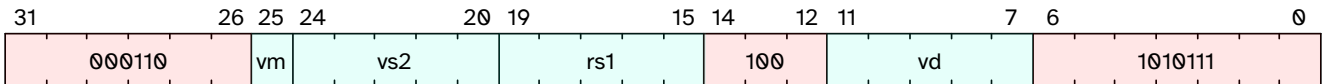
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.804. vmerge.vim

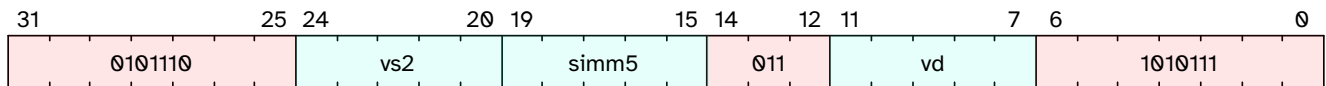
Synopsis

No synopsis available.

Mnemonic

vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.805. vmerge.vvm

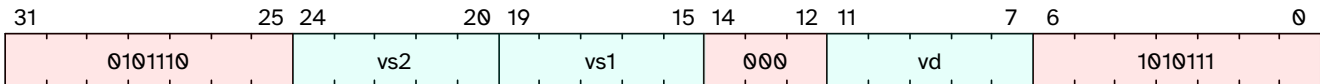
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.806. vmerge.vxm

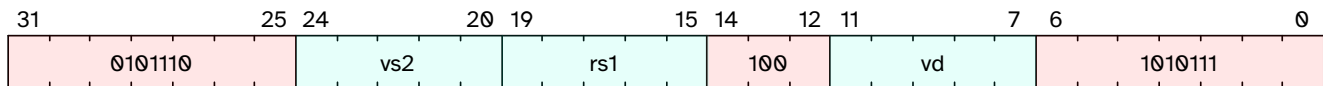
Synopsis

No synopsis available.

Mnemonic

vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.807. vmfeq.vf

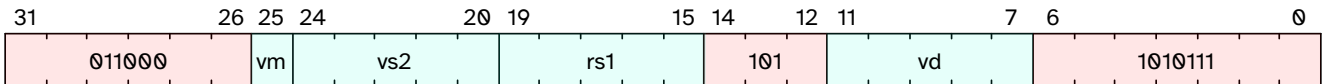
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.808. vmfeq.vv

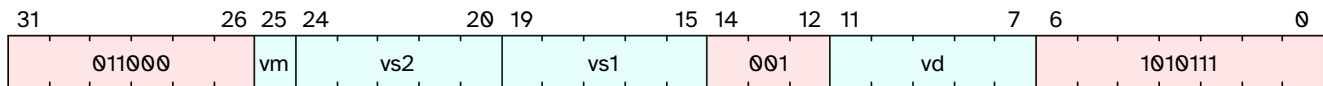
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.809. vmfge.vf

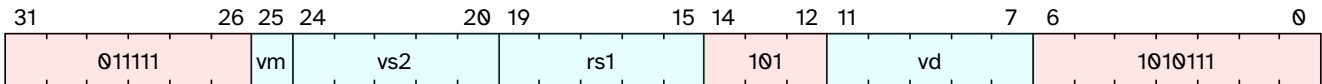
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.810. vmfgt.vf

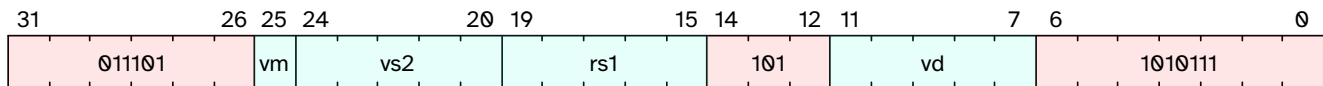
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.811. vmfle.vf

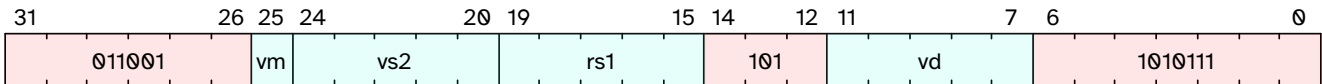
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.812. vmfle.vv

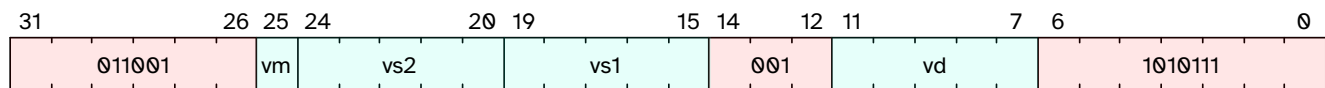
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.813. vmflt.vf

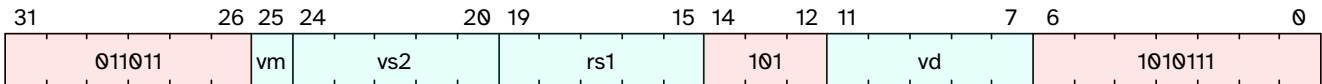
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.814. vmflt.vv

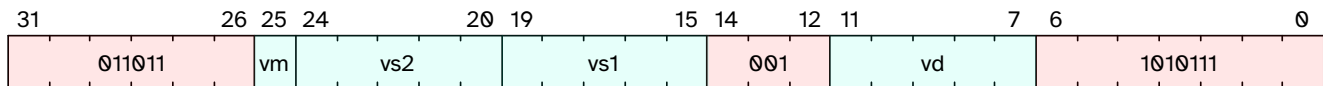
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.815. vmfne.vf

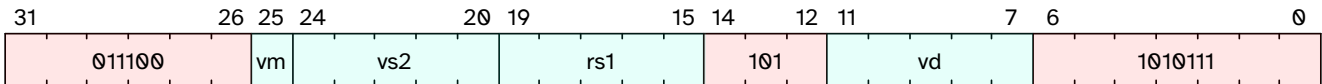
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.816. vmfne.vv

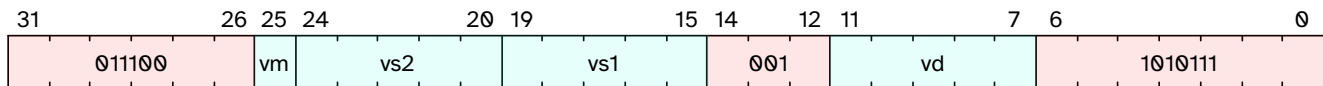
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.817. vmin.vv

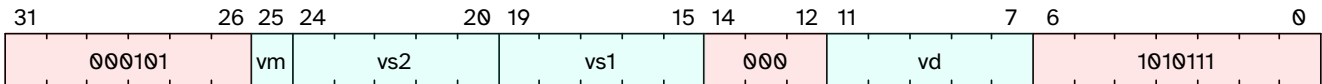
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.818. vmin.vx

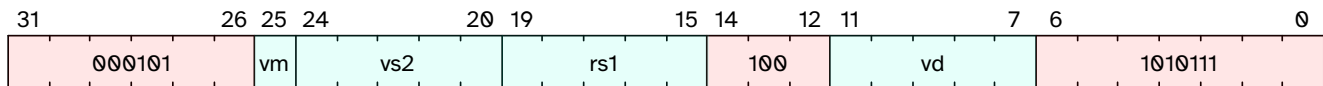
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.819. vminu.vv

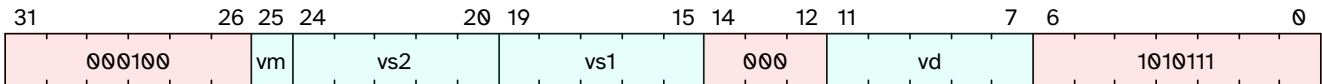
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.820. vminu.vx

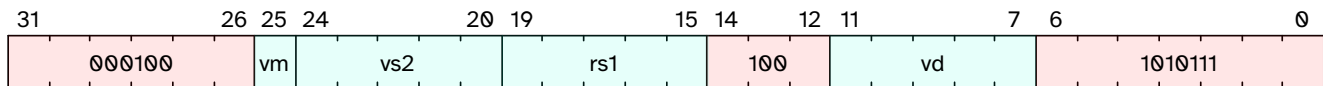
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.821. vmnand.mm

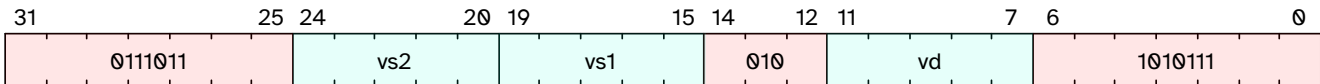
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.822. vmnor.mm

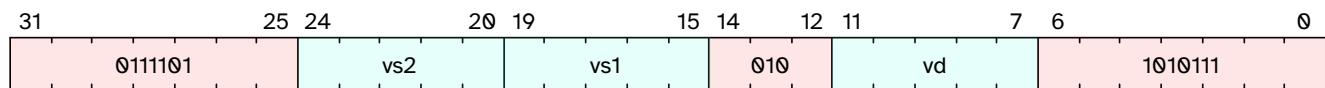
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.823. vmor.mm

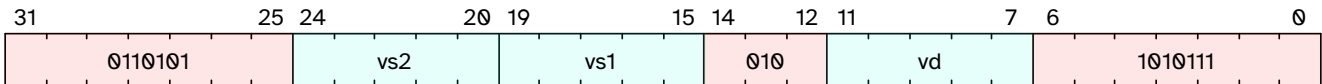
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.824. vmorn.mm

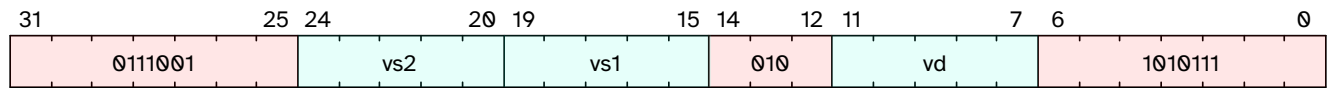
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.825. vmsbc.vv

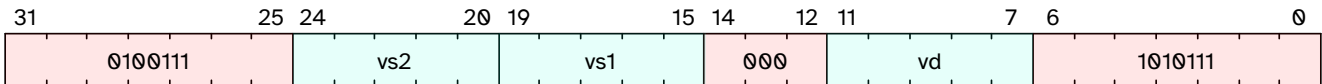
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.826. vmsbc.vvm

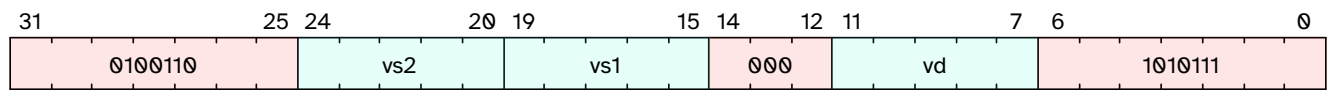
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.827. vmsbc.vx

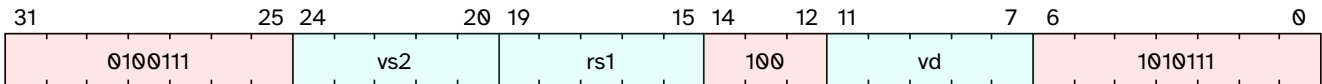
Synopsis

No synopsis available.

Mnemonic

vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.828. vmsbc.vxm

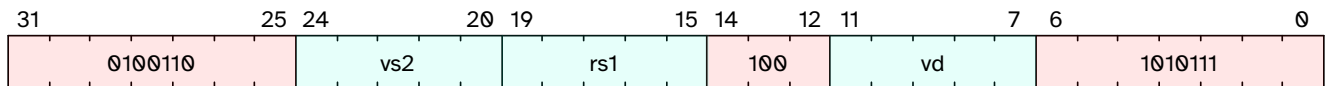
Synopsis

No synopsis available.

Mnemonic

vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.829. vmsbf.m

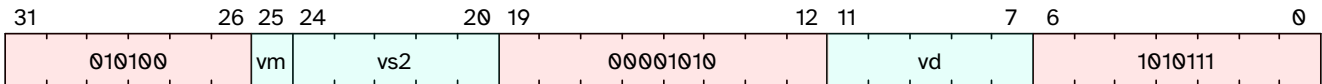
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.830. vmseq.vi

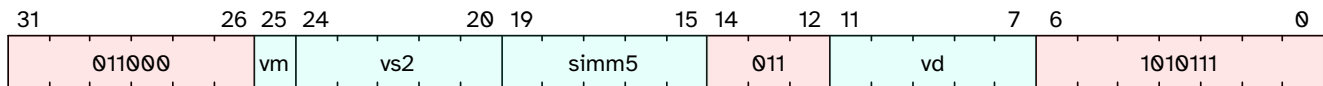
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.831. vmseq.vv

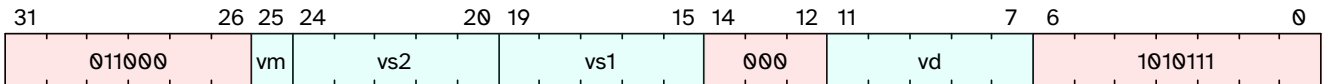
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.832. vmseq.vx

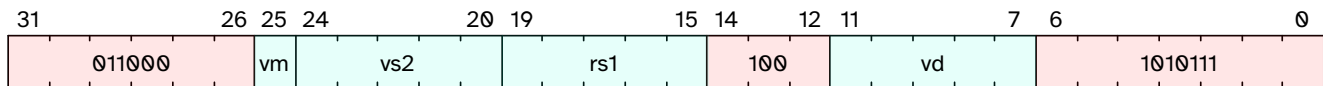
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.833. vmsgt.vi

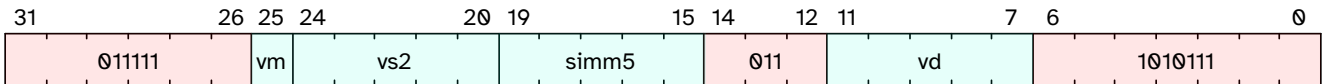
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.834. vmsgt.vx

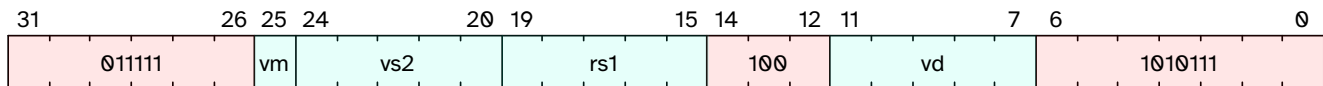
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.835. vmsgtu.vi

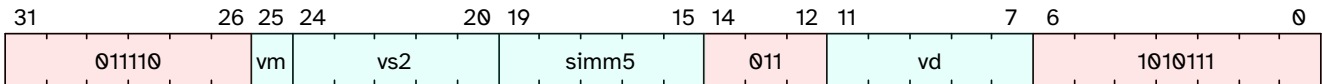
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.836. vmsgtu.vx

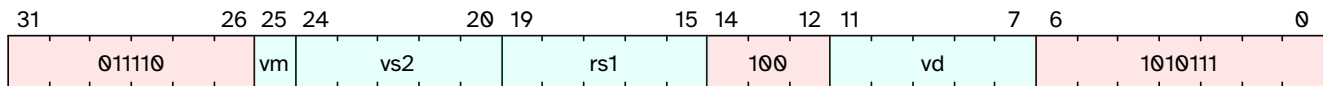
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.837. vmsif.m

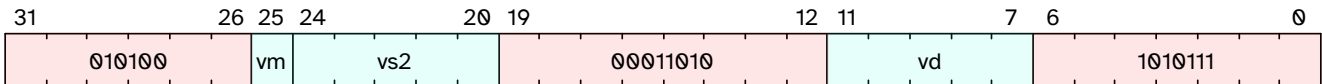
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.838. vmsle.vi

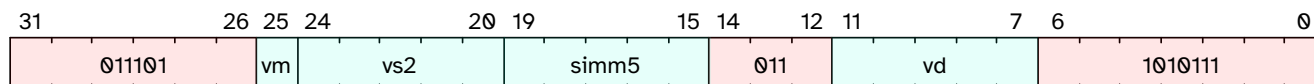
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.839. vmsle.vv

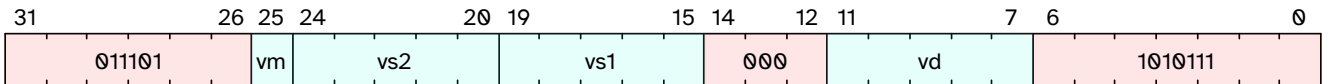
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.840. vmsle.vx

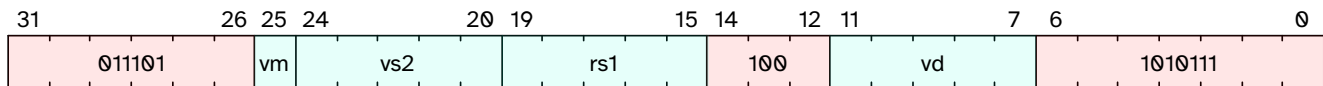
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.841. vmsleu.vi

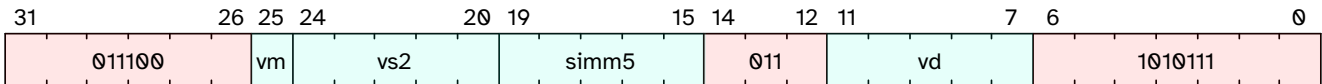
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.842. vmsleu.vv

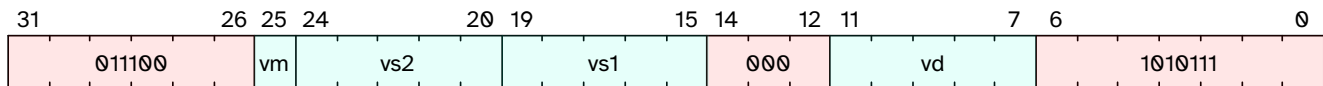
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.843. vmsleu.vx

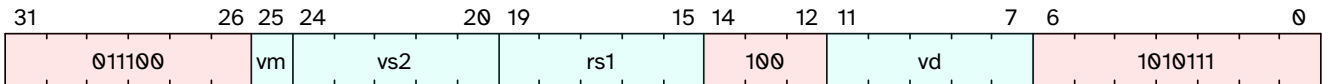
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.844. vmslt.vv

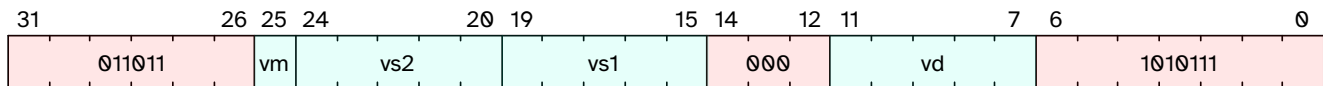
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.845. vmslt.vx

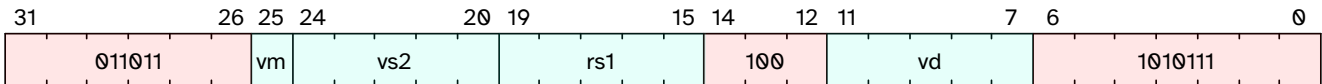
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.846. vmsltu.vv

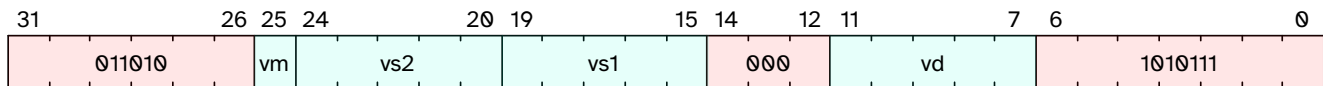
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.847. vmsltu.vx

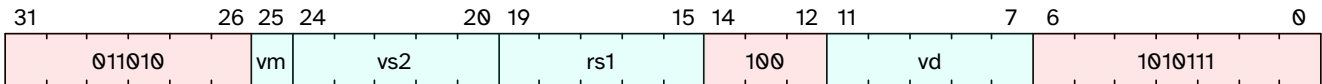
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.848. vmsne.vi

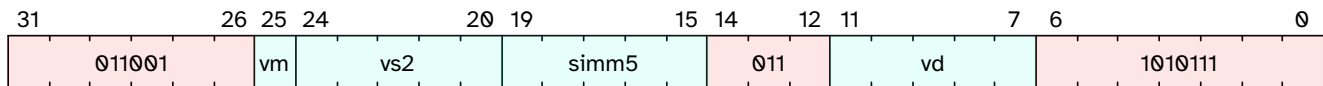
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.849. vmsne.vv

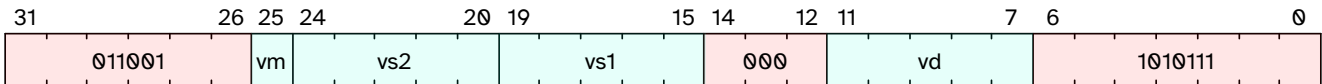
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.850. vmsne.vx

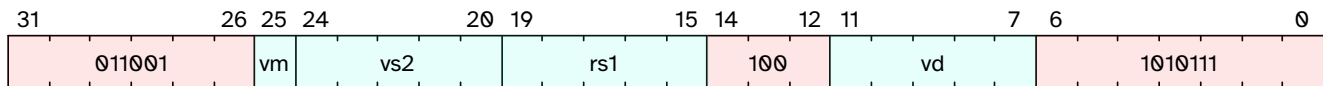
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.851. vmsof.m

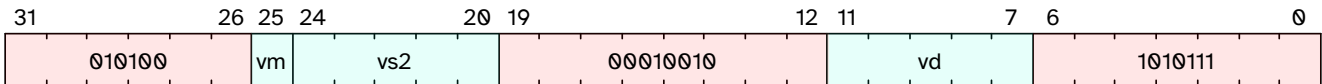
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.852. vmul.vv

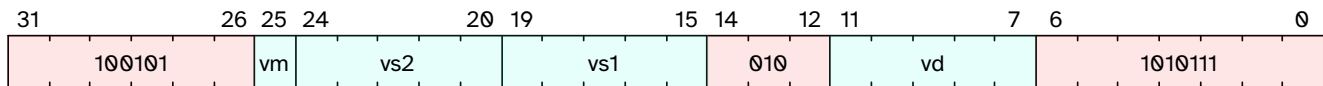
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.853. vmul.vx

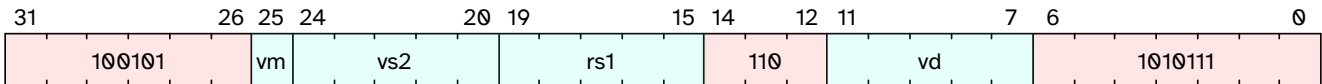
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.854. vmulh.vv

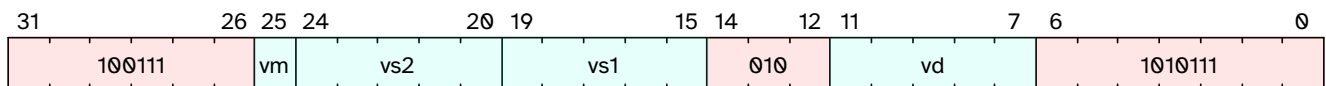
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.855. vmulh.vx

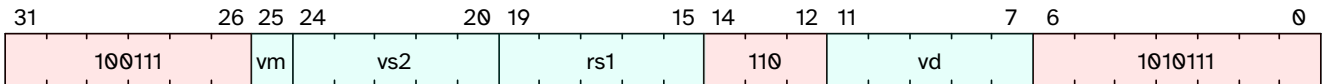
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.856. vmulhsu.vv

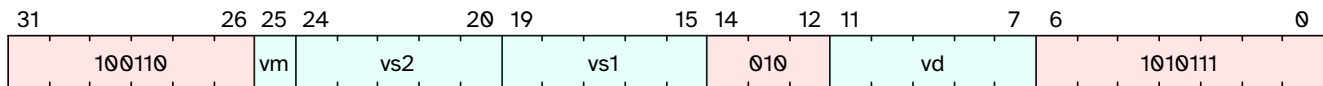
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.857. vmulhsu.vx

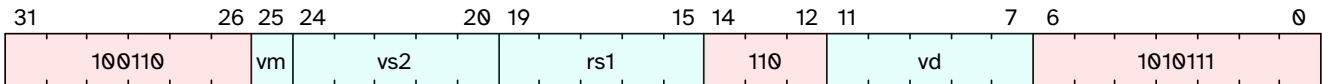
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.858. vmulhu.wv

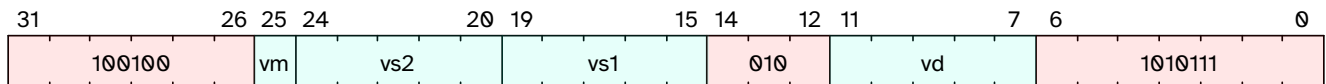
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.859. vmulhu.vx

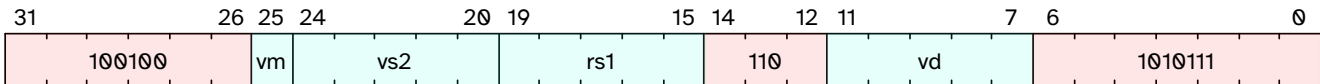
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.860. vmv.s.x

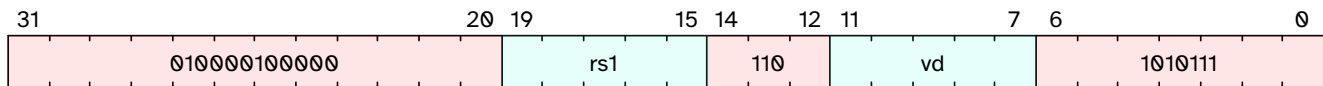
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.861. vmv.v.i

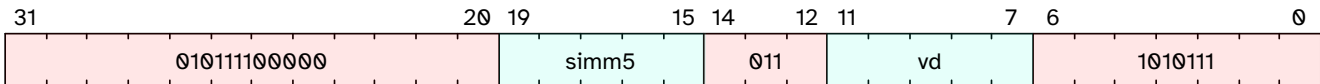
Synopsis

No synopsis available.

Mnemonic

vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.862. vmv.v.v

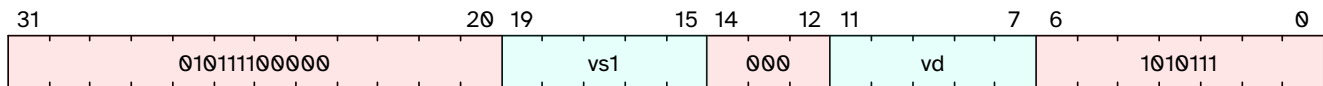
Synopsis

No synopsis available.

Mnemonic

vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.863. vmv.v.x

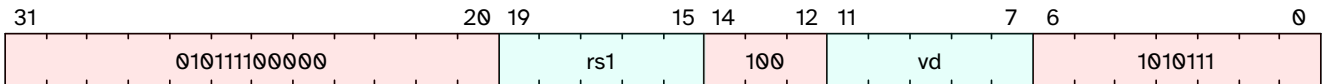
Synopsis

No synopsis available.

Mnemonic

rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.864. vmv.x.s

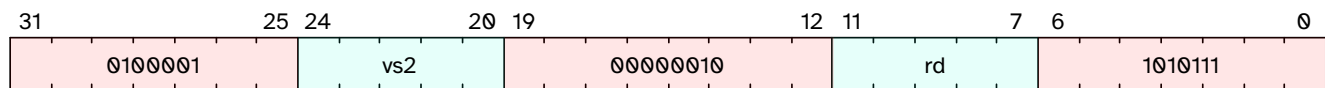
Synopsis

No synopsis available.

Mnemonic

vs2, rd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.865. vmv1r.v

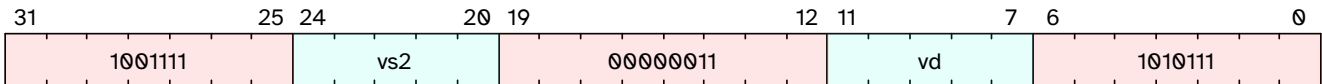
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.866. vmv2r.v

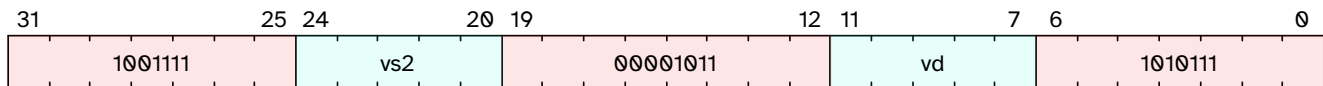
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.867. vmv4r.v

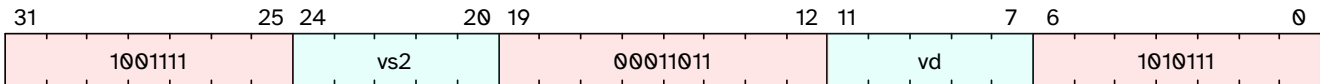
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.868. vmv8r.v

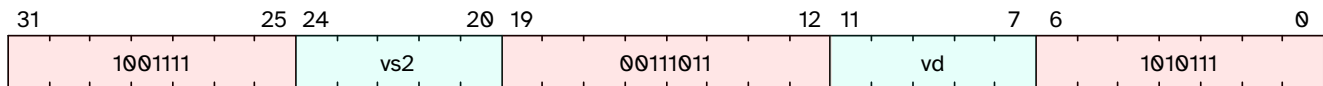
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.869. vmxnor.mm

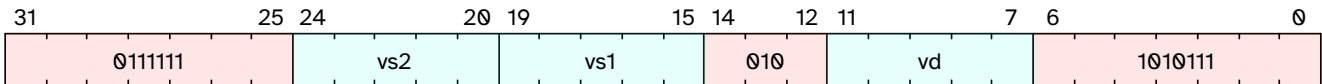
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.870. vmxor.mm

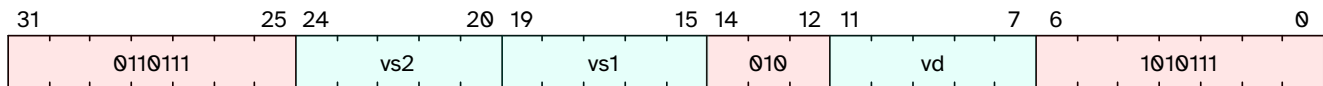
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.871. vnclip.wi

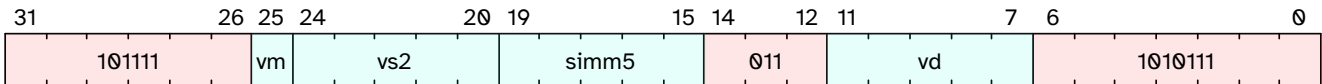
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.872. vnclip.wv

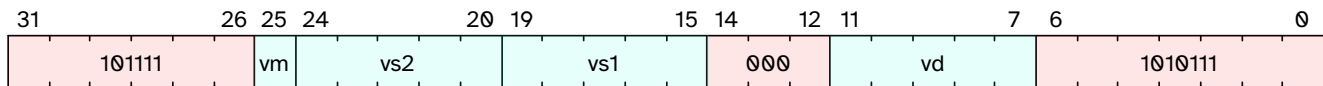
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.873. vnclip.wx

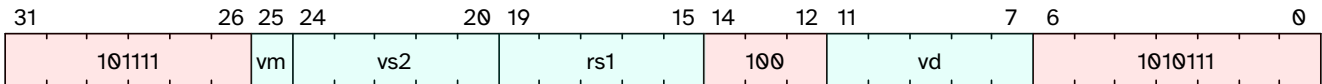
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.874. vnclipu.wi

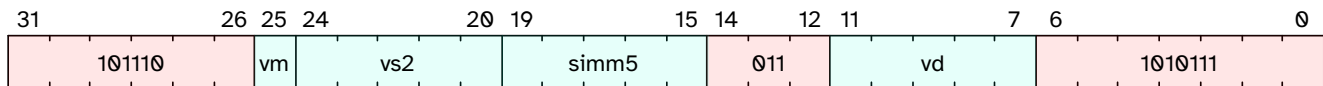
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.875. vnclipu.wv

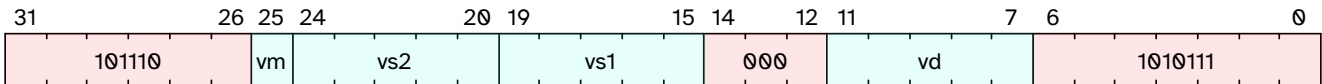
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.876. vnclipu.wx

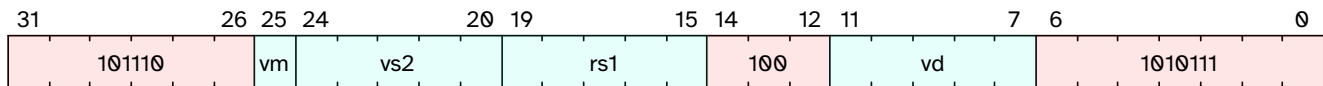
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.877. vnmsac.vv

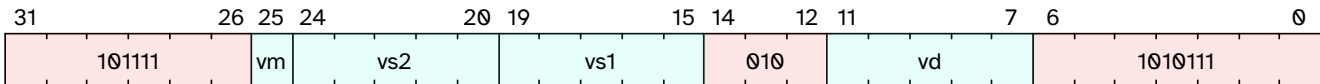
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.878. vnmsac.vx

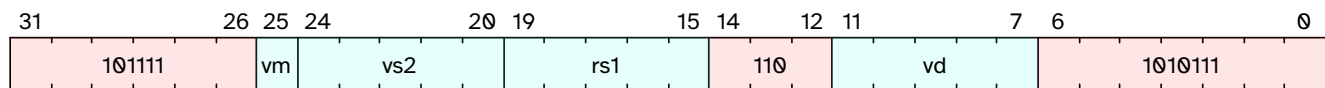
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.879. vnmsub.vv

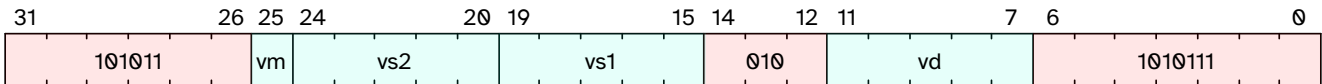
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.880. vnmsub.vx

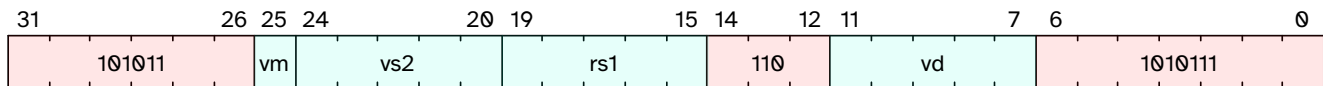
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.881. vnsra.wi

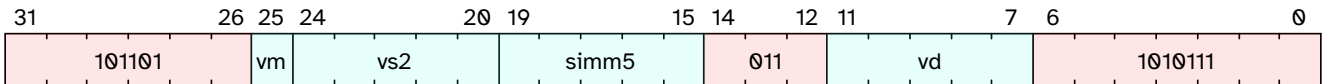
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.882. vnsra.wv

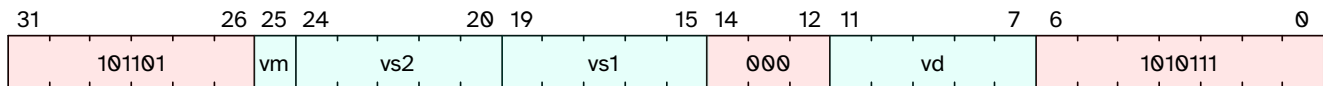
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.883. vnsra.wx

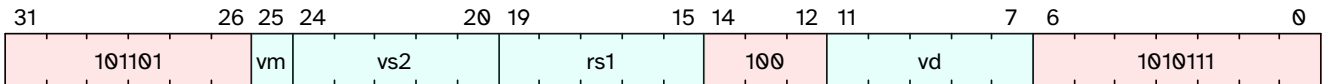
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.884. vnsrl.wi

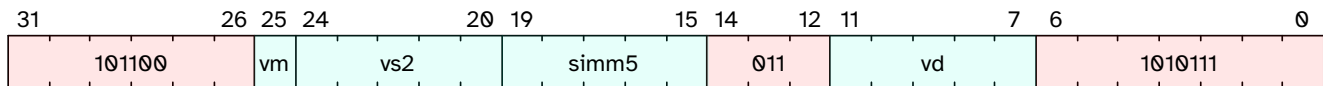
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.885. vnsrl.wv

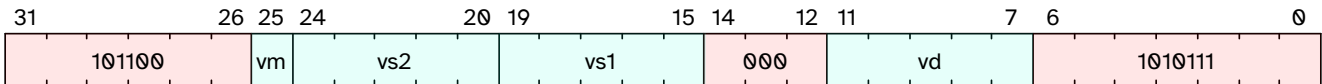
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.886. vnsrl.wx

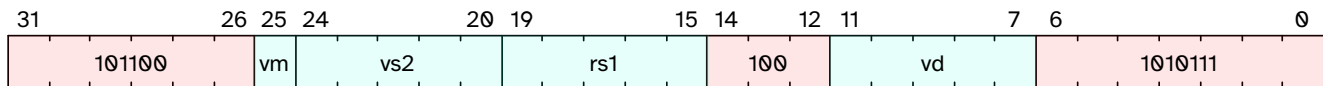
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.887. vor.vi

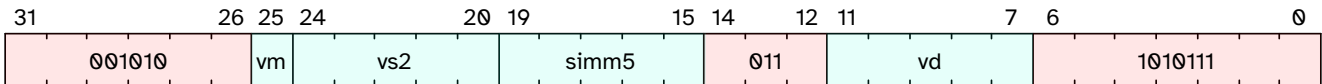
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.888. vor.vv

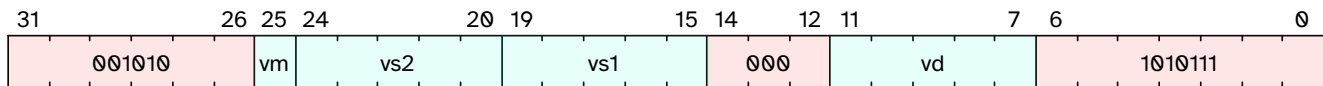
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.889. vor.vx

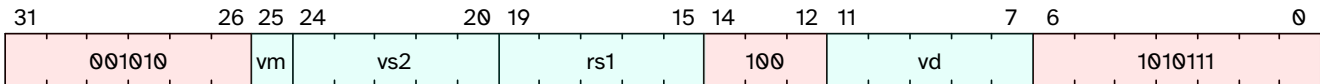
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.890. vredand.vs

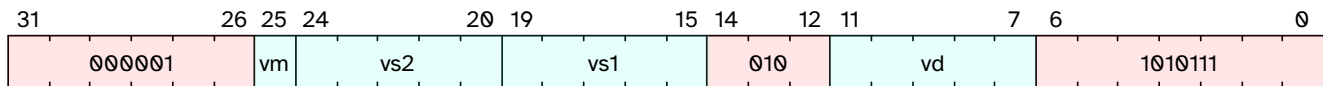
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.891. vredmax.vs

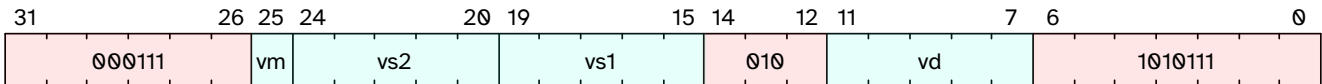
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.892. vredmaxu.vs

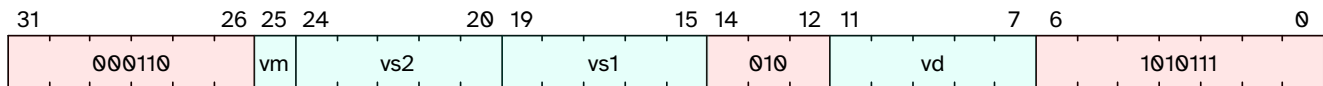
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.893. vredmin.vs

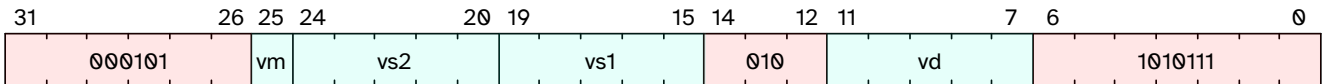
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.894. vredminu.vs

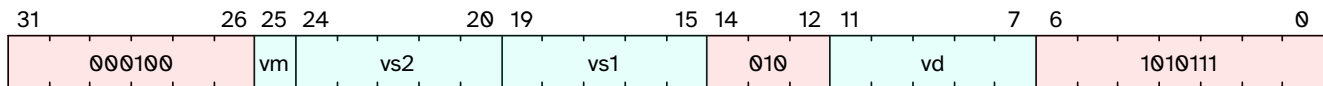
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.895. vredor.vs

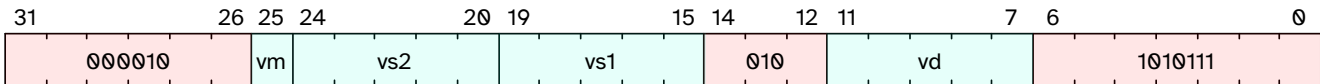
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.896. vredsum.vs

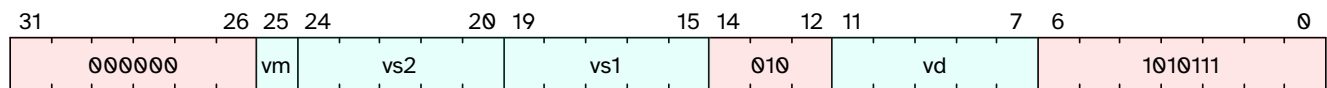
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.897. vredxor.vs

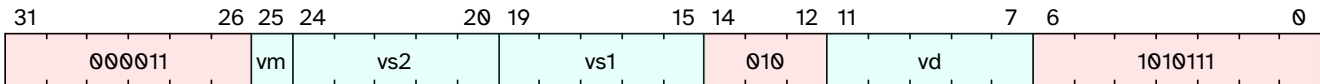
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.898. vrem.vv

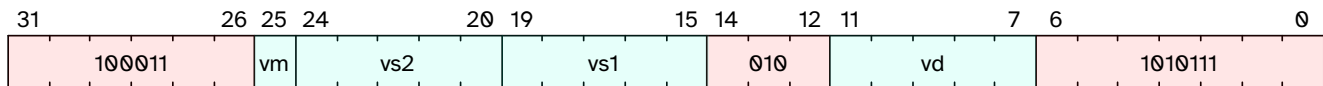
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.899. vrem.vx

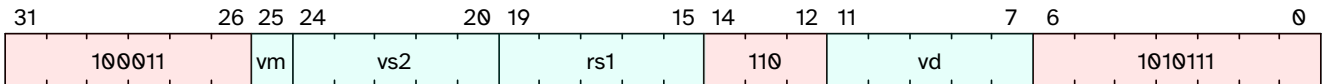
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.900. vremu.vv

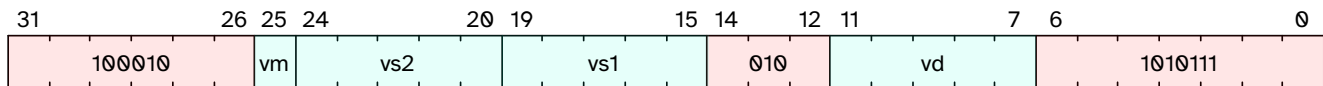
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.901. vremu.vx

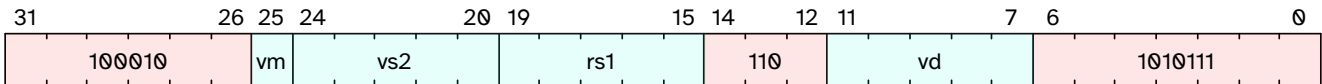
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.902. vrev8.v

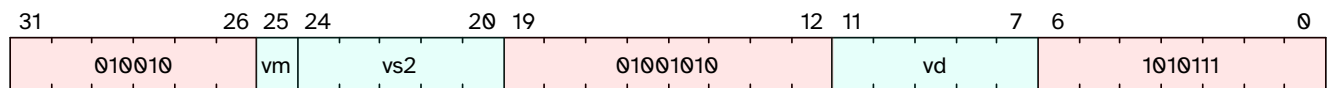
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.903. vrgather.vi

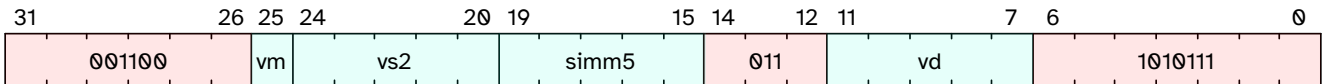
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.904. vrgather.vv

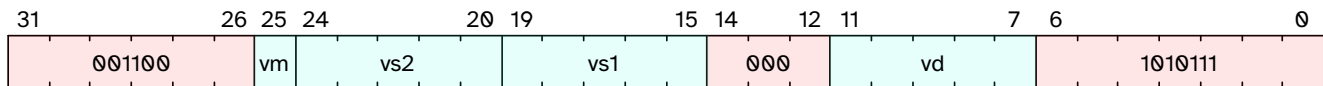
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.905. vrgather.vx

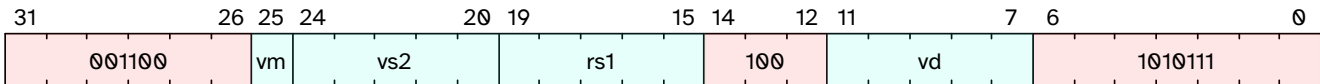
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.906. vrgatherei16.vv

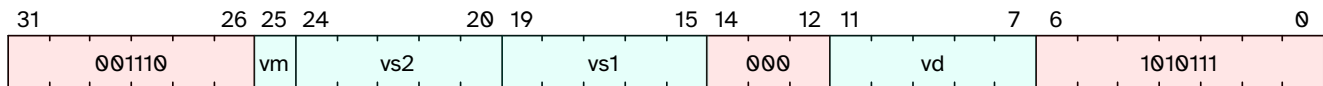
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.907. vrol.vv

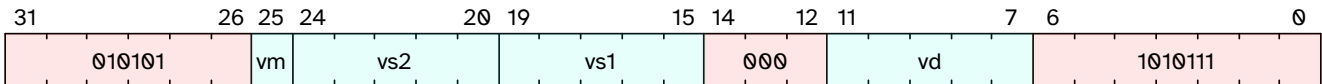
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.908. vrol.vx

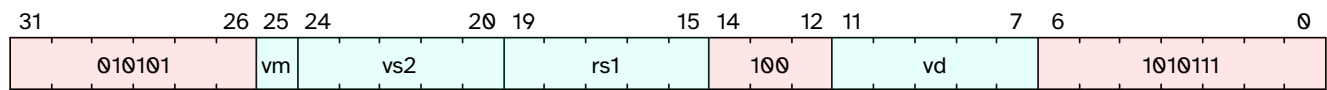
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.909. vrór.vi

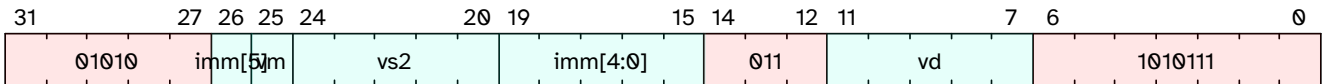
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.910. vrwr.vv

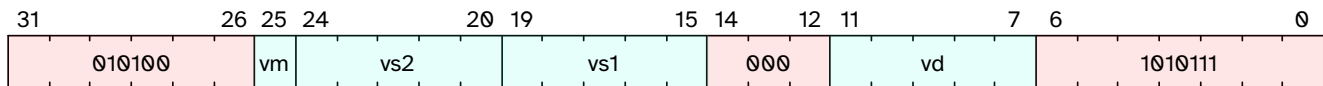
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.911. vrór.vx

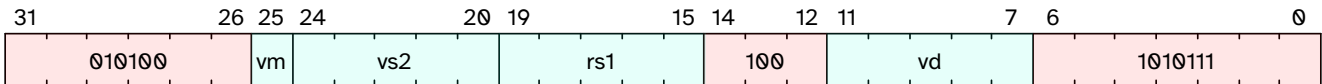
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvbb	~> 1.0.0
Zvkn	~> 1.0.0
Zvks	~> 1.0.0

41.912. vrsb.vi

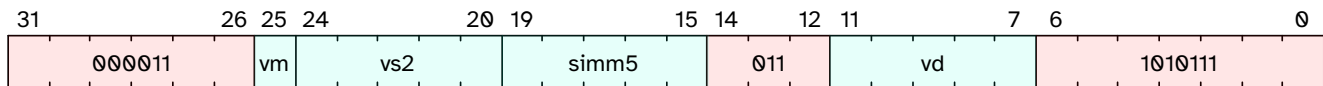
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.913. vrsb.vx

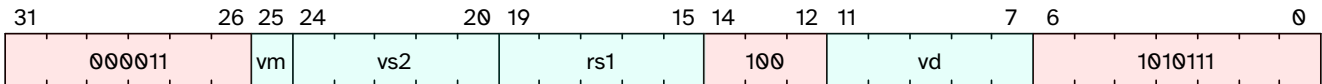
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.914. vs1r.v

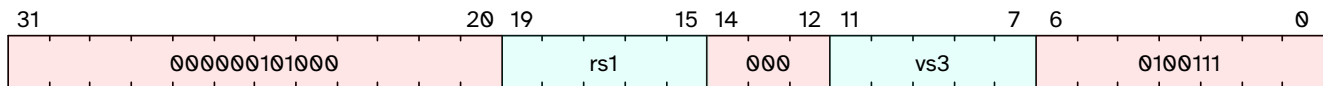
Synopsis

No synopsis available.

Mnemonic

rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.915. vs2r.v

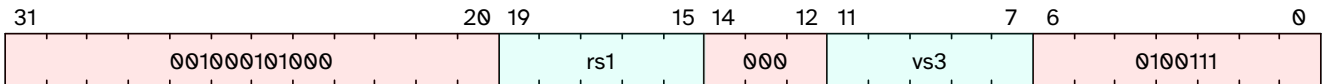
Synopsis

No synopsis available.

Mnemonic

rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.916. vs4r.v

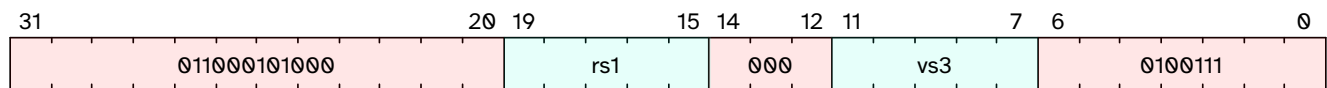
Synopsis

No synopsis available.

Mnemonic

rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.917. vs8r.v

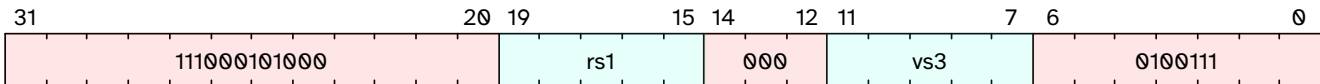
Synopsis

No synopsis available.

Mnemonic

rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.918. vsadd.vi

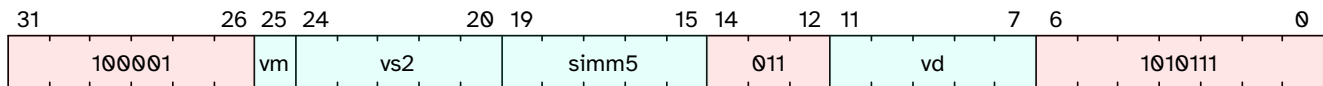
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.919. vsadd.vv

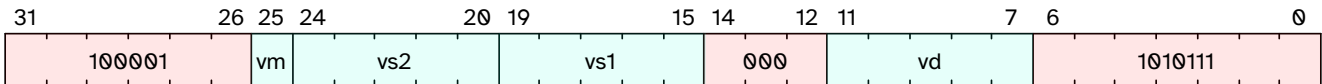
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.920. vsadd.vx

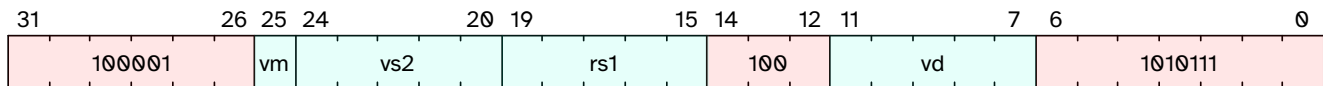
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.921. vsaddu.vi

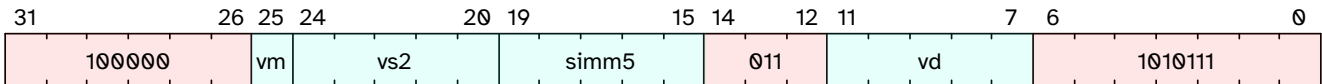
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.922. vsaddu.wv

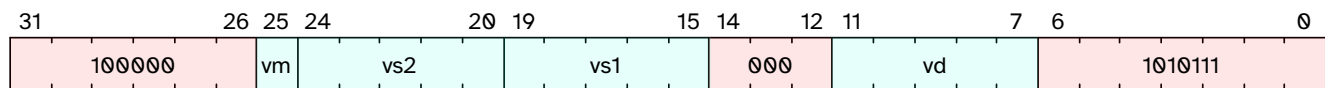
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.923. vsaddu.vx

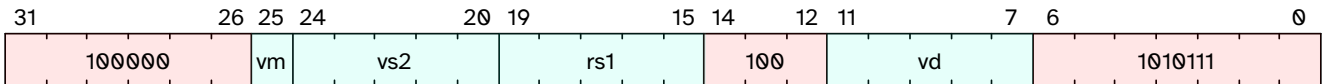
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.924. vsbc.vvm

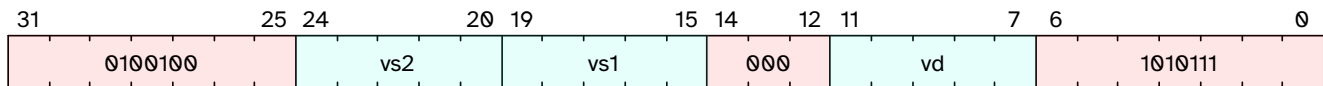
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.925. vsbc.vxm

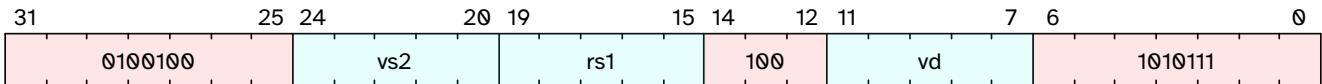
Synopsis

No synopsis available.

Mnemonic

vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.926. vse16.v

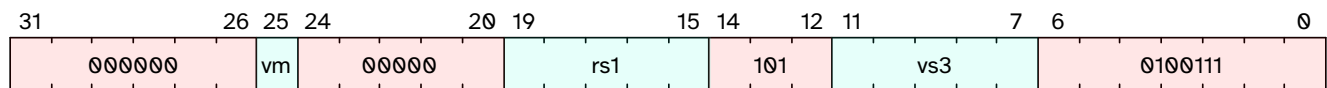
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.927. vse32.v

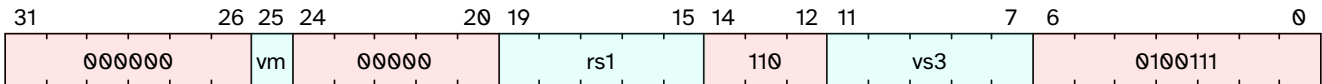
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.928. vse64.v

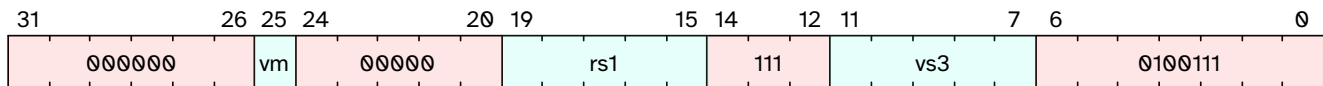
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.929. vse8.v

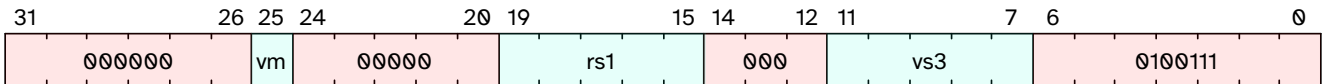
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.931. vsetvl

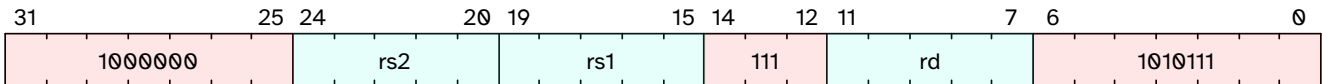
Synopsis

No synopsis available.

Mnemonic

rs2, rs1, rd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.932. vsetvli

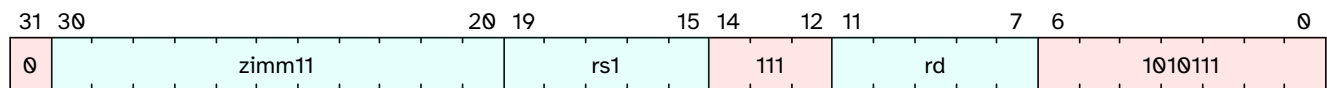
Synopsis

No synopsis available.

Mnemonic

rs1, rd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.933. vsex.vf2

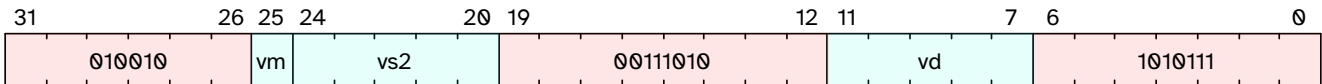
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.934. vsex.vf4

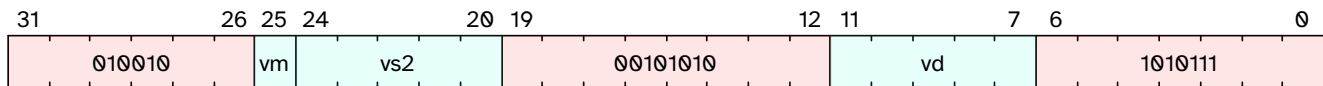
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.935. vsex.vf8

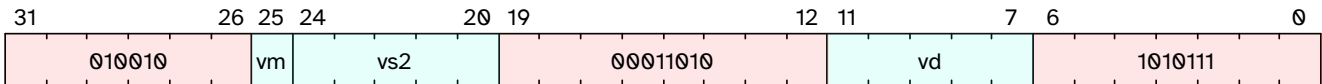
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.936. vsha2ch.vv

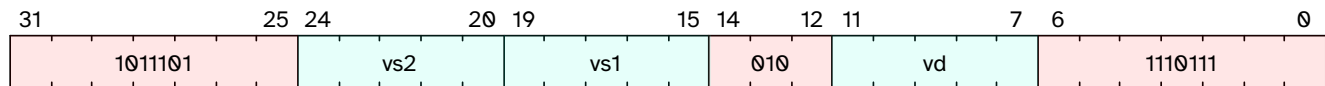
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvknha	~> 1.0.0
Zvknhb	~> 1.0.0

41.937. vsha2cl.vv

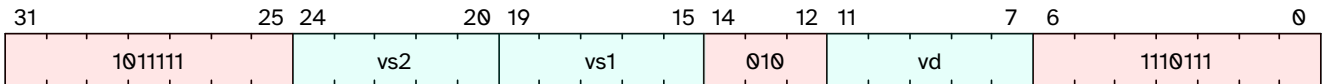
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvknha	~> 1.0.0
Zvknhb	~> 1.0.0

41.938. vsha2ms.vv

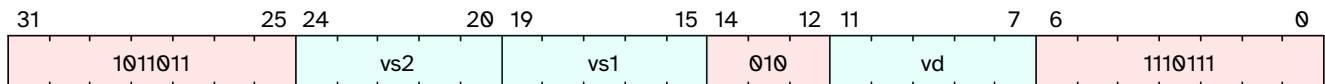
Synopsis

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvkn	~> 1.0.0
Zvknha	~> 1.0.0
Zvknhb	~> 1.0.0

41.939. vslide1down.vx

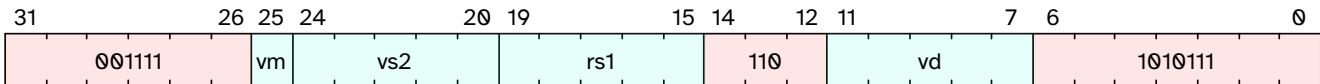
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.940. vslide1up.vx

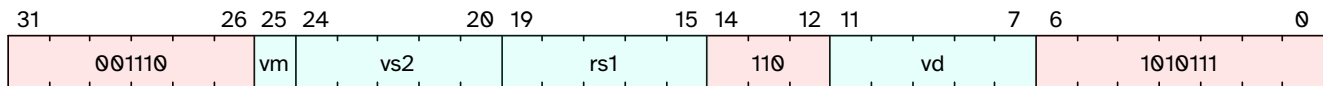
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.941. vslidedown.vi

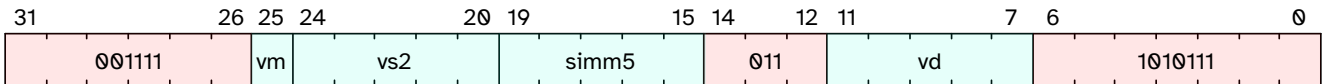
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.942. vslidedown.vx

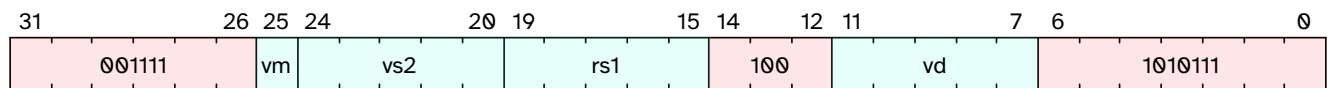
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.943. vslideup.vi

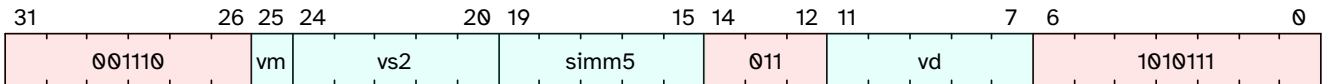
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.944. vslideup.vx

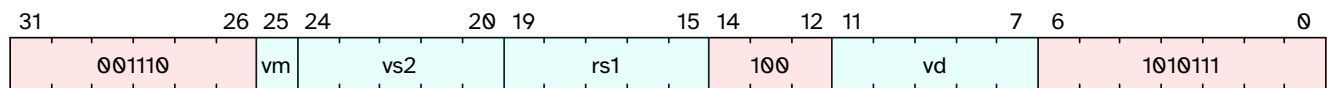
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.945. vsll.vi

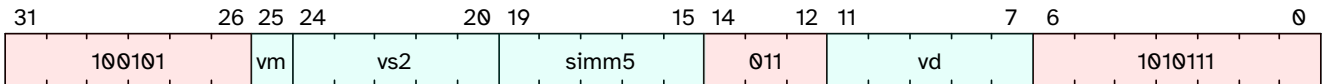
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.946. vsll.vv

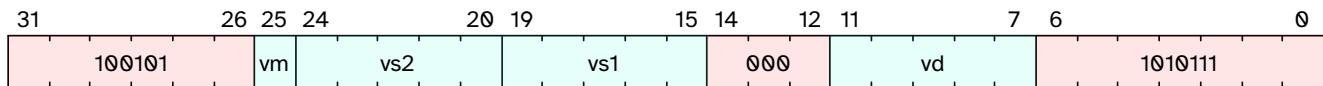
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.947. vsll.vx

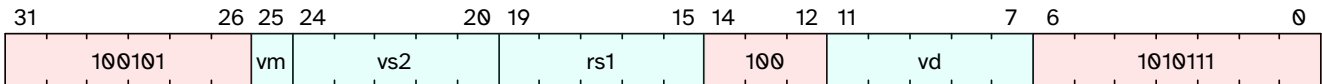
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.948. vsm.v

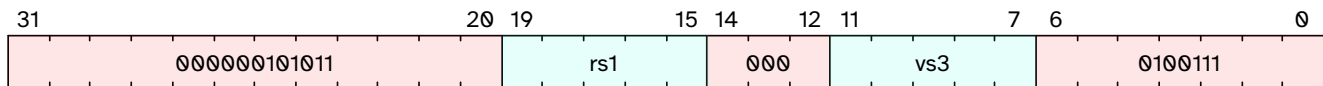
Synopsis

No synopsis available.

Mnemonic

rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.949. vsm3c.vi

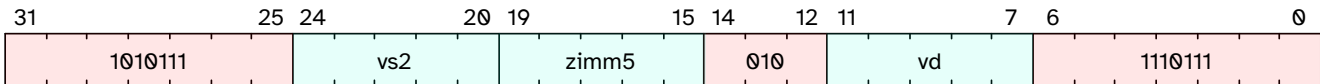
Synopsis

No synopsis available.

Mnemonic

vs2, vd, imm

Encoding



Description

No description available.

Included in

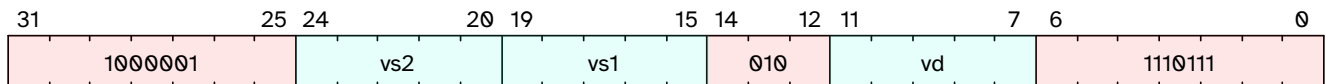
Extension	Version
Zvks	~> 1.0.0
Zvksh	~> 1.0.0

41.950. vsm3me.vv**Synopsis**

No synopsis available.

Mnemonic

vs2, vs1, vd

Encoding**Description**

No description available.

Included in

Extension	Version
Zvks	~> 1.0.0
Zvksh	~> 1.0.0

41.951. vsm4k.vi

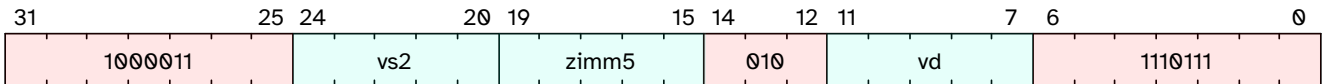
Synopsis

No synopsis available.

Mnemonic

vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
Zvks	~> 1.0.0
Zvkse	~> 1.0.0

41.952. vsm4r.vs

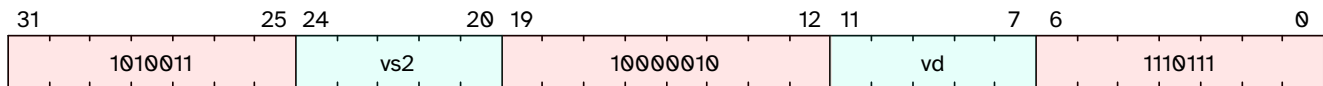
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvks	~> 1.0.0
Zvkse	~> 1.0.0

41.953. vsm4r.vv

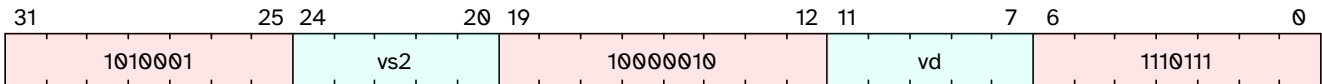
Synopsis

No synopsis available.

Mnemonic

vs2, vd

Encoding



Description

No description available.

Included in

Extension	Version
Zvks	~> 1.0.0
Zvkse	~> 1.0.0

41.954. vsmul.vv

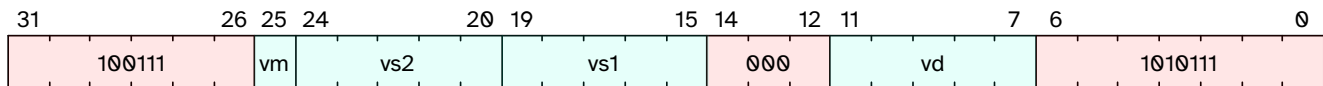
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.955. vsmul.vx

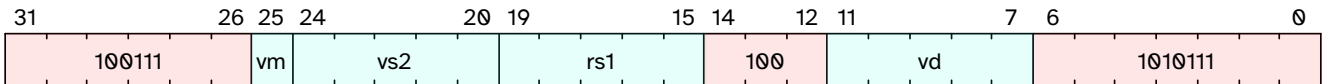
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.956. vsoxei16.v

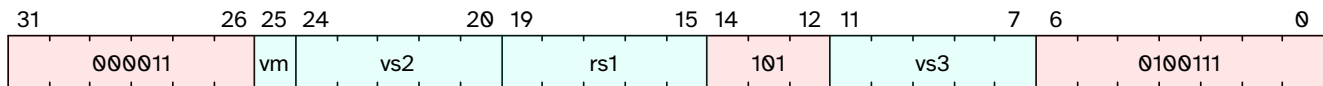
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.957. vsoxei32.v

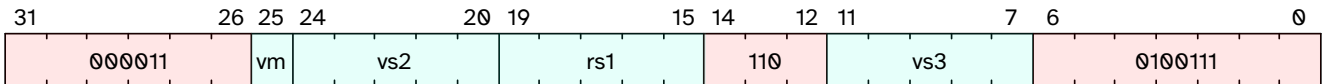
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.958. vsoxei64.v

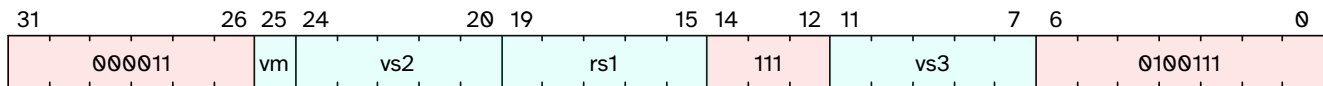
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.959. vsoxei8.v

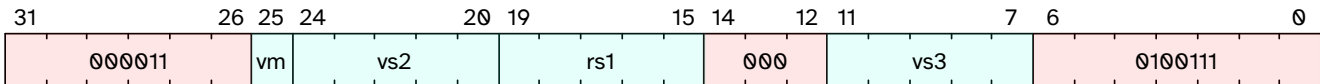
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.960. vsoxseg2ei16.v

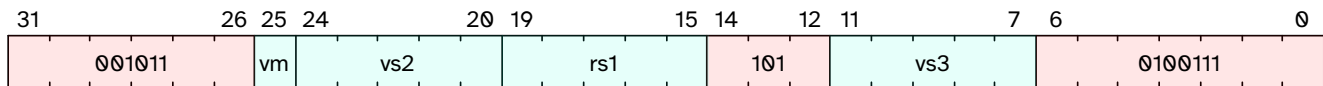
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.961. vsoxseg2ei32.v

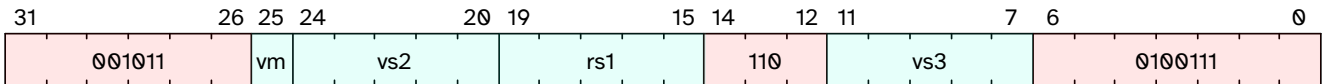
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.962. vsoxseg2ei64.v

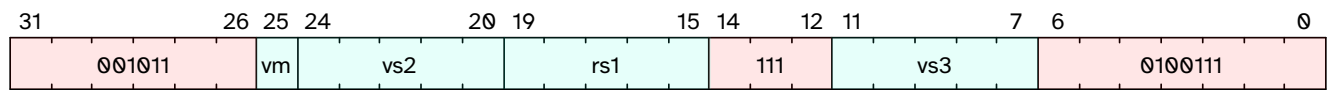
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.963. vsoxseg2ei8.v

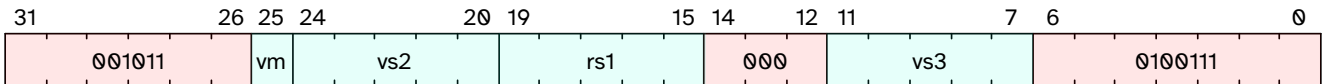
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.964. vsoxseg3ei16.v

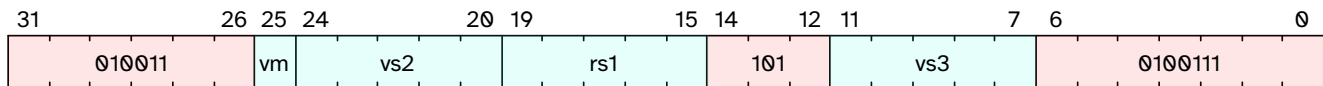
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.965. vsoxseg3ei32.v

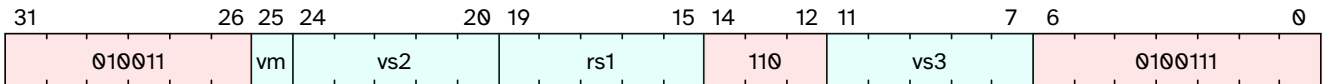
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.966. vsoxseg3ei64.v

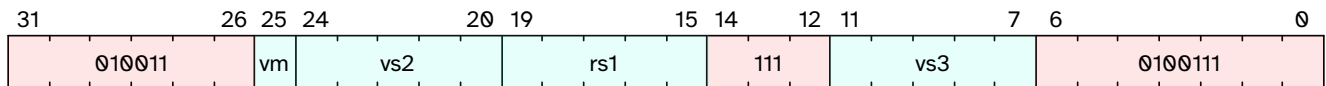
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.967. vsoxseg3ei8.v

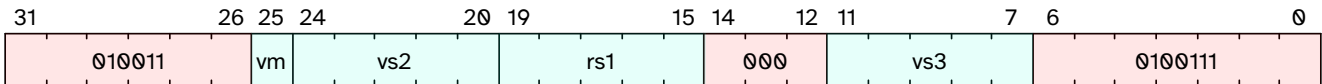
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.968. vsoxseg4ei16.v

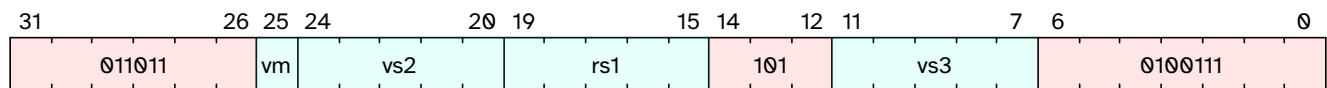
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.969. vsoxseg4ei32.v

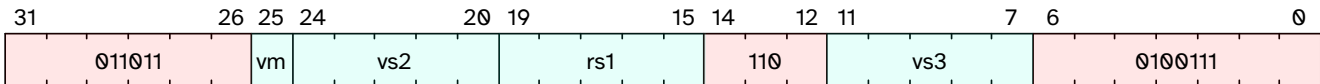
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.970. vsoxseg4ei64.v

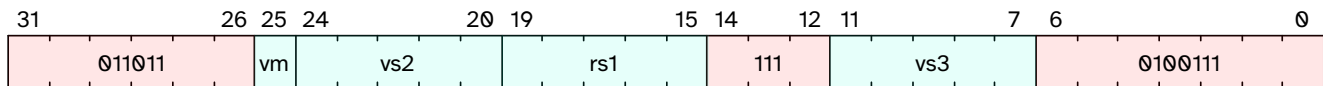
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.971. vsoxseg4ei8.v

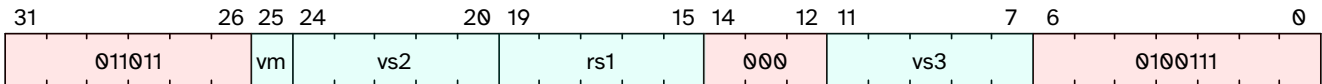
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.972. vsoxseg5ei16.v

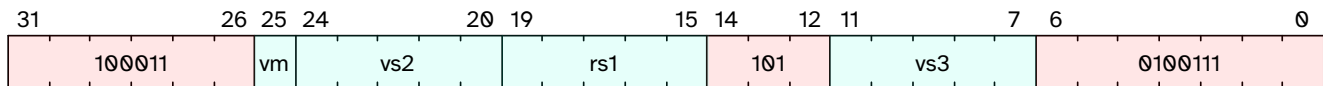
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.973. vsoxseg5ei32.v

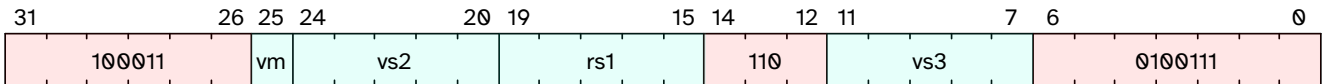
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.974. vsoxseg5ei64.v

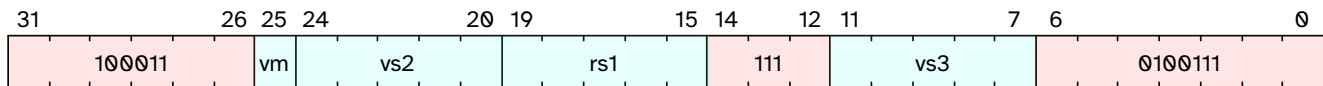
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.975. vsoxseg5ei8.v

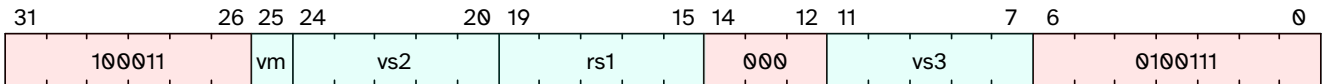
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.976. vsoxseg6ei16.v

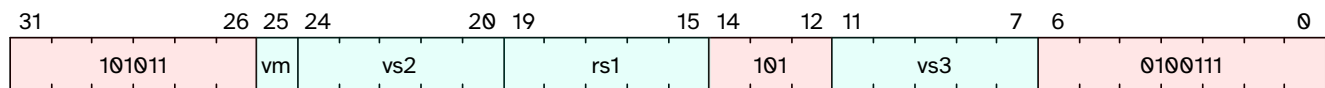
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.977. vsoxseg6ei32.v

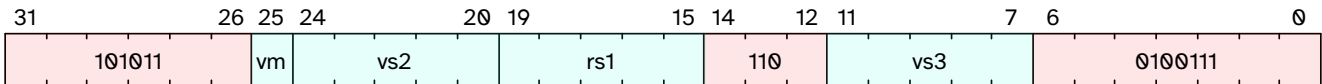
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.978. vsoxseg6ei64.v

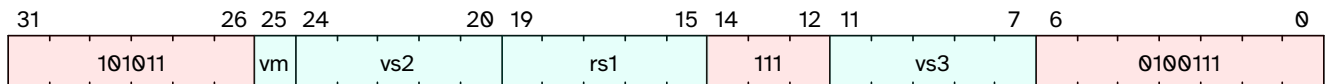
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.979. vsoxseg6ei8.v

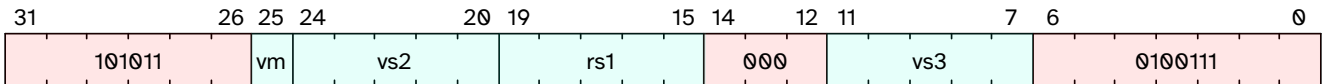
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.980. vsoxseg7ei16.v

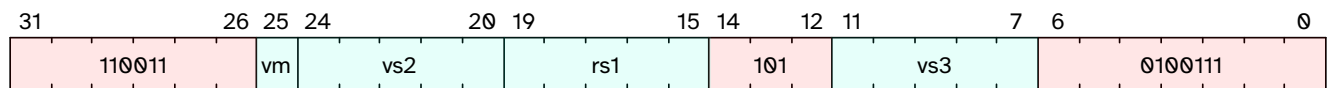
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.981. vsoxseg7ei32.v

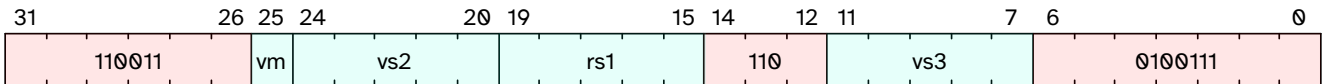
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.982. vsoxseg7ei64.v

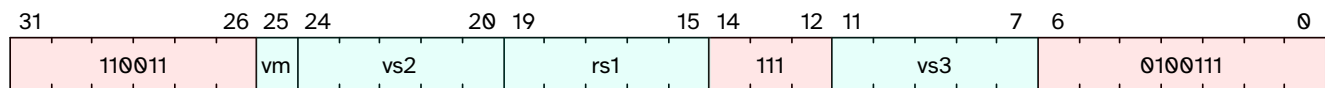
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.983. vsoxseg7ei8.v

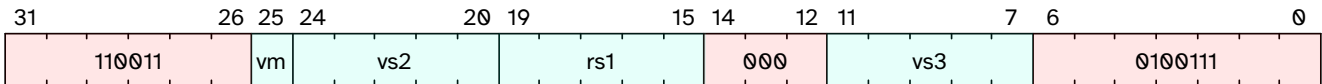
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.984. vsoxseg8ei16.v

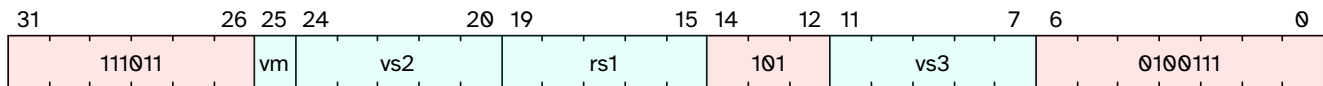
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.985. vsoxseg8ei32.v

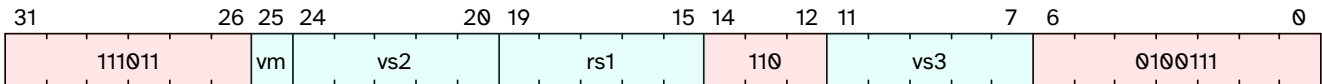
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.986. vsoxseg8ei64.v

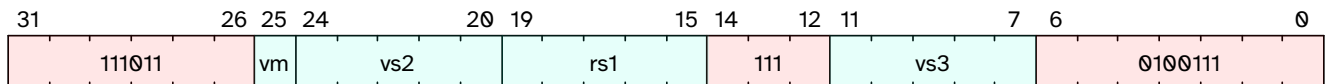
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.987. vsoxseg8ei8.v

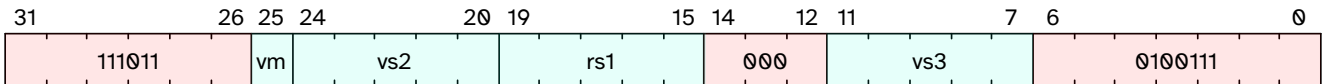
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.988. vsra.vi

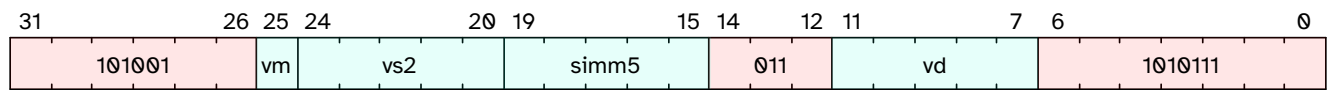
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.989. vsra.wv

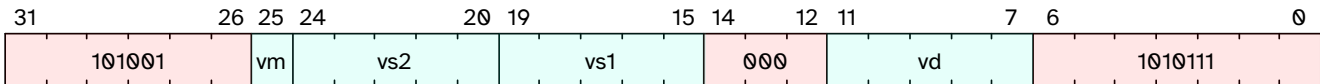
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.990. vsra.vx

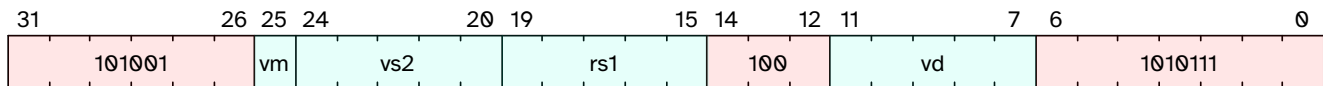
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.991. vsrl.vi

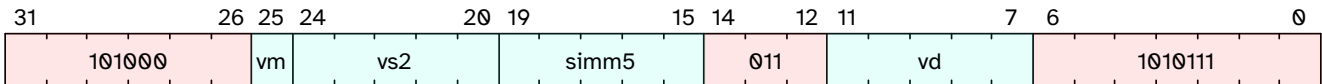
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.992. vsrl.vv

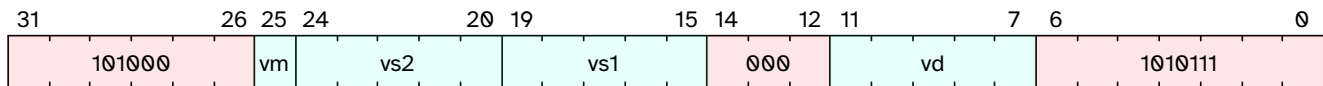
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.993. vsrl.vx

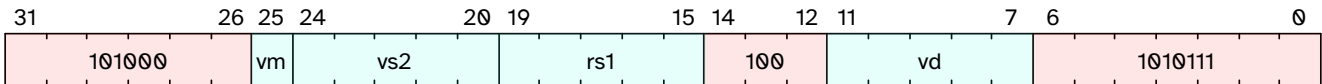
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.994. vsse16.v

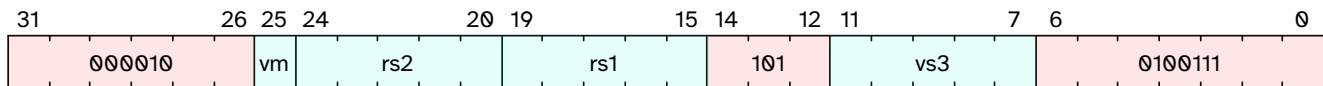
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.995. vsse32.v

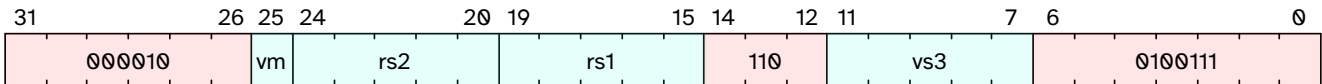
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.996. vsse64.v

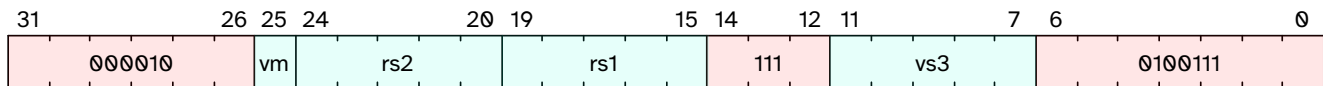
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.997. vsse8.v

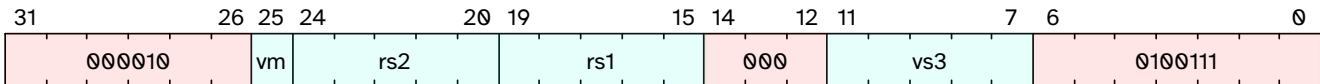
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.998. vsseg2e16.v

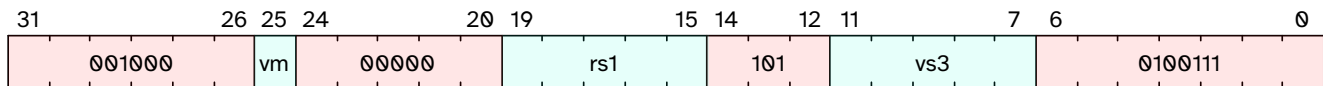
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.999. vsseg2e32.v

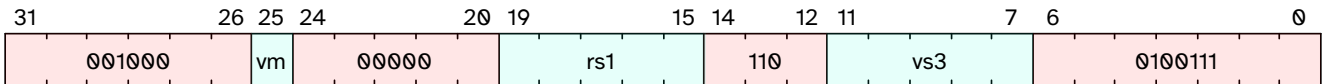
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1000. vsseg2e64.v

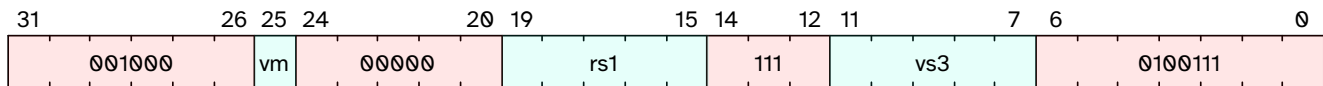
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1001. vsseg2e8.v

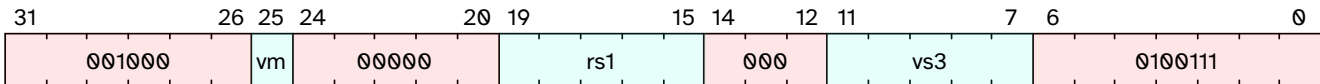
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1002. vsseg3e16.v

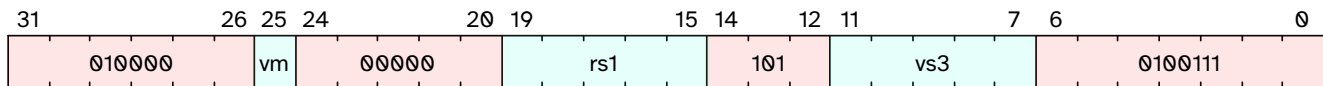
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1003. vsseg3e32.v

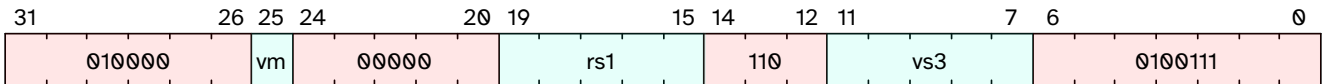
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1004. vsseg3e64.v

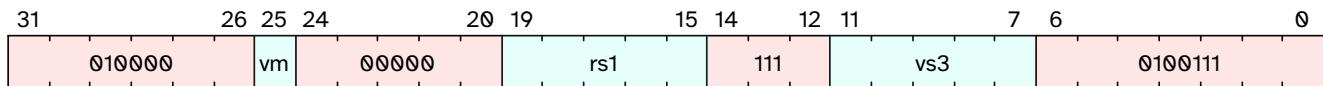
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1005. vsseg3e8.v

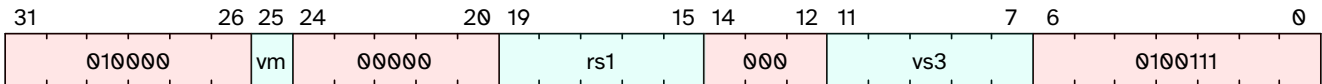
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1006. vsseg4e16.v

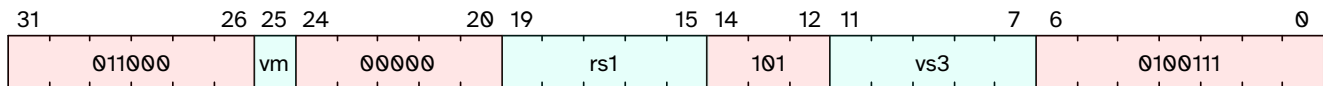
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1007. vsseg4e32.v

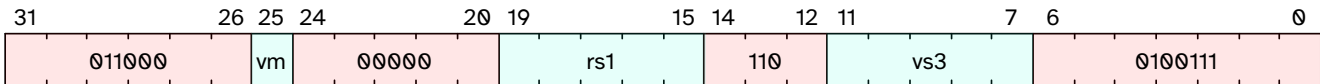
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1008. vsseg4e64.v

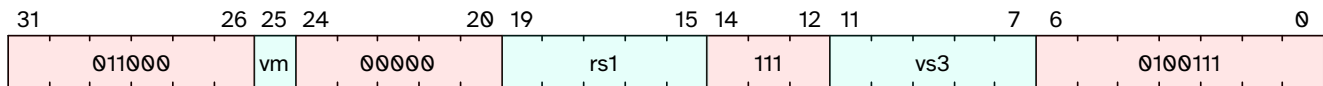
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1009. vsseg4e8.v

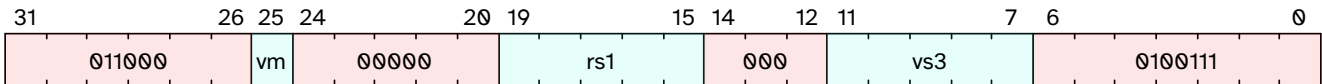
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1010. vsseg5e16.v

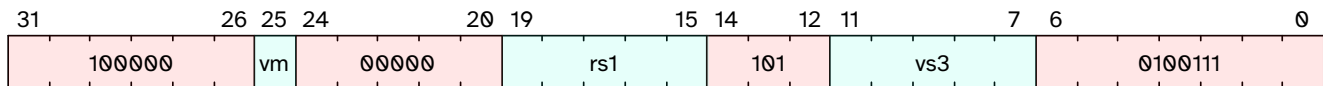
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1011. vsseg5e32.v

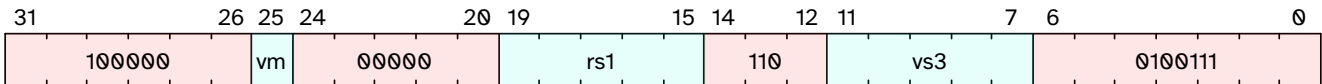
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1012. vsseg5e64.v

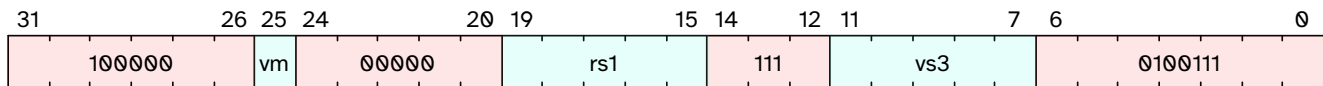
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1013. vsseg5e8.v

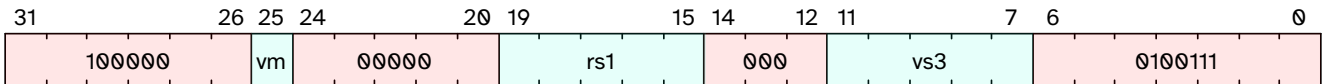
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1014. vsseg6e16.v

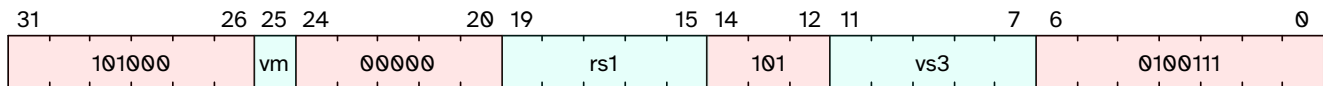
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1015. vsseg6e32.v

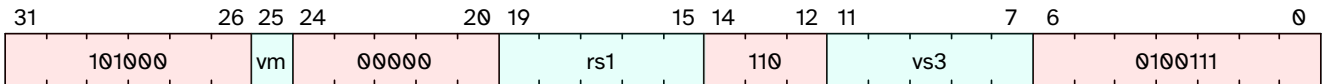
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1016. vsseg6e64.v

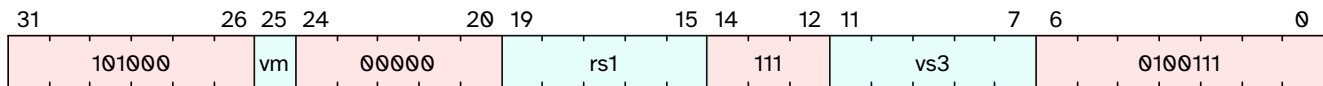
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1017. vsseg6e8.v

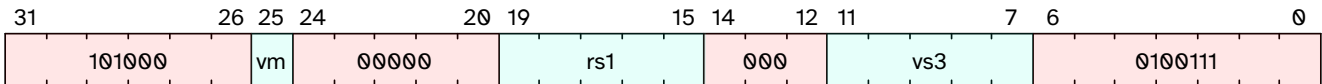
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1018. vsseg7e16.v

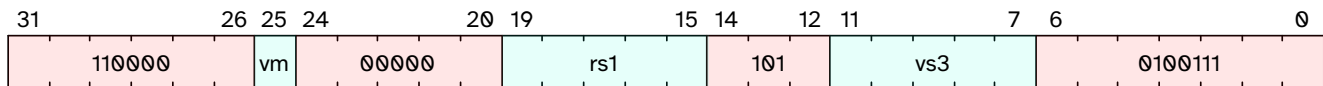
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1019. vsseg7e32.v

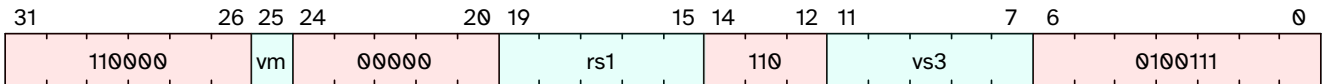
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1020. vsseg7e64.v

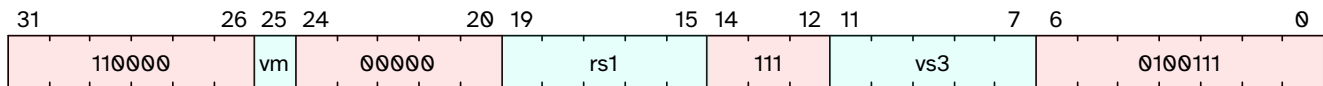
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1021. vsseg7e8.v

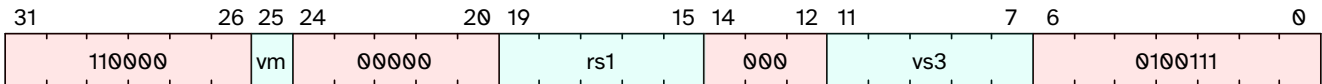
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1022. vsseg8e16.v

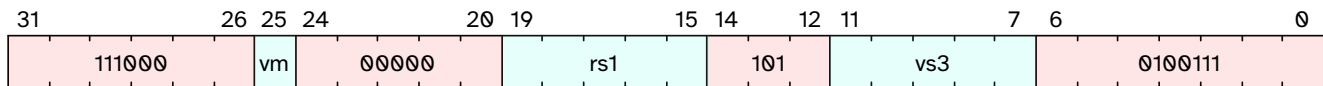
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1023. vsseg8e32.v

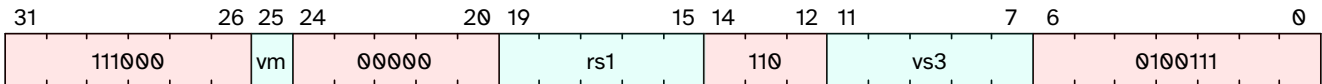
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1024. vsseg8e64.v

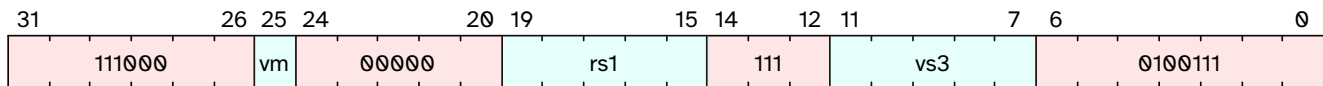
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1025. vsseg8e8.v

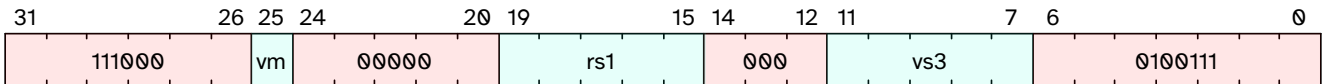
Synopsis

No synopsis available.

Mnemonic

vm, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1026. vssra.vi

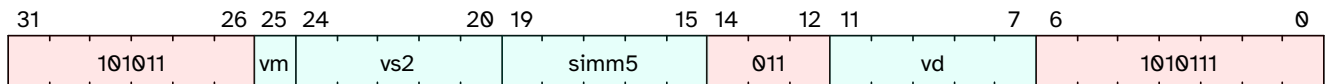
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1027. vssra.vv

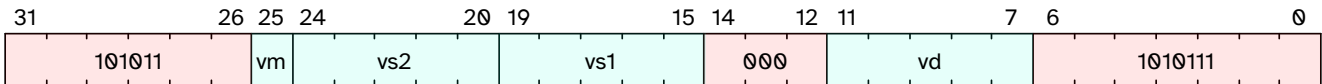
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1028. vssra.vx

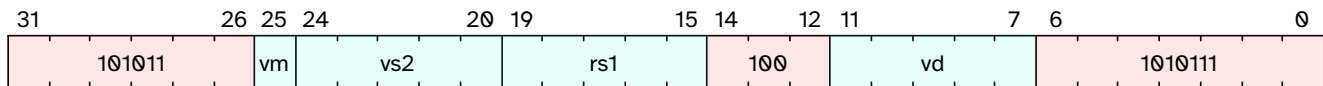
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1029. vssrl.vi

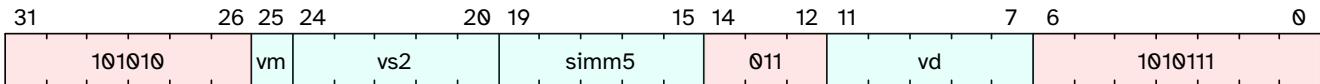
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vd, imm

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1030. vssrl.vv

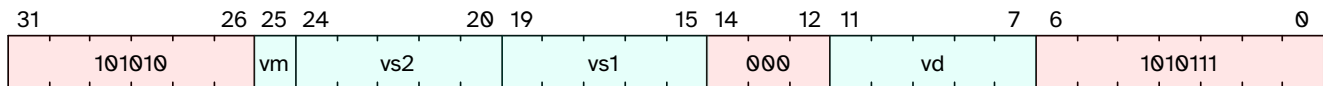
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1031. vssrl.vx

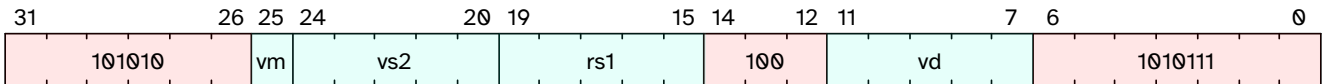
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1032. vssseg2e16.v

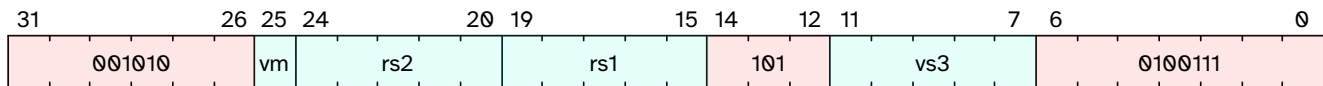
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1033. vssseg2e32.v

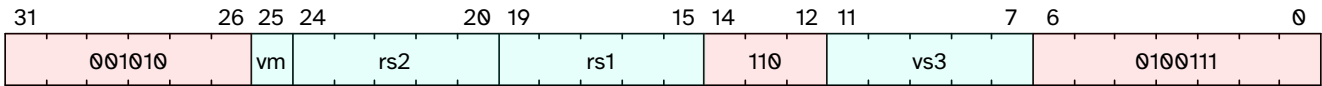
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1034. vssseg2e64.v

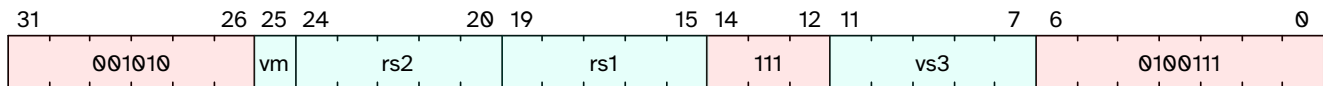
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1035. vssseg2e8.v

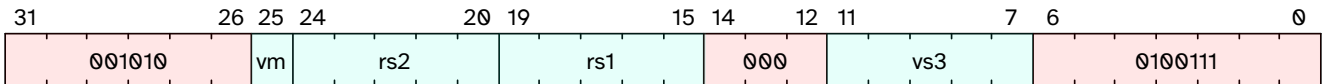
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1036. vssseg3e16.v

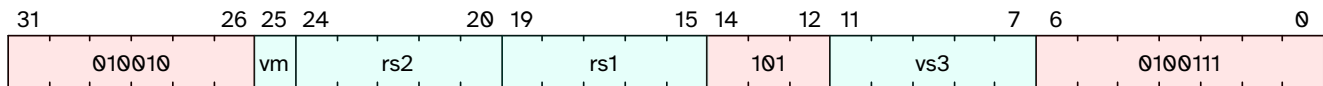
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1037. vssseg3e32.v

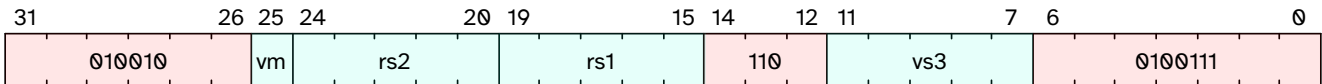
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1038. vssseg3e64.v

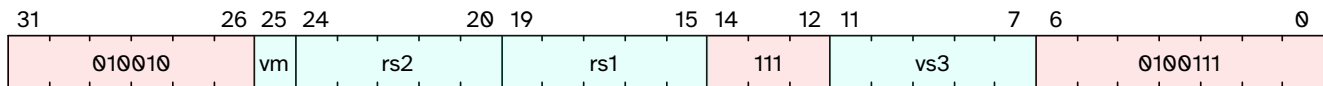
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1039. vssseg3e8.v

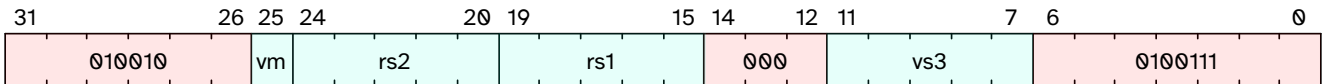
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1040. vssseg4e16.v

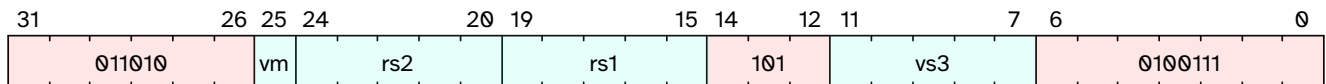
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1041. vssseg4e32.v

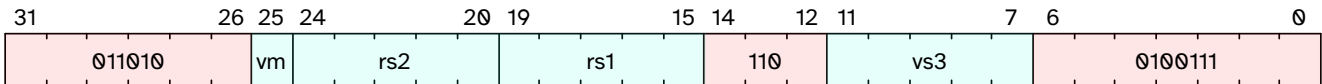
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1042. vssseg4e64.v

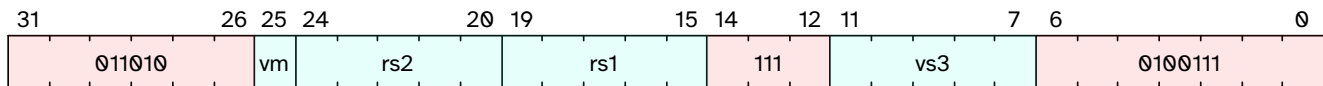
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1043. vssseg4e8.v

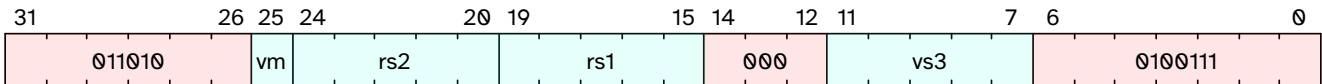
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1044. vssseg5e16.v

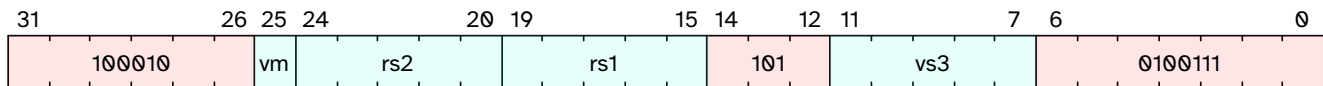
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1045. vssseg5e32.v

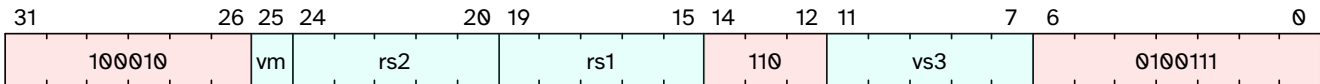
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1046. vssseg5e64.v

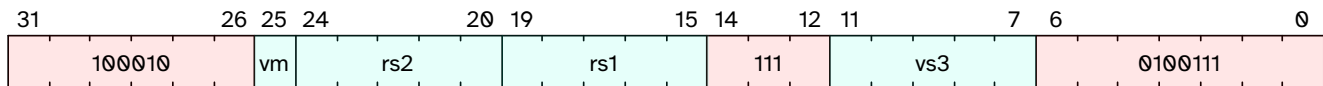
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1047. vssseg5e8.v

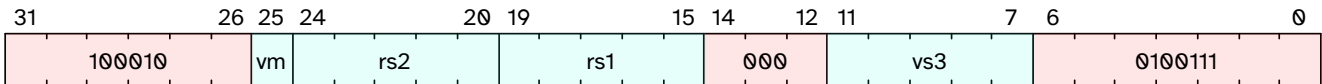
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1048. vssseg6e16.v

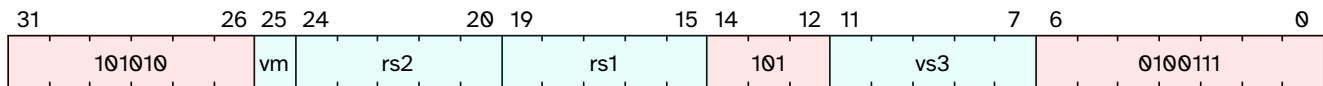
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1049. vssseg6e32.v

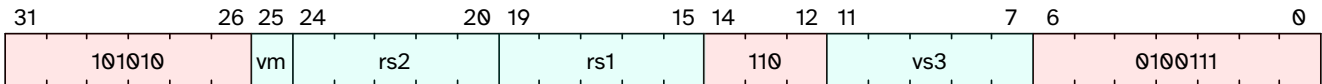
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1050. vssseg6e64.v

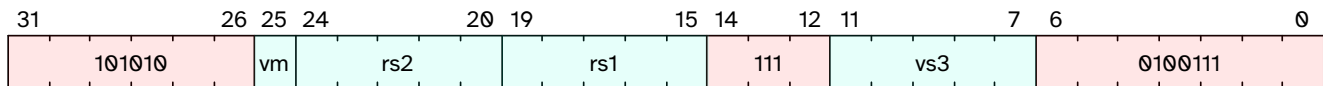
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1051. vssseg6e8.v

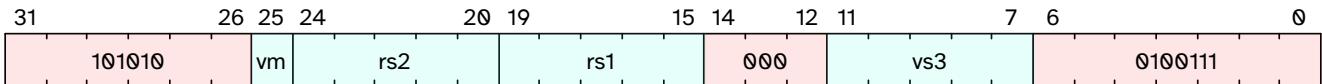
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1052. vssseg7e16.v

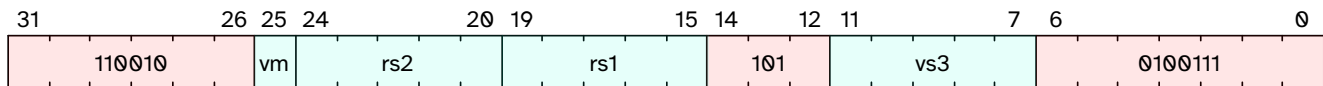
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1053. vssseg7e32.v

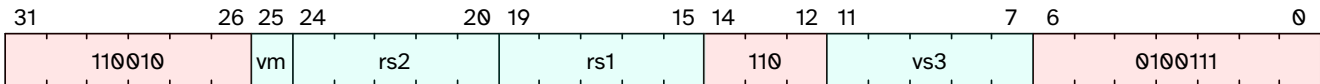
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1054. vssseg7e64.v

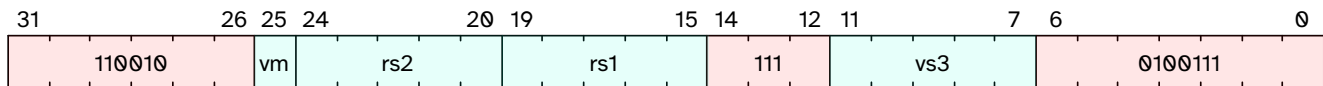
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1055. vssseg7e8.v

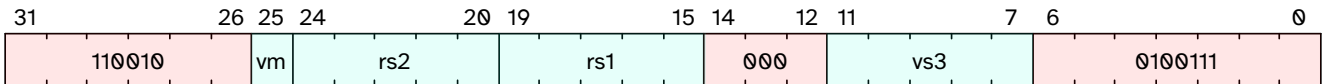
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1056. vssseg8e16.v

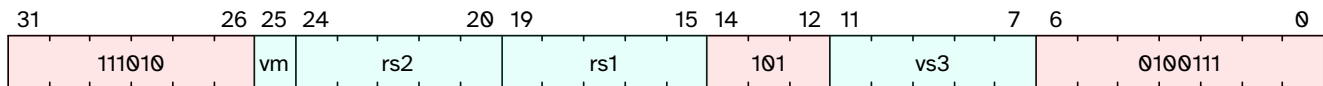
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1057. vssseg8e32.v

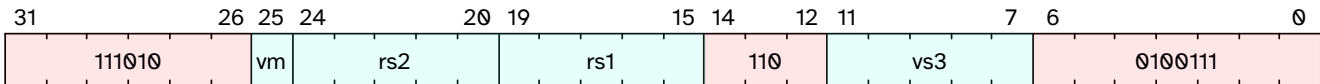
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1058. vssseg8e64.v

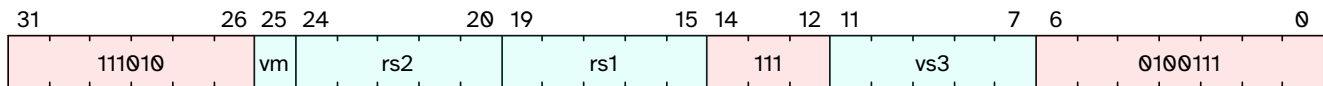
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1059. vssseg8e8.v

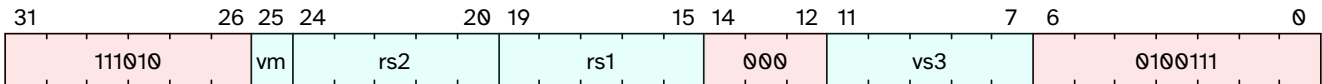
Synopsis

No synopsis available.

Mnemonic

vm, rs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1060. vssub.vv

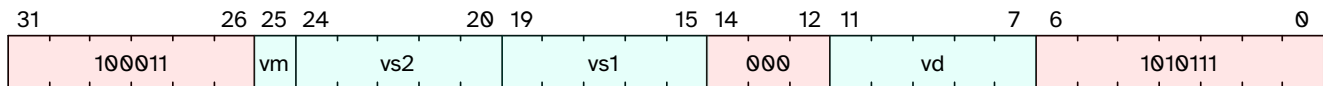
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1061. vssub.vx

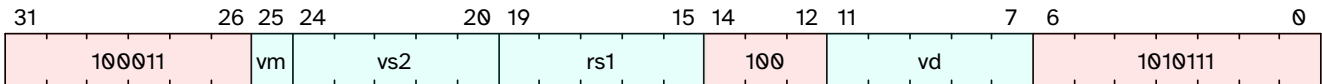
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1062. vssubu.vv

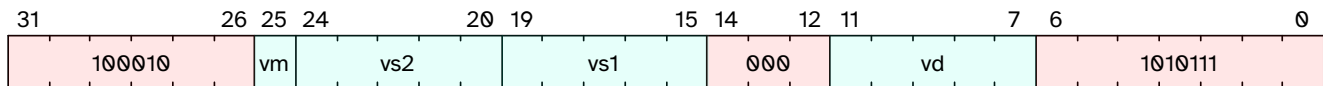
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1063. vssubu.vx

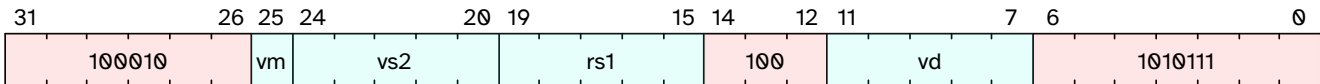
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1064. vsub.vv

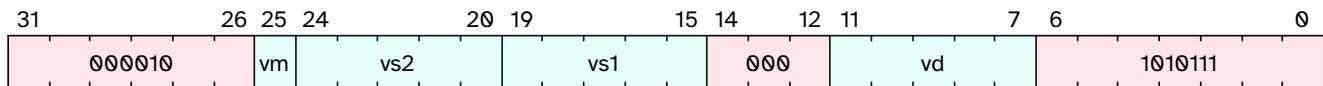
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1065. vsub.vx

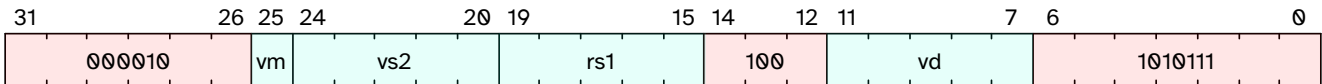
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1066. vsuxei16.v

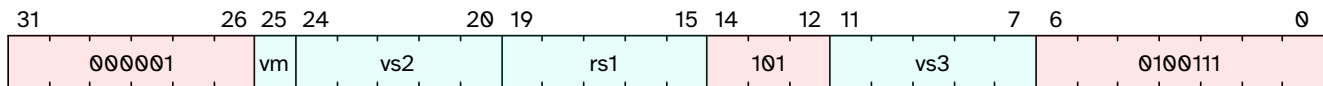
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1067. vsuxei32.v

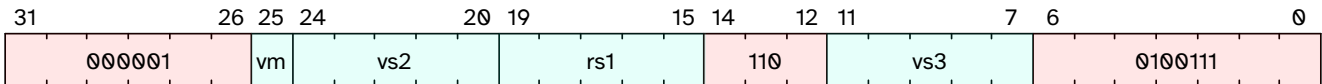
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1068. vsuxei64.v

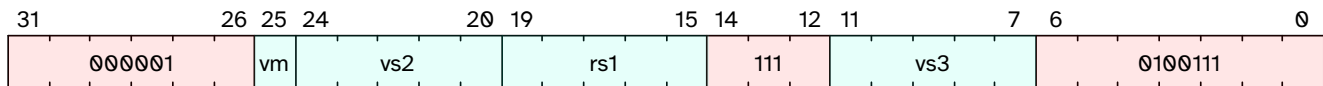
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1069. vsuxei8.v

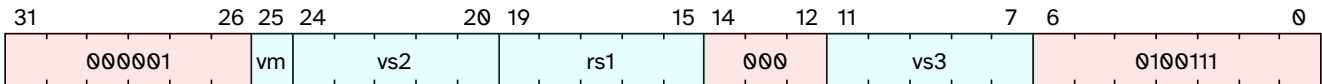
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1070. vsuxseg2ei16.v

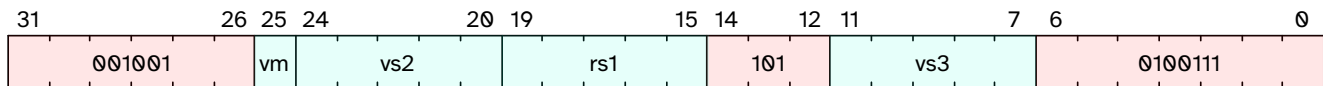
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1071. vsuxseg2ei32.v

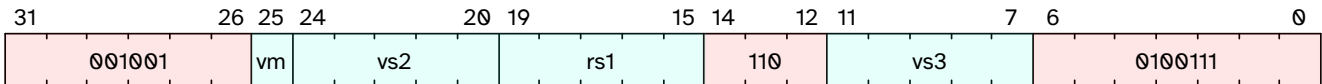
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1072. vsuxseg2ei64.v

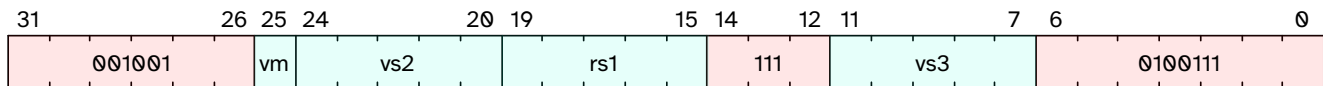
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1073. vsuxseg2ei8.v

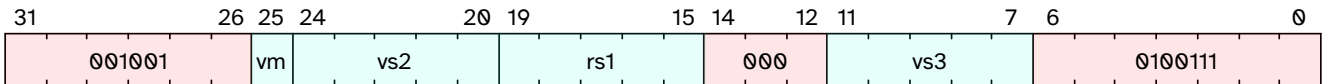
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1074. vsuxseg3ei16.v

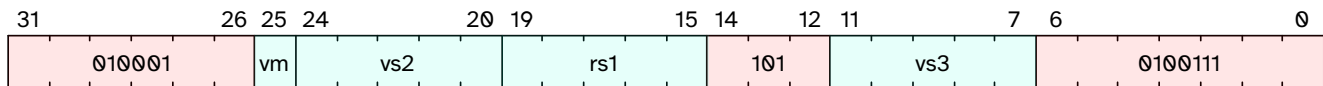
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1075. vsuxseg3ei32.v

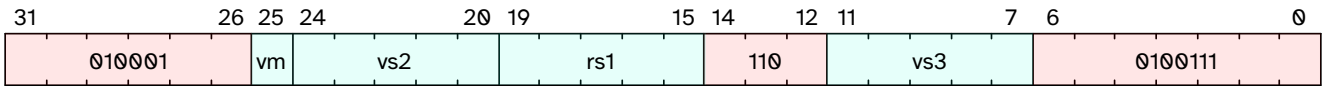
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1076. vsuxseg3ei64.v

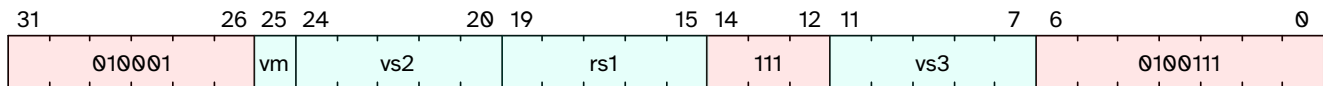
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1077. vsuxseg3ei8.v

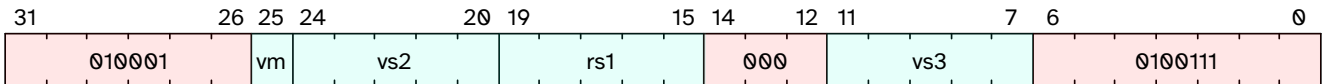
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1078. vsuxseg4ei16.v

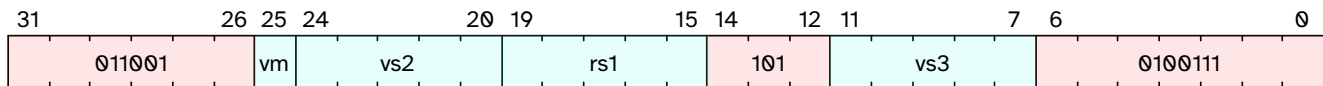
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1079. vsuxseg4ei32.v

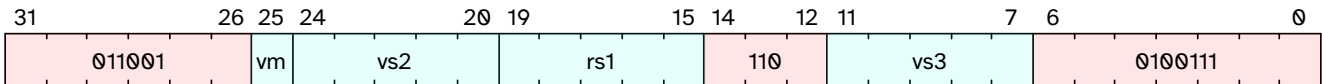
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1080. vsuxseg4ei64.v

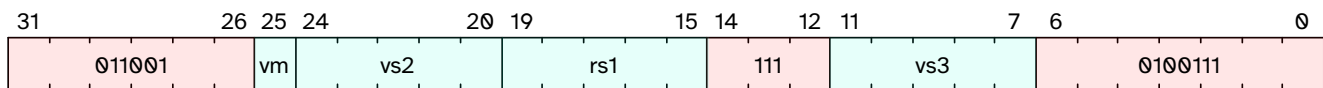
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1081. vsuxseg4ei8.v

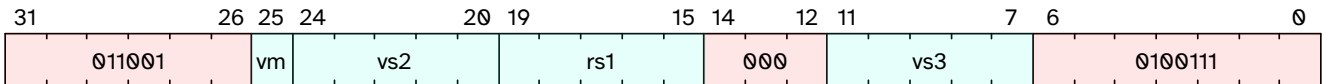
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1082. vsuxseg5ei16.v

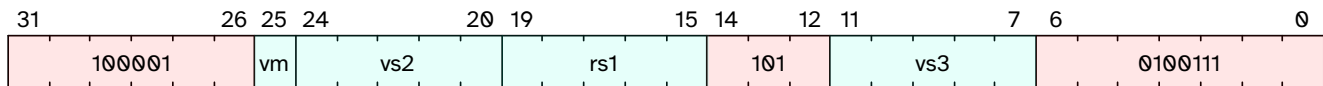
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1083. vsuxseg5ei32.v

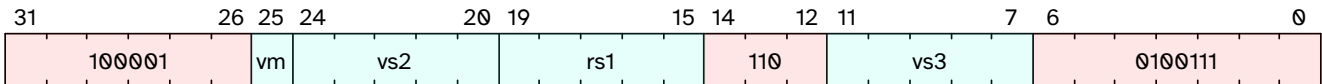
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1084. vsuxseg5ei64.v

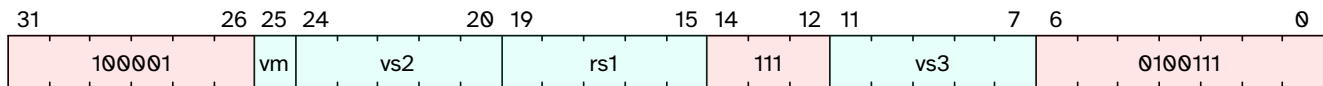
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1085. vsuxseg5ei8.v

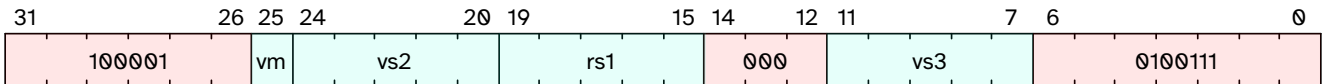
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1086. vsuxseg6ei16.v

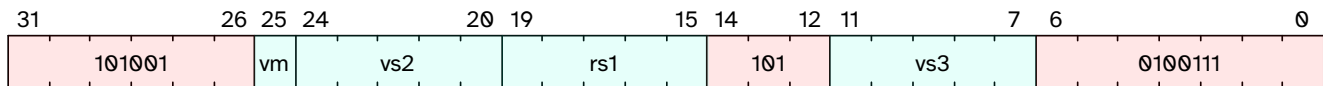
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1087. vsuxseg6ei32.v

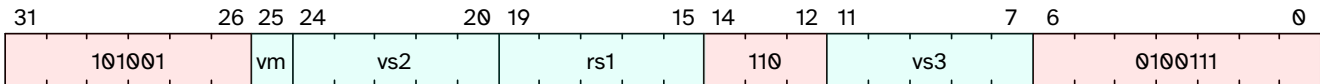
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1088. vsuxseg6ei64.v

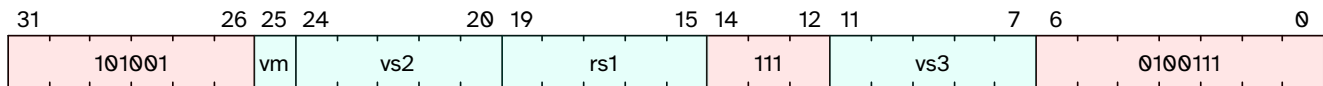
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1089. vsuxseg6ei8.v

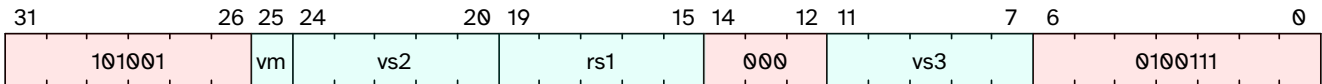
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1090. vsuxseg7ei16.v

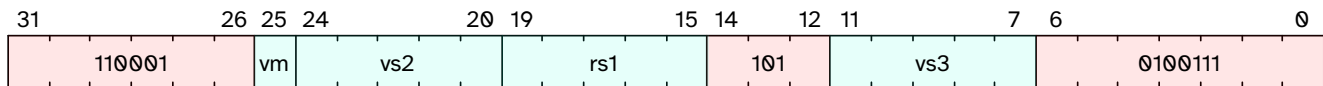
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1091. vsuxseg7ei32.v

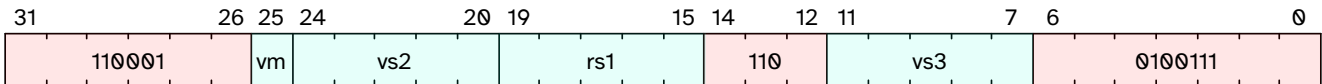
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1092. vsuxseg7ei64.v

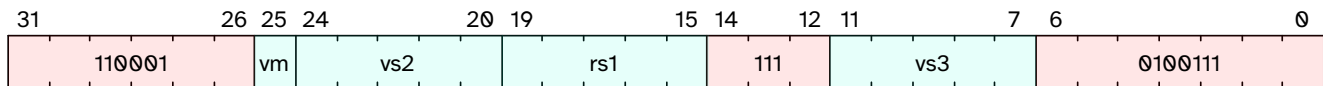
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1093. vsuxseg7ei8.v

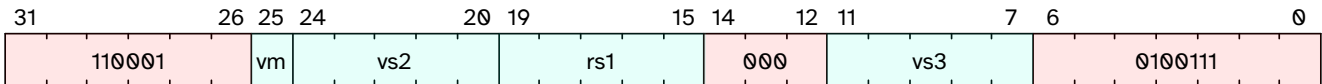
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1094. vsuxseg8ei16.v

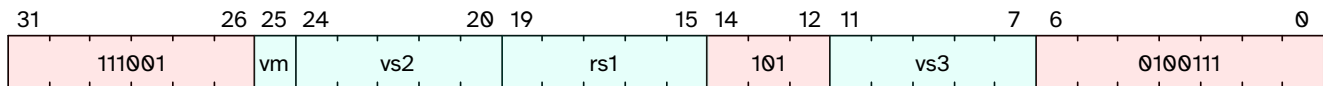
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1095. vsuxseg8ei32.v

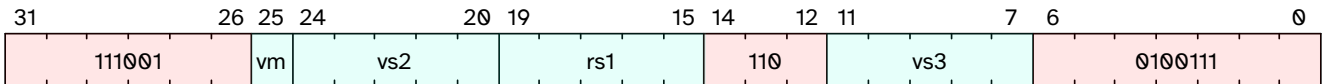
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1096. vsuxseg8ei64.v

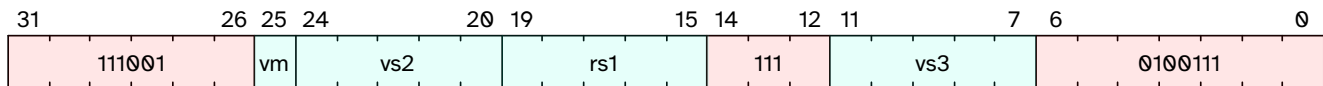
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

DRAFT

41.1097. vsuxseg8ei8.v

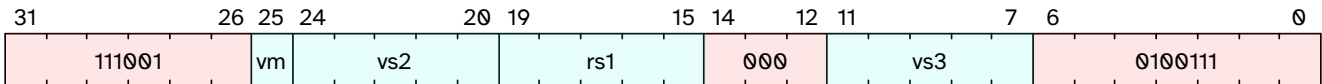
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vs3

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1098. vwadd.vv

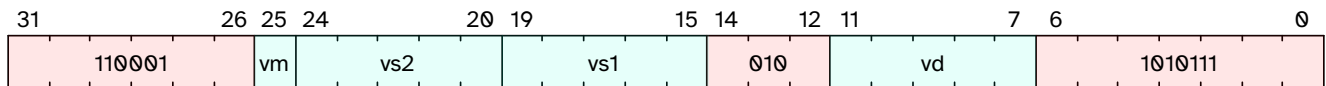
Synopsis

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding



Description

No description available.

Included in

Extension	Version
V	~> 1.0.0

41.1099. vwadd.vx

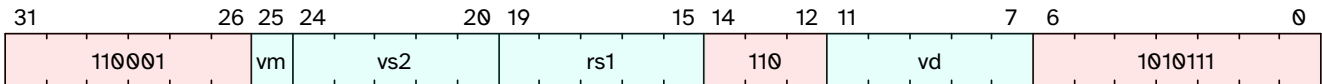
Synopsis

No synopsis available.

Mnemonic

vm, vs2, rs1, vd

Encoding



Description

No description available.

Included in

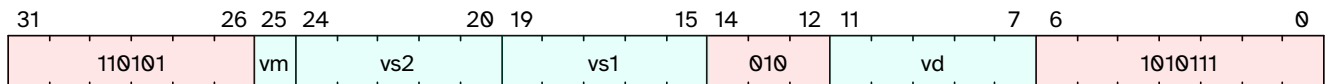
Extension	Version
V	~> 1.0.0

41.1100. vwadd.wv**Synopsis**

No synopsis available.

Mnemonic

vm, vs2, vs1, vd

Encoding**Description**

No description available.

Included in

Extension	Version
V	~> 1.0.0